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Notes on

STATISTICAL INFERENCE

M.Sc Statistics

PART II

PAPER I



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Statistical Inference

As the population is not completely known, we can obtain information about a population by use of samples drawn from that population. Such problems dealt with statistical inference. Statistical inference is the art of drawing conclusions or inferences about the population from the limited information contained in the sample.

Two Main aspects of Statistical Inference

- i) Estimation
- ii) Testing of Hypothesis

Estimation: A procedure of making judgement about population parameter is referred to as statistical estimation or simply estimation, which is further divided into Point estimation and Interval Estimation.

Point Estimation

The object of point estimation is to obtain a single number from the sample which will represent the unknown value of the parameter. Population parameter are estimated from corresponding sample statistic.

A statistic used to estimate a parameter is called a point estimator or simply an estimator. The actual numerical value which we obtain for an estimator in a given problem is called an estimate.

Let X_i be the r.v denoting the height of a student. A random sample of n students is drawn and the mean height of the sample $\bar{X}$ is computed. The statistic $\bar{X}$ is a point estimator of the population mean.

If for a specific sample of 100 students, the sample height is 66", we say 66" is an estimate of the mean μ .

we use the symbol μ for the population mean which is an unknown.

We draw a sample of size n , $x_1, x_2, \dots, x_n$ from the population, then we formulate a function of the sample obs $x_1, x_2, \dots, x_n$. We are looking for a statistic such that its value computed from the sample would represent the population parameter as closely as possible. The estimator of θ is denoted by $\hat{\theta}$. $\hat{\theta}$ is a random variable and has its own sampling distribution.

Properties (or Criteria) of a Good Estimator.

i) Unbiasedness ii) Consistency iii) Efficiency iv) Sufficiency

Unbiasedness:- An estimator is said to be unbiased if expected value or mean value is equal to corresponding population parameter.

Suppose a statistic $\hat{\theta}$ is used to estimate the pop parameter θ . When $\hat{\theta}$ is an unbiased estimator of θ .

If $E(\hat{\theta}) = \theta$ otherwise it is a biased estimator.

The bias of an estimator is defined as

$$E(\hat{\theta}) - \theta = \text{Bias}$$

In sampling theory it is known as selection bias.

In inference it is known as estimation bias.

$\bar{X}$ is an unbiased estimator of μ in $N(\mu, \sigma^2)$

$\bar{X}$ is an unbiased estimator of pop proportion in Bernoulli pop.

" " " " λ in Poisson Dist.

$$E(S^2) \neq \sigma^2$$

$$E(S^2) = \sigma^2$$

$$S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

a biased estimator of σ^2

unbiased estimator

biased estimator

$$E(\hat{p}) = \hat{p}$$

$$E(\hat{p}) = p$$

$$E(\bar{X}) = \mu$$

$$E(S^2) \neq \sigma^2$$

$$E(\hat{\sigma}^2) = \sigma^2$$

It is also possible to have more than one unbiased estimator for an unknown parameter.

Both sample mean and sample median are unbiased estimates of the population mean μ in normal population.

Example 1: Show that the sample mean $\bar{X}$ is an unbiased estimator of the population mean μ , i.e. $E(\bar{X}) = \mu$.

Sol: let $X_1, X_2, \dots, X_n$ be a random sample of size n from a population with mean μ .

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$E(\bar{X}) = \frac{1}{n} E[X_1 + X_2 + \dots + X_n]$$

$$E(\bar{X}) = \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)]$$

In a random sample, the $X_1, X_2, \dots, X_n$ are independently and identically distributed with population mean μ .

$$E(\bar{X}) = \frac{1}{n} [\mu + \mu + \dots + \mu]$$

$$E(\bar{X}) = \frac{n\mu}{n}$$

$$E(\bar{X}) = \mu \quad \text{Hence proved.}$$

Example 2: Let $X_1, X_2, \dots, X_n$ be a random sample from the pop with variance σ^2 . Then show that the sample variance $S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is a biased estimator of σ^2 .

i.e. $E(S^2) \neq \sigma^2$

and $\hat{S}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of σ^2

$$\text{i.e. } E(\hat{S}^2) = \sigma^2$$

Proof: Sample Variance is

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\begin{aligned}\sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \mu + \mu - \bar{X})^2 \\ &= \sum_{i=1}^n \left[(X_i - \mu) - (\bar{X} - \mu) \right]^2 \\ &= \sum_{i=1}^n \left[(X_i - \mu)^2 + (\bar{X} - \mu)^2 - 2(\bar{X} - \mu)(X_i - \mu) \right] \\ &= \sum_{i=1}^n (X_i - \mu)^2 + n(\bar{X} - \mu)^2 - 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \mu)\end{aligned}$$

$$\begin{aligned}\sum_{i=1}^n (X_i - \mu) &= n(\bar{X} - \mu) \\ &= \sum_{i=1}^n (X_i - \mu)^2 + n(\bar{X} - \mu)^2 - 2n(\bar{X} - \mu)^2 \\ \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2\end{aligned}$$

Applying Expectation

$$E \left[\sum_{i=1}^n (X_i - \bar{X})^2 \right] = E \left[\sum_{i=1}^n (X_i - \mu)^2 \right] - nE(\bar{X} - \mu)^2$$

In a random sample, the random variables $X_1, X_2, \dots, X_n$ are independently and identically distributed with mean μ and variance σ^2 .

$$\begin{aligned}E \left[\sum_{i=1}^n (X_i - \mu)^2 \right] &= E \left[(X_1 - \mu)^2 + (X_2 - \mu)^2 + \dots + (X_n - \mu)^2 \right] \\ &= E(X_1 - \mu)^2 + E(X_2 - \mu)^2 + \dots + E(X_n - \mu)^2 \\ &= \sigma^2 + \sigma^2 + \dots + \sigma^2 \\ E \left[\sum_{i=1}^n (X_i - \mu)^2 \right] &= n\sigma^2\end{aligned}$$

$$E(\bar{X} - \mu)^2 = E \left[\bar{X} - E(\bar{X}) \right]^2 = \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$$

Hence

$$E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = n\sigma^2 - n \frac{\sigma^2}{n}$$

$$E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = (n-1)\sigma^2$$

$$E(S^2) = \frac{1}{n} E\sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} (n-1)\sigma^2$$

$$\therefore E(S^2) \neq \frac{(n-1)}{n} \sigma^2 \quad E(S^2) = \frac{(n-1)}{n} \sigma^2$$

$$\therefore E(S^2) \neq \sigma^2$$

$\therefore S^2$ is a biased estimator of σ^2

and $E(S^2) = \frac{1}{(n-1)} E\sum_{i=1}^n (X_i - \bar{X})^2 = \frac{(n-1)}{(n-1)} \sigma^2$

$$E(S^2) = \sigma^2$$

Hence S^2 is an unbiased estimator of σ^2 .

Example 3; Show that the sample mean $\bar{X}$ of a random sample of size n from the density function

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \quad 0 < x < \infty$$

is an unbiased estimator of θ

Sol; We have to show that $E(\bar{X}) = \theta$

The mean of the density function is

$$\begin{aligned} \mu = E(X) &= \int x f(x; \theta) dx = \int_0^{\infty} \frac{x}{\theta} e^{-\frac{x}{\theta}} dx \\ &= \frac{1}{\theta} \int_0^{\infty} x e^{-x/\theta} dx = \frac{1}{\theta} \left[-\theta x e^{-x/\theta} \Big|_0^{\infty} + \theta \int_0^{\infty} e^{-x/\theta} dx \right] \\ &= \frac{1}{\theta} \left[0 + \theta (-\theta) \left| e^{-x/\theta} \right|_0^{\infty} \right] \end{aligned}$$

$$\mu = E(X) = -\sigma(0-1) = 0$$

$$\mu = E(X) = 0$$

$$E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = E(\bar{X}) = \frac{n \cdot 0}{n} = 0$$

$$E(\bar{X}) = 0$$

Hence $\bar{X}$ is an unbiased estimator of σ .

Example 4: Let $X_1, X_2, \dots, X_n$ denote a random sample from a normal population $N(0, \sigma)$. Show that $\frac{1}{n} \sum_{i=1}^n X_i^2$ is an unbiased estimator of σ .

Solution:- We have to show that

$$E\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \sigma$$

The density function of normal distribution with mean 0 and variance σ is

$$f(x, \sigma) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{x^2}{2\sigma}} \quad -\infty < x < \infty$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x, \sigma) dx = \int_{-\infty}^{\infty} \frac{x^2}{\sqrt{2\pi\sigma}} e^{-\frac{x^2}{2\sigma}} dx$$

Splitting up the integral into positive and negative regions of integration we get,

$$E(X^2) = \frac{1}{\sqrt{2\pi\sigma}} \left[\int_{-\infty}^0 x^2 e^{-\frac{x^2}{2\sigma}} dx + \int_0^{\infty} x^2 e^{-\frac{x^2}{2\sigma}} dx \right]$$

In the integral over negative region, we make the substitution

$$x = -u, \quad dx = -du$$

$$\text{We get } \int_{-\infty}^0 u^2 e^{-\frac{u^2}{2\sigma}} (-du) = - \int_{\infty}^0 u^2 e^{-\frac{u^2}{2\sigma}} du = \int_0^{\infty} u^2 e^{-\frac{u^2}{2\sigma}} du$$

$$E(X^2) = \frac{1}{\sqrt{2\pi\theta}} \left[\int_0^{\infty} u^2 e^{-\frac{u^2}{2\theta}} du + \int_0^{\infty} x^2 e^{-\frac{x^2}{2\theta}} dx \right]$$

$$= \frac{2}{\sqrt{2\pi\theta}} \int_0^{\infty} u^2 e^{-\frac{u^2}{2\theta}} du$$

$$= \frac{-2\theta}{\sqrt{2\pi\theta}} \int_0^{\infty} u \left(-\frac{u}{\theta} \right) e^{-\frac{u^2}{2\theta}} du$$

Now integrating by parts.

$$E(X^2) = \frac{-2\theta}{\sqrt{2\pi\theta}} \left[\left| u e^{-\frac{u^2}{2\theta}} \right|_0^{\infty} - \int_0^{\infty} e^{-\frac{u^2}{2\theta}} du \right]$$

$$= \frac{-2\theta}{\sqrt{2\pi\theta}} (0) + \frac{2\theta}{\sqrt{2\pi\theta}} \int_0^{\infty} e^{-\frac{u^2}{2\theta}} du$$

$$= \frac{\theta}{\sqrt{2\pi\theta}} \int_0^{\infty} e^{-\frac{u^2}{2\theta}} du$$

$$E(X^2) = \theta \left[\frac{1}{\sqrt{2\pi\theta}} \int_0^{\infty} e^{-\frac{u^2}{2\theta}} du \right]$$

Since the quantity in brackets is the area under the normal curve with mean 0 and variance θ and hence equal to 1,

Hence $E(X^2) = \theta$

and

$$E\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{n\theta}{n} = \theta$$

Hence $\frac{1}{n} \sum_{i=1}^n X_i^2$ is an unbiased estimator of θ .

Example 5; Given a random sample of size n from a population with known mean μ and finite variance σ^2 , show that

$\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ is an unbiased estimator of σ^2 .

Proof; we have to show that

$$E\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2\right) = \sigma^2$$

Because in a random sample, the random variables $X_1, X_2, \dots, X_n$ are independently and identically distributed with mean μ and variance σ^2 . Therefore

$$\begin{aligned} E\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2\right) &= \frac{1}{n} E\left[(X_1 - \mu)^2 + (X_2 - \mu)^2 + \dots + (X_n - \mu)^2\right] \\ &= \frac{1}{n} \left[E(X_1 - \mu)^2 + E(X_2 - \mu)^2 + \dots + E(X_n - \mu)^2\right] \\ &= \frac{1}{n} \left[\sigma^2 + \sigma^2 + \dots + \sigma^2\right] \\ &= \frac{n \sigma^2}{n} \end{aligned}$$

$$E\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2\right) = \sigma^2$$

Hence $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ is an unbiased estimator of σ^2 .

Example 6; If $\bar{X}_1$ and $\bar{X}_2$ are the means of independent random samples of size n_1 and n_2 from a normal population with the mean μ and the variance σ^2 , show that

$[\theta \bar{X}_1 + (1 - \theta) \bar{X}_2]$ where $0 < \theta < 1$ is an unbiased estimator of μ .

Sol;

We have to show that $E[\theta \bar{X}_1 + (1 - \theta) \bar{X}_2] = \mu$

Since $\bar{X}_1$ and $\bar{X}_2$ are the sample means of independent random samples from $N(\mu, \sigma^2)$, and the mean of the sampling dist of the mean is equal to the population mean, we have

$$E(\bar{X}_1) = \mu, \quad E(\bar{X}_2) = \mu$$

$$\begin{aligned} E[\theta \bar{X}_1 + (1-\theta) \bar{X}_2] &= \theta E(\bar{X}_1) + (1-\theta) E(\bar{X}_2) \\ &= \theta \mu + (1-\theta) \mu = \theta \mu + \mu - \theta \mu \end{aligned}$$

$$E[\theta \bar{X}_1 + (1-\theta) \bar{X}_2] = \mu$$

Hence $[\theta \bar{X}_1 + (1-\theta) \bar{X}_2]$ is an unbiased estimator of μ .

Example 7; A random sample of n observations, $x_1, x_2, \dots, x_n$ is chosen from a population with unknown mean μ and unknown variance σ^2 . From the sample we compute

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad s_x^2 = \frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2$$

Another random sample of m observations $y_1, y_2, \dots, y_m$ is chosen and

$$\bar{y} = \frac{1}{m} \sum_{i=1}^m y_i \quad s_y^2 = \frac{1}{(m-1)} \sum_{i=1}^m (y_i - \bar{y})^2$$

are computed. These sets of observations are pooled and μ (mean) is estimated by

$$\bar{x}_p = \frac{n\bar{x} + m\bar{y}}{n+m}$$

and variance σ^2 is estimated by

$$s_p^2 = \frac{(n-1)s_x^2 + (m-1)s_y^2}{n+m-2}$$

Show that the two estimators are unbiased estimators of μ and σ^2 respectively.

Sol; We know that for random sampling, the sample mean $\bar{x}$ and the sample variance s_x^2 are unbiased estimators of the population mean μ and the population variance σ^2 respectively. i.e. $E(\bar{x}) = \mu$ and $E(s_x^2) = \sigma^2$

Similarly $E(\bar{y}) = \mu$, $E(S_y^2) = \sigma^2$

Therefore

$$E(\bar{x}_p) = E\left[\frac{n\bar{x} + m\bar{y}}{n+m}\right] = \frac{1}{n+m} [nE(\bar{x}) + mE(\bar{y})]$$

$$= \frac{1}{(n+m)} (n\mu + m\mu) = \frac{(n+m)}{(n+m)} \mu$$

$$E(\bar{x}_p) = \mu$$

Hence $\bar{x}_p$ is an unbiased estimator of μ .

$$E(S_p^2) = E\left[\frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}\right]$$

$$= \frac{1}{(n+m-2)} [(n-1)E(S_x^2) + (m-1)E(S_y^2)]$$

$$= \frac{1}{(n+m-2)} [(n-1)\sigma^2 + (m-1)\sigma^2]$$

$$E(S_p^2) = \frac{(n+m-2)}{(n+m-2)} \sigma^2$$

$$E(S_p^2) = \sigma^2$$

Hence S_p^2 is an unbiased estimator of σ^2 .

For large sample
 $b_n(\theta)$

$$\text{Bias} = E(\hat{\theta}) - \theta = E(t_n) - \theta$$

often an estimator will not be unbiased, but its bias will be zero as n increases

For example $t_n = S_n + \frac{1}{n}$, where S_n is unbiased

t_n is biased estimator but asymptotically unbiased.

✓ Definition: We say t_n is asymptotically unbiased estimator of θ if

$$\lim_{n \rightarrow \infty} b_n(\theta) = 0 \quad \text{for all } \theta.$$

$$E(S^2) = \frac{(n-1)}{n} \sigma^2 = \left(1 - \frac{1}{n}\right) \sigma^2$$

$$b_n(\sigma^2) = E(S^2) - \sigma^2 = \sigma^2 - \frac{\sigma^2}{n} - \sigma^2$$

$$\lim_{n \rightarrow \infty} b_n(\sigma^2) = 0$$

② **Consistency** :- An estimator t_n , computed from a sample of n observations is said to be a consistent estimator of θ if and only if it converges in probability to θ as $n \rightarrow \infty$.

The statistic t_n is a consistent estimator of the parameter θ if and only if for each arbitrarily small positive number ϵ .

$$\lim_{n \rightarrow \infty} P(|t_n - \theta| \geq \epsilon) = 0$$

or equivalently

$$\lim_{n \rightarrow \infty} P(|t_n - \theta| < \epsilon) = 1$$

For example, the sample mean $\bar{X}$ is a consistent estimator of μ .

S_n^2 or S_{n-1}^2 are consistent estimators of σ^2 .

The sample median is a consistent estimator of the population mean only if the population distribution is symmetrical, otherwise the sample median would approach the population median, not the pop mean, as the sample size is increased. Similarly the sample proportion p is a consistent estimator of the parameter p of a binomial population.

For instance, the smallest observation, largest observation, or the mean of smallest and largest observation is not a consistent estimator of the population mean since they do not approach the pop values as the sample size is increased.

Consistency is an asymptotic property.

i.e. a limiting property of an estimator.

A consistent estimator is unbiased in the limit, but an unbiased estimator may or may not be a consistent estimator.

It is important to note that consistency is a limiting or large sample property i.e. it refers to the behaviour of $\hat{\theta}$ as n becomes large.

Theorem:- t_n is a consistent estimator of parameter θ of a given population if (1) t_n is unbiased and (2) $\text{Var}(t_n) \rightarrow 0$ as $n \rightarrow \infty$.
Proof. Suppose that X is a random variable with mean μ and variance σ^2 . If $X_1, X_2, \dots, X_n$ is a random sample of X , then we know that. $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$

i.e. $\bar{X}$ is unbiased and $\lim_{n \rightarrow \infty} \text{Var}(\bar{X}) = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0$

Example. Show that in random sampling from a normal population, the sample mean is a consistent estimator for the population mean.
Sol. let $X_1, X_2, \dots, X_n$ be a random sample drawn from $N(\mu, \sigma^2)$, then

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \text{ is S.N.V.}$$

$$P(|\bar{X} - \mu| < \epsilon) = P\left(\frac{|\bar{X} - \mu|}{\frac{\sigma}{\sqrt{n}}} < \frac{\sqrt{n}\epsilon}{\sigma}\right)$$

$$= P\left(|Z| < \frac{\sqrt{n}\epsilon}{\sigma}\right) = P\left(\frac{\sqrt{n}\epsilon}{\sigma} < Z < \frac{\sqrt{n}\epsilon}{\sigma}\right)$$

$$= \int_{-\frac{\sqrt{n}\epsilon}{\sigma}}^{\frac{\sqrt{n}\epsilon}{\sigma}} f(z) dz$$

$$= \int_{-\frac{\sqrt{n}\epsilon}{\sigma}}^{\frac{\sqrt{n}\epsilon}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$\lim_{n \rightarrow \infty} P(|\bar{X} - \mu| < \epsilon) = \lim_{n \rightarrow \infty} \int_{-\frac{\sqrt{n}\epsilon}{\sigma}}^{\frac{\sqrt{n}\epsilon}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz = 1$$

$$\lim_{n \rightarrow \infty} P(|\bar{x} - \mu| < \epsilon) = 1$$

Hence Sample Mean $\bar{x}$ is a consistent estimator for the pop mean in normal population.

Example:- Show that for random sampling from Cauchy pop

$$f(x, \mu) = \frac{1}{\pi (1 + (x - \mu)^2)} \quad -\infty \leq x \leq \infty$$

the sample mean is not a consistent estimator for the pop mean, but the sample median is a consistent estimator.

Sol; let $\bar{x}$ be the mean of the sample $x_1, x_2, \dots, x_n$ drawn from Cauchy population.

the sample mean $\bar{x}$ is distributed as x .

Its distribution is

$$f(\bar{x}, \mu) = \frac{1}{\pi [1 + (\bar{x} - \mu)^2]} \quad -\infty \leq \bar{x} \leq \infty$$

$$P(|\bar{x} - \mu| < \epsilon) = P(|y| < \epsilon) = P(-\epsilon < y < \epsilon)$$

$$= \int_{-\epsilon}^{\epsilon} f(y, \mu) dy$$

$$= \int_{-\epsilon}^{\epsilon} \frac{1}{\pi (1 + y^2)} dy = \frac{2}{\pi} \int_0^{\epsilon} \frac{1}{1 + y^2} dy$$

$$= \left| \frac{2}{\pi} \tan^{-1} y \right|_0^{\epsilon} = \frac{2}{\pi} \tan^{-1} \epsilon$$

$$P(|\bar{x} - \mu| < \epsilon) = \frac{2}{\pi} \tan^{-1} \epsilon$$

$\frac{2}{\pi} \tan^{-1} \epsilon$ does not include n , therefore it does not tends to 1, as $n \rightarrow \infty$, Hence $\bar{x}$ is not a consistent estimator of pop. mean.

For large n , the sample median is approximately normally distributed with mean μ , and variance $\frac{\pi^2}{4n}$:

$$Z = \frac{\bar{x} - \mu}{\frac{\pi}{2\sqrt{n}}} = \frac{2\sqrt{n}(\bar{x} - \mu)}{\pi} \text{ is a S.N.V.}$$

$$\begin{aligned} P(|\bar{x} - \mu| < \epsilon) &= P\left(\frac{|\bar{x} - \mu| 2\sqrt{n}}{\pi} < \epsilon \frac{2\sqrt{n}}{\pi}\right) \\ &= P\left(|Z| < \frac{2\sqrt{n}\epsilon}{\pi}\right) \\ &= \int_{-\frac{2\sqrt{n}\epsilon}{\pi}}^{\frac{2\sqrt{n}\epsilon}{\pi}} f(z) dz = \int_{-\frac{2\sqrt{n}\epsilon}{\pi}}^{\frac{2\sqrt{n}\epsilon}{\pi}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \\ &= \lim_{n \rightarrow \infty} \int_{-\frac{2\sqrt{n}\epsilon}{\pi}}^{\frac{2\sqrt{n}\epsilon}{\pi}} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \end{aligned}$$

$$P(|\bar{x} - \mu| < \epsilon) = 1$$

Hence sample median is a consistent estimator for the population mean in Cauchy population.

Definition: let $x_1, x_2, \dots, x_n$ be a random sample of X , whose probability distribution depends on the unknown parameter θ .

let $t_n(x_1, x_2, \dots, x_n)$ be any statistic then MSE of t_n as an estimator of θ is

$$MSE(t_n) = E(t_n - \theta)^2$$

$$= E\left[\{t_n - E(t_n)\} + \{E(t_n) - \theta\}\right]^2$$

$$= E\left[t_n^2 - 2t_n E(t_n) + E(t_n)^2 + 2E(t_n)\theta - 2\theta E(t_n) + \theta^2\right]$$

$$= Var(t_n) + [B(t_n)]^2 + 2[E(t_n) - \theta][E(t_n) - E(t_n)]$$

$$MSE(t_n) = Var(t_n) + [B(t_n)]^2$$

If t_n is unbiased estimator of θ then $B(t_n) = 0$

and

$$MSE(t_n) = Var(t_n)$$

Definition: We say t_n is consistent in MSE or consistent in quadratic mean if

$$E_\theta(t_n - \theta)^2 \rightarrow 0 \text{ as } n \rightarrow \infty \text{ for all } \theta.$$

Theorem: t_n is consistent in MSE if and only if

i) $Var(t_n) \rightarrow 0$ as $n \rightarrow \infty$

ii) t_n is asymptotically unbiased

Proof:

By definition of Variance

$$Var(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = Var(X) + [E(X)]^2$$

$$E_\theta(t_n - \theta)^2 = Var(t_n - \theta) + [E(t_n - \theta)]^2$$

$$E(t_n - \theta)^2 = \text{Var}(t_n) + [B(t_n)]^2$$

Which will $\rightarrow 0$ if both terms on RHS $\rightarrow 0$

Example: Let the p.d.f. of X be

$$f(x, \lambda) = \lambda e^{-\lambda x} \quad 0 \leq x < \infty$$

$$\mu = \frac{1}{\lambda}, \quad \sigma^2 = \frac{1}{\lambda^2}$$

If X_1, X_2 is a random sample of X , then $X_1, X_2, \bar{X} = \frac{X_1 + X_2}{2}$ are each unbiased estimator of the mean $\mu = \frac{1}{\lambda}$

$$E(X) = \mu = \frac{1}{\lambda}, \quad E(X_1) = \frac{1}{\lambda}, \quad E(X_2) = \frac{1}{\lambda}$$

$$\text{Var}(X_1) = \frac{1}{\lambda^2}, \quad \text{Var}(X_2) = \frac{1}{\lambda^2}$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{1}{2\lambda^2}$$

Geometric mean of $X_1, X_2, \sqrt{X_1 X_2}$ is a biased estimator.

$$\text{Var}(\sqrt{X_1 X_2}) = E(\sqrt{X_1 X_2}) - [E(\sqrt{X_1 X_2})]^2$$

$$\text{Var}(\sqrt{X_1 X_2}) = E(X_1) E(X_2) - [E(\sqrt{X_1}) E(\sqrt{X_2})]^2 \quad \text{--- (1)}$$

$$E(X^{1/2}) = \int_0^{\infty} x^{1/2} f(x, \lambda) dx = \int_0^{\infty} x^{1/2} \lambda e^{-\lambda x} dx$$

$$\text{putting } \lambda x = \mu$$

$$\lambda dx = d\mu \Rightarrow d$$

$$E(X^{1/2}) = \int_0^{\infty} \frac{\mu^{1/2}}{\lambda^{1/2}} e^{-\mu} d\mu$$

$$E(X^{1/2}) = \frac{1}{\lambda^{1/2}} \int_0^{\infty} \mu^{1/2-1} e^{-\mu} d\mu$$

$$= \frac{1}{\lambda^{1/2}} \int_0^{\infty} \mu^{1/2} e^{-\mu} d\mu = \frac{1}{2\lambda^{1/2}} \int_0^{\infty} \mu^{1/2} e^{-\mu} d\mu$$

$$E(X^{1/2}) = \frac{1}{2} \sqrt{\frac{\pi}{\lambda}}$$

Now from (1)

$$\text{Var}(\sqrt{X_1 X_2}) = \left(\frac{1}{\lambda}\right) \left(\frac{1}{\lambda}\right) - \left(\frac{1}{4} \frac{\pi}{\lambda}\right)^2$$

$$\text{Bias}(\sqrt{X_1 X_2}) = E[\sqrt{X_1 X_2}] - \frac{1}{\lambda}$$

$$= E\sqrt{X_1} E\sqrt{X_2} - \frac{1}{\lambda}$$

$$= \frac{\pi}{4\lambda} - \frac{1}{\lambda}$$

$$\text{Bias}(\sqrt{X_1 X_2}) = \frac{\pi - 4}{4\lambda}$$

$$\text{MSE}(\sqrt{X_1 X_2}) = \text{Var}(\sqrt{X_1 X_2}) + [\text{Bias}(\sqrt{X_1 X_2})]^2$$

$$= \frac{1}{\lambda^2} - \frac{\pi^2}{16\lambda^2} + \frac{(\pi - 4)^2}{16\lambda^2}$$

$$= \frac{16 - \pi^2 + \pi^2 - 8\pi + 16}{16\lambda^2}$$

$$= \frac{32 - 8\pi}{16\lambda^2} = \frac{4 - \pi}{2\lambda^2}$$

$$\text{MSE}(\sqrt{X_1 X_2}) = \frac{4 - \pi}{2\lambda^2} \quad (4 - \pi) < 1$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{1}{2\lambda^2}$$

$\sqrt{X_1 X_2}$ is a biased and $\bar{X}$ is an unbiased estimator.

$$MSE(\sqrt{X_1 X_2}) < Var(X)$$

$$ie \quad MSE(\text{Biased Estimator}) < Var(\text{unbiased estimator})$$

Efficiency

In general there will exist more than one consistent estimators of the parameter θ , even if we confine ourselves to unbiased estimators.

Consider the estimator of the mean of a normal population with variance σ^2 .

The sample mean is consistent and unbiased.

The sample median is also consistent and unbiased.

We must look for further criterion to choose between the estimators with common property of consistent. Such a criterion arises when we consider the sampling variance of the estimators.

Efficient Estimators

Let for large samples two consistent estimators t_n and t'_n be both distributed asymptotically normally about the true value of the parameter θ and with variances $\frac{\sigma^2}{n}$ and $\frac{\sigma'^2}{n}$ respectively (by virtue of central limit theorem CLT).

$$t_n \sim N(\theta, \frac{\sigma^2}{n})$$

and

$$t'_n \sim N(\theta, \frac{\sigma'^2}{n})$$

then t_n is said to be more consistent than t'_n if

$$\sigma^2 < \sigma'^2$$

If we can find a consistent estimator t_n , whose variance is less than what of all other consistent estimators for all n , then t_n is said to be the most efficient estimator and the efficiency E of any other estimator is defined as the ratio of the variance of most efficient estimator and

the variance of given estimator

Example: Show that in estimating the mean for random sampling from a normal population. Sample mean is more efficient than the sample median.

✓ Discuss the position of the two estimators for Cauchy population.
Solution: For a normal population with finite variance σ^2 the sample mean and sample variance are consistent estimators for the population mean and pop variance respectively.

$$\text{Var}(\text{mean}) = \frac{\sigma^2}{n} \quad (\text{For any } n)$$

$$\text{Var}(\text{Median}) = \frac{\sigma^2 \pi}{n^2} \quad (\text{for large } n)$$

Since $\text{Var}(\text{mean}) < \text{Var}(\text{Median})$, the mean is more efficient than the median.

Efficiency of the median relative to the mean is

$$\frac{\frac{\sigma^2/n}{\pi \sigma^2 / n^2}}{2} = \frac{2}{\pi} = 0.637$$

Example: If t is most efficient estimator and t' is less efficient estimator with efficiency E and if correlation coefficient between t and t' is P , then show by considering estimator t'' defined by

$$(1+E-2P\sqrt{E})t'' = (1-P\sqrt{E})t + (E-P\sqrt{E})t'$$

that $P = \sqrt{E}$

Solution:

$$E = \frac{\text{Var}(t)}{\text{Var}(t')}$$

We have estimator $\hat{t}''$ as

$$(1 + E - 2\rho\sqrt{E})\hat{t}'' = (1 - \rho\sqrt{E})t + (E - \rho\sqrt{E})\hat{t}'$$

Taking variance on both sides.

$$(1 + E - 2\rho\sqrt{E})^2 \text{Var}(\hat{t}'') = (1 - \rho\sqrt{E})^2 \text{Var}(t) + (E - \rho\sqrt{E})^2 \text{Var}(\hat{t}') \\ + 2(1 - \rho\sqrt{E})(E - \rho\sqrt{E})\text{Cov}(t, \hat{t}')$$

Now let

$$\text{Var}(t) = V$$

$$E = \frac{\text{Var}(t)}{\text{Var}(\hat{t}')} \quad \text{given}$$

$$\text{Var}(\hat{t}') = \frac{V}{E}$$

$$\rho = \frac{\text{Cov}(t, \hat{t}')}{\sqrt{\text{Var}(t) \text{Var}(\hat{t}')}} = \frac{\text{Cov}(t, \hat{t}')}{\sqrt{V \frac{V}{E}}} = \frac{\rho V}{\sqrt{E}}$$

$$\text{Cov}(t, \hat{t}') = \rho \sqrt{V \frac{V}{E}} = \frac{\rho V}{\sqrt{E}}$$

Now

$$(1 + E - 2\rho\sqrt{E})^2 \text{Var}(\hat{t}'') = (1 - \rho\sqrt{E})^2 V + E(\sqrt{E} - \rho)^2 \frac{V}{E} \\ + 2(1 - \rho\sqrt{E})(\sqrt{E} - \rho)\sqrt{E} \frac{\rho V}{\sqrt{E}}$$

$$= V \left[(1 - \rho\sqrt{E})^2 + (\sqrt{E} - \rho)^2 + 2(1 - \rho\sqrt{E})(\sqrt{E} - \rho)\rho \right]$$

$$= V \left[(1 - \rho\sqrt{E})^2 + (\sqrt{E} - \rho)^2 + 2(1 - \rho\sqrt{E})(\sqrt{E} - \rho)\rho \right]$$

$$= V \left[(1 - \rho\sqrt{E})^2 + (\sqrt{E} - \rho)^2 + 2(\sqrt{E} - \rho - \rho E + \rho\sqrt{E})\rho \right]$$

$$(1+E-2\rho\sqrt{E})^2 \text{Var}(t'') = V \left[(1-\rho\sqrt{E})^2 + (\sqrt{E}-\rho)^2 + 2(\rho\sqrt{E}-\rho^2-\rho^2E+\rho^3\sqrt{E}) \right]$$

$$= V \left[1+\rho^2E-2\rho\sqrt{E}+E+\rho^2-2\rho\sqrt{E}+2\rho\sqrt{E}-2\rho^2-2\rho^2E+2\rho^3\sqrt{E} \right]$$

$$= V \left[1-\rho^2-E\rho^2+E-2\rho\sqrt{E}+2\rho^3\sqrt{E} \right]$$

$$= V \left[(1-\rho^2)+E(1-\rho^2)-2\rho\sqrt{E}(1-\rho^2) \right]$$

$$= V(1-\rho^2)(1+E-2\rho\sqrt{E})$$

$$\text{Var}(t'') = \frac{V(1-\rho^2)(1+E-2\rho\sqrt{E})}{(1+E-2\rho\sqrt{E})^2}$$

$$= \frac{V(1-\rho^2)}{(1+E-2\rho\sqrt{E})^2}$$

$$= \frac{V(1-\rho^2)}{1-\rho^2+\rho^2+E-2\rho\sqrt{E}}$$

$$\text{Var}(t'') = \frac{V(1-\rho^2)}{(1-\rho^2)+(\rho-\sqrt{E})^2}$$

But $\text{Var}(t'') \gg V$
this holds when

$$\frac{(1-\rho^2)}{(1-\rho^2)+(\rho-\sqrt{E})^2} = 1$$

$$(1-\rho^2) = (1-\rho^2)+(\rho-\sqrt{E})^2$$

$$\rho-\sqrt{E} = 0 \quad \text{or} \quad \rho = \sqrt{E}$$

Hence proved.

Sufficiency:-

A statistic is said to be sufficient statistic for estimating population parameter θ , if it contains all the information in the sample about the parameter θ .

$\bar{X}$ is sufficient estimator of μ .

$$\sum_{i=1}^n X_i, \quad \mu.$$

$X_{(1)}, X_{(n)}$ is smallest and largest obs are not sufficient estimators.

Definition: (i) Sufficiency by Conditional density $\leftarrow$ see in lecture of 16/1/91

let $x_1, \dots, x_n$ be a random sample from the density $f(x, \theta)$ and let

$t = f_1(x_1, \dots, x_n)$ be a statistic (function of x only) let

$t^* = f_2(x_1, \dots, x_n)$ be another statistic. If for each of statistic t^* , the conditional density of t^* given t does not contain θ , then t is a sufficient statistic for θ .

$f(t^*/t)$ does not contain θ .

(ii) **The Factorization Criterion:-**

To determine whether a given statistic is sufficient, we use the following Criterion.

let $x_1, x_2, \dots, x_n$ be a random sample of size n from the density $f(x, \theta)$ and let the joint density of these n random variables be

$$L(x_1, \dots, x_n; \theta) = L_1(t, \theta) L_2(x_1, x_2, \dots, x_n)$$

where $L_2(x_1, \dots, x_n)$ does not involve θ , thus t is sufficient statistic for θ .

By Factorization

Example: Suppose $X_1, X_2, \dots, X_n$ is a random sample from the density function

$$f(x, \mu) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}[\lambda - \mu]^2} \quad -\infty < x < \infty$$

Show that $\sum X_i$ is a sufficient estimator for μ .

Sol; The joint density function is

$$f(x_1; \mu) f(x_2; \mu) \dots f(x_n; \mu)$$

$$= \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(x_1 - \mu)^2\right) \right) \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(x_2 - \mu)^2\right) \right) \dots \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(x_n - \mu)^2\right) \right)$$

$$= \left(\frac{1}{2\pi} \right)^{n/2} \exp\left[-\frac{1}{2} \sum (x_i - \mu)^2\right]$$

$$= \left(\frac{1}{2\pi} \right)^{n/2} \exp\left[-\frac{1}{2} \left(\sum x_i^2 - 2\mu \sum x_i + n\mu^2 \right)\right]$$

$$= \left(\frac{1}{2\pi} \right)^{n/2} \exp\left[\mu \sum x_i - \frac{n}{2} \mu^2\right] \exp\left[-\frac{1}{2} \sum x_i^2\right]$$

$$= L_1(\sum x_i; \mu) L_2(x_1, \dots, x_n)$$

$$= L_1(\mu; \mu) L_2(x_1, \dots, x_n)$$

Thus joint density function is factorized into two functions $L_1(\mu; \mu)$ and $L_2(x_1, x_2, \dots, x_n)$. The factorization criterion is satisfied.

Hence $\sum x_i$ is a sufficient estimator of μ . Since any one-to-one function of a sufficient estimator is also sufficient, the sample mean $\bar{x}$ is also sufficient.

Example: Let $x_1, x_2, \dots, x_n$ denote a random sample from the density function

$$f(x; p) = p x^{p-1} \quad 0 < x < 1, \quad p > 0$$

Show that the product $x_1 \cdot x_2 \cdot \dots \cdot x_n$ is a sufficient estimator for p .

Solution: The joint density function is

$$f(x_1; p) f(x_2; p) \dots f(x_n; p)$$

$$= p x_1^{p-1} p x_2^{p-1} \dots p x_n^{p-1}$$

$$\begin{aligned}
 &= p^n (x_1, x_2, \dots, x_n)^{p-1} \\
 &= p^n (x_1, x_2, \dots, x_n)^{\frac{p-1}{(x_1, x_2, \dots, x_n)}} \\
 &= L_1(x_1, x_2, \dots, x_n; p) L_2(x_1, x_2, \dots, x_n) \\
 &= L_1(\hat{p}, p) L_2(x_1, x_2, \dots, x_n)
 \end{aligned}$$

Thus the joint density function is factorized into two functions. The factorization criterion is satisfied. Hence product $x_1 \cdot x_2 \cdot \dots \cdot x_n$ is a sufficient estimator for p .

Example: let $x_1, x_2, \dots, x_n$ denote a random sample from the Bernoulli distribution.

$$f(x; p) = p^x (1-p)^{1-x} \quad x=0,1$$

Show that $\sum x_i$ is a sufficient estimator of p .

Solution: The joint probability function is

$$\begin{aligned}
 &f(x_1, p) f(x_2, p) \dots f(x_n, p) \\
 &= p^{x_1} (1-p)^{1-x_1} p^{x_2} (1-p)^{1-x_2} \dots p^{x_n} (1-p)^{1-x_n} \\
 &= p^{x_1+x_2+\dots+x_n} (1-p)^{n-(x_1+x_2+\dots+x_n)} \\
 &= p^{\sum x_i} (1-p)^{n-\sum x_i} \\
 &= L_1(\sum x_i; p) L_2(x_1, x_2, \dots, x_n) \\
 &= L_1(\hat{p}, p) L_2(x_1, x_2, \dots, x_n)
 \end{aligned}$$

Thus joint density function is factorized into two factors. The factorization criterion is satisfied. Hence $\sum x_i$ is a sufficient estimator for p .

Since any one-to-one function of sufficient estimator is also sufficient, the sample proportion $\sum x_i/n$ is also sufficient.

Example: If E_1 and E_2 are efficiencies of two unbiased estimators t_1 and t_2 and ρ is the correlation coefficient between t_1 and t_2 then show that

$$\sqrt{E_1 E_2} - \sqrt{(1-E_1)(1-E_2)} < \rho < \sqrt{E_1 E_2} + \sqrt{(1-E_1)(1-E_2)}$$

Sol, let

$$\text{Var}(t_1) = \frac{V}{E_1}, \quad \text{Var}(t_2) = \frac{V}{E_2}$$

where V is the minimum variance.

let $t = \lambda_1 t_1 + \lambda_2 t_2$ where λ_1 and λ_2 are constant such that $\lambda_1 + \lambda_2 = 1$, given t_1 and t_2 are unbiased

$$\text{Var}(t) = \text{Var}(\lambda_1 t_1 + \lambda_2 t_2)$$

$$= \lambda_1^2 \text{Var}(t_1) + \lambda_2^2 \text{Var}(t_2) + 2\lambda_1 \lambda_2 \text{Cov}(t_1, t_2)$$

$$= \lambda_1^2 \frac{V}{E_1} + \lambda_2^2 \frac{V}{E_2} + 2\lambda_1 \lambda_2 \rho \sqrt{\frac{V}{E_1 E_2}}$$

$$= V \left[\frac{\lambda_1^2}{E_1} + \frac{\lambda_2^2}{E_2} + \frac{2\rho \lambda_1 \lambda_2}{\sqrt{E_1 E_2}} \right]$$

Now $\text{Var}(t) \geq V$ where V is minimum variance

$$\therefore \frac{\lambda_1^2}{E_1} + \frac{\lambda_2^2}{E_2} + \frac{2\rho \lambda_1 \lambda_2}{\sqrt{E_1 E_2}} \geq 1$$

$$\geq (\lambda_1 + \lambda_2)^2 = \lambda_1^2 + \lambda_2^2 + 2\lambda_1 \lambda_2$$

$$\frac{\lambda_1^2}{E_1} + \frac{\lambda_2^2}{E_2} + \frac{2\rho \lambda_1 \lambda_2}{\sqrt{E_1 E_2}} - \lambda_1^2 - \lambda_2^2 - 2\lambda_1 \lambda_2 \geq 0$$

$$\lambda_1^2 \left(\frac{1}{E_1} - 1 \right) + 2\lambda_1 \lambda_2 \left(\frac{\rho}{\sqrt{E_1 E_2}} - 1 \right) + \lambda_2^2 \left(\frac{1}{E_2} - 1 \right) \geq 0$$

Now the discriminant of this quadratic equation is

$$4\lambda_2^2 \left(\frac{\rho}{\sqrt{E_1 E_2}} - 1 \right)^2 - 4\lambda_2^2 \left(\frac{1}{E_1} - 1 \right) \left(\frac{1}{E_2} - 1 \right) \geq 0$$

$$\frac{\rho^2}{E_1 E_2} - \frac{2\rho}{\sqrt{E_1 E_2}} + 1 - \frac{1}{E_1 E_2} + \frac{1}{E_1} + \frac{1}{E_2} - 1 \geq 0$$

$$P^2 - 2P\sqrt{E_1 E_2} + (E_1 + E_2 - 1) \geq 0$$

Solving this as a quadratic in P

$$P = \frac{2\sqrt{E_1 E_2} \pm \sqrt{4E_1 E_2 - (E_1 + E_2 - 1)}}{2} \geq 0$$

$$= \sqrt{E_1 E_2} \pm \sqrt{E_1 E_2 - E_1 - E_2 + 1}$$

$$= \sqrt{E_1 E_2} \pm \sqrt{(1-E_1)(1-E_2)}$$

$$\text{Hence } \sqrt{E_1 E_2} - \sqrt{(1-E_1)(1-E_2)} < P < \sqrt{E_1 E_2} + \sqrt{(1-E_1)(1-E_2)}$$

Example, Let $X_1, X_2, \dots, X_n$ be a random sample from the normal dist with mean zero and variance θ . Show that $\sum X_i^2$ is sufficient estimator for θ .

Sol;

$$f(x; \theta) = \frac{1}{\sqrt{2\pi\theta}} e^{-\frac{x^2}{\theta}} \quad -\infty < x < \infty$$

The joint density function is

$$f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta)$$

$$= \left(\frac{1}{\sqrt{2\pi\theta}} e^{-\frac{x_1^2}{\theta}} \right) \left(\frac{1}{\sqrt{2\pi\theta}} e^{-\frac{x_2^2}{\theta}} \right) \dots \left(\frac{1}{\sqrt{2\pi\theta}} e^{-\frac{x_n^2}{\theta}} \right)$$

$$= \left(\frac{1}{\sqrt{2\pi\theta}} \right)^{n/2} e^{-(x_1^2 + x_2^2 + \dots + x_n^2) \frac{1}{\theta}}$$

$$= \left(\frac{1}{\sqrt{2\pi\theta}} \right)^{n/2} e^{-\frac{\sum x_i^2}{\theta}} \quad (1)$$

$$= L_1(\sum x_i^2; \theta) L_2(x_1, \dots, x_n)$$

$$= L_1(\hat{\theta}; \theta) L_2(x_1, \dots, x_n)$$

where $\hat{\theta} = \sum x_i^2$ and $L_2(x_1, \dots, x_n) = 1$.

Thus the joint density function is factorized into two functions. The factorization criterion is satisfied. Hence $\sum X_i^2$ is a sufficient estimator for μ .

Example, let $X_1, X_2, \dots, X_n$ be a random sample of size n from the density function

$$f(x; \alpha) = \alpha(\alpha+1) x^{\alpha+1} (1-x) \quad 0 < x < 1$$

Show that the product $X_1 X_2 \dots X_n$ is sufficient estimator for α .
Sol;

$$f(x; \alpha) = \alpha(\alpha+1) x^{\alpha+1} (1-x) \quad 0 < x < 1$$

The joint density function is

$$\begin{aligned} f(x_1; \alpha) f(x_2; \alpha) \dots f(x_n; \alpha) &= [\alpha(\alpha+1) x_1^{\alpha+1} (1-x_1)] [\alpha(\alpha+1) x_2^{\alpha+1} (1-x_2)] \dots \\ &\quad [\alpha(\alpha+1) x_n^{\alpha+1} (1-x_n)] \\ &= \alpha^n (\alpha+1)^n (x_1 x_2 \dots x_n)^{\alpha+1} (1-x_1)(1-x_2) \dots (1-x_n) \\ &= L_1(x_1, x_2, \dots, x_n; \alpha) L_2(x_1, \dots, x_n) \\ &= L_1(\hat{\alpha}; \alpha) L_2(x_1, \dots, x_n) \end{aligned}$$

where $\hat{\alpha} = x_1 x_2 \dots x_n$ and

$$L_2(x_1, \dots, x_n) = (1-x_1)(1-x_2) \dots (1-x_n)$$

Thus the joint density function is factorized into two functions. The factorization criterion is satisfied. Hence the product $x_1 x_2 \dots x_n$ is a sufficient estimator of α .

Example; Prove that Sum of observations of a random sample of size n from a Poisson distribution with parameter λ is a sufficient estimator.

Sol; $f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots$

The joint probability function is

$$\begin{aligned}
 f(x_1; \lambda) f(x_2; \lambda) \dots f(x_n; \lambda) &= \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} \dots \frac{e^{-\lambda} \lambda^{x_n}}{x_n!} \\
 &= \frac{e^{-n\lambda} \lambda^{(x_1 + x_2 + \dots + x_n)}}{x_1! x_2! \dots x_n!} \\
 &= \frac{e^{-n\lambda} \lambda^{\sum x_i}}{x_1! x_2! \dots x_n!} \\
 &= L_1(\sum x_i; \lambda) L_2(x_1, \dots, x_n) \\
 &= L_1(\hat{\lambda}; \lambda) L_2(x_1, \dots, x_n)
 \end{aligned}$$

where

$\hat{\lambda} = \sum x_i$ and $L_2 = \frac{1}{x_1! \dots x_n!}$. Thus the joint probability function is factored into two functions.

The factorization criterion is satisfied. Hence $\sum x_i$ is a sufficient estimator of λ .

Example. Show that for a random sample of size n from the density function

$$f(x; \theta) = \frac{1}{\theta} e^{-x/\theta} \quad 0 < x < \infty$$

$\sum_{i=1}^n x_i$ is a sufficient estimator for θ .

Sol; The joint density function is

$$\begin{aligned}
 f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta) &= \left(\frac{1}{\theta} e^{-x_1/\theta} \right) \left(\frac{1}{\theta} e^{-x_2/\theta} \right) \dots \left(\frac{1}{\theta} e^{-x_n/\theta} \right) = \left(\frac{1}{\theta} \right)^n e^{-(x_1 + \dots + x_n)/\theta} \\
 &= \left(\frac{1}{\theta} \right)^n e^{-\sum x_i / \theta} \\
 &= L_1(\sum x_i; \theta) L_2(x_1, \dots, x_n) \\
 &\quad \hat{\theta} = \sum x_i, L_2(x_1, \dots, x_n) = 1
 \end{aligned}$$

$\sum x_i$ is a sufficient estimator for θ .

Minimum Variance Estimators

The sampling variance of an estimator is used as a criterion of its acceptability. Under fairly general conditions, there exist a bound, below which the variance of an estimator can not fall. In order to establish this bound, we first prove the following results.

If the p.d.f or p.f is $f(x, \theta)$, we define the likelihood function (L.F) of a sample of n independent observations by

$$L(x_1, \dots, x_n; \theta) = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta) \quad \text{--- (1)}$$

$L(x_1, \dots, x_n; \theta)$ is denoted by L .
 L is the L.F when considered as a function of θ and the prob of sample when it is considering as a function of x for fixed θ .
 Since L is the joint p.d.f of the observations.

$$\therefore \int \dots \int L \, dx_1 \dots dx_n = 1 \quad \text{--- (2)}$$

Suppose that first two observations derivatives of L , w.r.t θ exist for all θ . If we differentiate both sides of (2) w.r.t θ and if we interchange the operations of differentiation and integration on L.H.S.

$$\int \dots \int \frac{\partial L}{\partial \theta} \, dx_1 \dots dx_n = 0$$

$$\int \dots \int \frac{1}{L} \frac{\partial L}{\partial \theta} L \, dx_1 \dots dx_n = 0 \quad \text{--- (3)}$$

$$\int \dots \int \left(\frac{\partial \log L}{\partial \theta} \right) L \, dx_1 \dots dx_n = 0$$

$$E \left[\frac{\partial \log L}{\partial \theta} \right] = 0 \quad \text{--- (4)}$$

$$\therefore \int x f(x) dx = E(x)$$

If we differentiate (3) again, we get.

$$\int \dots \int \left\{ \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right) \left(\frac{\partial L}{\partial \theta} \right) + L \frac{\partial}{\partial \theta} \left[\frac{1}{L} \frac{\partial L}{\partial \theta} \right] \right\} dx_1 \dots dx_n = 0$$

$$\int \left\{ \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right)^2 L + L \frac{\partial}{\partial \theta} \left(\frac{\partial \log L}{\partial \theta} \right) \right\} dx_1, \dots, dx_n = 0$$

$$\int \left[\left(\frac{\partial \log L}{\partial \theta} \right)^2 + \frac{\partial^2 \log L}{\partial \theta^2} \right] L dx_1, \dots, dx_n = 0$$

$$\text{i.e. } E \left[\left(\frac{\partial \log L}{\partial \theta} \right)^2 + \frac{\partial^2 \log L}{\partial \theta^2} \right] = 0$$

$$E \left[\left(\frac{\partial \log L}{\partial \theta} \right)^2 \right] = -E \left[\frac{\partial^2 \log L}{\partial \theta^2} \right] \quad \text{--- (5)}$$

$$\text{Var} \left[\frac{\partial \log L}{\partial \theta} \right] = E \left(\frac{\partial \log L}{\partial \theta} \right)^2 - \left[E \left(\frac{\partial \log L}{\partial \theta} \right) \right]^2$$

but From Eq (4)

$$E \left(\frac{\partial \log L}{\partial \theta} \right) = 0$$

$$\therefore \text{Var} \left(\frac{\partial \log L}{\partial \theta} \right) = E \left(\frac{\partial \log L}{\partial \theta} \right)^2$$

$$\text{Var} \left(\frac{\partial \log L}{\partial \theta} \right) = -E \left(\frac{\partial^2 \log L}{\partial \theta^2} \right)$$

Cramer Rao Inequality (Information Inequality)

Let $X_1, X_2, \dots, X_n$ be independent random variables and each with p.d.f or (p.f) $f(x; \theta)$.

Let $t_n(x_1, \dots, x_n)$ be an estimator of some function of θ , $g(\theta)$.

Let $b_n(\theta)$ be the bias of t_n .

$$b_n(\theta) = E(t_n) - g(\theta).$$

Then the following results hold.

I C.R inequality for function of θ , $g(\theta)$

$$MSE = E_{\theta}(t_n - g(\theta))^2 \geq \frac{[g'(\theta) + b'(\theta)]^2}{E\left(\frac{\partial \log L}{\partial \theta}\right)^2}$$

$$L = L(x_1, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta)$$

where t_n is any estimator.

II C.R inequality for unbiased estimators of function of θ .
ie if t_n is an unbiased estimator of $g(\theta)$.

$$\text{Var}_{\theta}(t_n) \geq \frac{[g'(\theta)]^2}{E\left(\frac{\partial \log L}{\partial \theta}\right)^2}$$

III C.R inequality for unbiased estimator of θ .

ie if t_n is an unbiased estimator of θ

ie $g(\theta) = \theta$ and $g'(\theta) = 1$

$$\text{Var}_{\theta}(t_n) \geq \frac{1}{E\left(\frac{\partial \log L}{\partial \theta}\right)^2}$$

Proof:-

By definition of the bias of the t_n as an estimator of $g(\theta)$, we have

$$b_n(\theta) = E_{\theta}(t_n) - g(\theta)$$

$$b_n(\theta) + g(\theta) = E_{\theta}(t_n)$$

$$\text{ie } b_n(\theta) + g(\theta) = \int \dots \int t_n(x_1, \dots, x_n) L dx_1 \dots dx_n$$

$$\text{where } L = \prod_{i=1}^n f(x_i; \theta)$$

Now differentiating both sides w.r.t θ , we get.

$$g'(\theta) + b'_n(\theta) = \int \dots \int t_n(x_1, \dots, x_n) \frac{\partial L}{\partial \theta} dx_1 \dots dx_n$$

$$= \int \dots \int t_n(x_1, \dots, x_n) \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right) L dx_1 \dots dx_n$$

$$g'(\theta) + b'_n(\theta) = \int \dots \int t_n(x_1, \dots, x_n) \left(\frac{\partial \log L}{\partial \theta} \right) L dx_1 \dots dx_n \quad \text{--- ①}$$

Now

$$\int \dots \int f(x_1, \theta) \dots f(x_n, \theta) dx_1 \dots dx_n = 1$$

$$\int \dots \int L dx_1 \dots dx_n = 1$$

Differentiating both sides w.r.t θ , we get

$$\int \dots \int \frac{\partial L}{\partial \theta} dx_1 \dots dx_n = 0$$

Multiplying and dividing by L

$$\int \dots \int \left(\frac{\partial \log L}{\partial \theta} \right) L dx_1 \dots dx_n = 0$$

Multiplying and dividing both sides by $g(\theta)$

$$\int \dots \int g(\theta) \left(\frac{\partial \log L}{\partial \theta} \right) L dx_1 \dots dx_n = 0 \quad \text{--- ②}$$

Introducing the $g(\theta)$ term (which is equal to zero) in Eq ①
from (2)

$$g'(\theta) + b'_n(\theta) = \int \left\{ (t_n - g(\theta)) \left(\frac{\partial \log L}{\partial \theta} \right) \right\} L \, dx_1, \dots, dx_n$$

$$g'(\theta) + b'_n(\theta) = E_\theta \left[(t_n - g(\theta)) \left(\frac{\partial \log L}{\partial \theta} \right) \right]$$

Now by using Cauchy Schwarz's inequality.

For two random variables X and Y .

$$[E(XY)]^2 \leq E(X^2) \cdot E(Y^2)$$

$$g'(\theta) + b'_n(\theta) = E_\theta \left[(t_n - g(\theta)) \left(\frac{\partial \log L}{\partial \theta} \right) \right]$$

$$\left[E_\theta \left\{ (t_n - g(\theta)) \left(\frac{\partial \log L}{\partial \theta} \right) \right\} \right]^2 \leq \left[E_\theta \left\{ (t_n - g(\theta))^2 \right\} \right] \left[E_\theta \left\{ \left(\frac{\partial \log L}{\partial \theta} \right)^2 \right\} \right]$$

Changing sign on both sides,

$$E(Z^2) = E(-Z)^2$$

Similarly,

$$-g'(\theta) - b'_n(\theta) = E_\theta \left[(g(\theta) - t_n) \left(\frac{\partial \log L}{\partial \theta} \right) \right]$$

has the same upper bound

$$|g'(\theta) + b'_n(\theta)| \leq \left[E_\theta \left\{ (t_n - g(\theta))^2 \right\} \right]^{1/2} \left[E_\theta \left\{ \left(\frac{\partial \log L}{\partial \theta} \right)^2 \right\} \right]^{1/2}$$

Squaring both sides and dividing by $E_\theta \left(\frac{\partial \log L}{\partial \theta} \right)^2$

$$[g'(\theta) + b'_n(\theta)]^2 \leq \left(E_\theta \left[(t_n - g(\theta))^2 \right] \right) \left(E_\theta \left[\left(\frac{\partial \log L}{\partial \theta} \right)^2 \right] \right)$$

Now dividing both sides by $E_\theta \left(\frac{\partial \log L}{\partial \theta} \right)^2$

$$E_{\theta} (t_n - g(\theta))^2 \geq \frac{[g'(\theta) + b_n(\theta)]^2}{E_{\theta} \left(\frac{\partial \log L}{\partial \theta} \right)^2} \quad \text{I}$$

If t_n is unbiased estimator of $g(\theta)$ i.e. $b_n(\theta) = 0$

$$\text{Var}_{\theta}(t_n) \geq \frac{[g'(\theta)]^2}{E_{\theta} \left(\frac{\partial \log L}{\partial \theta} \right)^2} \quad \text{II}$$

If t_n is unbiased estimator of θ , i.e. $g(\theta) = \theta$ $g'(\theta) = 1$

$$\text{Var}_{\theta}(t_n) \geq \frac{1}{E_{\theta} \left(\frac{\partial \log L}{\partial \theta} \right)^2} \quad \text{III}$$

The C.R. inequality is of most use in form II and form III, when it gives a bound on the variance of an unbiased estimator. If an estimator achieves this lower bound, it is the best of the unbiased estimators in the sense of having smallest possible variance.

We shall call case II minimum variance bound (MVB) for the estimation of $g(\theta)$. An estimator which attains this bound is called MVB estimator.

The quantity

$$I = E_{\theta} \left[\left(\frac{\partial \log L}{\partial \theta} \right)^2 \right] = - E_{\theta} \left[\frac{\partial^2 \log L}{\partial \theta^2} \right]$$

is called amount of information in the sample.

Condition under which the MVB is Attained

We have

$$E[t - g(\theta)]^2 = \text{Var}(t) \geq \frac{[g'(\theta)]^2}{E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right]} \quad \text{--- II}$$

if t_n is an unbiased estimator of $g(\theta)$

This inequality is derived by the use of Schwarz's inequality. The necessary and sufficient condition that Schwarz's inequality becomes an equality is that $[t - g(\theta)]$ is proportional to $\left(\frac{\partial \ln L}{\partial \theta}\right)$.

$$\frac{\partial \ln L}{\partial \theta} = A[t - g(\theta)] \quad \text{--- ①}$$

Where A is independent of the observations but may be a function of θ i.e.

$$\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - g(\theta)] \quad \text{--- ②}$$

$$\text{Var}\left(\frac{\partial \ln L}{\partial \theta}\right) = E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right] - \left[E\left(\frac{\partial \ln L}{\partial \theta}\right)\right]^2$$

$$\text{Because } E\left(\frac{\partial \ln L}{\partial \theta}\right) = 0$$

Therefore

$$\begin{aligned} \text{Var}\left(\frac{\partial \ln L}{\partial \theta}\right) &= E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right] = E\left[\left(A(\theta)\{t_n - g(\theta)\}\right)^2\right] \\ &= [A(\theta)]^2 E[t_n - g(\theta)]^2 \end{aligned}$$

$$\text{Var}\left(\frac{\partial \ln L}{\partial \theta}\right) = [A(\theta)]^2 \text{Var}(t) \quad \text{--- ③}$$

Since II is an equality, i.e.

$$\text{Var}(t) = \frac{[g'(\theta)]^2}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2} \quad \text{--- ④}$$

Making substitution from ③ to ④

$$\text{Var}(t) = \frac{[g'(\theta)]^2}{[A(\theta)]^2 \text{Var}(t)}$$

$$[\text{Var}(t)]^2 = \frac{[g'(\theta)]^2}{[A(\theta)]^2} = \left[\frac{g'(\theta)}{A(\theta)} \right]^2$$

$$\boxed{\text{Var}(t) = \frac{g'(\theta)}{A(\theta)}} \quad \text{--- (5)}$$

We conclude that if Eq (4) is satisfied, t is MVB estimator of $g(\theta)$ with variance given by (5)

If $g(\theta) = \theta$, $g'(\theta) = 1$

$$\boxed{\text{Var}(t) = \frac{1}{A(\theta)}} \quad \text{is the condition under which } t \text{ is MVB estimator.}$$

Example 1, From the normal population having p.d.f

$$f(x; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\theta}{\sigma}\right)^2\right] \quad -\infty < x < \infty$$

where σ is known.

Estimate MVB estimator of θ .

Sol, From C.R. inequality we have

$$\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - g(\theta)] \quad \text{--- (1)}$$

likelihood function is

$$L(x_1, \dots, x_n; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x_1-\theta}{\sigma}\right)^2\right] \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x_2-\theta}{\sigma}\right)^2\right]$$

$$\dots \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x_n-\theta}{\sigma}\right)^2\right]$$

$$L = \frac{1}{(2\pi)^{n/2} \sigma^n} \prod_{i=1}^n \exp\left[-\frac{1}{2}\left(\frac{x_i - \theta}{\sigma}\right)^2\right]$$

$$\ln L = -\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{1}{2} \sum_{i=1}^n \left(\frac{x_i - \theta}{\sigma}\right)^2$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{1}{\sigma^2} \sum (x_i - \theta) = \frac{1}{\sigma^2} (\sum x_i - n\theta)$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\sigma^2} (\bar{x} - \theta), \text{ This is of the form } \textcircled{1}$$

Comparing with $\textcircled{1}$ $t = \bar{x}$, $g(\theta) = \theta$

Hence $\bar{x}$ is MVB estimator of θ with variance given by

$$\text{Var}(t) = \frac{g'(\theta)}{A(\theta)} = \frac{1}{A(\theta)}$$

$$\text{Var}(\bar{x}) = \text{Var}(t) = \frac{1}{n/\sigma^2} = \frac{\sigma^2}{n}$$

Example 2, Estimate θ in the Poisson distribution

$$f(x, \theta) = \frac{e^{-\theta} \theta^x}{x!} \quad x = 0, 1, 2, \dots, \infty$$

Sol, likelihood function is

$$L(x_1, \dots, x_n, \theta) = \frac{e^{-\theta} \theta^{x_1}}{x_1!} \cdot \frac{e^{-\theta} \theta^{x_2}}{x_2!} \cdots \frac{e^{-\theta} \theta^{x_n}}{x_n!}$$

$$L = \frac{e^{-n\theta} \theta^{\sum x_i}}{x_1! x_2! \cdots x_n!}$$

$$\ln L = -n\theta + \sum x_i \ln \theta - \ln x_1! - \cdots - \ln x_n!$$

$$\frac{\partial \ln L}{\partial \theta} = -n + \frac{\sum x_i}{\theta}$$

$$= \frac{n}{\theta} \left(\frac{\sum x_i}{n} - \theta \right) = \frac{n}{\theta} (\bar{x} - \theta)$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta} (\bar{x} - \theta)$$

Comparing with

$$\frac{\partial \ln L}{\partial \theta} = A(\theta) (\bar{x} - g(\theta))$$

$\bar{x}$ is MVB estimator of θ

with

$$\text{Var}(\bar{x}) = \frac{1}{A(\theta)} = \frac{\theta}{n}$$

Example 3; Estimate θ in the binomial Dist.

$$f(x, \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \quad x = 0, 1, 2, \dots, n$$

Sol;

If t is MVB estimate of $g(\theta)$ then

$$\frac{\partial \ln L}{\partial \theta} = A(\theta)(t - g(\theta)) \quad \text{--- ①}$$

In the binomial Dist. Same function is taken as likelihood function.

$$L = \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

$$\ln L = \ln \binom{n}{x} + x \ln \theta + (n-x) \ln (1-\theta)$$

$$\frac{\partial \ln L}{\partial \theta} = 0 + \frac{x}{\theta} + \frac{(n-x)(-1)}{(1-\theta)}$$

$$= \frac{x - x\theta - n\theta + x\theta}{\theta(1-\theta)} = \frac{x - n\theta}{\theta(1-\theta)}$$

$$= \frac{n}{\theta(1-\theta)} \left[\frac{x}{n} - \theta \right] \quad \text{--- ②}$$

Comparing ② with ①

$\frac{x}{n}$ is MVB estimate of θ , with variance.

$$\text{Var}\left(\frac{x}{n}\right) = \frac{1}{A(\theta)} = \frac{\theta(1-\theta)}{n}$$

Example 4; Estimate θ in the Normal pop.

$$f(x, \theta) = \frac{1}{\sqrt{2\pi}\theta} e^{-\frac{x^2}{2\theta^2}} \quad -\infty \leq x \leq \infty$$

Sol; If t is MVB estimate of $g(\theta)$, then we know

$$\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - g(\theta)] \quad \text{--- ①}$$

Likelihood function is

$$L(x_1, \dots, x_n, \theta) = \frac{1}{\sqrt{2\pi}\theta} e^{-\frac{x_1^2}{2\theta^2}} \frac{1}{\sqrt{2\pi}\theta} e^{-\frac{x_2^2}{2\theta^2}} \dots \frac{1}{\sqrt{2\pi}\theta} e^{-\frac{x_n^2}{2\theta^2}}$$

$$L = \frac{1}{(2\pi)^{n/2} \theta^n} e^{-\frac{1}{2\theta^2} \sum x_i^2}$$

$$\ln L = -\frac{n}{2} \ln(2\pi) - n \ln \theta - \frac{1}{2\theta^2} \sum x_i^2$$

$$\frac{\partial \ln L}{\partial \theta} = 0 - \frac{n}{\theta} + \frac{\sum x_i^2}{\theta^3}$$

$$= 0$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta^3} \left[\frac{\sum x_i^2}{n} - \theta^2 \right] \quad \text{--- (2)}$$

Comparing (1) and (2) $\frac{1}{n} \sum x_i^2$ is MVB estimator of θ^2 with $\rightarrow$

$$g(\theta) = \theta^2, \quad g'(\theta) = 2\theta$$

$$\rightarrow \text{Var} \left(\frac{1}{n} \sum x_i^2 \right) = \frac{g'(\theta)}{A(\theta)} =$$

$$\text{Var} \left(\frac{1}{n} \sum x_i^2 \right) = \frac{2\theta}{n/\theta^3} = \frac{2\theta^4}{n}$$

Example 5: Estimate θ in Cauchy pop.

$$f(x, \theta) = \frac{1}{\pi} \frac{1}{1+(x-\theta)^2} \quad -\infty \leq x \leq \infty$$

Sol: If t is MVB estimator of $g(\theta)$ then

$$\frac{\partial \ln L}{\partial \theta} = A(\theta) [t - g(\theta)] \quad \text{--- (1)}$$

likelihood function is

$$L = \left(\frac{1}{\pi} \frac{1}{1+(x_1-\theta)^2} \right) \times \left(\frac{1}{\pi} \frac{1}{1+(x_2-\theta)^2} \right) \dots \times \frac{1}{\pi} \frac{1}{1+(x_n-\theta)^2}$$

$$L = \frac{1}{\pi^n} \prod_{i=1}^n \frac{1}{1 + (x_i - \theta)^2}$$

$$\ln L = -n \ln \pi - \ln \sum [1 + (x_i - \theta)^2]$$

$$L = \frac{1}{\pi^n} \frac{1}{\sum [1 + (x_i - \theta)^2]}$$

$$\ln L = -n \ln \pi - \ln \sum [1 + (x_i - \theta)^2]$$

$$\frac{\partial \ln L}{\partial \theta} = 2 \sum \frac{(x_i - \theta)}{1 + (x_i - \theta)^2}$$

This can not be put in to required form, therefore there is no MVB estimator of θ in Cauchy pdf.

If $[t - g(\theta)]$ is proportional to $\frac{\partial \ln L}{\partial \theta}$ then

$$\frac{\partial \ln L}{\partial \theta} = A(\theta) [t - g(\theta)] \quad \text{--- ①}$$

Equation ① determines a condition in the p.d.f (or p.f) under which a MVB estimator of $g(\theta)$ exists. There may still be an estimator of $g(\theta)$, which has uniformly in θ smaller variance than another estimator, we call it a M.V estimator.

Integrating ① we get

$$\ln L = \int \{ A(\theta) t - A(\theta) g(\theta) \} d\theta = A(\theta) t + P(\theta) + R(x_1, \dots, x_n)$$

$$f(x, \theta) = \exp [A(\theta) B(x) + C(x) + D(\theta)] \quad \text{--- ②}$$

$$\text{Where } t = \sum B(x_i), \quad R(x_1, \dots, x_n) = \sum_{i=1}^n C(x_i)$$

$$P(\theta) = n D(\theta)$$

Equation ② is called exponential family of dist.

Generalization of the Cramér - Rao Inequality;

(Bhattacharyya Extension) (B.E)

$$\frac{\partial \ln L}{\partial \theta} = A(\theta) \{t - g(\theta)\}$$

This condition for Π to be an attainable bound is that there is an estimator t for which $(t - g(\theta))$ is a linear function of $\frac{\partial \ln L}{\partial \theta}$. But if no such estimator exist, there may still be an estimator for which $(t - g(\theta))$ is a linear function of $\frac{1}{L} \frac{\partial L}{\partial \theta}$ or $\frac{1}{L} \frac{\partial L}{\partial \theta^2}$ or in general of the higher derivatives of $\ln(L)$.

Statement of B.E;

If t is an unbiased estimator of $g(\theta)$ and if we write

$$L^{(z)} = \frac{\partial^z L}{\partial \theta^z}$$

$$g^{(z)} = \frac{\partial^z g(\theta)}{\partial \theta^z} \text{ and } J_{zp} = E\left(\frac{L^{(z)}}{L} \frac{L^{(p)}}{L}\right) \text{ is a matrix of coefficients}$$

and is non-singular. Then

$$\text{Var}(t) \geq \sum_{z=1}^r \sum_{p=1}^r g^{(p)} J_{zp}^{-1} g^{(z)}$$

Proof; Let us construct the function

$$D_s = t - g(\theta) - \sum_{z=1}^r \alpha_z \frac{L^{(z)}}{L} \quad \text{--- ①}$$

where α_z are the constants to be determined.

we have

$$\int \dots \int L dx_1 \dots dx_n = 1 \quad \text{Diff } r \text{ times}, \text{ differentiating } r \text{ times}$$

we obtain

$$\int \dots \int \frac{L^{(z)}}{L} L dx_1 \dots dx_n = 0$$

$$\int \dots \int \frac{\partial^z}{\partial \theta^z} L dx_1 \dots dx_n = 0$$

$$E\left(\frac{L^{(t)}}{L}\right) = 0 \quad \text{--- (2)}$$

From (1) and (2)

$$E(D_S) = E\left[t - g(\theta)\right] - \sum a_k E\left(\frac{L^{(t)}}{L}\right)$$

$$E(D_S) = E\left[t - g(\theta)\right] \quad \text{Since } t \text{ is an unbiased estimate of } g(\theta) \text{ i.e. } E(t) = g(\theta)$$

$$E(D_S) = 0 \quad \text{--- (3)}$$

The variance of D_S is

$$\begin{aligned} \text{Var}(D_S) &= E(D_S^2) = \int \dots \int D_S^2 L dx_1 \dots dx_n \\ &= \int \dots \int \left\{ \left[t - g(\theta) \right] - \sum_{k=1}^s a_k \frac{L^{(t)}}{L} \right\}^2 L dx_1 \dots dx_n \quad \text{--- (4)} \end{aligned}$$

Now we minimize (4) for variation of a_k .
diff w.r.t a_k

$$2 \int \dots \int \left\{ \left[t - g(\theta) \right] - \sum_{k=1}^s a_k \frac{L^{(t)}}{L} \right\} \frac{L^{(p)}}{L} L dx_1 \dots dx_n = 0 \quad \text{--- (5)}$$

$p = 1, 2, \dots, s$

$$\int \dots \int \left\{ \left[t - g(\theta) \right] \frac{L^{(p)}}{L} \right\} L dx_1 \dots dx_n = \sum_{k=1}^s a_k \int \dots \int \frac{L^{(t)}}{L} \frac{L^{(p)}}{L} L dx_1 \dots dx_n \quad \text{--- (6)}$$

By comparison with the results,

$$E(t) = \int \dots \int t L dx_1 \dots dx_n = g(\theta)$$

the L.H.S. of (6) is

$$\int \dots \int t L^{(p)} dx_1 \dots dx_n - \int \dots \int g(\theta) L^{(p)} dx_1 \dots dx_n = g^{(p)}(\theta)$$

$$\text{The R.H.S. of (6) is} = \sum_{k=1}^s a_k E\left(\frac{L^{(t)}}{L} \frac{L^{(p)}}{L}\right)$$

Making substitution in Eq (6)

$$g^{(p)} = \sum_{t=1}^s a_t E\left(\frac{L^{(t)}}{L} \frac{L^{(p)}}{L}\right) \quad \text{--- (7)} \quad p=1,2,\dots,s$$

$J_{tp} = E\left(\frac{L^{(t)}}{L} \frac{L^{(p)}}{L}\right)$ is non-singular and therefore its inverse exists.

Now $g^{(p)} = \sum_{t=1}^s a_t J_{tp}$, $p=1,2,\dots,s \rightarrow$ a set of linear equations

$$\sum_{p=1}^s g^{(p)} J_{tp}^{-1} = a_t \quad t=1,2,\dots,s \quad \text{--- (8)}$$

Substituting value of a_t in (1)

$$D_s = t - g(\theta) - \sum_{t=1}^s \sum_{p=1}^s g^{(p)} J_{tp}^{-1} \frac{L^{(t)}}{L} \quad \text{--- (9)}$$

Substituting value of a_t in (4)

$$\text{Var}(D_s) = E(D_s^2) = \int \dots \int \left[t - g(\theta) - \sum_{t=1}^s \sum_{p=1}^s g^{(p)} J_{tp}^{-1} \frac{L^{(t)}}{L} \right]^2 L dx_1 \dots dx_n \quad \text{--- (10)}$$

$$= \int \dots \int \left[t - g(\theta) \right]^2 L dx_1 \dots dx_n$$

$$- \int \dots \int t \sum_{t=1}^s \sum_{p=1}^s g^{(p)} J_{tp}^{-1} \frac{L^{(t)}}{L} L dx_1 \dots dx_n$$

$$= \text{Var}(t) - \sum_{t=1}^s \sum_{p=1}^s g^{(p)} J_{tp}^{-1} \int \dots \int t L^{(t)} dx_1 \dots dx_n$$

Because

$$g^{(t)}(\theta) = \int \dots \int t L^{(t)} dx_1 \dots dx_n$$

Therefore

$$\text{Var}(D_0) = \text{Var}(t_0) - \sum_{t=1}^b \sum_{p=1}^b g^{(p)} J_{tp}^{-1} g^{(t)}$$

Variance is always positive hence

$$\text{Var}(t_0) - \sum_{t=1}^b \sum_{p=1}^b g^{(t)} J_{tp}^{-1} g^{(p)} \geq 0$$

$$\text{Var}(t_0) \geq \sum_{t=1}^b \sum_{p=1}^b g^{(p)} J_{tp}^{-1} g^{(t)} \quad \text{Hence proved.}$$

Example; Suppose that $X_1, X_2, \dots, X_n$ are $(n \gg 2)$ independent random variables each $\sim (\mu, \sigma^2)$, where μ is known and $\sigma^2 (\sigma^2 > 0)$ is unknown. Find the CR bound on the variance of any unbiased estimator ϕ_n of σ^2 .

Solution; we have

$$f(x, \sigma^2) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Cramer Rao inequality for unbiased estimators of function of θ is

$$\text{Var}_{\theta}(t_n) \geq \frac{(g'(\theta))^2}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2}$$

If $g(\theta) = \theta$, then

$$\text{Var}_{\theta}(t_n) \geq \frac{1}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2}$$

Now we find $E\left(\frac{\partial \ln L}{\partial \theta}\right)^2$

$$f(x, \sigma^2) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} \quad i=1, 2, \dots, n$$

taking $\ln$

$$\ln f(x, \sigma^2) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (x_i - \mu)^2$$

minimizing w.r.t parameter σ^2 ,

$$\frac{\partial \ln f}{\partial \sigma^2} = -\frac{1}{2\sigma^2} - \frac{1}{2} (-1)(\sigma^2)^{-2} (x_i - \mu)^2 \quad i=1, 2, \dots, n$$

$$= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} (x_i - \mu)^2$$

$$\left(\frac{\partial \ln f}{\partial \sigma^2}\right)^2 = \frac{1}{4\sigma^4} + \frac{1}{4\sigma^8} (x_i - \mu)^4 - \frac{1}{2\sigma^6} (x_i - \mu)^2$$

$$E_{\sigma^2} \left(\frac{\partial \ln f}{\partial \sigma^2}\right)^2 = \frac{1}{4\sigma^4} + \frac{1}{4\sigma^4} E_{\sigma^2} \left(\frac{x_i - \mu}{\sigma}\right)^4 - \frac{1}{2\sigma^4} E_{\sigma^2} \left(\frac{x_i - \mu}{\sigma}\right)^2$$

$$E_{\sigma^2} \left(\frac{\partial \ln f}{\partial \sigma^2}\right)^2 = \frac{1}{4\sigma^4} + \frac{1}{4\sigma^4} E_{\sigma^2} (Z^4) - \frac{1}{2\sigma^4} E_{\sigma^2} (Z^2) \quad \text{--- (1)}$$

$$E(z^2) = \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z^2 e^{-\frac{1}{2}z^2} dz$$

putting $\frac{z^2}{2} = y$, $z dz = dy$

$$E(z^2) = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} \sqrt{2y} e^{-y} dy = \frac{2}{\sqrt{\pi}} \int_0^{\infty} y^{\frac{3}{2}-1} e^{-y} dy$$

$$= \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \frac{2}{\sqrt{\pi}} \cdot \frac{1}{2} \sqrt{\pi} = 1$$

$$\boxed{E(z^2) = 1}$$

$$E(z^4) = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z^4 e^{-\frac{1}{2}z^2} dz$$

after substitution $E(z^4) = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} (2y)^{3/2} e^{-y} dy$

$$E(z^4) = \frac{4}{\sqrt{\pi}} \int_0^{\infty} y^{3/2} e^{-y} dy = \frac{4}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 3$$

$$E(z^4) = \frac{4}{\sqrt{\pi}} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} = 3$$

Now from ① $E_{\sigma^2} \left(\frac{\partial \ln f}{\partial \sigma^2} \right)^2 = \frac{1}{4\sigma^4} + \frac{3}{4\sigma^4} - \frac{1}{2\sigma^4}$

with likelihood function $E_{\sigma^2} \left(\frac{\partial \ln f}{\partial \sigma^2} \right)^2 = \frac{1}{2\sigma^4}$

$$E_{\sigma^2} \left(\frac{\partial \ln L}{\partial \sigma^2} \right)^2 = \frac{n}{2\sigma^4}$$

By using CR inequality $\text{Var}_{\sigma^2}(\Phi_n) \geq E_{\sigma^2} \left(\frac{\partial \ln L}{\partial \sigma^2} \right)^2$

$$\text{Var}_{\sigma^2}(\Phi_n) \geq \frac{2\sigma^4}{n}$$

an unbiased estimator of σ^2 (when μ is known) is

$$\hat{\sigma}^2 = \frac{1}{(n-1)} \sum_{i=1}^n (x_i - \mu)^2$$

Hence $\text{Var}_{\sigma^2}(\Phi_n) = \frac{2\sigma^4}{n}$ is the minimum variance.

Uniqueness of M.V Estimators

We show that if a M.V estimator exists, it is essentially unique, irrespective of whether any bound is attained

Proof; Let t_1 and t_2 be M.V unbiased estimators of $g(\theta)$, each with variance V , consider the new estimator

$$t_3 = \frac{1}{2}(t_1 + t_2)$$

which also estimates $g(\theta)$ with variance

$$\text{Var}(t_3) = \frac{1}{4} [\text{Var}(t_1) + \text{Var}(t_2) + 2 \text{Cov}(t_1, t_2)] \quad \text{--- ①}$$

$$\text{Cov}(t_1, t_2) = \int \dots \int (t_1 - g)(t_2 - g) L dx_1 \dots dx_n$$

on R.H.S applying Schwarz inequality

$$(\sum XY)^2 \leq (\sum X^2)(\sum Y^2)$$

using inequality

$$\begin{aligned} & \left[\int \dots \int (t_1 - g)(t_2 - g) L dx_1 \dots dx_n \right]^2 \\ & \leq \left[\int \dots \int (t_1 - g)^2 L dx_1 \dots dx_n \int \dots \int (t_2 - g)^2 L dx_1 \dots dx_n \right]^{1/2} \\ & \leq [\text{Var}(t_1) \text{Var}(t_2)]^{1/2} \\ & \leq (V V)^{1/2} \end{aligned}$$

$$\text{Cov}(t_1, t_2) \leq V \quad \text{--- ②}$$

From ① and ②

$$\text{Var}(t_3) \leq \frac{1}{4} [V + V + 2V] = V$$

which contradicts the assumption that t_1, t_2 have M.V unless the equality holds. This implies that equality holds in ②. This can be so if

$$(t_1 - g) = K(\theta)(t_2 - g) \quad \text{--- ③}$$

That is the variables are proportional.

Multiplying both sides of ③ by $(t_2 - g)$

$$(t_1 - g)(t_2 - g) = K(\theta)(t_2 - g)^2$$

taking expectation on both sides

$$E(t_1 - g)(t_2 - g) = K(\theta) E(t_2 - g)^2$$

$$\text{Cov}(t_1, t_2) = K(\theta) V$$

Since the equality holds in ②

$$\text{ie } \text{Cov}(t_1, t_2) = V$$

$$\text{We have } V = K(\theta) V$$

$$\text{or } K(\theta) = 1$$

From ③ $t_1 = t_2$ identically

Thus MV estimator is unique.

Inequality for the Correlation Coefficient between any two Estimators;

If E_1 and E_2 are the efficiencies of two unbiased estimators of t_1 and t_2 and ρ is the correlation coefficient between t_1 and t_2 then

$$\sqrt{E_1 E_2} - \sqrt{(1-E_1)(1-E_2)} \leq \rho \leq \sqrt{E_1 E_2} + \sqrt{(1-E_1)(1-E_2)}$$

Proof; Suppose that t is the MV unbiased estimator of $g(\theta)$ with variance V and that t_1 and t_2 are any two unbiased estimators of $g(\theta)$ with finite variance.

Consider a new estimator

$$t_3 = a t_1 + (1-a) t_2$$

This will also estimate $g(\theta)$ with variance

$$\text{Var}(t_3) = a^2 \text{Var}(t_1) + (1-a)^2 \text{Var}(t_2) + 2a(1-a) \text{Cov}(t_1, t_2) \quad \text{--- ①}$$

$$\text{We take } \text{Var}(t_1) = K_1 V, \text{Var}(t_2) = K_2 V, \text{ where } K_1, K_2 > 1$$

Now making this substitution in ①

$$\text{Var}(t_3) = a^2 K_1 V + (1-a)^2 K_2 V + 2a(1-a) \text{Cov}(t_1, t_2)$$

Dividing both sides by V :

$$\frac{\text{Var}(t_3)}{V} = a^2 K_1 + (1-a)^2 K_2 + \frac{2a(1-a) \text{Cov}(t_1, t_2)}{V} \quad \text{--- ②}$$

Correlation coefficient between t_1 and t_2 is

$$\rho = \frac{\text{Cov}(t_1, t_2)}{\sqrt{\text{Var}(t_1) \text{Var}(t_2)}} = \frac{\text{Cov}(t_1, t_2)}{(K_1 K_2)^{1/2} V}$$

$$\text{Cov}(t_1, t_2) = \rho (K_1 K_2)^{1/2} V$$

Making substitution in ②

$$\frac{\text{Var}(t_3)}{V} = a^2 K_1 + (1-a)^2 K_2 + 2a(1-a) \rho (K_1 K_2)^{1/2}$$

$$\text{But } \text{Var}(t_3) \gg V \implies \frac{\text{Var}(t_3)}{V} \gg 1$$

because V is the variance of MV unbiased estimator.

Therefore

$$a^2 K_1 + (1-a)^2 K_2 + 2a(1-a) \rho (K_1 K_2)^{1/2} \gg 1 \quad \text{--- ③}$$

Equation ③ may be rearranged as a quadratic inequality in a .

$$\begin{aligned} & a^2 K_1 + K_2 + a^2 K_2 - 2a K_2 + 2a \rho (K_1 K_2)^{1/2} - 2a^2 \rho (K_1 K_2)^{1/2} - 1 \gg 0 \\ & (a^2 K_1 + a^2 K_2 - 2a^2 \rho (K_1 K_2)^{1/2}) + (2a \rho (K_1 K_2)^{1/2} - 2a K_2) + (K_2 - 1) \gg 0 \\ & a^2 (K_1 + K_2 - 2\rho \sqrt{K_1 K_2}) + 2a (\rho \sqrt{K_1 K_2} - K_2) + (K_2 - 1) \gg 0 \end{aligned}$$

Since the roots of the quadratic are complex or equal, the discriminant of LHS can not be positive.

So discriminant is

$$4 [\rho \sqrt{K_1 K_2} - K_2]^2 - 4 [K_1 + K_2 - 2\rho \sqrt{K_1 K_2}] (K_2 - 1) \leq 0$$

or

$$[\rho \sqrt{K_1 K_2} - K_2]^2 - [K_1 + K_2 - 2\rho \sqrt{K_1 K_2}] (K_2 - 1) \leq 0$$

$$\begin{aligned} & \rho^2 (K_1 K_2) - 2\rho \sqrt{K_1 K_2} K_2 + K_2^2 - K_1 K_2 - K_2^2 + 2\rho \sqrt{K_1 K_2} K_2 \\ & + K_1 + K_2 - 2\rho \sqrt{K_1 K_2} \leq 0 \end{aligned}$$

$$P^2(K_1 K_2) - 2P\sqrt{K_1 K_2} \leq K_1 K_2 - K_1 - K_2 + 1$$

Adding 1, on both sides

$$P^2(K_1 K_2) - 2P\sqrt{K_1 K_2} + 1 \leq K_1 K_2 - K_1 - K_2 + 1$$

$$(P\sqrt{K_1 K_2} - 1)^2 \leq (K_1 - 1)(K_2 - 1)$$

$$(P\sqrt{K_1 K_2} - 1) \leq \pm \sqrt{(K_1 - 1)(K_2 - 1)}$$

$$P\sqrt{K_1 K_2} \leq 1 \pm \sqrt{(K_1 - 1)(K_2 - 1)}$$

$$P \leq \frac{1}{\sqrt{K_1 K_2}} \pm \sqrt{\frac{(K_1 - 1)(K_2 - 1)}{K_1 K_2}}$$

$$\frac{1}{\sqrt{K_1 K_2}} - \sqrt{\frac{(K_1 - 1)(K_2 - 1)}{K_1 K_2}} \leq P \leq \frac{1}{\sqrt{K_1 K_2}} + \sqrt{\frac{(K_1 - 1)(K_2 - 1)}{K_1 K_2}}$$

$$\frac{1}{\sqrt{K_1 K_2}} - \sqrt{\left(1 - \frac{1}{K_1}\right)\left(1 - \frac{1}{K_2}\right)} \leq P \leq \frac{1}{\sqrt{K_1 K_2}} + \sqrt{\left(1 - \frac{1}{K_1}\right)\left(1 - \frac{1}{K_2}\right)}$$

$$E_1 = \frac{V}{\text{Var}(t_1)} = \frac{V}{K_1 V} = \frac{1}{K_1} \quad E_2 = \frac{V}{\text{Var}(t_2)} = \frac{V}{K_2 V}$$

$$E_1 = \frac{1}{K_1} \quad , \quad E_2 = \frac{1}{K_2}$$

$$K_1 = \frac{1}{E_1} \quad \sqrt{E_1 E_2} = \frac{1}{\sqrt{K_1 K_2}}$$

Hence

$$\sqrt{E_1 E_2} - \sqrt{(1 - E_1)(1 - E_2)} \leq P \leq \sqrt{E_1 E_2} + \sqrt{(1 - E_1)(1 - E_2)}$$

continue from the above.

If either E_1 or $E_2 = 1$, that is either t_1 or t_2 is the MV estimator of $g(\theta)$, then (1) becomes

$$\text{If } E_2 = 1 \quad \sqrt{E_1} \leq P \leq \sqrt{E_1}$$

$$\text{If } E_1 = 1, \quad \sqrt{E_1} \leq \rho \leq \sqrt{E_1}$$

$$\rho = \sqrt{E}$$

where E is the reciprocal of the relative variance of the other estimator and in large samples is usually called the efficiency.

Theorem: Show that if t is the M.V unbiased estimator and u is another unbiased estimator of θ , the covariance of t and $(u-t)$ is zero, i.e.

$$\text{Cov}(t, u-t) = 0$$

Proof: If u is any unbiased finite variance estimator and t is a M.V. estimator, then

$$E = \frac{\text{Var}(t)}{\text{Var}(u)} \quad \text{--- ①}$$

$$\sqrt{E} = \frac{\text{Cov}(u, t)}{\sqrt{\text{Var}(u) \text{Var}(t)}} \quad \text{--- ②}$$

From ① and ② $\text{Cov}(u, t) = \sqrt{E} \sqrt{\text{Var}(u) \text{Var}(t)}$

$$= \sqrt{\frac{\text{Var}(t)}{\text{Var}(u)}} \sqrt{\text{Var}(u) \text{Var}(t)}$$

$$\text{Cov}(u, t) = \text{Var}(t) \quad \text{--- ③}$$

NBW

$$\begin{aligned} \text{Cov}(t, u-t) &= E[t(u-t)] - E(t)E(u-t) \\ &= E(tu - t^2) - E(t)[E(u) - E(t)] \\ &= E(tu) - E(t^2) - E(t)E(u) + [E(t)]^2 \\ &= [E(tu) - E(t)E(u)] - [E(t^2) - (E(t))^2] \end{aligned}$$

$$\text{Cov}(t, u-t) = \text{Cov}(t, u) - \text{Var}(t)$$

Making substitution from ③

$$\text{Cov}(t, u-t) = \text{Var}(t) - \text{Var}(t)$$

$$\text{Cov}(t, u-t) = 0$$

Hence t has zero covariance with every finite variance function having zero expectation.

Conversely; if $\text{Cov}(t, u-t) = 0$ for all u with finite variance and $E(u) = E(t)$, then t is the M.V. estimator.

Proof: We are given

$$\text{Cov}(t, u-t) = 0$$

$$\text{or } E[t(u-t)] - E(t)E(u-t) = 0$$

$$E(tu) - E(t^2) - E(t)[E(u) - E(t)] = 0$$

$$\text{Because } E(u) = E(t) \quad \text{and} \quad \text{Cov}(t, u) = E(tu) - E(t)E(u)$$

$$\text{therefore } E(tu) - E(t^2) = 0$$

$$[\text{Cov}(t, u) + E(t)E(u)] - [E(t^2) + (E(t))^2] = 0$$

$$\text{Cov}(t, u) + (E(t))^2 - E(t^2) - (E(t))^2 = 0$$

$$\text{Cov}(t, u) = \text{Var}(t)$$

$$\text{Corr}(t, u) = \frac{\text{Cov}(t, u)}{\sqrt{\text{Var}(t)} \sqrt{\text{Var}(u)}}$$

$$\text{From (1)} \quad \text{Corr}(t, u) = \frac{\text{Var}(t)}{\sqrt{\text{Var}(t)} \sqrt{\text{Var}(u)}} = \sqrt{\frac{\text{Var}(t)}{\text{Var}(u)}} \leq 1$$

$$\sqrt{\frac{\text{Var}(t)}{\text{Var}(u)}} \leq 1 \quad \text{or} \quad \text{Var}(t) \leq \text{Var}(u)$$

i.e. t is M.V. estimator.

Sufficient Statistic (By Conditional Density)

A statistic T is said to be sufficient if it utilize all the information in a sample relevant to the estimation of the population parameters θ .

Definition; Let $x_1, x_2, \dots, x_n$ denote a random sample of size n from a distribution that has p.d.f. $f(x; \theta)$, $\theta \in \Omega$. Let

$T = u_1(x_1, \dots, x_n)$ be a statistic whose p.d.f is $g_1(t_1; \theta)$. Then T is a sufficient statistic for θ if and only if

$$f(x_1, \dots, x_n / t_1) = \frac{f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta)}{g_1[u_1(x_1, x_2, \dots, x_n); \theta]}$$

$$= h(x_1, x_2, \dots, x_n)$$

where $h(x_1, \dots, x_n)$ does not depend upon $\theta \in \Omega$ for every fixed value of $t_1 = u_1(x_1, \dots, x_n)$

Example; If $x_1, \dots, x_n$ are independent Bernoulli random variables with same parameter θ , show that $t = \sum_{i=1}^n x_i$ is a sufficient statistic for θ .

$$f(x, \theta) = \theta^{x_i} (1-\theta)^{1-x_i} \quad \text{for } x_i = 0, 1$$

Sol;

The joint distribution of $x_1, x_2, \dots, x_n$ is

$$f(x_1, \dots, x_n; \theta) = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta)$$

$$= \prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i}$$

$$f(x_1, \dots, x_n; \theta) = \theta^{\sum x_i} (1-\theta)^{n - \sum x_i}$$

$\sum x_i = X$ has the binomial dist.

$$b(x; n, \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

$$t = \frac{\sum x_i}{n}$$

$$g_1(t, \theta) = \binom{n}{nt} \theta^{nt} (1-\theta)^{n-nt}$$

$$t = 0, \frac{1}{n}, \dots, 1$$

$$f\left(\frac{x_1, \dots, x_n}{t}\right) = \frac{1}{\binom{n}{nt}}$$

Example: Let $Y_1 < Y_2 < \dots < Y_n$ denote the order statistics of a random sample $X_1, X_2, \dots, X_n$ from the dist that has p.d.f.

$$f(x; \theta) = \frac{1}{c} \cdot (x - \theta) \quad \theta < x < \infty, \quad -\infty < \theta < \infty$$

$$= 0 \text{ otherwise.}$$

Show that Y_1 is a sufficient statistic for θ

Sol;

Dist. of smallest observation is

$$g(y_i) = n[1 - F(y_i)]^{n-1} f(y_i)$$

$$F(y) = \int_0^y e^{-y} e^{+y} dy = e^0 \int_0^y e^{-y} dy$$

$$= e^0 \left| \frac{e^{-y}}{-1} \right|_0^y$$

$$F(y) = e^0 \left(-(\bar{e}^{-y} - \bar{e}^{-0}) \right) = 1 - \bar{e}^0 \bar{e}^{-y}$$

$$F(y_1) = 1 - e^{-(0-y_1)}$$

$$g(y_1) = n \left[1 - 1 + e^{-(y_1 - \theta)} \right]^{n-1} e^{-(y_1 - \theta)}$$

$$= n e^{n\theta} e^{-ny_1} e^{+\theta} e^{-y_1}$$

$$g(y_1) = n e^{-n(y_1 + \theta)} \quad 0 \leq y_1 < \infty$$

$$= 0 \text{ otherwise}$$

$$f\left(\frac{x_1, \dots, x_n}{t}\right) = \frac{f(x_1, \dots, x_n; \theta)}{g(t)}$$

$$\begin{aligned} F\left(\frac{x_1, \dots, x_n}{t}\right) &= \frac{e^{-(x_1 - \theta)} \dots e^{-(x_n - \theta)}}{n e^{-ny_1 + n\theta}} \\ &= \frac{e^{-\sum x_i + n\theta}}{n e^{-ny_1 + n\theta}} \\ &= \frac{e^{-\sum x_i}}{n e^{-ny_1}} \end{aligned}$$

it is independent of θ , therefore
 y_1 is sufficient statistic for θ .

Example: Let $x_1, \dots, x_n$ be independent random variables from the p.d.f

$$f(x; \theta) = \frac{1}{\theta} \quad 0 \leq x \leq \theta$$

Show that $Y_n = \max(x_i)$ is a sufficient statistic for θ , but $Y_1 = \min(x_i)$ is not a sufficient statistic for θ .

Sol: Joint dist of $x_1, \dots, x_n$ is

$$f(x_1, \dots, x_n, \theta) = \frac{1}{\theta^n}$$

2)

pdf of largest observation Y_n is

$$g(y_n) = n[F(y_n)]^{n-1} f(y_n)$$

$$f(y) = \frac{1}{\theta} \quad , \quad F(y) = \int_0^y \frac{1}{\theta} dy = \frac{1}{\theta} y$$

$$g(y_n) = n\left[\frac{1}{\theta} y_n\right]^{n-1} \frac{1}{\theta} = \frac{n y_n^{n-1}}{\theta^n}$$

$$f\left(\frac{x_1, \dots, x_n}{y_n}\right) = \frac{f(x_1, \dots, x_n; \theta)}{g(y_n)}$$

$$= \frac{1}{\theta^n} \frac{\theta^n}{n y_1^{n-1}}$$

$f\left(\frac{x_1, \dots, x_n}{y_1}\right) = \frac{1}{n y_1^{n-1}}$ It is independent of θ , therefore y_1 is a sufficient statistic for θ .

(ii) p.d.f of smallest observation is

$$g(y_1) = n[1-F(y_1)]^{n-1} f(y_1) \\ = n\left[1 - \frac{1}{\theta} y_1\right]^{n-1} \frac{1}{\theta}$$

$$g(y_1) = n[\theta - y_1]^{n-1} \frac{1}{\theta^n}$$

$$f\left(\frac{x_1, \dots, x_n}{y_1}\right) = \frac{1}{\theta^n} \frac{\theta^n}{n[\theta - y_1]^{n-1}} \\ = \frac{1}{n[\theta - y_1]^{n-1}}$$

It is not independent of θ , therefore y_1 is ^{not} a sufficient statistic for θ .

Neyman Factorization Theorem

Let $x_1, \dots, x_n$ denote a random sample from a distribution that has the p.d.f $f(x; \theta)$, $\theta \in \Omega$. The statistic $T_1 = U_1(x_1, \dots, x_n)$ is a sufficient statistic for θ if and only if we can find two non-negative functions h and k such that

$$f(x_1; \theta) \dots f(x_n; \theta) = h[U_1(x_1, \dots, x_n); \theta] K(x_1, \dots, x_n) \quad \text{--- (1)}$$

where, for every fixed value of $T_1 = U_1(x_1, \dots, x_n)$, $K(x_1, \dots, x_n)$ does not depend upon θ .

Proof; We shall prove the theorem when the random variables are of the continuous type. Assume the factorization in the theorem. We make the one to one transformation,

$$t_1 = U_1(x_1, \dots, x_n), \quad t_2 = U_2(x_1, \dots, x_n) \dots \dots \dots \\ \dots \dots \dots t_n = U_n(x_1, \dots, x_n)$$

Having Inverse functions

$$x_1 = w_1(t_1, \dots, t_n), x_2 = w_2(t_1, \dots, t_n), \dots, x_n = w_n(t_1, \dots, t_n)$$

$x_1 = w_1(t_1, \dots, t_n), x_2 = w_2(t_1, \dots, t_n) \dots x_n = w_n(t_1, \dots, t_n)$
and the Jacobian J .

The joint p.d.f of the statistic $t_1, \dots, t_n$ is

$$g(t_1, t_2, \dots, t_n; \theta) = h(t_1; \theta) K(w_1, \dots, w_n) |J|$$

where

$$w_i = w_i(t_1, t_2, \dots, t_n) \quad i = 1, 2, \dots, n$$

The p.d.f of t_1 is defined by

$$\begin{aligned} g_1(t_1; \theta) &= \int \dots \int g(t_1, \dots, t_n; \theta) dt_2 \dots dt_n \\ &= \int \dots \int h(t_1; \theta) K(w_1, \dots, w_n) |J| dt_2 \dots dt_n \\ &= h(t_1; \theta) \int \dots \int K(w_1, \dots, w_n) |J| dt_2 \dots dt_n \end{aligned}$$

The function K for every fixed value of $t_1 = u_1(x_1, \dots, x_n)$ does not depend on θ . So θ is neither introduced in the Jacobian nor in the limits of integration.

Hence the $(n-1)$ fold integral on the R.H.S is a function of t_1 , so this is $m(t_1)$. Thus

$$g_1(t_1; \theta) = h(t_1; \theta) m(t_1) \quad \text{--- (2)}$$

If $m(t_1) = 0$, then $g_1(t_1; \theta) = 0$.

If $m(t_1) > 0$ we can write

$$h(t_1; \theta) = \frac{g_1(t_1; \theta)}{m(t_1)}$$

$$h[u_1(x_1, \dots, x_n); \theta] = \frac{g_1[u_1(x_1, \dots, x_n); \theta]}{m[u_1(x_1, \dots, x_n)]} \quad \text{--- (3)}$$

Making substitutions from (3) into (1), then assumed factorization becomes

$$\begin{aligned} f(x_1, \theta) \dots f(x_n, \theta) &= h[u_1(x_1, \dots, x_n); \theta] K(x_1, \dots, x_n) \\ &= \frac{g_1[u_1(x_1, \dots, x_n); \theta]}{m[u_1(x_1, \dots, x_n)]} K(x_1, \dots, x_n) \end{aligned}$$

then

$$\frac{f(x_1, \theta) \dots f(x_n, \theta)}{g_1[u_1(x_1, \dots, x_n); \theta]} = \frac{K(x_1, \dots, x_n)}{m[u_1(x_1, \dots, x_n)]}$$

Neither the function K nor m depends upon θ , then in accordance with the definition of sufficiency, statistic t , is a sufficient statistic for the parameter θ .

Result, If t is a sufficient statistic for θ , any one to one function of t , $u(t)$ will also be sufficient.

In a normal density, $\sum x_i$ is sufficient for the pop. mean μ . $\bar{X} = \frac{1}{n} \sum x_i$ (a function of $\sum x_i$) is also sufficient for μ .

Example; let $x_1, \dots, x_n$ denote a random sample from the θ dist. with p.d.f

$$\begin{aligned} f(x; \theta) &= \theta x^{\theta-1} & 0 < x < 1 & \quad 0 < \theta \\ &= 0 & \text{otherwise} \end{aligned}$$

Show that Product $u_1(x_1, \dots, x_n) = x_1 \cdot x_2 \cdot \dots \cdot x_n$ is a sufficient statistic for θ .

Sol; The joint p.d.f. of $x_1, x_2, \dots, x_n$ is

$$\begin{aligned} f(x_1; \theta) \dots f(x_n; \theta) &= \theta x_1^{\theta-1} \theta x_2^{\theta-1} \dots \theta x_n^{\theta-1} \\ &= \theta^n (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{\theta-1} \left(\frac{1}{x_1 \cdot x_2 \cdot \dots \cdot x_n} \right) \end{aligned}$$

It

$$h(u_1(x_1, \dots, x_n); \theta) = \theta^n (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{\theta-1}$$

and

$$K(x_1, x_2, \dots, x_n) = \frac{1}{x_1 \cdot x_2 \cdot \dots \cdot x_n}$$

Since $K(x_1, x_2, \dots, x_n)$ does not depend upon θ , the product $x_1 \cdot x_2 \cdot \dots \cdot x_n$ is a sufficient statistic for θ .

Example; let $X_1, X_2, \dots, X_n$ be a random sample from the displaced exponential distribution

$$f(x; \theta) = e^{-(x-\theta)} \quad 0 < x < \infty, \quad -\infty < \theta < \infty$$

$$= 0 \text{ elsewhere}$$

Show that first order statistic Y_1 is a sufficient statistic for θ .

Sol; The joint p.d.f of $X_1, X_2, \dots, X_n$ is

$$f(x_1; \theta) \dots f(x_n; \theta) = e^{-(x_1-\theta)} \dots e^{-(x_n-\theta)} \prod_{i=1}^n I_{(\theta, \infty)}(x_i)$$

$$= e^{-\sum_{i=1}^n (x_i - \theta)} \prod_{i=1}^n I_{(\theta, \infty)}(x_i)$$

$$= e^{-\sum_{i=1}^n x_i} e^{n\theta} \prod_{i=1}^n I_{(\theta, \infty)}(x_i)$$

let

$$K(x_1, \dots, x_n) = e^{-\sum_{i=1}^n x_i}, \quad h(u, (x_1, \dots, x_n); \theta) = e^{n\theta} \prod_{i=1}^n I_{(\theta, \infty)}(x_i)$$

$\Rightarrow$

$$\text{Thus we have } \frac{e^{-\sum_{i=1}^n (x_i - \theta)}}{e^{-\sum_{i=1}^n x_i}} = \frac{e^{n\theta}}{e^{n \min(x_i)}} = \frac{e^{n\theta}}{e^{n Y_1}}$$

Which is free of θ for each fixed $Y_1 = \min(x_i)$, since $Y_1 \leq x_i$, $i=1, 2, \dots, n$. That is neither the formula nor the domain of the resulting ratio depends upon θ , and hence the first order statistic Y_1 is a sufficient statistic for θ .

Sufficient Statistics for Several Parameters.

Definition for Jointly Sufficient Statistics;

Let $X_1, \dots, X_n$ be a random sample from the p.d.f $f(x; \omega)$. The statistics $T_1, T_2, \dots, T_r$ are defined to be jointly sufficient if and only if the conditional distribution of $X_1, \dots, X_n$ given $T_1 = t_1, \dots, T_r = t_r$ does not depend on ω .

Theorem: If $T_1 = u_1(x_1, \dots, x_n), \dots, T_k = u_k(x_1, \dots, x_n)$ is a set of jointly sufficient statistics, then any set of one to one functions or transformations of $T_1, T_2, \dots, T_k$ is also jointly sufficient.

For example if $\sum x_i$ and $\sum x_i^2$ are jointly sufficient, then

$\bar{x}$ and $\sum (x_i - \bar{x})^2$ are also jointly sufficient.

$\bar{x}^2$ and $\sum (x_i - \bar{x})^2$ are not jointly sufficient.

Factorization Theorem (Jointly Sufficient Statistics)

Let $x_1, \dots, x_n$ be a random sample of size n from the p.d.f $f(x; \theta)$, where θ may be a vector. A set of jointly sufficient statistics

$T_1 = u_1(x_1, \dots, x_n), \dots, T_k = u_k(x_1, \dots, x_n)$ is jointly sufficient iff the joint density of $x_1, \dots, x_n$ can be factorized as

$$f(x_1, \dots, x_n; \theta) = h[u_1(x_1, \dots, x_n), \dots, u_k(x_1, \dots, x_n)] \cdot K(x_1, \dots, x_n)$$

Where the function K is non negative and does not involve the parameter θ .

The function h is non negative and depends on $x_1, \dots, x_n$ only through the functions $u_1(x_1, \dots, x_n), \dots, u_k(x_1, \dots, x_n)$.

Example: Let $x_1, x_2, \dots, x_n$ be a random sample from the density $N(\mu, \sigma^2)$. Then show that the statistics $T_1 = \sum x_i$, $T_2 = \sum x_i^2$ are jointly sufficient for (μ, σ^2) .

Sol: The joint density of $x_1, \dots, x_n$ is

$$\begin{aligned} f(x_1; \theta) \dots f(x_n; \theta) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x_1 - \mu}{\sigma}\right)^2\right] \dots \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x_n - \mu}{\sigma}\right)^2\right] \\ &= \frac{1}{(2\pi)^{n/2} \sigma^n} \exp\left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma}\right)^2\right] \\ &= \frac{\sigma^{-n}}{(2\pi)^{n/2}} \exp\left[-\frac{1}{2\sigma^2} \sum (x_i^2 - 2\mu x_i + \mu^2)\right] \\ &= \frac{\sigma^{-n}}{(2\pi)^{n/2}} \exp\left[-\frac{1}{2\sigma^2} (\sum x_i^2 - 2\mu \sum x_i + n\mu^2)\right] \times (1) \end{aligned}$$

where $h(u_1(x_1, \dots, x_n), \dots, u_k(x_1, \dots, x_n)) = \exp\left[-\frac{1}{2\sigma^2} (\sum x_i^2 - 2\mu \sum x_i + n\mu^2)\right]$ and $K(x_1, x_2, \dots, x_n) = 1$

Hence $T_1 = \sum X_i$, $T_2 = \sum X_i^2$ are jointly sufficient for (μ, σ^2) .

Therefore one to one functions...

$\bar{X} = \frac{1}{n} \sum X_i$ and $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ are also jointly sufficient.

Example; let $X_1, \dots, X_n$ be a random sample from a uniform dist over the interval (θ_1, θ_2) .

$$f(x; \theta_1, \theta_2) = \frac{1}{\theta_2 - \theta_1} \quad \theta_1 < x < \theta_2$$

$$= \frac{1}{(\theta_2 - \theta_1)} I_{(\theta_1, \theta_2)}(x)$$

Show that y_1 and y_n are jointly sufficient for θ_1 and θ_2 .

Sol: The joint density of $x_1, \dots, x_n$ is

$$f(x_1, \dots, x_n; \theta_1, \theta_2) = \prod_{i=1}^n \frac{1}{(\theta_2 - \theta_1)} I_{(\theta_1, \theta_2)}(x_i)$$

$$= \frac{1}{(\theta_2 - \theta_1)^n} \prod_{i=1}^n I_{(\theta_1, \theta_2)}(x_i)$$

$$= \frac{1}{(\theta_2 - \theta_1)^n} I_{(\theta_1, y_n)}(y_1) I_{(y_1, \theta_2)}(y_n) (1)$$

$$\theta_1 < x_1, \dots, x_n < \theta_2$$

$$\theta_1 < y_1, \dots, y_n < \theta_2$$

$$\theta_1 < y_1 < y_n < \theta_2$$

$$h[u_1(x_1, \dots, x_n), u_2(x_1, \dots, x_n)] \quad \text{where } K(x_1, \dots, x_n) = 1$$

Because K does not involve parameters θ_1, θ_2 and h depends on $x_1, \dots, x_n$, therefore function of $y_1, \dots, y_n$ hence y_1, y_n are jointly sufficient statistics for parameters.

Example; Let $X_1, \dots, X_n$ be a random sample from a uniform dist over the interval $(\theta, \theta+1)$.

then show that y_1, y_n are jointly sufficient.

Sol: $f(x; \theta, \theta+1) = \frac{1}{\theta+1 - \theta} = 1 \quad \theta < x < \theta+1$

$$f(x_1, \dots, x_n; \theta, \theta+1) = \frac{I_{(\theta, y_n)}(y_1)}{(\theta, y_n)} \frac{I_{(y_1, \theta+1)}(y_n)}{(y_1, \theta+1)}$$

$$\theta < y_1 < y_n < \theta+1$$

$$\text{Because } \frac{f(x_1; \theta, \theta+1) f(x_2; \theta, \theta+1) \dots f(x_n; \theta, \theta+1)}{g(u_1(x_1, \dots, x_n), u_2(x_1, \dots, x_n))} = 1$$

$$\text{where } K(x_1, \dots, x_n) = 1$$

Hence y_1, y_n are jointly sufficient for $\theta, \theta+1$.

Example; For uniform dist with integral over $(0, \theta)$. Show that y_n is sufficient for θ . $\theta_1 = \theta, \theta_2 = \theta$

Sol;
$$f(x; \theta_1, \theta_2) = \frac{1}{\theta} \quad 0 < x < \theta$$

$$= \frac{1}{\theta} \frac{I_{(0, \theta)}(x)}{I_{(0, \theta)}(x)}$$

The joint p.d.f of $x_1, \dots, x_n$ is

$$f(x_1, \dots, x_n; \theta) = \frac{1}{\theta^n} \prod_{i=1}^n I_{(0, \theta)}(x_i)$$

$$= \underbrace{\frac{1}{\theta^n} I_{(0, \theta)}(y_n)}_K \underbrace{I_{(0, y_n)}(y_1)}_K$$

Because $K(x_1, \dots, x_n) = I_{(0, y_n)}(y_1)$ does not depend on θ , therefore y_n is sufficient for parameter θ

Sufficiency and M.V.

$$L(x_1, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta)$$

$$= h(t, \theta) K(x_1, \dots, x_n)$$

Taking log of both sides and differentiating w.r.t. θ

$$\frac{\partial \ln L}{\partial \theta} = \frac{\partial \ln h(t, \theta)}{\partial \theta} \quad \text{--- ①} \quad t \text{ is sufficient for } \theta$$

Comparing ① with $\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - g(\theta)]$ --- ② t is MVB estimator of $g(\theta)$

(Eq ② is a condition that $\forall$ MVB estimator exists)

Such an estimator exist only if there is a sufficient statistic. In fact ② is a special case of ①, when

$$\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - g(\theta)] \quad \text{--- ③}$$

A sufficient statistic provides a MVB estimator where there is one.

Factorization Theorem;

Factorization theorem is not useful for proving that the statistic is not sufficient.

A statistic is not sufficient if any conditional distribution given to depend on θ .

Example; let $x_1, \dots, x_n$ be independent random variables each uniform in $[-\theta, \theta]$ for some $\theta > 0$. Find a sufficient statistic for θ .

Sol;



Minimal Sufficient Statistics

There are more than one set of sufficient statistics eg in sampling from a normal distribution with both mean and variance unknown, there are three set of joint sufficient statistics.

- 1, The sample $x_1, x_2, \dots, x_n$.
- 2, The ordered observations $y_1, y_2, \dots, y_n$.
- 3, $(\bar{x}, s^2)$ is sufficient for (μ, σ^2) .

Definition; A set of jointly sufficient statistics is defined to be minimal iff it is a function of every other set of sufficient statistics.

$$T(x_1, \dots, x_n) = K[t(x_1, \dots, x_n)]$$

Exponential Family OR Exponential Class;

Many of the parametric families of distributions are member of the exponential family.

Definition; A one parameter family (θ is unidimensional) of densities $f(x; \theta)$ that can be expressed as

$$f(x; \theta) = a(\theta) b(x) \exp[c(\theta) d(x)] \quad \text{for } -\infty < x < \infty$$

for all $\theta \in \Omega$ (parameter space) and for a smaller choice of functions $a(\cdot), b(\cdot), c(\cdot), d(\cdot)$ is defined to belong to the exponential family.

Example; $f(x; \theta) = \theta e^{-\theta x} I_{(0, \infty)}(x)$

it belong to exponential family.

where $a(\theta) = \theta$, $b(x) = I_{(0, \infty)}(x)$

$d(x) = x$, $c(\theta) = -\theta$

Poisson Density;

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} I_{(0, 1, 2, \dots)}(x)$$

$$= \frac{1}{x!} I_{(0, 1, 2, \dots)}(x) e^{-\lambda} e^{x \ln \lambda}$$

$$a(\lambda) = e^{-\lambda}, \quad b(x) = \frac{1}{x!} I_{(0,1,2,\dots)}^{(x)}, \quad c(\lambda) = \ln \lambda, \quad d(x) = x$$

It belongs to the exponential class.

$$If \quad f(x; \theta) = a(\theta) b(x) \exp[c(\theta) d(x)]$$

$$\prod_{i=1}^n f(x_i; \theta) = [a(\theta)]^n \prod_{i=1}^n b(x_i) \exp\left[c(\theta) \sum_{i=1}^n d(x_i)\right]$$

By Neyman Factorization Theorem (NFT) $\sum d(x_i)$ is a sufficient statistic for K -parameter Exponential Class;

A family of densities $f(x; \theta_1, \dots, \theta_K)$ that can be expressed as $f(x; \theta_1, \dots, \theta_K) = a(\theta_1, \dots, \theta_K) b(x) \exp\left[\sum_{j=1}^K c_j(\theta_1, \dots, \theta_K) d_j(x)\right]$ for a suitable choice of function $a(\cdot)$, $b(\cdot)$, $c(\cdot)$, $d(\cdot)$. $j=1, \dots, K$ is defined to belong to the exponential family.

e.g;

1, Normal density contains two parameters

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \quad -\infty < x < \infty$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2\sigma^2} \{x^2 - 2x\mu + \mu^2\}\right]$$

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2} \frac{\mu^2}{\sigma^2}\right] \exp\left(\frac{-x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2}\right)$$

$$\text{Where } a(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2} \frac{\mu^2}{\sigma^2}\right], \quad b(x) = 1$$

$$c_1(\mu, \sigma^2) = -\frac{1}{2\sigma^2}, \quad d_1(x) = x^2$$

$$c_2(\mu, \sigma^2) = \frac{\mu}{\sigma^2}, \quad d_2(x) = x$$

It means that normal density belongs to the exponential family.

2; β -distribution;

$$f(x; \theta_1, \theta_2) = \frac{1}{B(\theta_1, \theta_2)} x^{\theta_1-1} (1-x)^{\theta_2-1} I_{(0,1)}^{(x)}$$

$$f(x; \theta_1, \theta_2) = \frac{1}{B(\theta_1, \theta_2)} I_{(\theta_1)}(x) \exp[(\theta_1 - 1) \ln x + (\theta_2 - 1) \ln(1 - x)]$$

$$a(\theta_1, \theta_2) = \frac{1}{B(\theta_1, \theta_2)}, \quad b(x) = I_{(\theta_1)}(x)$$

$$c_1(\theta_1, \theta_2) = \theta_1 - 1, \quad d_1(x) = \ln x$$

$$c_2(\theta_1, \theta_2) = \theta_2 - 1, \quad d_2(x) = \ln(1 - x)$$

Hence it belongs to exponential family and $d_1(x), d_2(x)$ is a set of jointly sufficient statistics.

Uniformly Minimum Variance Unbiased Estimators (UMVUE)

Definition; let $x_1, \dots, x_n$ be a random sample from $f(x; \theta)$. An estimator

$T^* = t^*(x_1, \dots, x_n)$ of $g(\theta)$ is defined to be (UMVUE) of $g(\theta)$ if

i) $E_\theta(T^*) = g(\theta)$ i.e. T^* is unbiased

ii) $\text{Var}_\theta(T^*) \leq \text{Var}_\theta(T)$ for any other estimator $T = t(x_1, \dots, x_n)$ of $g(\theta)$ which satisfies $E_\theta(T) = g(\theta)$, which is also unbiased

Rao-Blackwell Theorem

An unbiased estimator which is function of sufficient statistic has smaller variance than an unbiased estimator which is not based on sufficient statistic.

Statement: let $x_1, \dots, x_n$ be a random sample from the density $f(x; \theta)$. let

$S_1 = S_1(x_1, \dots, x_n), \dots, S_k = S_k(x_1, \dots, x_n)$ be a set of jointly sufficient statistics. let $T = t(x_1, \dots, x_n)$ be an unbiased estimator of $g(\theta)$. we define

$$T^* = E[T/S_1, \dots, S_k], \text{ then}$$

i) T^* is a statistic and it is a function of the sufficient statistics, $S_1, \dots, S_k$.

we write $T^* = t^*(S_1, \dots, S_k)$

ii) $E(T^*) = g(\theta)$ i.e. T^* is an unbiased estimator of $g(\theta)$

iii) $\text{Var}_\theta(T^*) \leq \text{Var}_\theta(T)$ for every θ , and $\text{Var}_\theta(T^*) < \text{Var}_\theta(T)$ for some θ unless T^* is equal to T with probability 1.

Proof: $S_1, \dots, S_k$ are sufficient statistics, ^{so} the conditional distribution of any statistic, in particular, statistic T given $S_1, \dots, S_k$ is independent of θ . ~~$S_1, \dots, S_k$ is a statistic~~ Hence

$$T^* = E[T/S_1, \dots, S_k] \text{ --- (1) is independent of } \theta$$

So T^* is a statistic which is obviously a function of $S_1, \dots, S_k$.

(ii)

$$E_\theta[T^*] = E_\theta[E(T/S_1, \dots, S_k)] \\ = E_\theta(T)$$

$$E_\theta[T^*] = g(\theta) \text{ --- (2)}$$

ie T^* is an unbiased estimator of $g(\theta)$.

(iii)

$$\text{Var}_\theta(T) = E_\theta[T - E_\theta(T)]^2 \\ = E_\theta[T - g(\theta)]^2$$

$$\text{From (2) } g(\theta) = E_\theta(T^*)$$

$$\text{Var}_\theta(T) = E_\theta[T - E_\theta(T^*)]^2$$

$$= E_\theta[(T - T^*) + (T^* - E_\theta(T^*))]^2$$

$$= E_\theta[(T - T^*)^2 + 2(T - T^*)(T^* - E_\theta(T^*)) + (T^* - E_\theta(T^*))^2]$$

$$\text{Var}_\theta(T) = E_\theta(T - T^*)^2 + 2 E_\theta[(T - T^*)(T^* - E_\theta(T^*))] + E_\theta(T^* - E_\theta(T^*))^2 \quad (3)$$

Now we take 2nd term of (3)

$$E_\theta[(T - T^*)(T^* - E_\theta(T^*))] = E_\theta[E\{(T - T^*)(T^* - E_\theta(T^*)) / S_1, \dots, S_k\}]$$

$$\text{Term within bracket is } = E\{(T - T^*)(T^* - E_\theta(T^*)) / S_1 = s_1, \dots, S_k = s_k\}$$

$$= (T^* - E_\theta(T^*))$$

$$= (T^* - E_\theta(T^*)) E\{(T - T^*) / S_1, \dots, S_k\}$$

$$= [T^* - E_\theta(T^*)] [E_\theta\{T / S_1, \dots, S_k\} - E_\theta\{T^* / S_1, \dots, S_k\}]$$

$$= (T^* - E_\theta(T^*)) (T^* - T^*) = 0$$

Hence

$$E_\theta[(T - T^*)(T^* - E_\theta(T^*))] = 0$$

Now from (3)

$$\text{Var}_\theta(T) = E(T - T^*)^2 + \text{Var}(T^*) \geq \text{Var}(T^*)$$

$$\text{If } E_\theta(T - T^*)^2 = 0 \quad \text{Var}_\theta(T^*) = \text{Var}_\theta(T)$$

$\text{Var}(T) > \text{Var}(T^*)$ unless $T = T^*$ with prob 1.

Given an unbiased estimator T , we find another unbiased estimator which is function of sufficient statistic and that estimator will not have larger variance.

Example; Let $x_1, \dots, x_n$ be a random sample from

$$f(x; \theta) = \theta^x (1-\theta)^{1-x} \quad x=0,1$$

i.e. we have two estimators

$$T = x_1, \quad T^* = \frac{\sum x_i}{n} \quad \text{are unbiased estimators of } \theta$$

show that T^* is a UMVUE.

Sol; x_1 is an unbiased estimator of $g(\theta) = \theta$

$$T = t(x_1, \dots, x_n) = x_1$$

We know $S = \sum_{i=1}^n x_i$ is a sufficient statistic

We use $S = \sum_{i=1}^n x_i$ as ^{one} set (of one element) of sufficient statistic

According to the Theorem,

$$T^* = E(T/S)$$

$= E(T/\sum x_i)$ is an unbiased estimator of θ with no longer variance than

$$T = x_1$$

We first find the conditional distribution of x_1 given $\sum x_i = s$
 x_1 takes on the values 0, 1.

$$P[x_1 = x / \sum x_i = s] = \frac{P[x_1 = x, \sum x_i = s]}{P[\sum x_i = s]}$$

$$\begin{aligned} P[x_1 = 0 / \sum x_i = s] &= \frac{P[x_1 = 0, \sum_{i=2}^n x_i = s]}{P[\sum x_i = s]} \\ &= \frac{P[x_1 = 0, \sum_{i=2}^n x_i = s]}{P[\sum_{i=1}^n x_i = s]} \end{aligned}$$

$$\begin{aligned}
 P[X_1=0/\sum_{i=1}^n X_i=\delta] &= \frac{P(X_1=0) P(\sum_{i=2}^n X_i=\delta)}{P(\sum_{i=1}^n X_i=\delta)} \\
 &= \frac{(1-\theta) \binom{n-1}{\delta} \theta^\delta (1-\theta)^{n-1-\delta}}{\binom{n}{\delta} \theta^\delta (1-\theta)^{n-\delta}} \\
 &= \frac{\binom{n-1}{\delta}}{\binom{n}{\delta}} = \frac{(n-1)!}{\delta!(n-\delta-1)!} \cdot \frac{\delta!(n-\delta)!}{\delta!(n-\delta)(n-\delta-1)!}
 \end{aligned}$$

$$P[X_1=0/\sum_{i=1}^n X_i=\delta] = \frac{n-\delta}{n}$$

$$\begin{aligned}
 P[X_1=1/\sum_{i=1}^n X_i=\delta] &= P[X_1=1, \sum_{i=1}^n X_i=\delta] / P(\sum_{i=1}^n X_i=\delta) \\
 &= P[X_1=1, \sum_{i=2}^n X_i+\delta-1] / P(\sum_{i=1}^n X_i=\delta) \\
 &= P[X_1=1, \sum_{i=2}^n X_i=\delta-1] / P(\sum_{i=1}^n X_i=\delta) \\
 &= \frac{\theta \binom{n-1}{\delta-1} \theta^{\delta-1} (1-\theta)^{n-1-\delta+1}}{\binom{n}{\delta} \theta^\delta (1-\theta)^{n-\delta}} = \frac{\binom{n-1}{\delta-1}}{\frac{n}{\delta} \binom{n-1}{\delta-1}} \\
 &= \frac{\delta}{n}
 \end{aligned}$$

$$T^* = E(X_1/\sum_{i=1}^n X_i=\delta) = 0 \cdot \frac{n-\delta}{n} + 1 \cdot \frac{\delta}{n} = \frac{\delta}{n}$$

$$T^* = \frac{\delta}{n} = \frac{\sum_{i=1}^n X_i}{n}$$

$$Var(T) = \theta(1-\theta)$$

$$\begin{aligned}
 Var(T^*) &= Var\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{1}{n^2} Var\left(\sum_{i=1}^n X_i\right) \\
 &= \frac{1}{n^2} \sum_{i=1}^n Var(X_i) \\
 &= \frac{1}{n^2} n \theta(1-\theta)
 \end{aligned}$$

$$Var(T^*) = \frac{\theta(1-\theta)}{n}$$

$$For\ n > 1, \quad Var(T) > Var(T^*)$$

Hence T^* is UMVUE.

Completeness

The family of density function or probability function $f(x; \theta)$, $\theta \in \Omega$, is called complete, if for every function $u(x)$, the identity $E_{\theta}[u(x)] = 0$, for all $\theta \in \Omega$ implies $[u(x) = 0]$ at all x for which $f(x; \theta) > 0$ for some θ . Also the statistic $\{u(x)\}$ is said to be complete iff its family of density is complete. The only unbiased estimator of θ is θ itself.

Complete Sufficient Statistic :-

If a sufficient statistic T in some problem has a family of density or probability function $g(x; \theta)$ that is complete, then T is called complete sufficient statistic.

$$E_{\theta}[u(x)] = 0 \text{ implies } u(x) = 0$$

Example; Let $x_1, \dots, x_n$ be a random sample from

$$f(x; \theta) = \theta^x (1-\theta)^{1-x} \quad x = 0, 1$$

the statistic $T = \sum x_i$ is a sufficient statistic, then show that it is complete sufficient statistic.

Sol;

Let $u(T)$ be any statistic that is a function of T , for which

$$E_{\theta}[u(T)] = 0 \quad \text{for } \theta \in \Omega, 0 \leq \theta \leq 1.$$

To argue that T is complete, we must show that $u(T) = 0$ for $t = 0, 1, \dots, n$.

The p.f. of $T = \sum x_i$ is

$$g(t; \theta) = \binom{n}{t} \theta^t (1-\theta)^{n-t} \quad t = 0, 1, \dots, n$$

$$\begin{aligned} E_{\theta}[u(T)] &= \sum_{t=0}^n u(t) g(t; \theta) \\ &= \sum_{t=0}^n u(t) \binom{n}{t} \theta^t (1-\theta)^{n-t} = (1-\theta)^n \sum_{t=0}^n u(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t \end{aligned}$$

$E_{\theta}(u(T)) = 0$ for all $0 \leq \theta \leq 1$ implies that

$$(1-\theta)^n \sum_{t=0}^n u(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t = 0$$

$$\sum_{t=0}^n \binom{n}{t} \lambda^t u(t) = 0$$

where $\lambda = \frac{\theta}{1-\theta}$

In order for a polynomial in x to be identically zero, each coeff of x^t , $t = 0, 1, \dots, n$ must be zero i.e.

$$\binom{n}{t} u(t) = 0 \quad \text{for } t = 0, 1, 2, \dots, n$$

But $\binom{n}{t} \neq 0$, So $u(t) = 0$ for $t = 0, 1, 2, \dots, n$

$$\text{So } E_0(u(T)) = 0 \Rightarrow u(T) = 0$$

Hence T is Complete Sufficient Statistic

Example: Suppose that $x_1, \dots, x_n$ are independent random variables with each Poisson distribution with unknown parameter λ . Find a sufficient statistic for λ and check if it is complete.

Sol: The p.f of Poisson dist is

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots$$

The joint probability function is.

$$\begin{aligned} f(x_1; \lambda) f(x_2; \lambda) \dots f(x_n; \lambda) &= \frac{e^{-n\lambda} \lambda^{\sum x_i}}{x_1! x_2! \dots x_n!} \\ &= \frac{e^{-n\lambda} \lambda^{\sum x_i}}{x_1! x_2! \dots x_n!} \end{aligned}$$

Because joint density is factorized into two parts one containing parameter λ and other independent of λ . Therefore $\sum x_i$ is sufficient for λ .

The p.f of sufficient statistic $\sum x_i = x$ is

$$f(x; \lambda) = \frac{e^{-n\lambda} (n\lambda)^x}{x!} \quad x = 0, 1, 2, \dots$$

Let $u(\cdot)$ be any arbitrary function, then $E[u(T)] = 0$ means

$$\sum_{k=0}^{\infty} u(k) \frac{e^{-n\lambda} (n\lambda)^k}{k!} = 0$$

Since $e^{-n\lambda} > 0$ this holds iff $\sum_{k=0}^{\infty} u(k) \frac{(n\lambda)^k}{k!} = 0$

In order for a polynomial in x to be identically 0, each coefficient

of $(n\lambda)^k$, $k=0,1,2,\dots$, must be zero.

$$\text{i.e. } \frac{U(k)}{k!} = 0$$

$$\text{or } U(k) = 0 \text{ for } k=0,1,2,\dots \text{ since } k! \neq 0$$

$$\text{So } E(U(k)) = 0 \implies U(k) = 0$$

Hence $\sum X_i$ is complete sufficient statistic.

Example; Suppose $X_1, \dots, X_n$ are independent random variables each uniform on $[0, \theta]$ where $\theta > 0$. Find a sufficient statistic and check if it is complete.

Sol;

$$f(x, \theta) = \frac{1}{\theta} \quad 0 \leq x \leq \theta$$

$$= \frac{1}{\theta} I_{(0, \theta)}(x)$$

The joint dist of $x_1, x_2, \dots, x_n$ is

$$f(x_1, \dots, x_n; \theta) = \frac{1}{\theta^n}$$

The p.d.f of largest obs is $g(y_n) = n[F(y_n)]^{n-1} f(y_n)$

$$F(y) = \frac{1}{\theta} y$$

$$g(y_n) = \frac{n y_n^{n-1}}{\theta^n}$$

$$f(x_1, \dots, x_n / y_n) = \frac{f(x_1, \dots, x_n; \theta)}{g(y_n)} = \frac{1}{n y_n^{n-1}}$$

Because conditional dist is independent of θ , therefore y_n is sufficient statistic.

Let $u(\cdot)$ be any arbitrary function, then

$$E_\theta[u(y_n)] = 0 \text{ means}$$

$$\int_0^\theta u(x) n \frac{x^{n-1}}{\theta^n} dx = 0$$

$$\int_0^{\theta} u(x) x^{n-1} dx \equiv 0$$

Diff w.r.t θ

$$u(\theta) \theta^{n-1} \equiv 0$$

$$\text{or } u(\theta) = 0 \text{ or } u(x) = 0 \text{ for all } x > 0$$

So Y_n is Complete sufficient statistic.



Theorem; Let $x_1, x_2, \dots, x_n$ be a random sample from $f(x; \theta)$, $\theta \in \Omega$ where Ω is an interval (possibly infinite), if

$$f(x; \theta) = a(\theta) b(x) \exp[c(\theta) d(x)]$$

That is, the function $f(x; \theta)$ is a member of a one parameter exponential family then $\sum_{i=1}^n d(x_i)$ is a complete minimal sufficient statistic.

Lehmann-Scheffe's Theorem; Let $x_1, \dots, x_n$ be a random sample from the density $f(x; \theta)$. If $S = s(x_1, \dots, x_n)$ is a complete sufficient statistic and if $T^* = t^*(S)$ a function of S , is an unbiased estimator of $g(\theta)$, then T^* is an uniformly minimum variance unbiased estimator (UMVUE) of $g(\theta)$.

This theorem says that

- i) If a complete sufficient statistic exists and there is an unbiased estimator for $g(\theta)$, then there is a UMVUE for $g(\theta)$.
- ii) The UMVUE is the unique unbiased estimator of $g(\theta)$ which is a function of S .

B.R and LS Theorems (UMVUE)

Three Methods of Finding the (UMVUE's)

- i) We simply guess the correct form of the function of S that defines the desired estimator.
- ii) Guess or find any unbiased estimator of $g(\theta)$, and then calculate the conditional expectation of the unbiased estimator given the sufficient statistic.
- iii) Solve for $t^*(s)$ in the equation $E[t^*(s)] = g(\theta)$

Example:- Let $X_1, X_2, \dots, X_n$ be a random sample from the Poisson dist

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, 2, \dots$$

Find the UMVUE's of λ and $e^{-\lambda}$. (By Method (ii))

Sol,

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, 2, \dots$$

(i)

$$E(X) = \sum_{x=0}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \left[\lambda + \frac{\lambda^2}{2} + \frac{\lambda^3}{3} + \frac{\lambda^4}{4} + \dots \right]$$

$$= \lambda e^{-\lambda} e^{\lambda} = \lambda$$

$$E(X) = \lambda \implies E(\bar{X}) = \lambda$$

Hence $\bar{X}$ is an unbiased estimator of λ .

Since $\sum X_i$ is a sufficient statistic for λ ,

Because unbiased estimator of parameter λ , i.e. $\bar{X}$ is a function of (CSS) therefore $\bar{X}$ is a uniformly minimum variance unbiased estimator of λ .

(ii)

$$P(X=0) = e^{-\lambda}$$

$$I_{(0, \infty)}(x) = \begin{cases} 1 & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

$$I_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

$$I_{\{0\}}(x) = \begin{cases} 1 & x_1 = 0 \\ 0 & x_1 \neq 0 \end{cases}$$

We say that $I_{\{0\}}(x_1)$ is an unbiased estimator of $e^{-\lambda}$

$$E(I_{\{0\}}(x_1)) = (1) P[X_1=0] + (0) P[X_1 \neq 0] \\ = P(X_1=0) = e^{-\lambda}$$

$$E[I_{\{0\}}(x_1)] = e^{-\lambda} \quad \text{Hence } I_{\{0\}}(x_1) \text{ is an unbiased estimator of } e^{-\lambda}.$$

Now for UMVUE of $e^{-\lambda}$

By L.S theorem $E[I_{\{0\}}(x_1) / \sum X_i]$ is the UMVUE of $e^{-\lambda}$

The conditional expectation of $X_1 = 0$ given $\sum_{i=1}^n X_i$ is

$$\begin{aligned} P[X_1=0 / \sum_{i=1}^n X_i = s] &= \frac{P[X_1=0, \sum_{i=1}^n X_i = s]}{P[\sum_{i=1}^n X_i = s]} \\ &= \frac{P[X_1=0, \sum_{i=2}^n X_i = s]}{P[\sum_{i=1}^n X_i = s]} = \frac{P[X_1=0] P[\sum_{i=2}^n X_i = s]}{P[\sum_{i=1}^n X_i = s]} \\ &= \frac{e^{-\lambda} \frac{e^{-(n-1)\lambda} [(n-1)\lambda]^s}{s!}}{e^{-n\lambda} \frac{(n\lambda)^s}{s!}} \\ &= \left(\frac{n-1}{n}\right)^s, \quad n > 1 \end{aligned}$$

$$\begin{aligned} E[I_0(X) / \sum_{i=1}^n X_i = s] &= (1) P[X_1=0 / \sum_{i=1}^n X_i = s] + 0 P[X_1 \neq 0 / \sum_{i=1}^n X_i = s] \\ &= P[X_1=0 / \sum_{i=1}^n X_i = s] \\ &= \left(\frac{n-1}{n}\right)^s, \quad n > 1 \end{aligned}$$

Hence $\left(\frac{n-1}{n}\right)^{\sum_{i=1}^n X_i}$ is UMVUE of $e^{-\lambda}$

Example; Let $X_1, X_2, \dots, X_n$ be a random sample from

$$f(x; \theta) = \theta e^{-\theta x} \quad I_{(0, \infty)}(x)$$

Find the UMVUE of $\theta, \frac{1}{\theta}, e^{-\theta}$

Solution; (i) For $\frac{1}{\theta}$.

$$E(X) = \int_0^{\infty} x \theta e^{-\theta x} dx$$

$$\text{putting } x\theta = y \implies dx = \frac{1}{\theta} dy$$

$$E(X) = \int_0^{\infty} y e^{-y} dy = \frac{1}{\theta}$$

$$E(X) = \frac{1}{\theta} \implies E(\bar{X}) = \frac{1}{\theta}$$

It means that $\bar{X} = \frac{\sum X_i}{n}$ is unbiased estimator of $\frac{1}{\theta}$.

Since $\sum X_i$ is complete sufficient statistics

Because $\bar{X}$ is a function of CSS, therefore $\bar{X}$ is UMVUE of $\frac{1}{\theta}$.

$f(x; \theta)$ is member of exp family

with $a(\theta) = \theta, b(\eta) = I_{(0, \infty)}(x)$

$c(\theta) = -\theta, d(\eta) = x$

So $\sum d_i = \sum X_i$ is CSS.

(ii) For θ , we know that if X_i is an exponential variate, then $\sum X_i$ is a Gamma variate.

Now first we find unbiased estimator of θ

$$\text{let } S = \sum X_i$$

$$f(s) = \frac{\theta^n}{\Gamma(n)} e^{-\theta s} s^{n-1}$$

$$E\left(\frac{n}{s}\right) = \int_0^{\infty} \frac{n}{s} f(s) ds$$

$$E\left(\frac{n}{s}\right) = \int_0^{\infty} \frac{n}{s} \frac{\theta^n}{\Gamma(n)} e^{-\theta s} s^{n-1} ds$$

$$\text{putting } \theta s = u \quad ds = \frac{du}{\theta}$$

$$E\left(\frac{n}{s}\right) = \frac{1}{\Gamma(n)} \int_0^{\infty} \frac{n\theta}{u} \theta^n e^{-u} \left(\frac{u}{\theta}\right)^{n-1} \frac{du}{\theta}$$

$$= \frac{n\theta}{\Gamma(n)} \int_0^{\infty} e^{-u} u^{n-2} du = \frac{n\theta}{\Gamma(n)} \frac{1}{n-1}$$

$$= \frac{n\theta}{(n-1)\Gamma(n-1)} = \frac{\theta n}{(n-1)}$$

$$E\left(\frac{n}{s}\right) = \frac{\theta n}{n-1}$$

$$\frac{(n-1)}{n} E\left(\frac{n}{s}\right) = \theta$$

$$\text{or } E\left(\frac{n-1}{s}\right) = \theta$$

So $\frac{n-1}{s}$ is the unbiased estimator of θ .

By LS theorem $E\left[\frac{n-1}{s} / \sum X_i = s\right]$ is UMVUE

$$E\left(\frac{n-1}{s} / \sum X_i = s\right) = E\left(\frac{n-1}{s}\right)$$

Hence $\frac{n-1}{s}$ is UMVUE of θ for $n > 1$

(iii) For $e^{-k\theta}$

$$P(X > k) = \int_k^{\infty} \theta e^{-\theta x} dx = \left| -e^{-\theta x} \right|_k^{\infty} = e^{-k\theta}$$

$$E[I_{(K, \infty)}(X_1)] = 1 \cdot [P(X_1 > K)] + 0 \cdot [P(X_1 \leq K)]$$

$$= P[K < X_1 < \infty]$$

$$I_{(K, \infty)}(X_1) = \begin{cases} 1 & \text{if } K < X_1 < \infty \\ 0 & \text{otherwise} \end{cases}$$

$$E[I_{(K, \infty)}(X_1)] = P(X_1 > K) = e^{-K\theta}$$

Hence $I_{(K, \infty)}(X_1)$ is an unbiased estimator of $e^{-K\theta}$

By B.R and L.S theorems $E[I_{(K, \infty)}(X_1)/S = s]$ is UMVUE of $e^{-K\theta}$

$$E[I_{(K, \infty)}(X_1)/S = s] = 1 \cdot P[X_1 > K/S = s] + 0 \cdot P[I_{(K, \infty)}(X_1) \leq K/S = s]$$

$$= P[I_{(K, \infty)}(X_1) > K/S = s]$$

Now we find the conditional distribution of X_1 given $S = s$.

$$f(X_1/s) \Delta X_1 = \frac{f(X_1, s) \Delta X_1 \Delta s}{f(s) \Delta s}$$

$$= \frac{P[X_1 < X_1 < X_1 + \Delta X_1; s < \sum_{i=1}^n X_i < s + \Delta s]}{P[s < \sum_{i=1}^n X_i < s + \Delta s]}$$

$$= \frac{P[X_1 < X_1 < X_1 + \Delta X_1; s < \sum_{i=2}^n X_i + X_1 < s + \Delta s]}{P[s < \sum_{i=1}^n X_i < s + \Delta s]}$$

$$= \frac{P[X_1 < X_1 < X_1 + \Delta X_1] P[s - X_1 < \sum_{i=2}^n X_i < s - X_1 + \Delta s]}{P[s < \sum_{i=1}^n X_i < s + \Delta s]}$$

$$= \frac{(\theta e^{-\theta X_1}) \theta^{n-1} e^{-\theta(s-X_1)} (s-X_1)^{n-2} \Delta X_1 \Delta s}{(n-1)}$$

$$= \frac{1}{(n-1)} \theta^n e^{-\theta s} s^{n-2} \Delta s$$

$$f(x_1/s) \Delta x_1 = \frac{\cancel{n}}{n-1} \frac{(s-x_1)^{n-2}}{s^{n-1}} \Delta x_1$$

$$E[I_{(K, \infty)}(x_1)/s=s] = P[X_1 > K/s=s]$$

$$= \int_K^{\infty} f(x_1/s) dx_1$$

$$= \int_K^s \frac{\cancel{n}}{n-1} \frac{(s-x_1)^{n-2}}{s^{n-1}} dx_1$$

$$= \frac{(n-1)}{s^{n-1}} \int_K^s (s-x_1)^{n-2} dx_1 \quad \left. \begin{array}{l} \text{pulling } s-x_1=y \\ -dx_1=dy \end{array} \right\}$$

$$= \frac{(n-1)}{s^{n-1}} \int_{s-K}^0 y^{n-2} (-dy)$$

$$= \frac{(n-1)}{s^{n-1}} \int_0^{s-K} y^{n-2} dy$$

$$= \frac{(n-1)}{s^{n-1}} \left[\frac{y^{n-1}}{n-1} \right]_0^{s-K}$$

$$E[I_{(K, \infty)}(x_1)/s=s] = \left(\frac{s-K}{s} \right)^{n-1} \quad \text{for } K < s, n > 1$$

Hence $\left(\frac{\sum x_i - K}{\sum x_i} \right)^{n-1} I_{(K, \infty)}(\sum x_i)$ is UMVUE for $e^{-K\theta}$ for $n > 1$

Methods of Estimation

DIFFERENT METHODS OF ESTIMATION:-

There are several methods of estimation and each method leads to rather different estimates if used in same situation. In some of the common and important examples of estimation, all the methods lead to effectively the same results.

- i, Method of maximum Likelihood.
- ii, Method of Moments
- iii, Method of Least Squares
- iv, Bayesian Method.
- v, Inverse probability Method.

Likelihood Function:

Let $x_1, x_2, \dots, x_n$ be a random sample from the distribution that has probability function or probability density function $f(x; \theta)$. The joint p.f or p.d.f of the sample $x_1, x_2, \dots, x_n$ is $f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta)$.

The joint p.f or p.d.f denoted by $L(x_1, \dots, x_n; \theta)$ when considered as a function of θ , is called the likelihood function.

(i) METHOD OF MAXIMUM LIKELIHOOD (ML)

According to the principle of ML we should choose that estimator which makes the LF a maximum. That is t will be MLE if

$L(x_1, \dots, x_n; t) > L(x_1, \dots, x_n; t')$ for any other estimator t'
If the likelihood function $L(x_1, \dots, x_n; t)$ is differentiable function of θ then $L(x_1, \dots, x_n)$ is the solution (if any) of

$$\frac{\partial L}{\partial \theta} = 0 \text{ and } \frac{\partial^2 L}{\partial \theta^2} < 0$$

Since L is positive, the first equation is equivalent to $\frac{\partial \ln L}{\partial \theta} = 0$

The principle of ML can be applied to the case when the population is bivariate or multivariate.

Example; Suppose that a sample of 10 items is drawn from a lot with replacement. We find that 6 of these are non defective. Given the sample proportion $p = \frac{x}{n} = \frac{6}{10} = 0.6$.

What is a reasonable estimate of the population proportion of p .

Sol; If n items are selected WR, the distribution of x , the number of non defectives (success) is

$$f(x; p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x=0, 1, 2, \dots, n$$

where p is the probability of obtaining a non defective item.

The likelihood function is

$$L(p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x=0, 1, \dots, n$$

$$= \binom{10}{6} p^6 (1-p)^4$$

The likelihood function for all possible values of p is

p	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$L(p)$	0	0.0001	0.0055	0.0368	0.1115	0.2051	0.2506	0.2001	0.0881	0.0112	0

The likelihood function attains its maximum value at $p = 0.6$, which is the MLE.

PROPERTIES OF MLE

- MLE is consistent.
- MLE tends to normality as n increases.
- It is most efficient (M.V.B.E) and the variance of the estimate approaches the minimum variance for any consistent estimator of the given parameter.
- If a sufficient estimator exists it can always be found by the method of M.L.E.
- The maximum likelihood estimator is invariant under function transformation.

$$\lambda^2 = \sum x_i^2, \quad \lambda = \sqrt{\sum x_i^2}$$

(vi) M.L.E are not unbiased in general and in certain situation the bias may be serious even in the case of samples.

Example; let $x_1, x_2, \dots, x_n$ be a random sample of size n taken from the Binomial Distribution.

$$f(x; p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x=0,1,2, \dots, n \\ 0 \leq p \leq 1$$

Find Maximum Likelihood estimator of p .

Sol; The likelihood function of $x_1, x_2, \dots, x_n$ is

$$L(p) = f(x_1, \dots, x_n; p) = \prod_{i=1}^n \binom{n}{x_i} p^{x_i} (1-p)^{n-x_i}$$

$$\ln L(p) = \ln(n!) - \ln(x!) - \ln(n-x)! + x \ln(p) + (n-x) \ln(1-p)$$

diff w.r.t p and equating to zero

$$\frac{\partial \ln L}{\partial p} = \frac{x}{p} + \frac{(n-x)}{(1-p)} (-1) = 0$$

$$x - xp - np + xp = 0 \Rightarrow x = np$$

$$\hat{p} = \frac{x}{n} \text{ which is MLE of } p.$$

Example; Let $x_1, x_2, \dots, x_n$ be a random sample of size n taken from the Bernoulli Distribution

$$f(x) = p^x (1-p)^{1-x} \quad x=0,1 \quad 0 \leq p \leq 1$$

Find the MLE of p .

Solution; The LF of $x_1, x_2, \dots, x_n$ is

$$L(p) = L(x_1, \dots, x_n; p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$$

$$L(p) = p^{\sum x_i} (1-p)^{n - \sum x_i}$$

Taking natural logarithm

$$\ln L(p) = \sum x_i \ln(p) + (n - \sum x_i) \ln(1-p)$$

diff w.r.t p , and equating to zero

$$\frac{\partial \ln L(p)}{\partial p} = \frac{\sum x_i}{p} + (n - \sum x_i) \frac{(-1)}{(1-p)} = 0$$

$$(1-p) \sum x_i = np - p \sum x_i$$

$$\sum x_i - p \sum x_i = np - p \sum x_i$$

$$\hat{p} = \frac{\sum x_i}{n} = \bar{x} \quad \text{which is M.L.E of } p.$$

Example; let $x_1, x_2, \dots, x_n$ denote a random sample from the Poisson distribution

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots$$

Find the M.L.E of λ .

Solution; The L.F of $x_1, x_2, \dots, x_n$ is

$$L(\lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{x_1! x_2! \dots x_n!}$$

Taking natural logarithm

$$\ln L(\lambda) = -n\lambda + \sum x_i \ln \lambda - \ln(x_1! x_2! \dots x_n!)$$

Diff. w.r.t. λ and equating to zero

$$\frac{\partial [\ln L(\lambda)]}{\partial \lambda} = -n + \frac{\sum x_i}{\lambda} = 0$$

$$n = \frac{\sum x_i}{\lambda}$$

$$\hat{\lambda} = \frac{\sum x_i}{n} = \bar{x} \quad \text{is M.L.E of } \lambda$$

Example; let $x_1, x_2, \dots, x_n$ be a random sample from a normal population with parameter μ and σ^2 . Find M.L.E for

i) μ , when population variance σ^2 is known.

ii) σ^2 , when population mean μ is known.

iii) μ and σ^2 when both are unknown.

Solution; The density function of normal distribution is

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \quad -\infty < x < \infty$$

The likelihood function for a sample of size n with values $x_1, x_2, \dots, x_n$ is

$$L(\theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2\sigma^2}(x_i - \mu)^2}$$

$$L(\theta) = \left(\frac{1}{2\pi}\right)^{n/2} \left(\frac{1}{\sigma^2}\right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2}$$

Taking Natural logarithm

$$\ln[L(\theta)] = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \left[\sum (x_i - \mu)^2 \right]$$

(i) Here $\theta = \mu$ as σ^2 is known

$\therefore$ diff w.r.t μ and equating to zero

$$\frac{\partial [\ln L(\mu)]}{\partial \mu} = -\frac{1}{2\sigma^2} \left[2 \sum (x_i - \mu) \right] = 0$$

$$\sum (x_i - \mu) = 0$$

$$\hat{\mu} = \frac{\sum x_i}{n} = \bar{x}$$

Again

$$\frac{\partial^2 [\ln L(\mu)]}{\partial \mu^2} = -\frac{n}{\sigma^2} < 0$$

Thus MLE of μ is $\bar{x} = \frac{\sum x_i}{n}$

(ii) Here $\theta = \sigma^2$ as μ is known

Diff w.r.t σ^2 and equating to zero

$$\frac{\partial [\ln L(\sigma^2)]}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{(\sigma^2)^{-2}}{2} \sum (x_i - \mu)^2 = 0$$

$$\frac{n}{\sigma^2} = \frac{\sum (x_i - \mu)^2}{\sigma^4}$$

$$\hat{\sigma}^2 = \frac{\sum (x_i - \mu)^2}{n} \text{ is MLE of } \sigma^2$$

(iii) Here $\theta = (\mu, \sigma^2)$. To get the joint M.L estimators for μ and σ^2 we have to solve together the two equations.

$$\frac{\partial}{\partial \mu} [\ln(L(\mu, \sigma^2))] = 0 \text{ and } \frac{\partial}{\partial \sigma^2} [\ln(L(\mu, \sigma^2))] = 0$$

$$\text{now } \frac{\partial}{\partial \mu} [\ln(L(\mu, \sigma^2))] = 0 \Rightarrow -\frac{1}{2\sigma^2} \left[2 \sum (x_i - \mu) \right] = 0$$

and $\frac{\partial}{\partial \sigma^2} [\ln L(\mu, \sigma^2)] = 0 \Rightarrow -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 = 0$

Solving these two equations -

$$\hat{\mu} = \frac{\sum x_i}{n} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Thus the joint M.L. estimators for μ and σ^2 are the sample mean $\bar{x}$ and sample variance S^2 , which is not an unbiased estimator.

Example; Let $x_1, x_2, \dots, x_n$ be a random sample of size n taken from Gamma distribution

$$f(x; \theta) = \frac{1}{\theta^p \Gamma(p)} x^{p-1} e^{-x/\theta} \quad 0 < x < \infty$$

Find M.L.E of θ .

Solution; The L.F of $x_1, x_2, \dots, x_n$ is

$$L(\theta) = \left(\frac{1}{\theta^p \Gamma(p)} \right)^n x^{np-n} e^{-\frac{1}{\theta} \sum x_i}$$

$$\ln L(\theta) = -np \ln \theta - n \ln \Gamma(p) + (np-n) \ln(x) - \frac{1}{\theta} \sum x_i$$

diff w.r.t θ , and equating to zero.

$$\frac{\partial (\ln L(\theta))}{\partial \theta} = \frac{-np}{\theta} + \frac{\sum x_i}{\theta^2} = 0$$

$$\frac{\sum x_i}{\theta^2} = \frac{np}{\theta}$$

$$\hat{\theta} = \frac{\sum x_i}{np} \quad \text{Hence MLE of } \theta \text{ is } \frac{\sum x_i}{np}$$

Example; The p.d.f of a random variable x is given as

$$f(x; \alpha) = \alpha x^{\alpha-1} \quad 0 < x < 1$$

Find M.L.E of α .

Solution; The likelihood function is

$$L(\alpha) = \alpha^n (x_1 x_2 \dots x_n)^{\alpha-1}$$

$$\ln L(\alpha) = n \ln \alpha + (\alpha-1) \ln (x_1 \dots x_n)$$

diff w.r.t. α , and equating to zero

$$\frac{\partial (\ln L(\alpha))}{\partial \alpha} = \frac{n}{\alpha} + \ln (x_1 \dots x_n) = 0$$

$$\frac{n}{\hat{\alpha}} = -\ln(x_1 x_2 \dots x_n)$$

$$\hat{\alpha} = \frac{-n}{\ln(x_1 x_2 \dots x_n)} \quad \text{It is MLE of } \alpha.$$

Example; Find M.L.E of p in

$$f(x) = p(1-p)^{x-1} \quad x=1, 2, \dots$$

Sol; The L.F is

$$L(p) = p^{\sum x_i - n} (1-p)^n$$

$$\ln L(p) = n \ln p + (\sum x_i - n) \ln(1-p)$$

$$\frac{\partial [\ln L(p)]}{\partial p} = \frac{n}{p} + \frac{(\sum x_i - n)}{(1-p)} (-1) = 0$$

$$n - np - p \sum x_i + np = 0$$

$$\frac{1}{\hat{p}} = \frac{\sum x_i}{n} \Rightarrow \hat{p} = \frac{1}{\bar{x}}$$

Hence MLE of p is $1/\bar{x}$.

Example; Suppose that x is a random variable with the density function

$$f(x; \theta) = \theta e^{-\theta x} \quad x > 0, \theta > 0$$

Find M.L.E of θ .

Solution; Let $x_1, x_2, \dots, x_n$ be a random sample from the given density. The L.F of this sample is

$$\begin{aligned} L(\theta) &= (\theta e^{-\theta x_1})(\theta e^{-\theta x_2}) \dots (\theta e^{-\theta x_n}) \\ &= \theta^n e^{-\theta(x_1 + \dots + x_n)} \\ &= \theta^n e^{-\theta \sum x_i} \end{aligned}$$

$$\ln L(\theta) = n \ln \theta - \theta \sum x_i$$

$$\frac{d(\ln L(\theta))}{d\theta} = \frac{n}{\theta} - \sum x_i = 0$$

$$\frac{n}{\hat{\theta}} = \sum x_i$$

$$\hat{\theta} = \frac{1}{\bar{x}}$$

Hence MLE of θ is $1/\bar{x}$

Example; Find the Maximum likelihood estimators of the five parameters of the bivariate normal distribution.

$$f(x, y; \mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x-\mu_1}{\sigma_1} \right) \left(\frac{y-\mu_2}{\sigma_2} \right) + \left(\frac{y-\mu_2}{\sigma_2} \right)^2 \right\} \right] \quad - \text{a(x, y) \leq 0}$$

Solution; The likelihood function is

$$L(x, y, \dots) = \frac{1}{(2\pi)^n \sigma_1^n \sigma_2^n (1-\rho^2)^{n/2}} \exp \left[-\frac{1}{2(1-\rho^2)} \sum_{i=1}^n \left\{ \left(\frac{x_i - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x_i - \mu_1}{\sigma_1} \right) \left(\frac{y_i - \mu_2}{\sigma_2} \right) + \left(\frac{y_i - \mu_2}{\sigma_2} \right)^2 \right\} \right]$$

Taking natural logarithm

$$\begin{aligned} \ln L = & -n \ln(2\pi) - \frac{n}{2} \ln \sigma_1^2 - \frac{n}{2} \ln \sigma_2^2 - \frac{n}{2} \ln(1-\rho^2) \\ & - \frac{1}{2(1-\rho^2)} \left\{ \frac{1}{\sigma_1^2} \sum (x_i - \mu_1)^2 + \frac{1}{\sigma_2^2} \sum (y_i - \mu_2)^2 \right. \\ & \left. + \frac{2\rho}{\sigma_1\sigma_2(1-\rho^2)} \sum (x_i - \mu_1)(y_i - \mu_2) \right\} \quad \text{--- (A)} \end{aligned}$$

Diff. (A) w.r.t. ' μ_1 ' and equating to zero

$$\frac{\partial(\ln L)}{\partial \mu_1} = -\frac{1}{2(1-\rho^2)} \left(\frac{2}{\sigma_1^2} \sum (x_i - \mu_1)(-1) \right) + \frac{\rho(-1)}{\sigma_1\sigma_2(1-\rho^2)} \sum (y_i - \mu_2) = 0$$

$$\frac{1}{(1-\rho^2)\sigma_1} \left[\frac{\sum (x_i - \mu_1)}{\sigma_1} - \frac{\rho \sum (y_i - \mu_2)}{\sigma_2} \right] = 0$$

$$\frac{n}{(1-\rho^2)\sigma_1} \left[\frac{(\bar{x} - \mu_1)}{\sigma_1} - \frac{\rho(\bar{y} - \mu_2)}{\sigma_2} \right] = 0 \quad \text{--- (1)}$$

Similarly $\frac{\partial(\ln L)}{\partial \mu_2} = 0$, gives

$$\frac{n}{(1-\rho^2)\sigma_2} \left[\frac{(\bar{y} - \mu_2)}{\sigma_2} - \frac{\rho(\bar{x} - \mu_1)}{\sigma_1} \right] = 0 \quad \text{--- (2)}$$

Now diff. (A) w.r.t. ' σ_1^2 ', and equating to zero.

$$\begin{aligned} \frac{\partial \ln L}{\partial \sigma_1^2} &= -\frac{n}{2\sigma_1^2} - \frac{1}{2(1-\rho^2)} \left(\frac{(-1)}{\sigma_1^4} \sum (x_i - \mu_1)^2 \right) + \frac{(-1)(\sigma_1^2)^{-1/2}}{2\sigma_2(1-\rho^2)} \sum (x_i - \mu_1)(y_i - \mu_2) = 0 \\ &= -\frac{n}{2\sigma_1^2} + \frac{1}{2(1-\rho^2)\sigma_1^4} \sum (x_i - \mu_1)^2 - \frac{\rho}{2\sigma_1^3\sigma_2(1-\rho^2)} \sum (x_i - \mu_1)(y_i - \mu_2) = 0 \\ &= -\frac{1}{2\sigma_1^2(1-\rho^2)} \left[n(1-\rho^2) - \frac{\sum (x_i - \mu_1)^2}{\sigma_1^2} + \frac{\rho}{\sigma_1\sigma_2} \sum (x_i - \mu_1)(y_i - \mu_2) \right] = 0 \quad \text{--- ③} \end{aligned}$$

Similarly $\frac{\partial \ln L}{\partial \sigma_2^2} = 0$, gives

$$-\frac{1}{2\sigma_2^2(1-\rho^2)} \left[n(1-\rho^2) - \frac{\sum (y_i - \mu_2)^2}{\sigma_2^2} + \frac{\rho}{\sigma_1\sigma_2} \sum (x_i - \mu_1)(y_i - \mu_2) \right] = 0 \quad \text{--- ④}$$

now diff ③ w.r.t 'ρ' and equating to zero

$$\begin{aligned} \frac{\partial \ln L}{\partial \rho} &= -\frac{n(-2\rho)}{2(1-\rho^2)} - \frac{1}{2}(-1)(1-\rho^2)^{-1/2} \left[\frac{1}{\sigma_1^2} \sum (x_i - \mu_1)^2 + \frac{1}{\sigma_2^2} \sum (y_i - \mu_2)^2 \right] \\ &\quad + \frac{1}{\sigma_1\sigma_2} \left[\frac{(1-\rho^2)(1) - \rho(-2\rho)}{(1-\rho^2)^2} \right] \sum (x_i - \mu_1)(y_i - \mu_2) = 0 \\ \frac{n\rho}{(1-\rho^2)} + \frac{\rho}{(1-\rho^2)^2} \left[\frac{1}{\sigma_1^2} \sum (x_i - \mu_1)^2 + \frac{1}{\sigma_2^2} \sum (y_i - \mu_2)^2 \right] + \frac{(1+\rho^2)}{(1-\rho^2)} \frac{\sum (x_i - \mu_1)(y_i - \mu_2)}{\sigma_1\sigma_2} &= 0 \\ \frac{1}{(1-\rho^2)} \left[n\rho - \frac{1}{(1-\rho^2)} \left\{ \rho \left(\frac{\sum (x_i - \mu_1)^2}{\sigma_1^2} + \frac{\sum (y_i - \mu_2)^2}{\sigma_2^2} \right) - (1+\rho^2) \frac{\sum (x_i - \mu_1)(y_i - \mu_2)}{\sigma_1\sigma_2} \right\} \right] &= 0 \quad \text{--- ⑤} \end{aligned}$$

① Solving ③, ④, ⑤ to obtain M.L. estimates of σ_1^2 , σ_2^2 and ρ for μ_1, μ_2 known.

$$\begin{aligned} \text{From ③, ④} \quad \left. \begin{aligned} n(1-\rho^2) &= \frac{\sum (x_i - \mu_1)^2}{\sigma_1^2} - \frac{\rho \sum (x_i - \mu_1)(y_i - \mu_2)}{\sigma_1\sigma_2} \\ n(1-\rho^2) &= \frac{\sum (y_i - \mu_2)^2}{\sigma_2^2} - \frac{\rho \sum (x_i - \mu_1)(y_i - \mu_2)}{\sigma_1\sigma_2} \end{aligned} \right\} \quad \text{--- ⑥} \end{aligned}$$

From ⑤

$$n\rho(1-\rho^2) = \rho \frac{\sum (x_i - \mu_1)^2}{\sigma_1^2} + \rho \frac{\sum (y_i - \mu_2)^2}{\sigma_2^2} - \frac{(1+\rho^2)}{\sigma_1\sigma_2} \sum (x_i - \mu_1)(y_i - \mu_2)$$

$$n(1-\rho^2) = \frac{\sum (x_i - \mu_1)^2}{\sigma_1^2} + \frac{\sum (y_i - \mu_2)^2}{\sigma_2^2} - \frac{(1+\rho^2)}{\rho} \frac{\sum (x_i - \mu_1)(y_i - \mu_2)}{\sigma_1\sigma_2} \quad \text{--- ⑦}$$

Adding ⑥

$$2n(1-\rho^2) = \frac{\sum (x_i - \mu_1)^2}{\sigma_1^2} + \frac{\sum (y_i - \mu_2)^2}{\sigma_2^2} - \frac{2\rho}{\sigma_1\sigma_2} \sum (x_i - \mu_1)(y_i - \mu_2) \quad \text{--- ⑧}$$

Subtracting ⑦ from ⑧

$$n(1-\rho^2) = \left(-\frac{2\rho}{\sigma_1\sigma_2} + \frac{(1+\rho^2)}{\rho\sigma_1\sigma_2} \right) \sum (x_i - \mu_1)(y_i - \mu_2)$$

$$n(1-\rho^2) = \frac{(1-\rho^2)}{\sigma_1\sigma_2} \frac{1}{\rho} \sum_{i=1}^n (x_i - \mu_1)(y_i - \mu_2)$$

$$\hat{\rho} = \frac{\sum_{i=1}^n (x_i - \mu_1)(y_i - \mu_2)}{n\sigma_1\sigma_2} \quad \text{--- ⑨}$$

From ⑨ $n\sigma_1\sigma_2\rho = \sum_{i=1}^n (x_i - \mu_1)(y_i - \mu_2)$

putting in ⑥

$$n - n\rho^2 = \frac{\sum (x_i - \mu_1)^2}{\sigma_1^2} - n\rho^2$$

$$\hat{\sigma}_1^2 = \frac{\sum (x_i - \mu_1)^2}{n}, \quad \hat{\sigma}_2^2 = \frac{\sum (y_i - \mu_2)^2}{n}$$

Hence MLEs of $\sigma_1^2, \sigma_2^2, \rho$ when μ_1, μ_2 are known are

$$\boxed{\hat{\rho} = \frac{\sum (x_i - \mu_1)(y_i - \mu_2)}{n\hat{\sigma}_1\hat{\sigma}_2}}, \quad \boxed{\hat{\sigma}_1^2 = \frac{\sum (x_i - \mu_1)^2}{n}}$$

$$\boxed{\hat{\sigma}_2^2 = \frac{\sum (y_i - \mu_2)^2}{n}}$$

Now from ① and ②

$$\frac{(\bar{x} - \mu_1)}{\sigma_1} = \rho \frac{(\bar{y} - \mu_2)}{\sigma_2}$$

$$\frac{(\bar{y} - \mu_2)}{\sigma_2} = \rho \frac{(\bar{x} - \mu_1)}{\sigma_1}$$

The only solution is $\bar{x} - \mu_1 = \bar{y} - \mu_2 = 0$

$$\hat{\mu}_1 = \bar{x}, \quad \hat{\mu}_2 = \bar{y}$$

Example; Let $x_1, x_2, \dots, x_n$ be a random sample of size n from the p.d.f

$$f(x; \theta) = \begin{cases} e^{-(x-\theta)} & \theta < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

$$= e^{-(x-\theta)} I_{(\theta, \infty)}(x)$$

Find M.L.E of the parameter θ .

Sol; The L.F of the sample is

$$L(\theta) = \frac{e^{-(x_1-\theta)}}{e^{-(x_1-\theta)}} \frac{e^{-(x_2-\theta)}}{e^{-(x_2-\theta)}} \frac{e^{-(x_3-\theta)}}{e^{-(x_3-\theta)}} \dots \frac{e^{-(x_n-\theta)}}{e^{-(x_n-\theta)}} \prod_{i=1}^n I_{(\theta, \infty)}(x_i)$$

$$= e^{-\sum x_i + n\theta} \prod_{i=1}^n I_{(\theta, \infty)}(x_i)$$

The maximum of $L(\theta)$ can not found by differentiation.

$$\theta < x_1, \dots, x_n < \infty$$

If $y_1, y_2, \dots, y_n$ are the ordered observations, then

$$\theta < y_1 < y_2 < \dots < y_n < \infty$$

$$\theta < y_1 < \infty \quad \text{where } y_1 \text{ is the minimum of observations}$$

$$y_1 = \min(x_1, \dots, x_n)$$

The $L(\theta)$ is the product of an increasing function of θ and is maximized at $\hat{\theta} = y_1$

Hence MLE of θ is $\hat{\theta} = y_1 = \min(x_i) = \text{smallest sample value.}$

Example; Let $x_1, x_2, \dots, x_n$ be a random sample from the dist having

$$\text{p.d.f } f(x; \theta_1, \theta_2) = \begin{cases} \frac{1}{\theta_1} e^{-\frac{(x-\theta_2)}{\theta_1}} & x \geq \theta_2 \\ 0 & \text{otherwise} \end{cases}$$

where $-\infty < \theta_2 < \infty$ and $\theta_1 > 0$ are both unknown.

Find M.L.E's of θ_1, θ_2 .

Sol; The L.F of the sample is

$$L(\theta_1, \theta_2) = \frac{1}{\theta_1^n} e^{-\frac{1}{\theta_1} \sum (x_i - \theta_2)} \quad x_i \geq \theta_2$$

The max of $L(\theta_2)$ can not found by diff

$$x_1, x_2, \dots, x_n \geq \theta_2$$

If $y_1, y_2, \dots, y_n$ are ordered observations, then

$$\theta_2 \leq y_1 < y_2 < \dots < y_n < \infty$$

$L(\theta_1, \theta_2)$ is maximized by choosing θ_2 as large as possible

Such that $\hat{\theta}_2 = Y_1^{(n)}$ is MLE of θ_2

Now, $\ln L = -n \ln \theta_1 - \frac{1}{\theta_1} \sum x_i + \frac{n \theta_2}{\theta_1}$

$$\frac{\partial \ln L}{\partial \theta_1} = -\frac{n}{\theta_1} + \frac{\sum x_i}{\theta_1^2} - \frac{n \theta_2}{\theta_1^2} = 0$$

$$\frac{n}{\theta_1} = \frac{\sum x_i}{\theta_1^2} - \frac{n \theta_2}{\theta_1^2}$$

$$\theta_1 = \frac{\sum x_i}{n} - \theta_2$$

$$\hat{\theta}_1 = -\hat{\theta}_2 + \frac{\sum x_i}{n}$$

$$\hat{\theta}_1 = -Y_1^{(n)} + \bar{X} \quad \text{is MLE for } \theta_1$$

Ex; let the random variable X has uniform density given by

$$f(x; \theta) = 1 \quad \theta - \frac{1}{2} < x < \theta + \frac{1}{2}$$

$$= I_{(\theta - \frac{1}{2}, \theta + \frac{1}{2})}(x)$$

Where $-\infty < \theta < \infty$

Sol; The likelihood function for a sample of size n is

$$L(x_1, \dots, x_n; \theta) = \prod_{i=1}^n I_{(\theta - \frac{1}{2}, \theta + \frac{1}{2})}(x_i)$$

That is, $\prod_{i=1}^n I_{(\theta - \frac{1}{2}, \theta + \frac{1}{2})}(x_i)$ is unity if all the observations $x_1, \dots, x_n$ are in the interval $[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$

$$I_{(\theta - \frac{1}{2}, \theta + \frac{1}{2})}(x) = \begin{cases} 1 & \theta - \frac{1}{2} < x < \theta + \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

$$\theta - \frac{1}{2} \leq x_1, \dots, x_n \leq \theta + \frac{1}{2}$$

If $y_1, y_2, \dots, y_n$ are ordered observations, then

$$\theta - \frac{1}{2} \leq y_1 < y_2 < \dots < y_n < \theta + \frac{1}{2}$$

$$\theta - \frac{1}{2} \leq y_1 \leq y_n \leq \theta + \frac{1}{2}$$

$$\theta \leq y_1 + \frac{1}{2} \quad \theta \geq y_n - \frac{1}{2}$$

$$y_n - \frac{1}{2} \leq \theta \leq y_1 + \frac{1}{2}$$

$$L(x_1, \dots, x_n; \theta) = I_{(y_n - \frac{1}{2}, y_1 + \frac{1}{2})}(\theta)$$

Where y_1 is smallest obs and y_n is largest obs.

We see that the L.F is either 1 (for $y_n - \frac{1}{2} \leq \theta \leq y_1 + \frac{1}{2}$) or 0 otherwise.

Hence any statistic with value $\hat{\theta}$ satisfying $y_{n-\frac{1}{2}} \leq \hat{\theta} \leq y_{\frac{1}{2}}$ is a maximum likelihood estimate.

Ex; Suppose $x_1, x_2, \dots, x_n$ is a random sample from a population having the density function

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} & 0 < x < \theta \\ 0 & \text{otherwise} \end{cases}$$

Find M.L.E of θ .

Sol; $f(x; \theta) = \frac{1}{\theta} I_{(0, \theta)}(x)$

The likelihood function of the sample is

$$L(\theta) = \frac{1}{\theta^n} \prod_{i=1}^n I_{(0, \theta)}(x_i)$$

$L(\theta)$ is a decreasing function of θ . The maximum of such function cannot be found by differentiation, but by selecting θ as small as possible.

$$0 < x_1 < \dots < x_n < \theta$$

$$0 < y_1 < y_2 < \dots < y_n < \theta$$

$$0 < y_1 < y_n < \theta$$

$$0 \leq y_n \leq \theta$$

Thus the likelihood function $L(\theta)$ attains its maximum for $\theta = \max(x_1, \dots, x_n)$

Hence M.L.E of θ is maximum value in the sample

So M.L.E of θ is $y_n = \hat{\theta}$

Ex; Given a random sample of size n from a population having the density function

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta - \alpha} & \alpha < x < \beta, \beta > \alpha \\ 0 & \text{otherwise} \end{cases}$$

Find M.L.E of α and β .

Sol; The likelihood function of the sample is

$$L(\alpha, \beta) = \left(\frac{1}{\beta - \alpha} \right)^n \prod_{i=1}^n I_{(\alpha, \beta)}(x_i)$$

$$L(\alpha, \beta) = \frac{1}{(\beta - \alpha)^n} \prod_{i=1}^n I_{(\alpha, \beta)}(x_i)$$

$$\alpha \leq x_1, x_2, \dots, x_n \leq \beta$$

If $y_1, y_2, \dots, y_n$ are ordered observations ~~having~~

where $y_1 = \min(x_1, x_2, \dots, x_n)$ $y_n = \max(x_1, x_2, \dots, x_n)$

$$\alpha \leq y_1 < y_2 < \dots < y_n \leq \beta$$

$$\text{i.e. } \alpha \leq y_1, \beta \geq y_n$$

Thus $L(\alpha, \beta)$ attains its maximum for $\alpha = y_1 = \min(x_i)$ and

$$\beta = y_n = \max(x_i)$$

Hence MLE of α, β are

$$\hat{\alpha} = y_1, \hat{\beta} = y_n$$

Invariance Property of M.L. Estimators

Suppose we have

$$L(\theta) = f(x_1, \dots, x_n; \theta) \quad \theta \in \Omega$$

Suppose $\hat{\theta}$ is a M.L. estimator of θ . If $g(\theta)$ is a one to one function

$$\text{i.e. } g(\theta_1) = g(\theta_2) \iff \theta_1 = \theta_2$$

Then $g(\hat{\theta})$ is a MLE of $g(\theta)$. That is $g(\hat{\theta})$ is the value of $g(\theta)$ at which $L(\theta)$ is maximized.

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \hat{\sigma}^2 \text{ is the M.L.E of } \sigma^2.$$

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$g(\hat{\sigma}^2) = \hat{\sigma} \text{ is M.L.E of } g(\sigma^2) = \sigma$$

The result that $g(\hat{\theta})$ is a M.L.E of $g(\theta)$ holds even if $g(\cdot)$ is not a one to one function; and even if θ is a vector.

Example; Let $X_1, X_2, \dots, X_n$ be a random sample from normal population with parameters μ, σ^2 . Obtain M.L.E of σ .

Sol;

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$$

The L.F is

$$L(\mu, \sigma^2) = \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

$$\ln L = -\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

diff w.r.t. σ and equating to zero

$$\frac{\partial \ln L}{\partial \sigma} = -\frac{n}{\sigma} - \frac{(-2)}{2\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\frac{n}{\sigma} = \frac{\sum (x_i - \mu)^2}{\sigma^3}$$

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

$$\hat{\sigma} = \sqrt{\frac{\sum (x_i - \mu)^2}{n}} \quad \text{M.L.E of } \sigma \text{ is } \hat{\sigma}$$

Example; Let $X_1, \dots, X_n$ be independent random variables with the same log-normal distribution

$$Y_i = \log(X_i) \sim N(\mu, \sigma^2)$$

Find M.L.E of $E(X_i)$ and $\text{Var}(X_i)$

Solution; The p.d.f of log normal dist is

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{1}{2}\frac{(\log x - \mu)^2}{\sigma^2}} \quad 0 < x < \infty$$

$$Y = E(X_i) = E(e^{Y_i}) = \int_{-\infty}^{\infty} e^{y_i} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}} dy \quad \begin{cases} \log x = y \\ \frac{1}{x} dx = dy \end{cases}$$

$$E(X_i) = e^{\mu + \frac{\sigma^2}{2}}$$

$$\begin{aligned} t^2 = \text{Var}(X_i) &= \text{Var}(e^{Y_i}) \\ &= e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) \end{aligned}$$

If μ and σ^2 are unknown, then M.L.E is $\hat{\mu}, \hat{\sigma}^2$ with

$$\hat{\mu} = \bar{Y} \quad , \quad \hat{\sigma}^2 = \frac{1}{n} \sum (Y_i - \bar{Y})^2$$

The M.L.E of V is $\hat{V} = e^{\bar{Y} + \frac{\sum (Y_i - \bar{Y})^2}{2n}}$

$$\hat{\tau} = e^{\frac{2\bar{Y} + \sum (Y_i - \bar{Y})^2/n}{-1}} \left(e^{\frac{\sum (Y_i - \bar{Y})^2/n}{-1}} \right)$$



The Method of Moments Estimation

The M.M. exist of equating the first few moments of a population with the corresponding moments of a sample. In this way we obtain as many equations as are needed for the solution of the unknown parameters of the population. The k th sample moment of a set of n observations $x_1, x_2, \dots, x_n$ is

$$m'_k = \frac{1}{n} \sum_{i=1}^n x_i^k$$

The k th moment of a random variable X is

$$\mu'_k = E(X^k)$$

In the M.M. we use as many equations $m'_k = \mu'_k$, $k=1, 2, 3, \dots$ as are needed for the solution of the unknown parameters of the population. Thus the estimates are obtained by replacing the population moments by sample moments.

Example; Let $x_1, x_2, \dots, x_n$ be a random sample from a Poisson distribution with parameter λ . Find the method of moments estimator of λ .

Sol;

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, 2, \dots$$

Since only one parameter λ is to be estimated, we need to solve one equation, namely $m'_1 = \mu'_1$

Now for Poisson distribution $\mu'_1 = \lambda = E(X)$

Replacing the population moment μ'_1 by the sample moment m'_1

we get $m'_1 = \lambda$

Thus M.M. estimator of λ is $\hat{\lambda} = m'_1 = \frac{\sum x_i}{n} = \bar{x}$

Example; let $x_1, x_2, \dots, x_n$ be a random sample from a normal population with mean μ and variance σ^2 . Find the method of moments estimators of the parameters μ and σ .

Sol;

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$$

Since two parameters μ and σ^2 are to be estimated, we need two equations for solution. These equations are

$$m'_1 = \mu'_1, \quad m'_2 = \mu'_2$$

Now for normal distribution with mean μ and variance σ^2

$$\mu'_1 = \mu, \quad \text{and} \quad \mu'_2 = \mu_2 + \mu^2 = \sigma^2 + \mu^2$$

Replacing the population moments by the sample moments we have

$$m'_1 = \mu, \quad m'_2 = \sigma^2 + \mu^2 \quad \text{--- (1)}$$

Thus M.M estimator of μ is $\hat{\mu} = m'_1 = \bar{x}$

The method of moments estimator of σ^2 is

$$\hat{\sigma}^2 = m'_2 - \hat{\mu}^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$\Rightarrow \hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Example; The log normal distribution is given as

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} \quad 0 < x < \infty$$

Obtain the moment estimators for the unknown parameters μ and σ^2 .

Sol; First we evaluate mean and variance of the log normal dist

$$E(x) = \frac{1}{\sqrt{2\pi}\sigma x} \int_0^{\infty} x^2 e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} dx$$

$$\text{putting } z = \frac{\log x - \mu}{\sigma} \Rightarrow \sigma z = \log x - \mu$$

$$\sigma dz = \frac{1}{x} dx$$

$$dx = x \sigma dz$$

$$\text{where } \log x = \mu + \sigma z$$

$$x = e^{\mu + \sigma z}$$

$$\text{So } dx = \sigma e^{\mu + \sigma z} dz$$

$$x=0, z=-\infty$$

$$x=\infty, z=\infty$$

$$E(X) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \frac{-z^2/2}{e} \cdot e^{\mu+\sigma z} dz$$

$$= \frac{e^{\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2+\sigma^2-\sigma^2-2\sigma z)} dz$$

$$E(X) = \frac{e^{\mu+\frac{\sigma^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-\sigma)^2} dz$$

pulling $z-\sigma=t$, $dz=dt$

$$z=-\infty \Rightarrow t=-\infty$$

$$z=\infty \Rightarrow t=\infty$$

$$E(X) = \frac{e^{\mu+\frac{\sigma^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt$$

$$E(X) = e^{\mu+\frac{\sigma^2}{2}}$$

$$\therefore \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt = 1$$

$$E(X^2) = \frac{1}{\sqrt{2\pi}\sigma} \int_0^{\infty} x^2 e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} dx$$

pulling $\frac{\log x - \mu}{\sigma} = z \Rightarrow dx = \sigma e^{\mu+\sigma z} dz$

$$x = e^{\mu+\sigma z}$$

$$E(X^2) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{\mu+\sigma z} \frac{-z^2/2}{e} \cdot e^{\mu+\sigma z} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2+4\mu+4\sigma z+4\sigma^2+4\sigma^2)} dz$$

$$= \frac{e^{2\mu+2\sigma^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z^2+4\sigma^2-4\sigma z)} dz$$

$$= \frac{e^{2\mu+2\sigma^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-2\sigma)^2} dz$$

put $z-2\sigma=t$, $dz=dt$

$$E(X^2) = \frac{e^{2\mu+2\sigma^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt$$

$$E(X^2) = e^{2\mu + 2\sigma^2}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} \end{aligned}$$

$$\boxed{\text{Var}(X) = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1)}$$

Since two parameters are to be estimated, we need two equations for solution. $m'_1 = \mu'$, $m'_2 = \mu'_2$.

Now for log normal dist

$$\mu'_1 = e^{\mu + \frac{\sigma^2}{2}}, \mu'_2 = e^{\frac{z(\mu + \sigma^2)}{2}}$$

Replacing the population moments by sample moments.

$$m'_1 = e^{\mu + \frac{\sigma^2}{2}} = \bar{x}$$

$$\mu + \frac{\sigma^2}{2} = \ln \bar{x} \quad \text{--- ①}$$

$$m'_2 = e^{\frac{z(\mu + \sigma^2)}{2}} = \frac{1}{n} \sum x_i^z$$

$$2\mu + 2\sigma^2 = \ln \left(\frac{\sum x_i^z}{n} \right) \quad \text{--- ②}$$

Multiplying eq ① by 2 and subtracting from eq ②

$$2\mu + 2\sigma^2 = \ln \left(\frac{\sum x_i^z}{n} \right)$$

$$2\mu + \sigma^2 = \ln \bar{x}^2$$

$$\sigma^2 = \ln \left(\frac{\sum x_i^z}{n} \right) - \ln \bar{x}^2$$

$$\tilde{\sigma}^2 = \ln \frac{\sum x_i^z}{n} - \ln (\sum x_i)^2 + z \ln n$$

$$\tilde{\sigma}^2 = \ln \left(\frac{n \sum x_i^z}{(\sum x_i)^2} \right) = \ln \frac{M'_2}{M'_1{}^2}$$

From ①

$$\begin{aligned}\bar{\mu} &= \ln \bar{x} - \frac{\bar{\sigma}^2}{2} \\ &= \ln \bar{x} - \frac{1}{2} \ln \left(\frac{\sum x_i^2}{n} \right) + \frac{1}{2} \ln \bar{x}^2 \\ &= \ln \bar{x}^2 - \ln \sqrt{\sum x_i^2} + \ln \sqrt{n} \\ \bar{\mu} &= \ln \left(\frac{\sqrt{n} \bar{x}^2}{\sqrt{\sum x_i^2}} \right) = \ln \frac{M_1'^2}{M_2'}$$

Example; Use the method of moments to estimate α and β from the density function

$$f(x; \alpha, \beta) = \frac{1}{\beta - \alpha} \quad \alpha < x < \beta \quad \beta > \alpha$$

Solution; let $x_1, x_2, \dots, x_n$ be a random sample from the given density function. The method of moments equations are

$$m'_1 = \mu'_1, \quad m'_2 = \mu'_2$$

$$\begin{aligned}\mu'_1 &= \int_{\alpha}^{\beta} x f(x; \alpha, \beta) dx = \int_{\alpha}^{\beta} \frac{x}{\beta - \alpha} dx \\ &= \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} x dx = \frac{1}{\beta - \alpha} \left[\frac{x^2}{2} \right]_{\alpha}^{\beta}\end{aligned}$$

$$\mu'_1 = \frac{\alpha + \beta}{2}$$

$$\begin{aligned}\mu'_2 &= \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} x^2 dx = \frac{1}{\beta - \alpha} \left[\frac{x^3}{3} \right]_{\alpha}^{\beta} \\ &= \frac{\beta^3 - \alpha^3}{(\beta - \alpha) 3} = \frac{\beta^2 + \alpha^2 + \alpha\beta}{3}\end{aligned}$$

Replacing the ~~sample~~ population moments by sample moments

$$m'_1 = \frac{\alpha + \beta}{2} \quad \text{①} \quad m'_2 = \frac{\beta^2 + \alpha^2 + \alpha\beta}{3} \quad \text{②}$$

From ①

$$\alpha = 2m'_1 - \beta \quad \text{③} \quad \text{substituting in ②}$$

$$3m'_2 = \beta^2 + \beta(2m'_1 - \beta) + (2m'_1 - \beta)^2$$

$$3m'_2 = \beta^2 + 2m'_1\beta - \beta^2 + 4m_1'^2 + \beta^2 - 4m'_1\beta$$

$$\text{or} \quad \beta^2 - 2m'_1\beta + 4m_1'^2 - 3m_2'^2 = 0$$

Solving this quadratic form in β

$$\beta = \frac{2m'_1 \pm \sqrt{4m_1'^2 - 4(4m_1'^2 - 3m_2')}}{2}$$

$$= m'_1 \pm \sqrt{m_1'^2 - 4m_1'^2 + 3m_2'}$$

$$\beta = m'_1 \pm \sqrt{3(m_2' - m_1'^2)}$$

putting this in (3)

$$\alpha = m'_1 \mp \sqrt{3(m_2' - m_1'^2)}$$

Since $\alpha < \beta$

$$\therefore \tilde{\beta} = m'_1 + \sqrt{3(m_2' - m_1'^2)}, \quad \tilde{\alpha} = m'_1 - \sqrt{3(m_2' - m_1'^2)}$$

The method of moments is not applicable when the theoretical moments $\mu'_2, \mu'_3, \dots$, do not exist as in case of a Cauchy distribution. i.e. the distribution with density function

$$f(x; \theta) = \frac{\theta}{\pi(x^2 + \theta^2)} \quad -\infty < x < \infty, \theta > 0$$

Moments don't exist for this density function.

Example; Consider the Laplace distribution centred at the origin with sample parameters β

$$f(x; \beta) = \frac{1}{2\beta} e^{-|x|/\beta} \quad -\infty < x < \infty, \beta > 0$$

Find ML and M.M estimators of β .

Solution;

(i) The L.F is

$$L = \frac{1}{(2\beta)^n} e^{-\frac{1}{\beta} \sum_{i=1}^n |x_i|}$$

$$\ln L = -n \ln 2 - n \ln \beta - \frac{1}{\beta} \sum_{i=1}^n |x_i|$$

$$\frac{\partial \ln L}{\partial \beta} = -\frac{n}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n |x_i| = 0$$

$$\frac{n}{\beta} = \frac{\sum |x_i|}{\beta^2} \Rightarrow \hat{\beta} = \frac{\sum |x_i|}{n} \text{ is MLE of } \beta$$

ii) M.M. equations are

$$m'_1 = \mu'_1, \quad m'_2 = \mu'_2$$

$$\mu'_1 = \int x f(x; \beta) dx$$

$$= \int_{-\infty}^{\infty} \frac{x}{2\beta} e^{\frac{-|x|}{\beta}} dx$$

$$= \frac{1}{2\beta} \left[\int_{-\infty}^0 x e^{x/\beta} dx + \int_0^{\infty} x e^{-x/\beta} dx \right]$$

Replacing x by $(-x)$ in first integral

$$= \frac{1}{2\beta} \left[\int_{\infty}^0 -x e^{-x/\beta} (-dx) + \int_0^{\infty} x e^{-x/\beta} dx \right]$$

$$= \frac{1}{2\beta} \left[-\int_0^{\infty} x e^{-x/\beta} dx + \int_0^{\infty} x e^{-x/\beta} dx \right]$$

$$\mu'_1 = 0$$

Because first population moment is zero, therefore we proceed for second population moment.

$$\mu'_2 = E(x^2) = \frac{1}{2\beta} \int_{-\infty}^{\infty} x^2 e^{\frac{-|x|}{\beta}} dx$$

$$= \frac{1}{2\beta} \left[\int_{-\infty}^0 x^2 e^{x/\beta} dx + \int_0^{\infty} x^2 e^{-x/\beta} dx \right]$$

Replacing x by $(-x)$ in first integral

$$\mu'_2 = \frac{1}{2\beta} \left[\int_{\infty}^0 x^2 e^{-x/\beta} (-dx) + \int_0^{\infty} x^2 e^{-x/\beta} dx \right]$$

$$= \frac{1}{2\beta} \left[\int_0^{\infty} x^2 e^{-x/\beta} dx + \int_0^{\infty} x^2 e^{-x/\beta} dx \right]$$

$$\mu'_2 = \frac{1}{\beta} \left[\int_0^{\infty} x^2 e^{-x/\beta} dx \right]$$

putting $\frac{x-\alpha}{\beta} = u$, $\frac{1}{\beta} dx = du$

$$\begin{aligned}\mu'_2 &= \int_0^{\infty} (\beta u)^2 e^{-u} du \\ &= \beta^2 \int_0^{\infty} u^2 e^{-u} du\end{aligned}$$

$$\mu'_2 = 2\beta^2$$

Replacing pop moment by sample moment

$$m'_2 = 2\beta^2$$

$$\hat{\beta} = \sqrt{\frac{m'_2}{2}} \text{ is MME of } \beta$$

Example; Given the following distribution

$$f(x; \alpha, \beta) = \frac{1}{\beta} e^{-(x-\alpha)/\beta} \quad \alpha < x < \infty$$

Find M.M. estimators of α, β .

Solution; let $X_1, X_2, \dots, X_n$ be a random sample from the given density function. The method of equations are

$$m'_1 = \mu'_1, \quad m'_2 = \mu'_2$$

$$\begin{aligned}\mu'_1 &= \int_{\alpha}^{\infty} \frac{x}{\beta} e^{-(x-\alpha)/\beta} dx \\ &\quad \text{put } \frac{x-\alpha}{\beta} = t, \quad x = \alpha + \beta t \\ &\quad dx = \beta dt\end{aligned}$$

$$\begin{aligned}\mu'_1 &= \int_0^{\infty} (\alpha + \beta t) e^{-t} dt \\ &= \alpha \int_0^{\infty} e^{-t} dt + \beta \int_0^{\infty} t e^{-t} dt \\ &= \alpha \left| \frac{e^{-t}}{-1} \right|_0^{\infty} + \beta \cdot 1\end{aligned}$$

$$\mu'_1 = \alpha + \beta$$

$$\mu'_2 = \int_{\alpha}^{\infty} \frac{x^2}{\beta} e^{-(x-\alpha)/\beta} dx$$

$$\mu'_2 = \int_0^{\infty} (\alpha + \beta)^2 e^{-t} dt$$

$$= \alpha^2 \int_0^{\infty} e^{-t} dt + \beta^2 \int_0^{\infty} t^2 e^{-t} dt + 2\alpha\beta \int_0^{\infty} t e^{-t} dt$$

$$= \alpha^2 \left| \frac{e^{-t}}{-1} \right|_0^{\infty} + \beta^2 \int_0^{\infty} t^2 e^{-t} dt + 2\alpha\beta \int_0^{\infty} t e^{-t} dt$$

$$\mu'_2 = \alpha^2 + 2\beta^2 + 2\alpha\beta$$

Replacing population moments by the sample moments

$$m'_1 = \alpha + \beta \quad \text{--- ①} \quad m'_2 = \alpha^2 + 2\beta^2 + 2\alpha\beta \quad \text{--- ②}$$

$$\text{From ①} \quad \alpha = m'_1 - \beta \quad \text{--- ③}$$

Substituting ③ in ②

$$m'_2 = (m'_1 - \beta)^2 + 2\beta^2 + 2(m'_1 - \beta)\beta$$

$$= m_1'^2 + \beta^2 - 2\beta m'_1 + 2\beta^2 + 2\beta m'_1 - 2\beta^2$$

$$m'_2 = m_1'^2 + \beta^2$$

$$\beta^2 = m'_2 - (m'_1)^2$$

$$\tilde{\beta} = \sqrt{m'_2 - (m'_1)^2} \text{ is MME of } \beta$$

$$\text{From ①} \quad \tilde{\alpha} = m'_1 - \sqrt{m'_2 - (m'_1)^2} \text{ is MME of } \alpha.$$

Example; let $x_1, x_2, \dots, x_n$ be a random sample from the density function

$$f(x; \theta) = \theta e^{-\theta x} \quad 0 < x < \infty$$

Estimate θ by the method of moments.

Solution; The method of moments equation is

$$m'_1 = \mu'_1$$

$$\mu'_1 = \int_0^{\infty} x f(x; \theta) dx$$

$$= \int_0^{\infty} x \theta e^{-\theta x} dx$$

put $ex = t$, $dx = \frac{1}{e} dt$

$$\mu'_1 = \int_0^{\infty} \frac{t^{1-1}}{e} e^{-t} dt$$

$$\mu'_1 = \frac{1}{e} \Gamma_2 = 1/e$$

Replacing pop moments by sample moments

$$m'_1 = 1/e$$

$$\hat{\theta} = \frac{1}{m'_1} = \frac{1}{\bar{x}} \text{ is M.M.E of } \theta.$$

Example; let X be a binomial random variable with parameters n and p , both unknown. Given a random sample of n observations of X , compute M.M.E of n and p .

Solution; The method of moments equations are

$$m'_1 = \mu'_1, \quad m'_2 = \mu'_2$$

$$\mu'_1 = np, \quad \mu'_2 = np(1-p) + n^2 p^2$$

Replacing the population moments by the sample moments

$$m'_1 = np \quad \text{--- (1)} \quad m'_2 = np(1-p) + n^2 p^2 \quad \text{--- (2)}$$

From (1) $p = \frac{m'_1}{n}$ putting in (2)

$$\begin{aligned} m'_2 &= n \left(\frac{m'_1}{n} \right) \left(1 - \frac{m'_1}{n} \right) + n^2 \left(\frac{m'_1}{n} \right)^2 \\ &= m'_1 - \frac{m'^2_1}{n} + m'^2_1 \end{aligned}$$

$$\text{or } \frac{m'^2_1}{n} = m'_1 + m'^2_1 - m'_2 \implies \tilde{n} = \frac{m'^2_1}{m'_1 + m'^2_1 - m'_2}$$

$$\tilde{p} = \frac{m'_1 + m'^2_1 - m'_2}{m'_1}$$

$\tilde{n}, \tilde{p}$ are MME of n and p respectively.

Example; Suppose $x_1, x_2, \dots, x_n$ is a random sample of size n from a population having the probability function

$$f(x; p) = p(1-p)^{x-1} \quad x = 1, 2, 3, \dots$$

Find M.M. for p .

Sol; The method of moments equation is $m'_i = \mu'_i$

$$\begin{aligned} \mu'_1 &= \sum_{x=1}^{\infty} x p(1-p)^{x-1} \\ &= p + 2p(1-p) + 3p(1-p)^2 + 4p(1-p)^3 + \dots \\ &= p[1 + 2(1-p) + 3(1-p)^2 + 4(1-p)^3 + \dots] \end{aligned}$$

We know

$$\begin{aligned} (1+x)^n &= 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots \\ (1-x)^n &= 1 - nx + \frac{n(n-1)}{2!} x^2 - \frac{n(n-1)(n-2)}{3!} x^3 + \dots \end{aligned}$$

$$\mu'_1 = p[1 - (1-p)^{-2}] = p(1-p)^{-2} = \left(\frac{1}{p^2}\right) p$$

$$\mu'_1 = \frac{1}{p}$$

Replacing population moment by sample moments

$$m'_1 = \frac{1}{p}$$

$$\hat{p} = \frac{1}{m'_1} = \frac{1}{\bar{x}} \quad \text{is M.M.E for } p.$$

Example; Assume that x is a continuous random variable with density function

$$f(x; \theta) = e^{\theta - x} \quad x > \theta$$

Find M.M. for θ .

Sol;

$$m'_1 = \mu'_1$$

$$\mu'_1 = e^{\theta} \int_{\theta}^{\infty} x e^{-x} dx$$

$$= e^{\theta} \left[-x e^{-x} \Big|_{\theta}^{\infty} + \int_{\theta}^{\infty} e^{-x} dx \right] = e^{\theta} \left[-0 + \theta e^{-\theta} + \left[-e^{-x} \right]_{\theta}^{\infty} \right]$$

$$= e^{\theta} \left[\theta e^{-\theta} + e^{-\theta} \right]$$

$$\mu'_1 = \theta + 1$$

$$m'_1 = \theta + 1 \quad \therefore \hat{\theta} = m'_1 - 1$$

Thus method of moment estimator of θ is $\hat{\theta} = \bar{x} - 1$

Examples



Properties of THE Maximum Likelihood Estimators;

i) M.L.E are invariant. That is, the M.L.E of a function of parameters is the same function of the M.L.E of parameters.

If $\hat{\theta}$ is a M.L.E of θ , then $g(\hat{\theta})$ is also a M.L.E of $g(\theta)$.

If $\hat{\sigma}^2$ is a M.L.E of σ^2 then $\hat{\sigma}$ is also a M.L.E of σ .

ii) Maximum likelihood estimators are generally biased. However, as the sample size increases, the bias becomes negligible. Thus, in the limit, the maximum likelihood estimators are approximately normally distributed unbiased.

iii) If a single sufficient statistic exists for θ , it is a function of M.L.E of θ .

Proof; If the statistic t is a sufficient statistic for θ , then by N.F.T.,

$$L(x; \theta) = g(t; \theta) h(x)$$

where $h(x)$ is independent of θ .

Choosing $\hat{\theta}$ to maximize $L(x; \theta)$ is equivalent to choosing $\hat{\theta}$

to maximize $g(t; \theta)$.

Therefore $\hat{\theta}(x)$ will be a function of t alone. ✓

iv) Under fairly general conditions, M.L.E are consistent.

Proof; let n independent observations be taken from a dist $f(x; \theta)$ and for each n we choose the M.L.E $\hat{\theta}$ such that

$$\ln L(x; \hat{\theta}) \geq \ln L(x; \theta) \quad \text{--- (1)}$$

We denote true value of θ by θ_0 , and let E_0 denote the operation of taking expectations when the true value θ_0 holds.

Consider random variable $\frac{L(x; \hat{\theta})}{L(x; \theta_0)}$

In virtue of the fact that the G.M of a distribution cannot exceed its A.M, we have for all $\theta^* \neq \theta_0$

$$E_0 \left[\ln \frac{L(x; \theta^*)}{L(x; \theta_0)} \right] < \ln E_0 \left[\frac{L(x; \theta^*)}{L(x; \theta_0)} \right] \quad \text{--- (2)}$$

G.M A.M

The expectation on R.H.S. of ② is

$$\int \frac{L(x; \theta^*)}{L(x; \theta_0)} L(x; \theta_0) dx = 1$$

$$E_0 \left[\frac{\ln L(x; \theta^*)}{\ln L(x; \theta_0)} \right] < \ln 1.$$

$$E_0 \left[\frac{\ln L(x; \theta^*)}{\ln L(x; \theta_0)} \right] < 0, \quad E_0 [\ln L(x; \theta^*) - \ln L(x; \theta_0)] < 0$$

$$E_0 [\ln L(x; \theta^*)] < E_0 [\ln L(x; \theta_0)] \quad \text{--- ③}$$

For any value of θ

$\frac{1}{n} \ln L(x; \theta) = \frac{1}{n} \sum \ln f(x; \theta)$ is the mean of a set of n independent variates with expectation

$$E_0 [\ln f(x; \theta)] = E_0 \left[\frac{1}{n} \ln L(x; \theta) \right]$$

By the law of large numbers $\frac{1}{n} \ln L(x; \theta)$ converges with probability 1 to its expectation as n increases.

Thus as $n \rightarrow \infty$, we have from ③

$$\frac{1}{n} \ln L(x; \theta^*) < \frac{1}{n} \ln L(x; \theta_0)$$

$$\lim_{n \rightarrow \infty} P \left[\frac{1}{n} \ln L(x; \theta^*) < \frac{1}{n} \ln L(x; \theta_0) \right] = 1, \quad \theta^* \neq \theta_0 \quad \text{--- ④}$$

From ④, with $\theta = \theta_0$ gives $\ln L(x; \hat{\theta}^*) \geq \ln L(x; \theta_0) \quad \text{--- ⑤}$

From ④ and ⑤, as $n \rightarrow \infty$

$L(x; \hat{\theta})$ can not take any other value than $L(x; \theta_0)$

If $L(x; \theta)$ is a one-to-one function of θ , this implies that

$$P \left[\lim_{n \rightarrow \infty} \hat{\theta} = \theta_0 \right] = 1$$

(v) The maximum likelihood estimator $\hat{\theta}$ is asymptotically normally distributed about mean θ_0 and variance

$$\frac{1}{R^2(\theta_0)} = \frac{1}{E \left(\frac{\partial \ln L}{\partial \theta} \right)^2}_{\theta_0}$$

Hence if a M.L.E exists, it is efficient.

Proof; We assume that the first two derivatives of $\ln(L(x; \theta))$ exist

$$\begin{aligned} \int \dots \int L dx_1 \dots dx_n &= 1, & \int \dots \int \frac{\partial L}{\partial \theta} dx_1 \dots dx_n &= 0 \\ \int \dots \int \frac{1}{L} \frac{\partial L}{\partial \theta} L dx_1 \dots dx_n &= 0 \\ \int \dots \int \left(\frac{\partial \ln L}{\partial \theta} \right) L dx_1 \dots dx_n &= 0 \end{aligned}$$

$$E\left(\frac{\partial \ln L}{\partial \theta}\right) = 0 \quad \text{--- (1)}$$

$$R^2(\theta_0) = -E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right) = E\left(\frac{\partial \ln L}{\partial \theta}\right)^2 \quad \text{--- (2)}$$

where $R^2(\theta_0) > 0$

Using Taylor's theorem we would expand this

$$\left(\frac{\partial \ln L}{\partial \theta}\right)_{\hat{\theta}} = \left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} + (\hat{\theta} - \theta_0) \left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\theta^*} \quad \text{--- (3)}$$

where θ^* is some value between $\hat{\theta}$ and θ_0 .

$\hat{\theta}$ is the solution of $\frac{\partial \ln L}{\partial \theta} = 0$

The L.H.S of (3) is zero. On R.H.S both $\frac{\partial \ln L}{\partial \theta}$ and $\frac{\partial^2 \ln L}{\partial \theta^2}$ are the sum of independent identical variates and as $n \rightarrow \infty$ each converges to its expectation by law of large numbers.

i.e. $E\left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} = 0$ and second term $\frac{\partial^2 \ln L}{\partial \theta^2} \neq 0$

Therefore $\hat{\theta} - \theta_0$ must converges to zero as $n \rightarrow \infty$, $\hat{\theta}$ is a consistent estimator.

We write Eq (3) in the form

$$-\left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} / \left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\theta^*} = \hat{\theta} - \theta_0$$

$$(\hat{\theta} - \theta_0) R(\theta_0) = \frac{\left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} / R(\theta_0)}{\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\theta^*} / (-R^2(\theta_0))} \quad \text{--- (4)}$$

In the denominator on the R.H.S of (4), we have since $\hat{\theta}$ is consistent for θ_0 . So from eq (2)

$$\lim_{n \rightarrow \infty} P\left[\frac{\partial^2 \ln L}{\partial \theta^2}\right]_{\hat{\theta}} = -R^2(\theta_0) = A \quad \text{--- (5)}$$

So the denominator converges to 1.

The numerator on the R.H.S of (4) is the ratio to $R(\theta_0)$ of sum of independent and identical variates $\frac{\partial \ln L}{\partial \theta}$.

$$E\left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} = 0 \text{ and } \text{Var}\left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} = \frac{1}{E\left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0}^2} = \frac{1}{R^2(\theta_0)} = R^2(\theta_0) \quad *$$

From (4)
$$\frac{\hat{\theta} - \theta_0}{\frac{1}{R(\theta_0)}} = \left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} / \frac{\frac{\partial \ln L}{\partial \theta}}{R(\theta_0)}$$

By Central limit theorem the ratio $\left(\frac{\partial \ln L}{\partial \theta}\right)_{\theta_0} / R(\theta_0)$ is asymptotically S.N.V, the same is true of L.H.S

$\therefore (\hat{\theta} - \theta_0)$ is asymptotically S.N.

Hence the M.L.E $\hat{\theta}$ is asymptotically normally distributed with mean θ_0 and variance $\frac{1}{R^2(\theta_0)}$.

(i)

$$\begin{aligned} \text{Var}(\hat{\theta}) &\geq \frac{1}{E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)} \\ &\geq \frac{1}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2} \end{aligned}$$

The asymptotic variance of M.L.E $\hat{\theta}$ is equal to the M.V.B. This implies that the M.L.E $\hat{\theta}$ is efficient.

The Method of Least Squares (LS)

As a general principle, it states that if we wish to estimate the parameter vector θ from a set of n observations of $j = 1, 2, \dots, n$ which related to θ through their expectations

$$E(y_j) = \beta_j(\theta)$$

Then we should choose our estimator $\hat{\theta}$ to minimize the sum of squared deviations between y_j and their expectations i.e we minimize

$$S = \sum_{j=1}^n (y_j - \beta_j(\hat{\theta}))^2$$

If the distribution of the y_j about their expectations are independently normal with same variance σ^2 and the L.S.E will be identical with maximum likelihood estimator of θ .

Like the M.L. method, the L.S method has some asymptotic optimum properties. In a certain class of situations it has the property, even in small samples, it provides unbiased estimators linear in the observations, which have minimum variance (MV). This situation is usually described as the linear model (L.M), in which the y_j are distributed with constant variance about mean values which are linear function of the unknown parameters.

The Least Square Estimator in the Linear Model

We write the linear model in the form

$$y = X\theta + \epsilon \quad \text{--- (1)}$$

where y is a $(n \times 1)$ vector of the observations, X is a $(n \times k)$ matrix of known coefficients (with $n > k$), θ is a $(k \times 1)$ vector of parameters and ϵ is an $(n \times 1)$ vector of "error" random variables ϵ_i with

$$E(\epsilon) = 0 \quad \text{--- (2)}$$

and dispersion matrix

$$\text{Var}(\epsilon) = E(\epsilon\epsilon') = \sigma^2 I \quad \text{--- (3)}$$

where I is an $(n \times n)$ identity matrix.

Eq (2) and (3) contain the assumption that the ϵ_i are uncorrelated and all have zero mean and same variance σ^2 .

The least square method requires that we minimize the SS

$$S = (y - X\theta)'(y - X\theta) \quad \text{--- (4)}$$

For variation in the components of θ , Differentiating and equating to 0, we get.

$$2X'(y - X\theta) = 0$$

$$X'y - X'X\theta = 0$$

Thus L.S.E is

$$\hat{\theta} = (X'X)^{-1}X'y \quad \text{--- (5)}$$

~~we assume that $X'X$ is constant~~

we assume that $(X'X)$ is non singular

The L.S.E given by Eq (5) is unbiased.

Proof: Using Eq (1) and (5)

$$\begin{aligned} \hat{\theta} &= (X'X)^{-1}X'(X\theta + \epsilon) \\ &= (X'X)^{-1}(X'X)\theta + (X'X)^{-1}X'\epsilon \\ \hat{\theta} &= \theta + (X'X)^{-1}X'\epsilon \quad \text{--- (6)} \end{aligned}$$

Since X is constant, then using the assumption (2)

$$E(\hat{\theta}) = \theta \quad \text{--- (7)}$$

Hence $\hat{\theta}$ is an unbiased estimator of θ

Now Dispersion matrix of $\hat{\theta}$ is

$$\text{Var}(\hat{\theta}) = E[(\hat{\theta} - \theta)(\hat{\theta} - \theta)'] \quad \text{--- (8)}$$

Making substitution from (6) we have

$$\text{Var}(\hat{\theta}) = E\left[\{(X'X)^{-1}X'\epsilon\}\{(X'X)^{-1}X'\epsilon\}'\right]$$

$$= E[(X'X)^{-1}X'\epsilon\epsilon'X(X'X)^{-1}]$$

$$= (X'X)^{-1}X'\sigma^2 X(X'X)^{-1}$$

$$\text{Var}(\hat{\theta}) = (X'X)^{-1}\sigma^2 \quad \text{--- (9)}$$

Optimum Properties of The L.S Method

We show that the minimum variance unbiased linear estimator of any sets of linear function of parameter θ , is given by the L.S method
 Proof: let $\underline{t}$ be any vector of estimator which is linear in the observed y . i.e. of the form

$$\underline{t} = \underline{I} y \quad \text{--- (10)}$$

If $\underline{t}$ is to be unbiased for a set of linear function of the parameters say, $\underline{c}\theta$, we have

$$E(\underline{t}) = E(\underline{I} y)$$

$$E(\underline{t}) = \underline{c}\theta$$

where $\underline{c}$ is a known matrix of coefficients.

Now using equation (1)

$$E[\underline{I}(\underline{X}\theta + \epsilon)] = \underline{c}\theta \quad \text{--- (11)}$$

using Eq (2), (11) becomes

$$\underline{I}\underline{X}\theta + \underline{I}E\epsilon = \underline{c}\theta$$

or

$$\underline{I}\underline{X}\theta = \underline{c}\theta$$

$$\underline{I}\underline{X} = \underline{c} \quad \text{--- (12)}$$

Equation (12) is a necessary and sufficient condition for $\underline{c}\theta$ to be unbiased, estimable by $\underline{I}y$.

The dispersion matrix of $\underline{t}$ is

$$\text{Var}(\underline{t}) = E[(\underline{t} - \underline{c}\theta)(\underline{t} - \underline{c}\theta)'] \quad \text{--- (13)}$$

$$\underline{t} = \underline{I}y$$

$$= \underline{I}(\underline{X}\theta + \epsilon)$$

$$\underline{t} = \underline{I}\underline{X}\theta + \underline{I}\epsilon$$

$$\underline{t} = \underline{c}\theta + \underline{I}\epsilon$$

$$\because \underline{I}\underline{X} = \underline{c}$$

$$\underline{t} - \underline{c}\theta = \underline{I}\epsilon$$

Now Eq (13) becomes

$$\text{Var}(\underline{t}) = E[\underline{I}\epsilon][\underline{I}\epsilon]'$$

$$= E[\underline{I}\epsilon\epsilon'\underline{I}']$$

$$\text{Var}(\underline{t}) = \underline{I}\underline{I}'\sigma^2 \quad \text{--- (14)}$$

We wish to minimize the diagonal elements of (14) which are the variances of a set of estimators.

For this we use the identity.

$$T T' = [C(X'X)^{-1}X'] [C(X'X)^{-1}X']' + [I - C(X'X)^{-1}X'] [I - C(X'X)^{-1}X']' \quad (15)$$

it is verified ~~and~~ by multiplying out R.H.S

The R.H.S of (15) is the sum of two terms of the form AA' , each of these terms has therefore non negative diagonal elements. Only the second term on the R.H.S. of (15) is a function of T . Therefore the sum of the two terms has strictly minimum diagonal elements when the second term has all zero diagonal elements. So this occurs when

$$T = C(X'X)^{-1}X' \quad (16)$$

So that the M.V.U. vector of estimators of β is

$$t = T y$$

$$t = C(X'X)^{-1}X' y \quad (17)$$

In which vector β is replaced by its L.S.E $\hat{\beta}$, i.e.

$$t = C \hat{\beta} \quad (18)$$

From (17) and (18) we have

$$\begin{aligned} \text{Var}(t) &= \sigma^2 T T' \\ &= \sigma^2 C(X'X)^{-1}X'X(X'X)^{-1}C' \\ \text{Var}(t) &= \sigma^2 C(X'X)^{-1}C' \quad (19) \end{aligned}$$

If $C = I$, so that we are estimating β itself, and in this case Eq (17), (18) reduces to (5) and (9), i.e.

$$\text{Var}(t) = \sigma^2 (X'X)^{-1} \quad t = (X'X)^{-1}X'y$$

Estimation of Variance;

The second part of the Gauss's theorem on L.S is concerned with the estimation of the variance σ^2 .

Consider the set of residuals in the L.S estimate

$$y - X\hat{\beta} = [y - X\beta + \epsilon] - X(X'X)^{-1}X'y$$

$$\begin{aligned}
 \underline{y} - \underline{X}\hat{\underline{\theta}} &= \underline{X}\underline{\theta} + \underline{\epsilon} - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'(\underline{X}'\underline{X}\underline{\theta} + \underline{\epsilon}) \\
 &= \underline{X}\underline{\theta} + \underline{\epsilon} - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{X}\underline{\theta} - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{\epsilon} \\
 &= \underline{X}\underline{\theta} + \underline{\epsilon} - \underline{X}\underline{\theta} - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{\epsilon} \\
 &= \underline{\epsilon} - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'\underline{\epsilon}
 \end{aligned}$$

$$\underline{y} - \underline{X}\hat{\underline{\theta}} = \left(\underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \right) \underline{\epsilon} \quad \text{--- (1)}$$

where $\underline{I}_n$ is the identity matrix of order n .

Now the matrix in square bracket of (1) is symmetric and idempotent.

$$\left[\underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \right] \left[\underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \right] = \underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'$$

The sum of squared residual is

$$\begin{aligned}
 (\underline{y} - \underline{X}\hat{\underline{\theta}})'(\underline{y} - \underline{X}\hat{\underline{\theta}}) &= \underline{\epsilon}' \left(\underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \right) \left(\underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \right) \underline{\epsilon} \\
 &= \underline{\epsilon}' \left[\underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \right] \underline{\epsilon} \quad \text{--- (2)}
 \end{aligned}$$

$$\begin{aligned}
 E \left[(\underline{y} - \underline{X}\hat{\underline{\theta}})'(\underline{y} - \underline{X}\hat{\underline{\theta}}) \right] &= E(\underline{\epsilon}' \underline{B} \underline{\epsilon}) \\
 &= E(\text{tr } \underline{\epsilon}' \underline{B} \underline{\epsilon}) \\
 &= E(\text{tr } \underline{B} \underline{\epsilon} \underline{\epsilon}') \\
 &= \text{tr } \underline{B} E(\underline{\epsilon} \underline{\epsilon}') \\
 &= \sigma^2 \text{tr } \underline{B} \quad \text{--- (3)}
 \end{aligned}$$

$$\begin{aligned}
 \text{tr } \underline{B} &= \text{tr} \left[\underline{I}_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \right] \\
 &= \text{tr } \underline{I}_n - \text{tr } \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}' \\
 &= n - \text{tr } (\underline{X}'\underline{X})(\underline{X}'\underline{X})^{-1} \\
 &= n - \text{tr } \underline{I}_K
 \end{aligned}$$

$$\text{tr } \underline{B} = n - K$$

Now from (3)

$$E \left[(\underline{y} - \underline{X}\hat{\underline{\theta}})'(\underline{y} - \underline{X}\hat{\underline{\theta}}) \right] = \sigma^2 (n - K)$$

Thus unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{(n-K)} \left[(\underline{y} - \underline{X}\hat{\underline{\theta}})'(\underline{y} - \underline{X}\hat{\underline{\theta}}) \right]$$

The Normality Assumption

If we wish to test the hypothesis concerning the parameters, distributional assumptions are necessary. If we postulate that the ϵ_i are normally distributed, the fact that they are uncorrelated implies their independence.

We shall use the result that an idempotent quadratic form in independent standardized normal variables has a chi-square variate with d.f. given by the rank of the quadratic form

$$Q.F. = \epsilon' B \epsilon \quad \epsilon \text{ is } n \times 1 \quad B \text{ is } n \times n$$

Applying this result to the sum of squared residual

$$(\underline{y} - \underline{X}\hat{\underline{\theta}})'(\underline{y} - \underline{X}\hat{\underline{\theta}}) = \epsilon' [I_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'] \epsilon$$

$$(n-k) s^2 = \epsilon' [I_n - \underline{X}(\underline{X}'\underline{X})^{-1}\underline{X}'] \epsilon$$

$\frac{(n-k) s^2}{\sigma^2}$ is chi square variate with $(n-k)$ d.f.

We have the identity $\underline{y}'\underline{y} = (\underline{y} - \underline{X}\hat{\underline{\theta}})'(\underline{y} - \underline{X}\hat{\underline{\theta}}) + (\underline{X}\hat{\underline{\theta}})'(\underline{X}\hat{\underline{\theta}})$ — (1)

which can be verified by satisfying

$$\hat{\underline{\theta}} = (\underline{X}'\underline{X})^{-1}\underline{X}'\underline{y}$$

The second term on the R.H.S of (1) is

$$\begin{aligned} \hat{\underline{\theta}}' \underline{X}' \underline{X} \hat{\underline{\theta}} &= \underline{y}' \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}' \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}' \underline{y} \\ &= \underline{y}' \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}' \underline{y} \end{aligned}$$

$$\hat{\underline{\theta}}' \underline{X}' \underline{X} \hat{\underline{\theta}} = (\underline{\theta}' \underline{X}' + \epsilon') \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}' (\underline{X} \underline{\theta} + \epsilon) \quad \text{--- (2)}$$

From (2) if $\underline{\theta} = 0$, then

$$\hat{\underline{\theta}}' \underline{X}' \underline{X} \hat{\underline{\theta}} = \epsilon' \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}' \epsilon \quad \text{--- (3)}$$

Making substitution from (3) in (1)

$$\underline{y}'\underline{y} = (\underline{y} - \underline{X}\hat{\underline{\theta}})'(\underline{y} - \underline{X}\hat{\underline{\theta}}) + \epsilon' \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}' \epsilon \quad \text{--- (4)}$$

$$= \epsilon' [I_n - \underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}'] \epsilon + \epsilon' [\underline{X} (\underline{X}' \underline{X})^{-1} \underline{X}'] \epsilon \quad \text{--- (5)}$$

The rank of matrix in the first term on R.H.S is $(n-k)$

and that of second term is k . Thus the ranks on R.H.S is added to n , the ranks on the L.H.S and Cochran theorem implies. Thus the two Q.F.s on R.H.S are independently distributed.

After dividing by σ^2 .

like χ^2 with $n-k$ and k df if $\theta = 0$

The χ^2 distribution of $(y - X\hat{\theta})'(y - X\hat{\theta})$ holds whatever the true value of θ .

whereas

$(X\hat{\theta})'(X\hat{\theta})$ is chi-square distribution if the true value of $\theta = 0$

$$\begin{aligned} E[(X\hat{\theta})'(X\hat{\theta})] &= E[(\epsilon + \theta'X)'X(X'X)^{-1}X(\epsilon + \theta'X)] \\ &= E[\epsilon'X(X'X)^{-1}X\epsilon] + \theta'X'X\theta \\ &= k\sigma^2 + (X\theta)'(X\theta) \end{aligned}$$

which exceeds $k\sigma^2$ unless $\theta = 0$

Under H_0 , $\theta = 0$, we use the ratio to test the null hypothesis

$$\frac{(X\hat{\theta})'(X\hat{\theta})}{k\sigma^2} \sim F(k, n-k)$$

Successive Approximations to M.L Estimators

Sometimes the likelihood is so complicated that it can not be solved explicitly. In that case we use iterative methods to find a root starting from some trial value.

We expand $\frac{\partial \ln L}{\partial \theta}$ in a Taylor series about its value at t and obtain

$$\left(\frac{\partial \ln L}{\partial \theta}\right)_{\hat{\theta}} = \left(\frac{\partial \ln L}{\partial \theta}\right)_t + (\hat{\theta} - t) \left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\theta^*}$$

where θ^* lies between $\hat{\theta}$ and t .

$$\hat{\theta} = t - \frac{\left(\frac{\partial \ln L}{\partial \theta}\right)_t}{\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\theta^*}} \quad \text{--- (1)}$$

If we can choose t so that it is likely to be in the neighbourhood of $\hat{\theta}$, we can replace θ^* in (1) by t and we get

$$\hat{\theta} = t - \frac{\left(\frac{\partial \ln L}{\partial \theta}\right)_t}{\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_t} \quad \text{--- (2)}$$

which will give a closer approximation to $\hat{\theta}$. This process is repeated until no further correction is achieved.

The most common method for the choice of t is to take it as the value of some consistent estimator of θ .

As $n \rightarrow \infty$ $\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\theta^*}$ in (1) can be replaced approximately by $E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_t$

This gives the alternative iterative procedure.

$$\hat{\theta} = t - \frac{\left(\frac{\partial \ln L}{\partial \theta}\right)_t}{E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_t}$$

$$\hat{\theta} = t + \left(\frac{\partial \ln L}{\partial \theta}\right)_t \text{Var}(\hat{\theta}) \quad \text{--- (3)}$$

$\text{Var}(\hat{\theta})$ being the asymptotic variance

Eq ② is known as Robson iterative procedure.

Eq ③ is known as the method of scoring for parameters and is due to Fisher.

Example: To estimate the location parameter θ in the Cauchy population

$$f(x; \theta) = \frac{1}{\pi} \frac{1}{1+(x-\theta)^2} \quad -\infty < x < \infty$$

The likelihood function is

$$L(x_1, \dots, x_n; \theta) = \frac{1}{\pi^n} \prod_{i=1}^n \frac{1}{1+(x_i-\theta)^2}$$

$$\ln L = -n \ln \pi - \sum_{i=1}^n \ln [1+(x_i-\theta)^2]$$

$$\frac{\partial \ln L}{\partial \theta} = 2 \sum \frac{(x_i - \theta)}{[1+(x_i - \theta)^2]} = 0$$

$$\text{Var}(\hat{\theta}) = -1/E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)$$

$$f(x; \theta) = \frac{1}{\pi} \frac{1}{1+(x-\theta)^2}, \quad \ln f = -\ln \pi - \ln(1+(x-\theta)^2)$$

$$\frac{\partial \ln f}{\partial \theta} = \frac{2(x-\theta)}{[1+(x-\theta)^2]}$$

$$\frac{\partial^2 \ln f}{\partial \theta^2} = \frac{(1+(x-\theta)^2)(-2) - 2(x-\theta)\{2(x-\theta)(-1)\}}{[1+(x-\theta)^2]^2}$$

$$= \frac{-2 - 2(x-\theta)^2 + 4(x-\theta)^2}{[1+(x-\theta)^2]^2}$$

$$\frac{\partial^2 \ln f}{\partial \theta^2} = \frac{2(x-\theta)^2 - 2}{(1+(x-\theta)^2)^2}$$

By iterative procedure

$$\hat{\theta} = t + \left(\frac{\partial \ln L}{\partial \theta} \right)_t \text{Var}(\hat{\theta})$$

$$\text{Var}(\hat{\theta}) = -1/E \left(\frac{\partial^2 \ln L}{\partial \theta^2} \right)$$

$$E \left(\frac{\partial^2 \ln f}{\partial \theta^2} \right) = \int_{-\infty}^{\infty} \left\{ \frac{z(x-\theta)^2 - 2}{(1+(x-\theta)^2)^2} \right\} \left(\frac{1}{\pi} \frac{1}{1+(x-\theta)^2} \right) dx$$

$$= \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{((x-\theta)^2 - 1)}{(1+(x-\theta)^2)^3} dx$$

put $x - \theta = y$ $\therefore dx = dy$

$$E \left(\frac{\partial^2 \ln f}{\partial \theta^2} \right) = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{y^2 - 1}{(1+y^2)^3} dy$$

$$= \frac{4}{\pi} \int_0^{\infty} \frac{y^2 - 1}{(1+y^2)^3} dy$$

put $y = \tan z$, $dy = \sec^2 z dz$

$$E \left(\frac{\partial^2 \ln f}{\partial \theta^2} \right) = \frac{4}{\pi} \int_0^{\pi/2} \frac{(\tan^2 z - 1)}{(1 + \tan^2 z)^3} \sec^2 z dz$$

$$= \frac{4}{\pi} \int_0^{\pi/2} \frac{\tan^2 z - 1}{(\sec^2 z)^2} dz$$

$$= \frac{4}{\pi} \left[\int_0^{\pi/2} \sin z \cos z dz - \int_0^{\pi/2} \cos^3 z dz \right]$$

we know that

$$2 \int_0^{\pi/2} \sin^{p-1} \theta \cos^{q-1} \theta d\theta = \frac{\Gamma(p/2) \Gamma(q/2)}{\Gamma\left(\frac{p+q}{2}\right)}$$

Therefore

$$E \left(\frac{\partial^2 \ln f}{\partial \theta^2} \right) = \frac{2}{\pi} \left[2 \int_0^{\pi/2} \sin^{2-1} \theta \cos^{3-1} \theta dz - 2 \int_0^{\pi/2} \sin^{1-1} \theta \cos^{5-1} \theta dz \right]$$

$$\begin{aligned}
 E\left(\frac{\partial^2 \ln f}{\partial \theta^2}\right) &= \frac{2}{\pi} \left[\frac{\frac{3}{2} \sqrt{\frac{1}{2}}}{\sqrt{\frac{1}{2}}} - \frac{\sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{\sqrt{\frac{1}{2}}} \right] \\
 &= \frac{2}{\pi} \left[\frac{\frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{\frac{2}{2}} - \frac{\frac{3}{2} \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{\frac{2}{2}} \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{4} - \frac{3\pi}{4} \right] = -\frac{2\pi}{\pi 4}
 \end{aligned}$$

$$E\left(\frac{\partial^2 \ln f}{\partial \theta^2}\right) = -1/2$$

$$E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right) = -1/2$$

$$\text{Var}(\hat{\theta}) = 2/\pi$$

$$\left(\frac{\partial \ln L}{\partial \theta}\right)_t = \sum \frac{z(x_i - t)}{1 + (x_i - t)^2}$$

$$\hat{\theta} = t + \frac{4}{\pi} \sum \frac{(x_i - t)}{1 + (x_i - t)^2}$$

This is first approximation to $\hat{\theta}$.

Result: For a distribution admitting a single sufficient statistic for the parameter θ .

$$E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right) = \left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\hat{\theta}=\theta} \quad \text{--- ①}$$

So that there is no need to evaluate the expectation in this case. The minimum variance bound becomes simply

$$-\frac{1}{\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)_{\hat{\theta}=\theta}}$$

Generalization; The generalization of ① for the multiparameter case is for a distribution admitting a set of jointly sufficient statistics for the parameters $\theta_1, \dots, \theta_k$. In this case again no need to evaluate expectation.

$$E\left(\frac{\partial^2 \ln L}{\partial \theta_i \partial \theta_j}\right) = \left(\frac{\partial^2 \ln L}{\partial \theta_i \partial \theta_j}\right)_{\hat{\theta}=\theta}$$

Example: If $X \sim N(0, \sigma^2)$, find MLE estimator of σ and also find variance of sampling of σ .

Sol,

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{x^2}{\sigma^2}}$$

$n(x) < \infty$

$$L = \frac{1}{(2\pi)^n \sigma^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2}$$

$$\ln L = -n \ln(2\pi) - n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2$$

$$\frac{\partial \ln L}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum x_i^2$$

Estimating equation is $\frac{n}{\sigma} = \frac{\sum x_i^2}{\sigma^3}$

$$\hat{\sigma}^2 = \frac{\sum x_i^2}{n} \quad \hat{\sigma} = \sqrt{\frac{\sum x_i^2}{n}}$$

$$\text{Var}(\hat{\sigma}) = -1 / \left(\frac{\partial^2 \ln L}{\partial \sigma^2} \right)_{\hat{\sigma} = \sigma}$$

$$\frac{\partial^2 \ln L}{\partial \sigma^2} = \frac{+n}{\sigma^2} - \frac{3}{\sigma^4} \sum x_i^2 = \frac{n}{\sigma^2} - \frac{3n\hat{\sigma}^2}{\sigma^4}$$

$$= \frac{n}{\sigma^2} \left(1 - \frac{3\hat{\sigma}^2}{\sigma^2} \right) = \frac{-2n}{\sigma^2}$$

$$\text{Var}(\hat{\sigma}) = -1 / \left(\frac{-2n}{\sigma^2} \right) = \frac{\hat{\sigma}^2}{2n} \quad \text{This is asymptotic or limiting variance of } \hat{\sigma}.$$

Properties of MLE estimator

The joint MLE estimators tend, under regularity conditions, with dispersion matrix whose inverse is given by

$$\underline{V}^{-1} = -E \left(\frac{\partial^2 \ln L}{\partial \theta_i \partial \theta_j} \right) = E \left(\frac{\partial \ln L}{\partial \theta_i} \frac{\partial \ln L}{\partial \theta_j} \right)$$

If there is a set of jointly sufficient statistics we may use the last result, to obtain inverse of dispersion matrix of MLE

$$\underline{V}^{-1} = - \left(\frac{\partial^2 \ln L}{\partial \theta_i \partial \theta_j} \right)_{\hat{\theta}} = \left(\frac{\partial \ln L}{\partial \theta_i} \frac{\partial \ln L}{\partial \theta_j} \right)_{\hat{\theta}}$$

Example; The M.L estimators of μ and σ^2 in $N(\mu, \sigma^2)$ are

$$\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = s^2$$

As the M.L estimators $\bar{x}$ and s^2 are jointly sufficient for μ and σ^2 , Therefore we obtain inverse of dispersion matrix as

$$V^{-1} = \frac{\partial^2 \ln L}{\partial \theta_i \partial \theta_j}$$

$$V^{-1} = - \begin{pmatrix} \frac{\partial^2 \ln L}{\partial \mu^2} & \frac{\partial^2 \ln L}{\partial \mu \partial \sigma^2} \\ \frac{\partial^2 \ln L}{\partial \sigma^2 \partial \mu} & \frac{\partial^2 \ln L}{\partial (\sigma^2)^2} \end{pmatrix} \quad \mu = \bar{x}, \sigma^2 = s^2$$

The likelihood function is

$$L = \frac{1}{(2\pi)^{n/2} \sigma^n} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right]$$

$$\ln L = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

$$\frac{\partial \ln L}{\partial \mu} = -\frac{1}{\sigma^2} \sum (x_i - \mu)(-1) = \frac{\sum (x_i - \mu)}{\sigma^2} = \frac{\sum x_i - n\mu}{\sigma^2}$$

$$\frac{\partial^2 \ln L}{\partial \mu^2} = -\frac{n}{\sigma^2}$$

$$\frac{\partial^2 \ln L}{\partial \mu \partial \sigma^2} = \frac{-\sum (x_i - \mu)}{\sigma^4} = \frac{-n(\bar{x} - \mu)}{\sigma^4}$$

by replacing $\hat{\mu} = \bar{x}$

$$\frac{\partial \ln L}{\partial \mu \partial \sigma^2} = 0 = \frac{\partial^2 \ln L}{\partial \sigma^2 \partial \mu}$$

$$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2$$

$$\frac{\partial^2 \ln L}{\partial (\sigma^2)^2} = \frac{n}{2\sigma^4} - \frac{\sum (x_i - \mu)^2}{\sigma^6}$$

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}, \quad n\sigma^2 = \sum (x_i - \mu)^2$$

$$\frac{\partial^2 \ln L}{\partial (\sigma^2)^2} = \frac{n}{2\sigma^4} - \frac{n\sigma^2}{\sigma^6} = \frac{n}{2\sigma^4} - \frac{n}{\sigma^4}$$

$$\frac{\partial^2 \ln L}{\partial (\sigma^2)^2} = -\frac{n}{2\sigma^4}$$

$$\underline{V}^{-1} = \begin{pmatrix} n/\sigma^2 & 0 \\ 0 & n/2\sigma^4 \end{pmatrix} \quad \mu = \hat{\mu}, \quad \sigma^2 = \hat{\sigma}^2$$

$$\underline{V} = \begin{pmatrix} \sigma^2/n & 0 \\ 0 & 2\sigma^4/n \end{pmatrix}$$

Thus we have asymptotically $\text{Var}(\hat{\mu}) = \sigma^2/n$

$$\text{Var}(\hat{\sigma}^2) = 2\sigma^4/n, \quad \text{Cov}(\hat{\mu}, \hat{\sigma}^2) = 0$$

Example; let $x_1, x_2, \dots, x_n$ be a random sample of size n from $N(\mu, \sigma^2)$.

Find the MLE estimator and UMVUE of point A such that

$$\int_A^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 0.05$$

$$-\infty < x < \infty$$

Sol;

$$P(Z > z) = 0.05$$

0.05 area is at $z = 1.645$

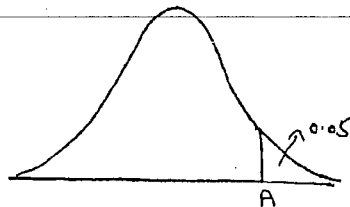
A in SN units is

$$\frac{A - \mu}{\sigma} = 1.645$$

$$A = \mu + 1.645\sigma$$

We know that MLE of μ and σ are

$$\hat{\mu} = \bar{x}, \quad \hat{\sigma} = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2} = s$$



M.L.E of A is

$$\hat{A} = \bar{x} + 1.645 \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

The minimum-variance estimator of A is given by a function of $\bar{x}$, $\frac{\sum (x_i - \bar{x})^2}{n}$ and its expected value is $A = \mu + 1.645\sigma$

We have

$$E(\hat{A}) = E(\bar{x}) + (1.645) E(\sqrt{\hat{\sigma}^2})$$

$$E(\hat{A}) = \mu + 1.645 E(\hat{\sigma}) \quad \text{--- (1)}$$

Now

$$E(\hat{\sigma}) = E\left(\frac{\sum (x_i - \bar{x})^2}{n}\right)^{1/2}$$

$$= \frac{\sigma}{\sqrt{n}} E\left(\frac{\sum (x_i - \bar{x})^2}{\sigma^2}\right)^{1/2}$$

$$E(\hat{\sigma}) = \frac{\sigma}{\sqrt{n}} E(\chi^2)^{1/2} = \frac{\sigma}{\sqrt{n}} E(\chi) \quad \text{--- (2)}$$

$\therefore$ we know that $\frac{\sum (x_i - \bar{x})^2}{\sigma^2} = \frac{nS^2}{\sigma^2}$ is a χ^2 variate with $(n-1)$ d.f.

The p.d.f of χ with $(n-1)$ d.f. is

$$dF(\chi) = \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} e^{-\frac{\chi^2}{2}} (\chi)^{n-2} d\chi \quad 0 < \chi < \infty$$

$$E(\chi_{n-1}) = \int_0^{\infty} \chi \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} e^{-\frac{\chi^2}{2}} (\chi)^{n-2} d\chi$$

$$= \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} \int_0^{\infty} \chi^{n-1} e^{-\frac{\chi^2}{2}} d\chi$$

$$\text{put } \frac{1}{2}\chi^2 = y \Rightarrow \chi = (2y)^{1/2}$$

$$\chi d\chi = dy$$

$$d\chi = \frac{1}{(2y)^{1/2}} dy$$

$$E(\chi_{n-1}) = \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} \int_0^{\infty} (2y)^{\frac{n-1}{2}} e^{-y} \frac{1}{(2y)^{1/2}} dy$$

$$= \frac{2^{\frac{n-1}{2}}}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} \int_0^{\infty} y^{\frac{n-1}{2}-1} e^{-y} dy$$

$$E(Y) = \frac{\sqrt{Z}}{\sqrt{\frac{n-1}{Z}}} \int_0^{\infty} y^{\frac{n-1}{Z}-1} e^{-y} dy$$

$$E(Y) = \frac{\sqrt{Z} \sqrt{\frac{n-1}{Z}}}{\sqrt{\frac{n-1}{Z}}}$$

$$\text{but } Y = \left(\frac{n S^2}{\sigma^2} \right)^{1/2}$$

$$\therefore E(Y) = E\left(\frac{\sqrt{n} S}{\sigma}\right) = \frac{\sqrt{Z} \sqrt{\frac{n-1}{Z}}}{\sqrt{\frac{n-1}{Z}}}$$

$$E(\sqrt{n} S) \frac{\sqrt{\frac{n-1}{Z}}}{\sqrt{Z} \sqrt{\frac{n-1}{Z}}} = \sigma$$

$$E\left(\frac{\sqrt{n} \sqrt{\frac{n-1}{Z}} S}{\sqrt{Z} \sqrt{\frac{n-1}{Z}}}\right) = \sigma$$

$$E\left(\frac{\sqrt{\frac{n-1}{Z}} \sqrt{\sum (x_i - \bar{x})^2}}{\sqrt{Z} \sqrt{\frac{n-1}{Z}}}\right) = \sigma$$

Y is a χ^2 variate with n d.f

So $\bar{x} + 1.645 \frac{\sqrt{\frac{n-1}{Z}} \sqrt{\sum (x_i - \bar{x})^2}}{\sqrt{Z} \sqrt{\frac{n-1}{Z}}}$ is UMVUE of A .

Example; If the M.L estimator $\hat{\theta}$ is a root of $\frac{\partial \ln L}{\partial \theta} = 0$, so that the most general form of the dist differentiable in θ for which $\hat{\theta} = \bar{x}$, the sample mean, is

$$f(x; \theta) = \exp[A(\theta) + A'(\theta)(x - \theta) + B(x)]$$

Sol; We see that a solution of

$$\frac{\partial \ln L}{\partial \theta} = \sum \frac{\partial \ln f}{\partial \theta} = 0 \quad \text{is}$$

$$\hat{\theta} = \bar{x} = \frac{1}{n} \sum x$$

or $\sum (x - \hat{\theta}) = 0$ this is true for all x and θ , and

hence

$$\frac{\partial \ln f}{\partial \theta} = K(x - \hat{\theta})$$

where K is independent of x but may depend on θ .

If we take

$$K = A''(\theta)$$

Then integrating

$$\begin{aligned}\log f &= \int (x-\theta) A''(\theta) d\theta \\ &= (x-\theta) A'(\theta) - \int A'(\theta) (-1) d\theta \\ &= (x-\theta) A'(\theta) + A(\theta) + B(x)\end{aligned}$$

Hence we have

$$f = \exp[A(\theta) + A'(\theta)(x-\theta) + B(x)]$$

Example; Show that the most general form continuous dist. for which the M.L estimator of a parameter θ is the geometric mean of the sample, is

$$f(x; \theta) = \left(\frac{x}{\theta}\right)^{\theta A'(\theta)} \exp[A(\theta) + B(x)]$$

Sol;

$$\frac{\partial \ln L}{\partial \theta} = \sum \frac{\partial \ln f}{\partial \theta} = 0$$

$$\ln \hat{\theta} = \frac{1}{n} \sum \ln x_i$$

$$\text{or } \sum \ln(x-\theta) = 0 \quad \sum (\ln x_i - \ln \theta) = 0$$

$$\sum \ln(x/\theta) = 0$$

$$\text{Hence we have } \frac{\partial \ln f}{\partial \theta} = K \ln\left(\frac{x}{\theta}\right)$$

where K is chosen such that

$$\begin{aligned}\int K d\theta &= \theta A'(\theta) \quad \text{or} \\ K &= \theta A''(\theta) + A'(\theta)\end{aligned}$$

Then

$$\begin{aligned}\ln f &= \int (\theta A''(\theta) + A'(\theta)) \ln\left(\frac{x}{\theta}\right) d\theta \\ &= \int \theta \ln \frac{x}{\theta} A''(\theta) d\theta + \int A'(\theta) \ln \frac{x}{\theta} d\theta \\ &= \left(\theta \ln \frac{x}{\theta}\right) A'(\theta) - \int A'(\theta) \frac{d}{d\theta} \left(\theta \ln \frac{x}{\theta}\right) d\theta \\ &\quad + \int A'(\theta) \ln \frac{x}{\theta} d\theta\end{aligned}$$

$$\ln f = \theta A'(\theta) \ln \frac{x}{\theta} - \int A'(\theta) \frac{d}{d\theta} (\theta \ln \frac{x}{\theta}) d\theta + \int A'(\theta) \ln \frac{x}{\theta} d\theta$$

$$\begin{aligned} \frac{d}{d\theta} \theta \ln \frac{x}{\theta} &= \frac{d}{d\theta} [\theta \ln x - \theta \ln \theta] \\ &= \ln x - 1 - \ln \theta = \ln \frac{x}{\theta} - 1 \end{aligned}$$

$$\begin{aligned} \ln f &= \theta A'(\theta) \ln \frac{x}{\theta} - \int A'(\theta) \ln \frac{x}{\theta} d\theta + \int A'(\theta) d\theta \\ &\quad + \int A'(\theta) \ln \frac{x}{\theta} d\theta \\ &\quad + B(x) \end{aligned}$$

$$\ln f = \ln \frac{x}{\theta}^{\theta A'(\theta)} + A(\theta) + B(x)$$

Hence

$$f = \left(\frac{x}{\theta}\right)^{\theta A'(\theta)} \exp[A(\theta) + B(x)]$$

Example; Show that the most general continuous dist of which the M.L.E. of θ is the H.M. of the sample, is

$$f(x; \theta) = \exp\left[\frac{1}{x} \{ \theta A'(\theta) - A(\theta) \} - A'(\theta) + B(x)\right]$$

Sol;

$$\sum \frac{\partial \ln f}{\partial \theta} = 0$$

$$\text{we have } \frac{1}{\theta} = \frac{1}{n} \sum x_i$$

$$\text{or } \sum \left(\frac{\theta}{x} - 1\right) = 0$$

Hence we have

$$\frac{\partial \ln f}{\partial \theta} = K \left(\frac{\theta}{x} - 1\right)$$

where K is chosen such that $K = A''(\theta)$

$$\ln f = \int A''(\theta) \left(\frac{\theta}{x} - 1\right) d\theta$$

$$= \int A''(\theta) \frac{\theta}{x} d\theta - \int A''(\theta) d\theta$$

$$= \frac{\theta}{x} A'(\theta) - \int A'(\theta) \frac{1}{x} d\theta - A'(\theta) + B(x)$$

$$\ln f = \frac{\theta}{x} A'(\theta) - \frac{A(\theta)}{x} - A'(\theta) + B(x)$$

$$\ln f = \frac{1}{x} \{ \theta A'(\theta) - A(\theta) \} - A'(\theta) + B(x)$$

$$f = \exp \left[\frac{1}{x} \{ \theta A'(\theta) - A(\theta) \} - A'(\theta) + B(x) \right]$$



A MORE General Linear Model

In GL model $E\epsilon = 0$

$$\text{Var}(\epsilon) = E(\epsilon\epsilon') = \sigma^2 I$$

i.e. the error terms are uncorrelated and have constant variance. We can generalize the linear model to the solution where the dispersion matrix of errors is $\sigma^2 V$, V being positive definite and we find that

$$\hat{\beta} = C\hat{\theta} = C(X'X)^{-1}X'Y \text{ generalizes to}$$

$$\hat{\beta} = C(X'V^{-1}X)^{-1}X'V^{-1}Y \quad \text{--- ①}$$

and this is the minimum variance unbiased linear estimator of $C\theta$.

Similarly

$$\begin{aligned} \text{Var}(\hat{\beta}) &= \sigma^2 C(X'X)^{-1}C' \text{ becomes} \\ &= \sigma^2 C(X'V^{-1}X)^{-1}C' \quad \text{--- ②} \end{aligned}$$

In particular, if V is diagonal, but not equal to I , so that the errors are uncorrelated but unequal variances. Eq ① provides the required set of estimates.

Ordered L.S. Estimation of Location and Scale Parameters;

A particular situation in which ① and ② are applicable is in the estimation of location and scale parameters from the ordered statistics. The results are due to Lloyd and Downton (Lloyd's Method).

We denote the ordered statistics by $y_1, y_2, \dots, y_n$. We write μ and σ for the location and scale parameters to be estimated (which are not necessarily mean and SD of the distribution).

$$Z_z = \frac{y_z - \mu}{\sigma}, \quad z = 1, 2, \dots, n \quad \text{③}$$

$$E(Z) = 0, \quad \text{Var}(Z) = V \quad \text{--- ④}$$

Where $\underline{Z}$ is the $(n \times 1)$ vector of the Z_i . Since $\underline{Z}$ has been standardized by ③, $\underline{z}$ and $\underline{V}$ are independent of μ and σ .

From ③ and ④

$$y_i = \mu + \sigma Z_i$$

$$E(y_i) = \mu + \sigma E(Z_i)$$

$$\text{Var}(y_i) = \sigma^2 \text{Var}(Z_i)$$

Generalizing these results

$$E \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{pmatrix} + \sigma E \begin{pmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_n \end{pmatrix}$$

$$\text{or } E(\underline{Y}) = \mu \underline{1} + \sigma \underline{z} \quad \text{--- ⑤}$$

Where $\underline{Y}$ is the vector of y_i and $\underline{1}$ is a vector of units.

$$\begin{aligned} \text{Var}(\underline{Y}) &= \sigma^2 \text{Var}(\underline{Z}) \\ &= \sigma^2 \underline{V} \quad \text{--- ⑥} \end{aligned}$$

We now apply ① and ② to find the L.S estimators of μ and σ .

$$E(\underline{Y}) = \mu \underline{1} + \sigma \underline{z}$$

$$E(\underline{Y}) = \begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix} \begin{pmatrix} \mu \\ \sigma \end{pmatrix}$$

$$\begin{pmatrix} \hat{\mu} \\ \hat{\sigma} \end{pmatrix} = \left[\begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix}' \underline{V}^{-1} \begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix} \right]^{-1} \begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix}' \underline{V}^{-1} \underline{Y} \quad \text{--- ⑦}$$

$$\text{Var} \begin{pmatrix} \hat{\mu} \\ \hat{\sigma} \end{pmatrix} = \sigma^2 \left[\begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix}' \underline{V}^{-1} \begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix} \right]^{-1} \quad \text{--- ⑧}$$

$$\begin{aligned} \left[\begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix}' \underline{V}^{-1} \begin{pmatrix} \underline{1} & \underline{z} \end{pmatrix} \right]^{-1} &= \begin{pmatrix} \underline{1}' \underline{V}^{-1} \underline{1} & \underline{1}' \underline{V}^{-1} \underline{z} \\ \underline{z}' \underline{V}^{-1} \underline{1} & \underline{z}' \underline{V}^{-1} \underline{z} \end{pmatrix}^{-1} \\ &\quad \begin{matrix} n \times n & n \times n & n \times 1 \end{matrix} \end{aligned}$$

$$= \frac{1}{\Delta} \begin{pmatrix} \underline{z}' \underline{V}^{-1} \underline{z} & -\underline{1}' \underline{V}^{-1} \underline{z} \\ -\underline{1}' \underline{V}^{-1} \underline{z} & \underline{1}' \underline{V}^{-1} \underline{1} \end{pmatrix} \quad \text{--- ⑨}$$

Where $\Delta = [(1'V^{-1}1)(\alpha'V^{-1}\alpha) - (1'V^{-1}\alpha)^2]$ ——— ⑩

From ⑦ and ⑨

$$\begin{aligned} \begin{pmatrix} \hat{\mu} \\ \hat{\sigma} \end{pmatrix} &= \frac{1}{\Delta} \begin{pmatrix} \alpha'V^{-1}\alpha & -1'V^{-1}\alpha \\ -1'V^{-1}\alpha & 1'V^{-1}1 \end{pmatrix} \begin{pmatrix} 1 \\ \alpha \end{pmatrix}' V^{-1}Y \\ &= \frac{1}{\Delta} \begin{pmatrix} \alpha'V^{-1}\alpha 1' - 1'V^{-1}\alpha\alpha' \\ -1'V^{-1}\alpha 1' + 1'V^{-1}\alpha\alpha' \end{pmatrix} V^{-1}Y \end{aligned}$$

$$\hat{\mu} = \frac{1}{\Delta} [-\alpha'V^{-1}(1\alpha' - \alpha 1') V^{-1}Y]$$

$$\hat{\sigma} = \frac{1}{\Delta} [1'V^{-1}(1\alpha' - \alpha 1') V^{-1}Y]$$

From ⑩ and ⑨

$$\text{Var}(\hat{\mu}) = \sigma^2 \alpha' V^{-1} \alpha / \Delta$$

$$\text{Var}(\hat{\sigma}) = \sigma^2 1' V^{-1} 1 / \Delta$$

$$\text{Cov}(\hat{\mu}, \hat{\sigma}) = -\sigma^2 1' V^{-1} \alpha / \Delta$$

Relative Efficiency of THE Sample Mean and L.S estimated $\hat{\mu}$;

$$Z_h = \frac{y_h - \mu}{\sigma} \quad h = 1, 2, \dots, n \quad ①$$

$$E(Z) = \alpha, \quad \text{Var}(Z) = V \quad ②$$

Since V and V^{-1} are positive definite, we may write

$$V = T T'$$

$$V^{-1} = (T T')^{-1} = (T^{-1})' (T^{-1}) \quad ③$$

For any arbitrary vector b

$$\begin{aligned} b' V b &= b' T T' b \\ &= (T' b)' (T' b) \end{aligned}$$

$$\underline{b}' \underline{V} \underline{b} = \sum_{i=1}^n h_i^2$$

Where h_i is the i th row element of the vector $\underline{T}'\underline{b}$

Similarly for any vector $\underline{c}$

$$\begin{aligned}\underline{c}' \underline{V}' \underline{c} &= \underline{c}' (\underline{T}')' (\underline{T}') \underline{c} \\ &= (\underline{T}' \underline{c})' (\underline{T}' \underline{c})\end{aligned}$$

$$\underline{c}' \underline{V}' \underline{c} = \sum_{i=1}^n k_i^2$$

where k_i being i th row element of $\underline{T}'\underline{c}$

By Cauchy Inequality

$$(\sum h_i^2)(\sum k_i^2) \geq (\sum h_i k_i)^2$$

$$\begin{aligned}(\underline{b}' \underline{V} \underline{b})(\underline{c}' \underline{V}' \underline{c}) &\geq [(\underline{T}' \underline{b})' (\underline{T}' \underline{c})]^2 \\ &\geq [\underline{b}' \underline{T} \underline{T}' \underline{c}]^2\end{aligned}$$

$$(\underline{b}' \underline{V} \underline{b})(\underline{c}' \underline{V}' \underline{c}) \geq (\underline{b}' \underline{c})^2 \quad \text{--- (4)}$$

In (4), put

$$\underline{b} = (\underline{V}' - \underline{I}) \underline{1}, \quad \underline{c} = \underline{x} \quad \text{--- (5)}$$

$$[\underline{1}' (\underline{V}' - \underline{I}) \underline{V} (\underline{V}' - \underline{I}) \underline{1}] (\underline{x}' \underline{V}' \underline{x}) \geq [\underline{1}' (\underline{V}' - \underline{I}) \underline{x}]^2 \quad \text{--- (6)}$$

If μ and σ are the mean and S.D, then

$$E(\underline{1}' \underline{Z}) = \underline{1}' \underline{x} = 0$$

$$\text{Var}(\underline{1}' \underline{Z}) = \underline{1}' \text{Var}(\underline{Z}) \underline{1}$$

$$= \underline{1}' \underline{V} \underline{1}$$

$$= \underline{1}' \underline{I} \underline{1}$$

$$\text{Var}(\underline{1}' \underline{Z}) = \underline{1}' \underline{1} = n \quad \text{--- (7)}$$

Now from (6)

$$(\underline{1}' \underline{V}' - \underline{1}' \underline{I}) \underline{V} (\underline{V}' - \underline{I}) \underline{1} \underline{x}' \underline{V}' \underline{x} \geq [\underline{1}' \underline{V}' \underline{x} - \underline{1}' \underline{I} \underline{x}]^2$$

$$(\underline{1}' \underline{V}' \underline{V} - \underline{1}' \underline{I} \underline{V}) (\underline{V}' - \underline{I}) \underline{1} \underline{x}' \underline{V}' \underline{x} \geq [\underline{1}' \underline{V}' \underline{x} - \underline{1}' \underline{x}]^2$$

$$(1'V'1 - 1'11 - 1'1V'V'1 + 1'1V'11) \alpha'V'\alpha \gg (1'V'\alpha)^2 \quad \because 1'\alpha = 0$$

Using ⑦

$$(1'V'1 - 1'1 - 1'1 + 1'1) \alpha'V'\alpha \gg (1'V'\alpha)^2$$

$$(1'V'1 - 1'1) \alpha'V'\alpha \gg (1'V'\alpha)^2$$

$$(1'V'1)(\alpha'V'\alpha) - n \alpha'V'\alpha \gg (1'V'\alpha)^2$$

$$(1'V'1)(\alpha'V'\alpha) - (1'V'\alpha)^2 \gg n \alpha'V'\alpha$$

We know that $\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\Delta} (\alpha'V'\alpha)$

where $\Delta = (1'V'1)(\alpha'V'\alpha) - (1'V'\alpha)^2$

Thus

$$\Delta \gg n(\alpha'V'\alpha)$$

$$\geq \frac{n\Delta}{\sigma^2} \text{Var}(\hat{\beta})$$

$$\frac{\sigma^2}{n} \gg \text{Var}(\hat{\beta})$$

$$\text{or } \text{Var}(\hat{\beta}) \leq \frac{\sigma^2}{n} = \text{Var}(\bar{y}) \quad \text{--- ⑧}$$

Since $\bar{y}$, the sample mean is a linear estimator and can not have variance less than the Minimum variance estimator $\hat{\beta}$.

The inequality ⑧ becomes strict equality, when then the ratio of h_i and k_i is a constant

i.e. $h_i = \lambda k_i$ for some constant λ

$$\text{or } T'b = \lambda T'c$$

From ③ $b = (V^{-1} - I)1$

$$T'b = T'(V^{-1} - I)1 = \lambda T'c \quad \because c = 1$$

$$\text{or } TT'(V^{-1} - I)1 = \lambda 1 \quad \text{--- ⑨}$$

but we know

$$TT' = V$$

so $V(V^{-1} - I)1 = \lambda 1$

$$(I - V)I = \lambda d \quad (10)$$

It is the condition under which

$$\text{Var}(\hat{\mu}) = \text{Var}(\bar{y}) = \frac{\sigma^2}{n}$$

If (10) holds then

$$\hat{\mu} = \bar{y} \quad (11)$$

If the parent distribution is symmetrical, the vector of expectations

$$d = \begin{pmatrix} E(Z_1) \\ \vdots \\ E(Z_n) \end{pmatrix} = \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} \quad (12)$$

has $d_i = -d_{n+1-i}$, all i

$$\text{Hence } d'V^{-1}I = I'V^{-1}d = 0 \quad (13)$$

$$\begin{aligned} d_1 &= -d_n \\ d_2 &= -d_{n-1} \\ &\vdots \\ d_n &= -d_1 \end{aligned}$$

we have

$$\hat{\mu} = -d'V^{-1}(Id - dI)V^{-1}\bar{y} / [(I'V^{-1}I)(d'V^{-1}d) - (I'V^{-1}d)^2]$$

$$= (-d'V^{-1}Id + d'V^{-1}dI)V^{-1}\bar{y} / (I'V^{-1}I)(d'V^{-1}d)$$

$$\hat{\mu} = (d'V^{-1}d)(I'V^{-1}\bar{y}) / (d'V^{-1}d)(I'V^{-1}I) \quad \because d'V^{-1}I = 0$$

$$\hat{\mu} = \frac{(I'V^{-1}\bar{y})}{(I'V^{-1}I)}$$

similarly

$$\hat{\sigma}^2 = I'V^{-1}(Id - dI)V^{-1}\bar{y} / [(I'V^{-1}I)(d'V^{-1}d) - (I'V^{-1}d)^2]$$

$$= [I'V^{-1}Id - I'V^{-1}dI]V^{-1}\bar{y} / (I'V^{-1}I)(d'V^{-1}d)$$

$$\hat{\sigma}^2 = \frac{(I'V^{-1}I)d'V^{-1}\bar{y}}{(I'V^{-1}I)(d'V^{-1}d)}$$

$$\hat{\sigma}^2 = \frac{d'V^{-1}\bar{y}}{d'V^{-1}d}$$

Now

$$\begin{aligned} \text{Var}(\hat{\beta}) &= \frac{\sigma^2 \underline{d}' \underline{V}^{-1} \underline{d}}{(\underline{1}' \underline{V}^{-1} \underline{1})(\underline{d}' \underline{V}^{-1} \underline{d}) - (\underline{1}' \underline{V}^{-1} \underline{d})^2} \\ &= \frac{\sigma^2 (\underline{d}' \underline{V}^{-1} \underline{d})}{(\underline{1}' \underline{V}^{-1} \underline{1})(\underline{d}' \underline{V}^{-1} \underline{d})} \end{aligned}$$

$$\text{Var}(\hat{\rho}) = \frac{\sigma^2}{(\underline{1}' \underline{V}^{-1} \underline{1})}$$

$$\text{Var}(\hat{\sigma}) = \frac{\sigma^2 \underline{1}' \underline{V}^{-1} \underline{1}}{(\underline{1}' \underline{V}^{-1} \underline{1})(\underline{d}' \underline{V}^{-1} \underline{d})}$$

$$\text{Var}(\hat{\sigma}) = \frac{\sigma^2}{\underline{d}' \underline{V}^{-1} \underline{d}}$$

$$\text{Cov}(\hat{\mu}, \hat{\sigma}) = -\sigma^2 (\underline{1}' \underline{V}^{-1} \underline{d}) / \Delta = 0$$

Thus the ordered L.S estimators $\hat{\mu}, \hat{\sigma}$ are uncorrelated if the parent distribution is symmetrical.

Example: A sample of size n is taken from the distribution whose density function is

$$f(x) = \frac{1}{\sigma} \quad \mu - \frac{1}{2}\sigma \leq x \leq \mu + \frac{1}{2}\sigma$$

if $Z_t = \frac{y_t - \mu}{\sigma}$ where y_t are ordered observations

Then show that $d_t = E(Z_t) = \frac{t}{n+1} - \frac{1}{2}$ and

$$V_{Z_t} = \frac{t(n-t+1)}{(n+1)^2(n+2)} \quad t < n$$

Given that the variance-covariance matrix is obtain BLUE of μ, σ also obtain their variances.

Sol; As we can prove that

See example 20
order statistics

$$d_t = \frac{t}{n+1} - \frac{1}{2} \quad \text{--- (1)} \quad V_{Z_t} = \frac{t(n-t+1)}{(n+1)^2(n+2)}$$

$$V_{11} = \frac{1(n-1+1)}{(n+1)^2(n+2)} = \frac{n}{(n+1)^2(n+2)}$$

$$V_{12} = \frac{1(n-2+1)}{(n+1)^2(n+2)} = \frac{n-1}{(n+1)^2(n+2)}$$

$$V_{13} = \frac{1(n-3+1)}{(n+1)^2(n+2)} = \frac{n-2}{(n+1)^2(n+2)}$$

$$\vdots$$

$$V_{1n} = \frac{1(n-n+1)}{(n+1)^2(n+2)} = \frac{1}{(n+1)^2(n+2)}$$

$$V_{21} = 0 \quad \text{since } 2 < n$$

$$V_{22} = \frac{2(n-2+1)}{(n+1)^2(n+2)} = \frac{2(n-1)}{(n+1)^2(n+2)}$$

$$\vdots$$

$$V_{2n} = \frac{2(n-n+1)}{(n+1)^2(n+2)} = \frac{2}{(n+1)^2(n+2)}$$

$$\vdots$$

$$V_{nn} = \frac{n(n-n+1)}{(n+1)^2(n+2)} = \frac{n}{(n+1)^2(n+2)}$$

$$V = \frac{1}{(n+1)^2(n+2)} \begin{bmatrix} n & n-1 & n-2 & \dots & 1 \\ 0 & 2(n-1) & 2(n-2) & \dots & 2 \\ 0 & 0 & 3(n-3) & \dots & 3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & n \end{bmatrix}$$

and the inverse is

$$V^{-1} = (n+1)(n+2) \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -1 \end{bmatrix}$$

Now $\alpha_t = \frac{t}{n+1} - \frac{1}{2}$

$$\alpha_1 = \frac{1}{n+1} - \frac{1}{2} = \frac{2 - (n+1)}{2(n+1)} = \frac{(1-n)}{2(n+1)}$$

$$\alpha_2 = \frac{2}{n+1} - \frac{1}{2} = \frac{4 - (n+1)}{2(n+1)} = \frac{(3-n)}{2(n+1)}$$

$$\vdots$$

$$\alpha_{n-1} = -\frac{(3-n)}{2(n+1)}$$

$$\alpha_n = \frac{n}{n+1} - \frac{1}{2} = -\frac{(1-n)}{2(n+1)}$$

$$\underline{\alpha} = \begin{bmatrix} -\frac{(n-1)}{2(n+1)} \\ -\frac{(n-3)}{2(n+1)} \\ -\frac{(n-5)}{2(n+1)} \\ \vdots \\ -\frac{(3-n)}{2(n+1)} \\ -\frac{(1-n)}{2(n+1)} \end{bmatrix} = -\frac{1}{2(n+1)} \begin{bmatrix} n-1 \\ n-3 \\ n-5 \\ \vdots \\ 3-n \\ 1-n \end{bmatrix}$$

$$\underline{\alpha}' \underline{V}^{-1} = -\frac{(n+1)(n+2)}{2(n+1)} \begin{bmatrix} (n+1) & 0 & 0 & \dots & 0 & -(n+1) \end{bmatrix}$$

$$\underline{\alpha}' \underline{V}^{-1} = -\frac{(n+1)(n+2)}{2} \begin{bmatrix} 1 & 0 & 0 & \dots & -1 \end{bmatrix}$$

$$\underline{\alpha}' \underline{V}^{-1} \underline{y} = \frac{(n+1)(n+2)}{2} \begin{bmatrix} -1 & 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$\underline{\alpha}' \underline{V}^{-1} \underline{y} = \frac{(n+1)(n+2)}{2} (y_n - y_1)$$

$$\alpha' \bar{V} \alpha = -\frac{(n+1)(n+2)}{2} \begin{bmatrix} 1 & 0 & 0 & \dots & -1 \end{bmatrix} \cdot \frac{-1}{2(n+1)} \begin{pmatrix} n-1 \\ n-3 \\ \vdots \\ 1-n \end{pmatrix}$$

$$= \frac{1}{4} (n+2) [n-1 - (1-n)]$$

$$= \frac{1}{4} (n+2) (2n-2) = \frac{2(n+2)(n-1)}{4} = \frac{(n+2)(n-1)}{2}$$

$$\alpha' \bar{V} \alpha = \frac{(n+2)(n-1)}{2}$$

$$\hat{\sigma} = \frac{\alpha' \bar{V} y}{\alpha' \bar{V} \alpha}$$

$$= \frac{\frac{(n+1)(n+2)}{2} (y_n - y_1)}{\frac{(n+2)(n-1)}{2}}$$

$$\hat{\sigma} = \frac{(n+1)}{(n-1)} (y_n - y_1)$$

$$\hat{\mu} = \frac{1' \bar{V} y}{1' \bar{V} 1}$$

$$1' \bar{V} = \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & & & \\ 0 & -1 & 2 & & \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & & & -1 & 2 \end{bmatrix} (n+1)(n+2)$$

$$= (n+1)(n+2) \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

$$1' \bar{V} 1 = (n+1)(n+2) \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

$$1' \bar{V} 1 = 2(n+1)(n+2)$$

$$1' \bar{V} y = (n+1)(n+2) \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$\underline{1}' \underline{V}' \underline{d} = (n+1)(n+2)(y_1 + y_n)$$

$$\hat{\mu} = \frac{(n+1)(n+2)(y_1 + y_n)}{2(n+1)(n+2)}$$

$$\hat{\mu} = \frac{y_1 + y_n}{2}$$

$$\text{Var}(\hat{\mu}) = \frac{\sigma^2}{\underline{1}' \underline{V}' \underline{1}} = \frac{\sigma^2}{2(n+1)(n+2)}$$

$$\text{Var}(\hat{\mu}) = \frac{\sigma^2}{2(n+1)(n+2)}$$

$$\text{Var}(\hat{\sigma}^2) = \frac{\sigma^2}{\underline{d}' \underline{V}' \underline{d}}$$

$$\text{Var}(\hat{\sigma}^2) = \frac{2\sigma^2}{(n+1)(n+2)}$$

$$\text{Cov}(\hat{\mu}, \hat{\sigma}^2) = 0$$

So far we have assumed that our random sample came from the density $f(x; \theta)$ where the function $f(x; \theta)$ was assumed known. We have also assumed that θ was some fixed point though unknown to us. In some problems there is often additional information about θ (previous assumption about θ is that $\theta \in \Omega$).

For example, the estimator may have evidence that θ itself acts as a random variable for which he postulates a realistic density function.

Question: How can this additional information about θ be used in our estimation problem.

Sol: we assume that the unknown parameter θ is the value of a random variable say Θ , we are still interested in estimating some function of θ , $T(\theta)$. If Θ is a random variable, it has a distribution.

We let $G(\theta) = G_{\theta}(\theta)$ denote the C.d.f of Θ and $g(\theta) = g_{\theta}(\theta)$ denote the density function of Θ . We assume that these functions contain no unknown parameters.

Posterior Distribution:

Whenever we want to indicate that the parameter θ is the value of a random variable Θ we shall write the density of x as $f(x/\theta)$ instead of $f(x; \theta)$.

Now $f(x/\theta)$ is a conditional density, it is the density of x given $\theta = \theta$

$$f(x/\theta) = f_{x/\theta=\theta}(x/\theta)$$

Let $x_1, \dots, x_n$ be a random sample from the density $f(x/\theta)$ where θ is the value of random variable Θ . Assume that the density of θ , $g(\theta)$, is known and contains no unknown parameters. Suppose we want to estimate $T(\theta)$.

The L.F includes the observed sample $(x_1, \dots, x_n)$ as the form of the density $f(x; \theta)$. Now we need an expression that contains all the information in the L.F plus the added information of the known density $g(\theta)$. $g(\theta)$ is called the prior distribution of the random variable θ . It summarizes what we know about θ prior to taking a random sample. We seek an expression that summarizes what we know about θ after we have a random sample. That is we seek the posterior dist of θ , given

$x_1 = x_1, x_2 = x_2, \dots, x_n = x_n$

Definition: The density $g(\theta)$ is called the prior distribution of θ . The conditional density of θ given $x_1 = x_1, \dots, x_n = x_n$ denoted by $f(\theta/x_1, \dots, x_n) = f_{\theta/x_1, \dots, x_n}(\theta/x_1, \dots, x_n)$

is called the posterior distribution of θ .

$$\begin{aligned} f(\theta/x_1, \dots, x_n) &= \frac{f(x_1, \dots, x_n/\theta) g(\theta)}{\int f(x_1, \dots, x_n/\theta) g(\theta) d\theta} \\ &= \frac{\left[\prod_{i=1}^n f(x_i/\theta) \right] g(\theta)}{\int \left[\prod_{i=1}^n f(x_i/\theta) g(\theta) d\theta \right]} \end{aligned}$$

Unlike the L.F (as a function of θ), the posterior dist is a density function. We shall use the mean of the posterior distribution as an estimate of θ , and in general we can estimate $T(\theta)$ as the mean of $T(\theta)$ given $x_1 = x_1, \dots, x_n = x_n$.

That is take

$$E[T(\theta)/x_1 = x_1, \dots, x_n = x_n] \text{ as our}$$

estimate of $T(\theta)$

Definition: let $x_1, \dots, x_n$ be a random sample from a density $f(x/\theta)$. where θ is the value of a random variable θ with known density $g(\theta)$. The posterior Bayes estimator of $\tau(\theta)$ with respect to the prior $g(\theta)$ is defined as $E[\tau(\theta)/x_1, \dots, x_n]$

$$\begin{aligned} E[\tau(\theta)/x_1=x_1, \dots, x_n=x_n] &= \int \tau(\theta) f(\theta/x_1, \dots, x_n) d\theta \\ &= \frac{\int \tau(\theta) \left[\prod_{i=1}^n f(x_i/\theta) \right] g(\theta) d\theta}{\int \left[\prod_{i=1}^n f(x_i/\theta) \right] g(\theta) d\theta} \end{aligned}$$

In general a posterior Bayes estimator is not unbiased.

Example: let $x_1, \dots, x_n$ denote a random sample from the Bernoulli density

$$f(x/\theta) = \theta^x (1-\theta)^{1-x} \quad x=0,1$$

Assume that the prior distribution of θ is given by

$$g(\theta) = 1 \quad 0 \leq \theta \leq 1$$

$$g(\theta) = I_{(0,1)}(\theta)$$

Find the posterior Bayes estimator of θ and $\tau(\theta) = \theta(1-\theta)$

Solution:

$$f(\theta/x_1, \dots, x_n) = \frac{\theta^{\sum x_i} (1-\theta)^{n-\sum x_i} I_{(0,1)}(\theta)}{\int_0^1 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d\theta}$$

Posterior Bayes estimator of θ with respect to the uniform prior dist is

$$\begin{aligned} E[\theta/x_1=x_1, \dots, x_n=x_n] &= \int_0^1 \theta f(\theta/x_1, \dots, x_n) d\theta \\ &= \frac{\int_0^1 \theta \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d\theta}{\int_0^1 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d\theta} \end{aligned}$$

$$E(\theta / x_1 = x_1, \dots, x_n = x_n) = \frac{\beta(\sum x_i + 2, n - \sum x_i + 1)}{\beta(\sum x_i + 1, n - \sum x_i + 1)}$$

$$= \frac{\frac{\sqrt{\sum x_i + 2} \sqrt{n - \sum x_i + 1}}{\sqrt{n+3}} \times \frac{\sqrt{n+2}}{\sqrt{\sum x_i + 1} \sqrt{n - \sum x_i + 1}}$$

$$= \frac{(\sum x_i + 1) \sqrt{\sum x_i + 1} \sqrt{n+2}}{(n+2) \sqrt{n+2} \sqrt{\sum x_i + 1}}$$

$$E(\theta / x_1 = x_1, \dots, x_n = x_n) = \frac{\sum x_i + 1}{n+2}$$

Hence the posterior Bayes estimator of θ w.r.t the uniform prior dist is $\sum x_i + 1 / n+2$.

Now to obtain Posterior Bayes estimator of $T(\theta) = \theta(1-\theta)$ we calculate

$$E[T(\theta) / x_1 = x_1, \dots, x_n = x_n] = \int_0^1 \theta(1-\theta) f(\theta / x_1, \dots, x_n) d\theta$$

$$= \frac{\int_0^1 \theta(1-\theta) \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d\theta}{\int_0^1 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} d\theta}$$

$$= \frac{\beta(\sum x_i + 2, n - \sum x_i + 2)}{\beta(\sum x_i + 1, n - \sum x_i + 1)}$$

$$= \frac{\frac{\sqrt{\sum x_i + 2} \sqrt{n - \sum x_i + 2}}{\sqrt{n+4}} \times \frac{\sqrt{n+2}}{\sqrt{\sum x_i + 1} \sqrt{n - \sum x_i + 1}}$$

$$= \frac{(\sum x_i + 1)(n - \sum x_i + 1)}{(n+3)(n+2)}$$

So the posterior Bayes estimator of $\theta(1-\theta)$ w.r.t a uniform prior dist is

$$\frac{(\sum x_i + 1)(n - \sum x_i + 1)}{(n+3)(n+2)}$$

Loss Function Approach

Let $T = t(x_1, \dots, x_n)$ be an estimator of $\tau(\theta)$. The mean squared error (MSE) of the estimator $T = t(x_1, \dots, x_n)$ is defined as

$$MSE_t(\theta) = E_\theta \{ \{T - \tau(\theta)\}^2 \}$$

$$= \text{Var}(T) + \{ \tau(\theta) - E_\theta(T) \}^2$$

$$= \text{Var}(T) + \text{Bias}^2$$

If $E_\theta(T) = \tau(\theta)$, then

$$MSE(\theta) = \text{Var}(T)$$

MSE of an estimator is used as measure of the closeness of the estimator to $\tau(\theta)$.

Mean absolute deviation is $E_\theta \{ |T - \tau(\theta)| \}$.

Decision Theory Approach

On the basis of observed random sample from the same density function, we call the value of some estimate a decision, and call the estimator itself a decision function.

Now the estimate t of $\tau(\theta)$ might be in error. The word loss is used in place of error and loss function is used as a measure of the error.

Def; Loss Function:

Let t denote an estimate of $\tau(\theta)$. The loss function denoted by $l(t; \theta)$ is defined to be a real-valued function, satisfying

i) $l(t; \theta) \geq 0$ for all possible estimates t and all θ

ii) $l(t; \theta) = 0$ for $t = \tau(\theta)$

$l(t; \theta)$ represents the loss incurred if we estimate $\tau(\theta)$

to be t when θ is true parameter value.

Some possible loss functions are

- (1) $l_1(t; \theta) = [t - T(\theta)]^2$ is called squared error loss function.
- (2) $l_2(t; \theta) = |t - T(\theta)|$ is called absolute error loss function.
- (3) $l_3(t; \theta) = \rho(\theta) |t - T(\theta)|^c$ for $\rho(\theta) \geq 0$ and $c > 0$.

Risk Function

For any estimator $T = t(x_1, \dots, x_n)$ we take average loss. ~~loss~~ loss depends on the sample $x_1, \dots, x_n$. We can remove the dependence of the loss on the sample by choosing that estimator that makes the average loss small. This average loss is called risk function.

For a given loss function $l(t; \theta)$ the risk function denoted by $R_t(\theta)$ of an estimator $T = t(x_1, \dots, x_n)$ is defined as

$$R_t(\theta) = E_\theta [l(T; \theta)] \quad \text{--- (1)}$$

The risk function in (1) can be carried out in two ways.

First, if the density $f(x; \theta)$ from which we are sampling is a p.d.f then

$$\begin{aligned} E_\theta [l(T; \theta)] &= E_\theta [l(t(x_1, \dots, x_n); \theta)] \\ &= \int \dots \int l(t(x_1, \dots, x_n); \theta) f(x_1; \theta) \dots f(x_n; \theta) dx_1 \dots dx_n \end{aligned}$$

Second, we can consider the random variable T and the density of T .

$$E_\theta [l(T; \theta)] = \int l(t; \theta) f(t) dt$$

where $f(t)$ is the density of estimator t .

Some loss functions and their corresponding risk functions are.

Loss Function	Risk Function
1) $(T - \tau(\theta))^2$	$E_\theta[(T - \tau(\theta))^2]$ MSE
2) $ T - \tau(\theta) $	$E_\theta T - \tau(\theta) $ Mean absolute error
3) $\rho(\theta) T - \tau(\theta) ^2$	$E_\theta[\rho(\theta) T - \tau(\theta) ^2]$

Our object is now to select an estimator that make the average loss (risk function) small and ideally to select an estimator that has the smallest risk.

Admissible Estimator

For two estimators $T_1 = t_1(x_1, \dots, x_n)$, $T_2 = t_2(x_1, \dots, x_n)$, estimator T_1 is defined to be a better estimator than T_2 if

$$R_{t_1}(\theta) \leq R_{t_2}(\theta) \quad \text{for all } \theta \in \Omega$$

$$R_{t_2}(\theta) < R_{t_1}(\theta) \quad \text{for at least one } \theta \text{ in } \Omega$$

An estimator $T = t(x_1, \dots, x_n)$ is defined to be admissible if there is no better estimator.

In general, given two estimators T_1 and T_2 neither is better than the other. That is the risk functions as function of θ , cross. Since θ is unknown, it is difficult to make a choice between the two estimators. This difficulty is caused by the dependence of the risk function on θ . We average out θ just as we average out the dependence on $x_1, \dots, x_n$.

(1) One way of removing the dependence of risk function on θ is to replace the risk function by its maximum value and compare estimators by looking at their respective maximum risks and selecting that

estimator with smallest maximum risk. Such an estimator is called minimax estimator.

Minimax Estimator: An estimator T^* is defined to be a minimax estimator iff

$$\sup_{\theta} R_{T^*} \leq \sup_{\theta} R_T(\theta)$$

for every value of estimator T .

(2) The problem of averaging out θ is also dealt with by Bayes estimators.

Let $x_1, \dots, x_n$ be a random sample from density $f(x/\theta)$, $\theta \in \Omega$ where function $f(x/\theta)$ is assumed known except for θ . We assume that the unknown θ is the value of some random variable θ and that the distribution of θ is known. On the basis of random sample $x_1, \dots, x_n$ we want to estimate $T(\theta)$, some function of θ . In addition we assume that a loss function $l(T; \theta)$ is specified.

For any estimator $T = t(x_1, \dots, x_n)$ let $R_T(\theta) = E_{\theta}[l(T; \theta)]$ be the risk function. The two estimators T_1 and T_2 can be compared by looking at their respective risks $R_{T_1}(\theta)$ and $R_{T_2}(\theta)$, selecting the estimator with smallest risk.

The dependence of the risk function on θ can be removed by averaging out θ , by using the density of θ .

Bayes Risk

Let $x_1, \dots, x_n$ be a random sample from a density $f(x/\theta)$, where θ is the value of a random variable θ with c.d.f $G(\theta)$ and corresponding p.d.f $g(\theta)$. In estimating $T(\theta)$, let $l(T; \theta)$ be the loss function. The risk of the estimator T is denoted by $R_T(\theta)$. The Bayes risk of estimator $T = t(x_1, \dots, x_n)$ w.r.t the loss function $l(T; \theta)$ and prior c.d.f $G(\theta)$, denoted by $r(T)$, is defined as

$$r(t) = \int R_t(\theta) g(\theta) d\theta$$

This risk is called Bayes risk.

Bayes Estimator:

The Bayes estimator of $T(\theta)$ denoted by T^* w.r.t the loss function $l(t; \theta)$ and prior c.d.f $G(\theta)$, is defined to be the estimator with smallest Bayes risk. i.e. the Bayes estimator of $T(\theta)$ is that estimator T^* satisfying

$$r(t^*) \leq r(t)$$

for every other estimator

$$T = t(x_1, \dots, x_n) \text{ of } T(\theta)$$

The posterior Bayes estimator of $T(\theta)$ was defined without regard to a loss function, whereas this definition requires the specification of a loss function.

Theorem:

Let $x_1, \dots, x_n$ be a random sample from the density $f(x/\theta)$ and let $g(\theta)$ be the density of θ .

Further let $l(t; \theta)$ be the loss function for estimating $T(\theta)$.

The Bayes estimator of $T(\theta)$ is that estimator T^* which minimizes

$$\int \int l(t(x_1, \dots, x_n); \theta) f(x_1, \dots, x_n/\theta) d\theta$$

as a function of t .

Proof:

For a general loss function we seek that estimator which minimizes

$$\int R_t(\theta) g(\theta) d\theta$$

$$\int R_t(\theta) g(\theta) d\theta = \int \left[\int l(t(x_1, \dots, x_n); \theta) f(x_1, \dots, x_n/\theta) dx_1, \dots, dx_n \right] g(\theta) d\theta$$

$$\int_{\Omega} R_t(\theta) g(\theta) d\theta = \int_{\Omega} \left[\int_{\Omega} l(t(x_1, \dots, x_n); \theta) f(x_1, \dots, x_n / \theta) g(\theta) d\theta \right] \\ \times dx_1 \dots dx_n$$

$$= \int_{\Omega} \left[\int_{\Omega} l(t(x_1, \dots, x_n); \theta) \frac{f(x_1, \dots, x_n / \theta) g(\theta)}{f(x_1, \dots, x_n)} d\theta \right]$$

$$\text{But } f(\theta / x_1, \dots, x_n) = \left\{ \frac{f(x_1, \dots, x_n / \theta) g(\theta)}{f(x_1, \dots, x_n)} \right\} \times f(x_1, \dots, x_n) dx_1 \dots dx_n$$

Therefore

$$= \int_{\Omega} \left[\int_{\Omega} l(t(x_1, \dots, x_n); \theta) f(\theta / x_1, \dots, x_n) d\theta \right] f(x_1, \dots, x_n) dx_1 \dots dx_n$$

Minimizing the double integral is equivalent to minimizing the expression within the square brackets, which is called posterior risk.

So in general, the Bayes estimator of $T(\theta)$ w.r.t the loss function $l(t; \theta)$ and prior density $g(\theta)$ is that function estimator which minimizes the posterior risk.

Corollary: Under the assumptions of the theorem, the Bayes estimator of $T(\theta)$ for a squared error loss function is given by

$$E[T(\theta) / x_1, \dots, x_n] = \int T(\theta) f(\theta / x_1, \dots, x_n) d\theta$$

which is identical with the posterior Bayes estimator.

Proof: We see that estimator T^* which minimizes the expression $\int_{\Omega} R_t(\theta) g(\theta) d\theta$

$$\begin{aligned}
 \int_{\Omega} R_t(\theta) g(\theta) d\theta &= \int_{\Omega} \left[\int_{\mathcal{X}} E_{\theta} \{ t(x_1, \dots, x_n) - \tau(\theta) \}^2 g(\theta) d\theta \right] \\
 &= \int_{\Omega} \left[\int_{\mathcal{X}} \left\{ t(x_1, \dots, x_n) - \tau(\theta) \right\}^2 f(x_1, \dots, x_n | \theta) \prod_{i=1}^n dx_i \right] g(\theta) d\theta \\
 &= \int_{\mathcal{X}} \left[\int_{\Omega} \left(t(x_1, \dots, x_n) - \tau(\theta) \right)^2 \frac{f(x_1, \dots, x_n | \theta) g(\theta)}{f(x_1, \dots, x_n)} d\theta \right] f(x_1, \dots, x_n) \prod_{i=1}^n dx_i \\
 &= \int_{\mathcal{X}} \left[\int_{\Omega} \left(t(x_1, \dots, x_n) - \tau(\theta) \right)^2 f(\theta | x_1, \dots, x_n) d\theta \right] f(x_1, \dots, x_n) \prod_{i=1}^n dx_i
 \end{aligned}$$

Since the integrand is non negative, the double integral can be minimized if the expression within the brackets is minimized for each $x_1, \dots, x_n$. But the expression within the square brackets is the conditional expectation of $[t(x_1, \dots, x_n) - \tau(\theta)]^2$ with respect to the posterior distribution of θ given $x_1 = x_1, \dots, x_n = x_n$, which is minimized as a function of t for t^* equal to the conditional expectation of $\tau(\theta)$ w.r.t the posterior distribution of θ given $x_1 = x_1, \dots, x_n = x_n$. $\left\{ \begin{array}{l} E(Z-a)^2 \text{ is minimized as a function of} \\ a \text{ for } a^* = E(Z) \end{array} \right.$

Corollary. Under the assumptions of the theorem, the Bayes estimator of θ is given by the median of the posterior distribution of θ for a loss function equal to the absolute deviation.

Proof: We seek that estimator t^* which minimizes the expression $\int_{\Omega} R_t(\theta) g(\theta) d\theta$

$$\begin{aligned}
 \int R_t(\theta) g(\theta) d\theta &= \int_{\Omega} \left[E_0 |t(x_1, \dots, x_n) - \tau(\theta)| g(\theta) \right] d\theta \\
 &= \int_{\Omega} \left[\int_K |t(x_1, \dots, x_n) - \tau(\theta)| f\left(x_1, \dots, x_n / \theta\right) \prod_{i=1}^n dx_i \right] g(\theta) d\theta \\
 &= \int_K \left[\int_{\Omega} |t(x_1, \dots, x_n) - \tau(\theta)| \frac{f(x_1, \dots, x_n / \theta) g(\theta)}{f(x_1, \dots, x_n)} d\theta \right] f(x_1, \dots, x_n) \prod_{i=1}^n dx_i \\
 &= \int_K \left[\int_{\Omega} |t(x_1, \dots, x_n) - \tau(\theta)| f(\theta / x_1, \dots, x_n) d\theta \right] f(x_1, \dots, x_n) \prod_{i=1}^n dx_i
 \end{aligned}$$

The expression within the square bracket is minimized as a function of t for t^* equal to the median of the posterior distribution of θ . ($E|Z-a|$ is minimized as a function of a for $a^* = \text{median}(Z)$)

Example: let $x_1, \dots, x_n$ be a random sample from the density $N(\theta, 1)$. Assume that the r.v. θ has a $N(\mu_0, 1)$. Find the Bayes estimator with squared error loss function.

Solution: The posterior distribution of θ is

$$\begin{aligned}
 f(\theta / x_1, \dots, x_n) &= \frac{f(x_1, \dots, x_n / \theta) g(\theta)}{f(x_1, \dots, x_n)} \\
 &= \frac{\prod_{i=1}^n f(x_i / \theta) g(\theta)}{\int_{\Omega} \prod_{i=1}^n f(x_i / \theta) g(\theta) d\theta}
 \end{aligned}$$

$$f(x/\theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\theta)^2}, \quad g(\theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(\theta-\mu_0)^2}$$

$$f(\theta/x_1, \dots, x_n) = \frac{\left(\frac{1}{2\pi}\right)^{n/2} \exp\left[-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2\right] \left(\frac{1}{2\pi}\right)^{1/2} \exp\left[-\frac{1}{2} (\theta - \mu_0)^2\right]}{\left(\frac{1}{2\pi}\right)^{n/2} e^{-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2} \left(\frac{1}{2\pi}\right)^{1/2} e^{-\frac{1}{2} (\theta - \mu_0)^2} d\theta}$$

Assuming $\mu_0 = x_0$

$$= \frac{\exp\left[-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2\right]}{\int \exp\left[-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2\right] d\theta}$$

$$= \frac{\exp\left[-\frac{1}{2} \left\{ \sum_{i=1}^n x_i^2 + (n+1)\theta^2 - 2\theta \sum_{i=1}^n x_i \right\}\right]}{\int \exp\left[-\frac{1}{2} \left\{ \sum_{i=1}^n x_i^2 + (n+1)\theta^2 - 2\theta \sum_{i=1}^n x_i \right\}\right] d\theta}$$

$$= \frac{\exp\left[-\frac{(n+1)}{2} \left\{ \frac{\sum_{i=1}^n x_i^2}{(n+1)} + \theta^2 - 2\theta \frac{\sum_{i=1}^n x_i}{(n+1)} + \left(\frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2 - \left(\frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2 \right\}\right]}{\int \exp\left[-\frac{(n+1)}{2} \left\{ \frac{\sum_{i=1}^n x_i^2}{(n+1)} + \theta^2 - 2\theta \frac{\sum_{i=1}^n x_i}{(n+1)} + \left(\frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2 - \left(\frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2 \right\}\right] d\theta}$$

$$= \frac{\exp\left[-\frac{(n+1)}{2} \left\{ \left(\theta - \frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2 + \left(\frac{\sum_{i=1}^n x_i^2}{(n+1)} - \left(\frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2\right) \right\}\right]}{\int \exp\left[-\frac{(n+1)}{2} \left\{ \left(\theta - \frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2 + \left(\frac{\sum_{i=1}^n x_i^2}{(n+1)} - \left(\frac{\sum_{i=1}^n x_i}{(n+1)}\right)^2\right) \right\}\right] d\theta}$$

$$= \frac{1}{\sqrt{2\pi/(n+1)}} \exp\left[-\frac{(n+1)}{2} \left\{ \theta - \frac{\sum_{i=1}^n x_i}{(n+1)} \right\}^2\right]$$

$$= \frac{1}{\sqrt{2\pi/(n+1)}} \int \exp\left[-\frac{(n+1)}{2} \left\{ \theta - \frac{\sum_{i=1}^n x_i}{(n+1)} \right\}^2\right] d\theta$$

The denominator is unity since it is the integral of a complete density.

Therefore

$$= \frac{1}{\sqrt{2\pi/(n+1)}} \exp\left[-\frac{(n+1)}{2} \left\{ \theta - \frac{\sum_{i=1}^n x_i}{(n+1)} \right\}^2\right]$$

Hence posterior distribution of θ is normal with mean $\frac{\sum_{i=1}^n x_i}{n+1}$ and variance $\frac{1}{n+1}$.

Thus Bayes estimator of θ with respect to squared error loss is

$$\frac{x_0 + \sum_{i=1}^n x_i}{n+1} = \frac{\mu_0 + \sum_{i=1}^n x_i}{n+1}$$

With loss function as absolute deviation, median of the posterior distribution is the Bayes estimator which is given by

$$\frac{\mu_0 + \sum_{i=1}^n x_i}{n+1}$$

Because for normal dist. Mean = Median

Example: Let $x_1, \dots, x_n$ be a random sample from the density

$$f(x|\theta) = \frac{1}{\theta} \quad 0 \leq x \leq \theta$$

$$\text{or} \quad = \frac{1}{\theta} I_{(0,\theta)}(x)$$

Estimate θ with the loss function

$$l(t;\theta) = \frac{(t-\theta)^2}{\theta^2}$$

Assume that θ has a density given by $g(\theta) = I_{(0,1)}(\theta)$
 $= 1 \quad 0 < \theta < 1$

Let Y_n denote $\max\{x_1, \dots, x_n\}$. Find posterior distribution of θ .

Solution:

$$\begin{aligned} f(\theta/x_1, \dots, x_n) &= \frac{f(x_1, \dots, x_n/\theta) g(\theta)}{\int f(x_1, \dots, x_n/\theta) g(\theta) d\theta} \\ &= \frac{\left(\frac{1}{\theta}\right)^n \prod_{i=1}^n [I_{(0,\theta)}(x_i)] I_{(0,1)}(\theta)}{\int_0^1 \left(\frac{1}{\theta}\right)^n \left[\prod_{i=1}^n I_{(0,\theta)}(x_i) \right] I_{(0,1)}(\theta) d\theta} \end{aligned}$$

Since $0 \leq \text{all } x_i \leq \theta$

$$0 \leq Y_n \leq \theta \leq 1$$

$$\begin{aligned}
 \therefore f(\theta/x_1, \dots, x_n) &= \frac{(\frac{1}{\theta})^n I_{(y_n,1)}(\theta)}{\int_{y_n}^1 \frac{1}{\theta^n} d\theta} \\
 &= \frac{\frac{1}{\theta^n} I_{(y_n,1)}(\theta)}{\left| \frac{\theta^{-n+1}}{-n+1} \right|_{y_n}^1} = \frac{\frac{1}{\theta^n} I_{(y_n,1)}(\theta)}{\left[\frac{1}{-n+1} - \frac{y_n^{-n+1}}{-n+1} \right]} \\
 &= \frac{\frac{1}{\theta^n} I_{(y_n,1)}(\theta)}{-\frac{1}{(n-1)} + \frac{y_n^{-(n-1)}}{(n-1)}}
 \end{aligned}$$

$$f(\theta/x_1, \dots, x_n) = \frac{\frac{1}{\theta^n} I_{(y_n,1)}(\theta)}{\left(\frac{1}{(n-1)} \left[\frac{1}{y^{n-1}} - 1 \right] \right)}$$

We seek the estimator which minimizes posterior risk

$$\begin{aligned}
 &\int_{\Omega} l(t, \theta) f(\theta/x_1, \dots, x_n) d\theta \\
 &= \frac{1}{\left(\frac{1}{(n-1)} \left[\frac{1}{y^{n-1}} - 1 \right] \right)} \int_{\Omega} \frac{(t - \theta)^2}{\theta^2} \frac{1}{\theta^n} I_{(y_n,1)}(\theta) d\theta \\
 &= \frac{1}{\left(\frac{1}{(n-1)} \left[\frac{1}{y^{n-1}} - 1 \right] \right)} \int_{y_n}^1 \frac{(t(y_n) - \theta)^2}{\theta^{n+2}} d\theta \\
 &= \frac{(n-1)}{\left[\frac{1}{y^{n-1}} - 1 \right]} \left[\int_{y_n}^1 \frac{(t(y_n))^2}{\theta^{n+2}} d\theta + \int_{y_n}^1 \frac{1}{\theta^n} d\theta - 2t(y_n) \int_{y_n}^1 \frac{d\theta}{\theta^{n+1}} \right]
 \end{aligned}$$

The above equation is a quadratic in $t(y_n)$, this quadratic assumes its minimum when

$$\begin{aligned}
 2[t(y_n)] \int_{y_n}^1 \frac{d\theta}{\theta^{n+2}} - 2 \int_{y_n}^1 \frac{d\theta}{\theta^{n+1}} &= 0 \\
 t(y_n) &= \frac{\int_{y_n}^1 \frac{d\theta}{\theta^{n+1}}}{\int_{y_n}^1 \frac{d\theta}{\theta^{n+2}}}
 \end{aligned}$$

$$\begin{aligned}
 t(y_n) &= \frac{\left| \frac{\theta^{-n-1+1}}{-n-1+1} \right| y_n}{\left| \frac{\theta^{-n-2+1}}{-n-2+1} \right| y_n} = \frac{\left| \frac{\theta^{-n}}{-n} \right| y_n}{\left| \frac{\theta^{-(n+1)}}{-(n+1)} \right| y_n} \\
 &= \frac{(n+1) \left(1 - \frac{1}{y_n^n} \right)}{n \left(1 - \frac{1}{y_n^{n+1}} \right)} = \frac{(n+1) (y_n^n - 1)}{n (y_n^{n+1} - 1)} \frac{y_n^{n+1}}{y_n} \\
 t(y_n) &= \frac{(n+1)}{n} y_n \frac{(1 - y_n^n)}{(1 - y_n^{n+1})}
 \end{aligned}$$

Minimax Estimator: Minimax estimator is an estimator whose maximum risk is less than or equal to the maximum risk of any other estimator. The following theorem is useful for finding a minimax estimator.

Theorem: If $T^* = t^*(x_1, \dots, x_n)$ is a Bayes estimator having constant risk, that $R_{T^*}(\theta) = \text{Constant}$, then T^* is a minimax estimator.

Example: Find minimax estimator of θ in sampling from the Bernoulli distribution using a squared error loss function. The family of Beta distribution is a family of possible prior distribution.

Solution: The posterior dist of θ is

$$f(\theta/x_1, \dots, x_n) = \frac{f(x_1, \dots, x_n/\theta) g(\theta)}{f(x_1, \dots, x_n)}$$

$$g(\theta) = \frac{1}{B(a, b)} \theta^{a-1} (1-\theta)^{b-1}$$

$$f(x_1, \dots, x_n/\theta) = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}$$

$$\begin{aligned}
 f(\theta/x_1, \dots, x_n) &= \frac{\theta^{\sum x_i} (1-\theta)^{n-\sum x_i} \left(1/p(a,b)\right) \theta^{a-1} (1-\theta)^{b-1}}{\frac{1}{p(a,b)} \int_0^1 \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} \theta^{a-1} (1-\theta)^{b-1} d\theta} \\
 &= \frac{\theta^{\sum x_i + a - 1} (1-\theta)^{n - \sum x_i + b - 1}}{\int_0^1 \theta^{\sum x_i + a - 1} (1-\theta)^{n - \sum x_i + b - 1} d\theta} \\
 &= \frac{\theta^{\sum x_i + a - 1} (1-\theta)^{n - \sum x_i + b - 1}}{p(\sum x_i + a, n - \sum x_i + b)}
 \end{aligned}$$

Bayes estimator of θ is

$$\begin{aligned}
 E[\theta/x_1, \dots, x_n] &= \frac{\int_0^1 \theta \theta^{\sum x_i + a - 1} (1-\theta)^{n - \sum x_i + b - 1} d\theta}{p(\sum x_i + a, n - \sum x_i + b)} \\
 &= \frac{p(\sum x_i + a + 1, n - \sum x_i + b)}{p(\sum x_i + a, n - \sum x_i + b)} \\
 &= \frac{\frac{(\sum x_i + a + 1)! (n - \sum x_i + b)!}{(n + a + b + 1)!}}{\frac{(\sum x_i + a)! (n - \sum x_i + b)!}{(n + a + b)!}} \\
 &= \frac{(\sum x_i + a + 1)}{(n + a + b + 1)} \times \frac{(n + a + b)!}{(\sum x_i + a)! (n - \sum x_i + b)!}
 \end{aligned}$$

$$E[\theta/x_1, \dots, x_n] = \frac{\sum x_i + a}{n + a + b}$$

Thus the Bayes estimator w.r.t p prior distribution is

$$\frac{\sum x_i + a}{n + a + b}$$

We evaluate the risk of $\frac{\sum x_i + a}{n + a + b}$ and select a and b so that risk is constant.

$$\begin{aligned}
 t^*(x_1, \dots, x_n) &= \frac{\sum x_i}{n + a + b} + \frac{a}{n + a + b} \\
 &= \frac{\sum x_i + a}{n + a + b}
 \end{aligned}$$

$$R_{t^*}(\theta) = E[l(t^*, \theta)] = E[A \sum x + B - \theta]^2$$

$$= E[A \sum x - nA\theta + B - \theta + nA\theta]^2$$

$$= E[A(\sum x - n\theta) + (B - \theta + nA\theta)]^2$$

$$= A^2 E[(\sum x - n\theta)^2] + (B - \theta + nA\theta)^2 + 2A(B - \theta + nA\theta) E(\sum x - n\theta)$$

For Bernoulli Dist $E(\sum x) = n\theta$

$$\text{Var}(\sum x) = n\theta(1-\theta)$$

$$= A^2 \text{Var}(\sum x) + (B - \theta + nA\theta)^2 + 2A(B - \theta + nA\theta)(0)$$

$$= A^2 n\theta(1-\theta) + B^2 + \theta^2 + n^2 A^2 \theta^2 + 2n\theta AB - 2nA\theta^2 - 2B\theta$$

$$= A^2 n\theta - A^2 n\theta^2 + B^2 + \theta^2 + n^2 A^2 \theta^2 + 2n\theta AB - 2nA\theta^2 - 2B\theta$$

$$= \theta^2 [n^2 A^2 - 2nA + 1 - nA^2] + \theta [nA^2 + 2nAB - 2B] + B^2$$

$$= \theta^2 [(nA-1)^2 - nA^2] + \theta [nA^2 + 2(nA-1)B] + B^2$$

This is a constant if

$$(nA-1)^2 - nA^2 = 0 \text{ --- (1) and } nA^2 + 2(nA-1)B = 0 \text{ --- (2)}$$

From (1)

$$(nA-1)^2 = nA^2$$

$$nA-1 = \pm \sqrt{n} A$$

$$(n \pm \sqrt{n})A = 1 \implies A = \frac{1}{\sqrt{n}(\sqrt{n} \pm 1)} \text{ putting in (2)}$$

From (2) $B = \frac{-nA^2}{2(nA-1)}$

putting $A = \frac{1}{\sqrt{n}(\sqrt{n}+1)}$ in above equation

$$\begin{aligned}
 B &= \frac{-n \left(\frac{1}{\sqrt{n}(\sqrt{n}+1)} \right)^2}{2 \left[\frac{n}{\sqrt{n}(\sqrt{n}+1)} + 1 \right]} \\
 &= \frac{-\frac{n}{\cancel{2}} \left(\frac{1}{\sqrt{n}+1} \right)^2}{2 \left[\frac{\sqrt{n} + \sqrt{n} + 1}{(\sqrt{n}+1)} \right]} = \frac{+\frac{1}{(\sqrt{n}+1)^2}}{2 \frac{1}{(\sqrt{n}+1)}} \\
 B &= +\frac{1}{2(\sqrt{n}+1)}
 \end{aligned}$$

now

$$\begin{aligned}
 A \sum x_i + B &= \frac{\sum x_i}{n+a+b} + \frac{a}{n+a+b} \\
 \frac{\sum x}{\sqrt{n}(\sqrt{n}+1)} + \frac{1}{2(\sqrt{n}+1)} &= \frac{\sum x_i}{n+a+b} + \frac{a}{n+a+b}
 \end{aligned}$$

Comparing both sides

$$n+a+b = \sqrt{n}(\sqrt{n}+1) \quad \text{and} \quad \frac{a}{n+a+b} = \frac{1}{2(\sqrt{n}+1)}$$

$$a = n + \sqrt{n} - n - b$$

$$a = \sqrt{n} - b$$

putting this value of 'a' in 2nd eq.

$$\frac{\sqrt{n} - b}{n + \sqrt{n} - b + b} = \frac{1}{2(\sqrt{n}+1)}$$

$$\sqrt{n} - b = \frac{\sqrt{n}(\sqrt{n}+1)}{2(\sqrt{n}+1)} = \frac{\sqrt{n}}{2}$$

$$b = \sqrt{n} - \frac{\sqrt{n}}{2} = \frac{\sqrt{n}}{2}$$

$$\text{Thus } b = \frac{\sqrt{n}}{2}, \quad a = \frac{\sqrt{n}}{2}$$

Thus the Bayes estimator with constant risk is

$$\frac{\sum x + \sqrt{n}/2}{n + \sqrt{n}} \quad \text{which is minimax estimator.}$$



Minimum Chi-Square (MCS) Estimator.

Let $x_1, \dots, x_n$ be a random sample from the density given by $f(x; \theta)$ and let $S_1, S_2, \dots, S_K$ be a partition of the range of X . The probability that an observation falls in any cell S_j , $j = 1, 2, \dots, K$, denoted by $p_j(\theta)$, can be found. If $f(x; \theta)$ is that pdf of a r.v., then

$$p_j(\theta) = P[X \text{ falls in cell } S_j] = \int_{S_j} f(x; \theta) dx$$

Note that $\sum_{j=1}^K p_j(\theta) = 1$

Let the r.v. N_j denote the number of X_i 's in the sample which falls in cell S_j , $j = 1, 2, \dots, K$, then $\sum_{j=1}^K N_j = n$, the sample size. Form the following summation

$$\chi^2 = \sum_{j=1}^K \frac{[N_j - np_j(\theta)]^2}{np_j(\theta)}$$

where n_j is a value of N_j .

The MCS estimator of θ is that estimator $\hat{\theta}$ which minimizes χ^2 . It is that θ among all possible θ 's which makes the expected number of observations in cells, nearest the observed number.

Example; let $x_1, \dots, x_n$ be a random sample from a Bernoulli distribution

$$f(x; \theta) = \theta^x (1-\theta)^{1-x} \quad x = 0, 1$$

Take $N_j =$ no of observations equal to j for $j = 0, 1$.

The range of X is partitioned into two sets consisting of the numbers 0 and 1 respectively.

Now
$$\chi^2 = \sum_{j=0}^1 \frac{[n_j - np_j(\theta)]^2}{np_j(\theta)}$$

$$= \frac{1}{n\theta} [n_1 - n\theta]^2 + \frac{1}{n(1-\theta)} [n_0 - n(1-\theta)]^2$$

$$n_0 + n_1 = n \Rightarrow n_0 = n - n_1$$

So

$$\chi^2 = \frac{1}{n\theta} [n_1 - n\theta]^2 + \frac{1}{n(1-\theta)} [n - n_1 - n + n\theta]^2$$

$$= \frac{(n_1 - n\theta)^2}{n\theta} + \frac{(n_1 - n\theta)^2}{n(1-\theta)}$$

$$= \frac{(n_1 - n\theta)^2}{n} \left[\frac{\theta + 1 - \theta}{\theta(1-\theta)} \right]$$

$$\chi^2 = \frac{(n_1 - n\theta)^2}{n\theta(1-\theta)}$$

χ^2 will be zero when $(n_1 - n\theta)^2 = 0$
 $n_1 = n\theta$
 $\theta = n_1/n$

The MCSE of θ is

$$\hat{\theta} = n_1/n$$

Modified MCS Estimator

Often it is difficult to locate that $\hat{\theta}$ which minimizes χ^2 . In this situation the denominator $np_j(\theta)$ is sometimes changed to n_j (if $n_j = 0$, unity is used) forming a modified χ^2

$$\hat{\chi}^2 = \sum_{j=1}^K \frac{(n_j - np_j(\theta))^2}{n_j}$$

The modified MCS estimator of θ is that $\hat{\theta}$ which minimizes $\hat{\chi}^2$.

In case of empty cells

$$\chi^2 = \left(\sum_{j=1}^k (n_{ij} - np_{j(0)})^2 / n_{ij} \right) + 2M$$

where M is sum of hypothetical frequency of the empty cells.



Testing of Hypothesis

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STATISTICAL HYPOTHESIS:

A statistical hypothesis is an assertion or conjecture about the distribution of one or more random variables.

SIMPLE HYPOTHESIS:

If the statistical hypothesis completely specifies the dist. then it is called simple hypothesis.

Example: Let $x_1, \dots, x_n$ be a random sample from

$$f(x; \theta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta)^2}{2\sigma^2}}$$

$$= N(\theta, \sigma^2)$$

The statistical hypothesis that the mean of normal distribution is equal to 17 is

$$H : \theta = 17$$

Such a hypothesis is called simple because it completely specifies the distribution.

COMPOSITE HYPOTHESIS:

If the statistical hypothesis does not completely specify the distribution, then it is called composite hypothesis.

In the above example, the statistical hypothesis that mean of normal distribution is less than or equal to 17 is

$$H : \theta \leq 17$$

Such a hypothesis is composite; it does not completely specify the distribution.

NULL HYPOTHESIS:

A hypothesis which is to be tested for possible rejection or nullification under the assumption that it is true is called null hypothesis. It is denoted by H_0 . It may be simple or composite.

Alternative Hypothesis:

An alternative hypothesis is any other hypothesis which we are willing to accept when H_0 is rejected. It is denoted by H_1 . (When H_0 is not rejected we say H_0 is accepted)

LEVEL OF SIGNIFICANCE (α): Probability of rejecting a true hypothesis.

TYPE I ERROR (α -error): Rejection of true hypothesis H_0 when it is true.

TYPE II ERROR (β -error): Acceptance of false hypothesis H_0 .

Decision	H_0 True	H_0 false
Accept H_0	correct	wrong type II error
Reject H_0	wrong type I error	correct

$$P(\text{Type I error}) = \alpha, \quad P(\text{Type II error}) = \beta$$

POWER OF A TEST:

The probability of rejecting a null hypothesis when it is actually false. The power is complement of β , type II error.

$$\text{Power} = P(\text{reject } H_0 / H_0 \text{ is false}) \\ = 1 - \beta$$

TEST OF A STATISTICAL HYPOTHESIS:

A test of a statistical hypothesis H is a rule or procedure for deciding whether to reject H (either we reject the hypothesis or do not reject). A test will be denoted by Y .

For example we reject

$$H: 0 \leq 17 \quad \text{iff} \quad \bar{X} > 17 + \text{constant} \quad (\text{capital up})$$

$$\text{For } f(x; \theta) = N(0, 25)$$

$$\text{we reject } H: 0 \leq 15 \quad \text{iff} \quad \bar{X} > 15 + \frac{5}{\sqrt{n}}$$

A test can be either randomized or non randomized. The test Y given in above example is a non-randomized test.

Another possible test Y' of H in example given above is the following.

"Toss a coin and reject H iff a head appears. Such a test

is an example of a randomized test.

We let $\mathcal{X}$ denote the sample space of observations, that is

$$\mathcal{X} = \{(x_1, \dots, x_n) : (x_1, \dots, x_n) \text{ is a possible value of } (x_1, \dots, x_n)\}$$

NON-RANDOMIZED TEST AND CRITICAL REGION:

Let a test Y of a statistical hypothesis H be defined as follows

"Reject H iff $(x_1, \dots, x_n) \in C_Y$, where C_Y is a subset of the sample space $\mathcal{X}$, then Y is called a non randomized test, and C_Y is called the critical region of the test Y ."

Example: let $x_1, \dots, x_n$ be a random sample from $N(0, 25)$, $\mathcal{X}$ is $(x_1, \dots, x_n)$. Consider

$H : \theta \leq 17$ and the test Y :

Reject H iff $\bar{x} > 17 + \frac{5}{\sqrt{n}}$. Then Y is non randomized test and

$$C_Y = \{(x_1, \dots, x_n) : \bar{x} > 17 + \frac{5}{\sqrt{n}}\}$$

Remark: A non-randomized test Y of H is a decomposition of $\mathcal{X}$ into C_Y and $\bar{C}_Y$ such that if $(x_1, \dots, x_n) \in C_Y$ H is rejected.

RANDOMIZED TEST AND CRITICAL FUNCTION:

A test Y of H is defined to be a randomized test if Y is defined by the function

$$\psi_Y(x_1, \dots, x_n) = P[H \text{ is rejected} / (x_1, \dots, x_n) \text{ is observed}]$$

The function $\psi_Y(x_1, \dots, x_n)$ is called the critical function of test Y .

In non randomized test, we observe the random sample and see whether the observed sample falls in the CR and reject H when it is so.

on the other hand to perform a randomized test, we observe a random sample $x_1, \dots, x_n$, then evaluate $\psi_Y(x_1, \dots, x_n)$ and finally observe the result of some Bernoulli trial that has ψ_Y as its probability of success, and if the Bernoulli trial results in a success, then H is rejected.

Example: let $x_1, \dots, x_n$ be a random sample of size 10 from

$$f(x; \theta) = \theta^x (1-\theta)^{1-x} \quad x=0,1$$

Suppose we want to test the hypothesis

$$H: \theta < 1/2$$

A possible test Y is the following: Reject H if $\sum_{i=1}^{10} x_i > 5$, accept H if $\sum_{i=1}^{10} x_i < 5$, and decide between rejecting and accepting by tossing a fair coin if $\sum_{i=1}^{10} x_i = 5$.

Such a test Y partitions $\mathcal{X}$ in to three regions, say A , B and C , where

$$A = \{(x_1, \dots, x_{10}) : \sum_{i=1}^{10} x_i < 5\}$$

$$B = \{(x_1, \dots, x_{10}) : \sum_{i=1}^{10} x_i = 5\}$$

$$C = \{(x_1, \dots, x_{10}) : \sum_{i=1}^{10} x_i > 5\}$$

The critical function of test Y is given by

$$\psi_Y(x_1, \dots, x_{10}) = \begin{cases} 1 & \text{if } (x_1, \dots, x_{10}) \in C \\ 1/2 & \text{if } (x_1, \dots, x_{10}) \in B \\ 0 & \text{if } (x_1, \dots, x_{10}) \in A \end{cases}$$

Neyman - Pearson Theory

In classical theory we restrict ourselves to CR's of size less than or equal to α .

For the practical purposes, we hold the probability of type I error fixed and look for the test statistic which minimizes the probability of type II error equivalently, which maximizes $1-\beta$. When testing the NH $H_0: \theta = \theta_0$ against the AH $H_1: \theta = \theta_1$, the quantity $1-\beta$ is represented to as the power of the test at $\theta = \theta_1$.

MP Test: A CR for testing a simple NH $\theta = \theta_0$ against a simple AH $\theta = \theta_1$ is to be best or most powerful if the power of the test at $\theta = \theta_1$ is a maximum.

To construct a MP CR, we refer to the likelihood of a random sample of size n from the population under consideration when $\theta = \theta_0$ and $\theta = \theta_1$. Denoting the likelihoods by L_0 and L_1 , we have

$$L_0 = \prod_{i=1}^n f(x; \theta_0) \quad \text{and} \quad L_1 = \prod_{i=1}^n f(x; \theta_1)$$

$\frac{L_0}{L_1}$ should be small for sample points inside the CR, which leads to type I error when $\theta = \theta_0$ and to correct decision when $\theta = \theta_1$. That is reject $\theta = \theta_0$ if it is more likely that the sample come from $\theta = \theta_1$ than from $\theta = \theta_0$.

Similarly $\frac{L_0}{L_1}$ should be large for sample points outside the CR, which leads to correct decision when $\theta = \theta_0$ and type II error when $\theta = \theta_1$. The fact that this argument guarantees a MP CR is proved by the following lemma.

Neyman - Pearson Lemma

Statement: If C is a CR of size α and K is a constant such that

$$\frac{L_0}{L_1} \leq K \quad \text{inside } C$$

and

$$\frac{L_0}{L_1} \geq K \quad \text{outside } C$$

then C is a MP CR of size α for testing $\theta = \theta_0$ against $\theta = \theta_1$.

Proof:

Suppose C is a CR of size α satisfying the conditions of the theorem and D is some other CR of size α .

Thus

$$\int_C \dots \int L_0 dx_1 \dots dx_n = \int_D \dots \int L_0 dx_1 \dots dx_n = \alpha$$

where the multiple integrals are over the respective n dimensional regions C and D

using these sets

$$[C \cap D] \cup [C \cap D^c] = C$$

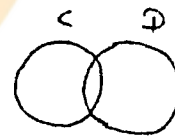
$$[C^c \cap D] \cup [C \cap D] = D$$

We can write

$$\begin{aligned} \int_{C \cap D} \dots \int L_0 dx_1 \dots dx_n + \int_{C \cap D^c} \dots \int L_0 dx_1 \dots dx_n \\ = \int_{C \cap D} \dots \int L_0 dx_1 \dots dx_n + \int_{C \cap D} \dots \int L_0 dx_1 \dots dx_n \end{aligned}$$

or

$$\int_{C \cap D} \dots \int L_0 dx_1 \dots dx_n = \int_{C \cap D} \dots \int L_0 dx_1 \dots dx_n \quad \text{--- (E)}$$



Since

$$L_1 \gg L_0/K \text{ inside } C \text{ and } L_1 \leq L_0/K \text{ outside } C, \quad \text{--- (2)}$$

then it follows from (1) and (2)

$$\begin{aligned} \int_{CND'} \dots \int L_1 dx_1 \dots dx_n &\gg \int_{CND'} \dots \int \frac{L_0}{K} dx_1 \dots dx_n \\ &= \int_{CND} \dots \int \frac{L_0}{K} dx_1 \dots dx_n \gg \int_{CND} \dots \int L_1 dx_1 \dots dx_n \end{aligned}$$

$$\int_{CND'} \dots \int L_1 dx_1 \dots dx_n \gg \int_{CND} \dots \int L_1 dx_1 \dots dx_n$$

Adding $\int_{CND} \dots \int L_1 dx_1 \dots dx_n$ on both sides

$$\begin{aligned} \int_{CND} \dots \int L_1 dx_1 \dots dx_n + \int_{CND'} \dots \int L_1 dx_1 \dots dx_n &\gg \\ \int_{CND} \dots \int L_1 dx_1 \dots dx_n + \int_{CND} \dots \int L_1 dx_1 \dots dx_n \end{aligned}$$

For the CR C , the probability of not committing type II error is greater than or equal to the corresponding probability of any other CR D . Hence C is MP CR.

Calculation of α and β

Example 1: Suppose we want to test the null hypothesis that the mean of a normal population with $\sigma^2 = 1$ is μ_0 against the H_1 that it is μ_1 , where $\mu_1 > \mu_0$. Find the value of K such that $\bar{x} > K$ provided a critical region of size $\alpha = 0.05$ for a random sample of size n .

Sol: CR for testing $H_0: \mu = \mu_0$
against $H_1: \mu = \mu_1$

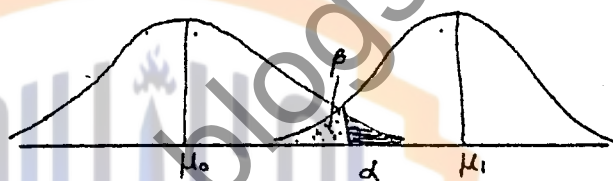
$$Z_{0.05} = 1.645$$

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$1.645 = \frac{K - \mu_0}{\sigma/\sqrt{n}}$$

$$K = \mu_0 + \frac{1.645}{\sqrt{n}} \quad \sigma = 1$$

$$\bar{x} > \mu_0 + \frac{1.645}{\sqrt{n}}$$



Example 2: A random sample of size n from a normal population with $\sigma^2 = 1$ is to be used to test the $H_0: \mu = \mu_0$ against $H_1: \mu = \mu_1$ where $\mu_1 > \mu_0$. Use the N.P lemma to find the MP CR of size α .

Solution:

$$L_0 = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2} \sum (x_i - \mu_0)^2}$$

$$L_1 = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2} \sum (x_i - \mu_1)^2}$$

$$\frac{L_0}{L_1} = \frac{e^{-\frac{1}{2} \sum (x_i - \mu_0)^2}}{e^{-\frac{1}{2} \sum (x_i - \mu_1)^2}}$$

$$\frac{L_0}{L_1} = \frac{e^{-\frac{1}{2} n \mu_0^2 + \mu_0 \sum x_i}}{e^{-\frac{1}{2} n \mu_1^2 + \mu_1 \sum x_i}}$$

$$= e^{\frac{1}{2} n (\mu_1^2 - \mu_0^2) + \sum x_i (\mu_0 - \mu_1)}$$

We must find the constant K and a CR C of the sample space such that

$$e^{\frac{n}{2} (\mu_1^2 - \mu_0^2) + \sum x_i (\mu_0 - \mu_1)} \leq K \quad \text{inside } C$$

$$e^{\frac{n}{2} (\mu_1^2 - \mu_0^2) + \sum x_i (\mu_0 - \mu_1)} \geq K \quad \text{out side } C$$

Taking natural log on both sides

$$\frac{n}{2} (\mu_1^2 - \mu_0^2) + \sum x_i (\mu_0 - \mu_1) \leq \ln K$$

now

$$\frac{n}{2} (\mu_1^2 - \mu_0^2) + \sum x_i (\mu_0 - \mu_1) \geq \ln K$$

now

$$(\mu_0 - \mu_1) \sum x_i \leq \ln K - \frac{n}{2} (\mu_1^2 - \mu_0^2) \quad \text{inside } C$$

$$(\mu_0 - \mu_1) \sum x_i \geq \ln K - \frac{n}{2} (\mu_1^2 - \mu_0^2) \quad \text{out side } C$$

dividing by the negative quantity $n(\mu_0 - \mu_1)$, the above inequalities becomes

$$-\frac{\sum x_i}{n} \leq \left[\ln K - \frac{n}{2} (\mu_1^2 - \mu_0^2) \right] / -n(\mu_0 - \mu_1)$$

$$+\frac{\sum x_i}{n} \geq \left[\ln K - \frac{n}{2} (\mu_1^2 - \mu_0^2) \right] / +n(\mu_0 - \mu_1)$$

$$\text{let } \left[\ln K - \frac{n}{2} (\mu_1^2 - \mu_0^2) \right] / n(\mu_0 - \mu_1) = R \quad (\text{some constant})$$

$$\frac{\sum x_i}{n} \geq R$$

$$\frac{\sum x_i}{n} \leq R$$

$$\bar{x} \geq R \quad \text{inside } C$$

$$\bar{x} < R \quad \text{outside } C$$

where R depends on μ_0, μ_1, n and α, K

Thus the most powerful critical region of size α for testing the null hypothesis $\mu = \mu_0$ against the A.H $\mu = \mu_1$ (with $\mu_1 > \mu_0$) for given normal population

$$\bar{x} \geq \mu_0 + Z_{\alpha} \sigma / \sqrt{n}$$

It does not depend on μ_1

$$\text{If } \mu_1 < \mu_0$$

$$\text{Then } \bar{x} = \mu_0 - \frac{\sigma}{\sqrt{n}} Z_{\alpha} \quad \text{for } \sigma^2 = 1$$

Example: Find which one is simple and which one is composite hypothesis.

(1) The hypothesis that a r.v has a gamma dist with

$$\alpha = 0.3 \quad \text{and} \quad \beta = 2$$

As Gamma dist has two parameters α and β and both are given, so it is simple hypothesis.

(2) A hypothesis that a r.v has a exponential density.

Composite hypothesis (As no parameter is specified)

(3) A random variable has β -dist with mean $\mu = 0.5$

Composite

(4) A poisson dist with mean $\lambda = 0.05$ is simple hypothesis

If $\lambda > 0.05$ (Composite hypothesis)

(5) A normal dist with mean $\mu = 100$ (Composite)

(6) A, r.v has -ve binomial dist with $k = 3$ and $p < 0.6$

so it is Composite.

Example: Let $x_1, \dots, x_n$ be a random sample from

$$f(x; \theta) = \theta e^{-\theta x} \quad 0 < x < \infty$$

Use NP Lemma To test

$$H_0: \theta = \theta_0 \quad \text{vs} \quad H_1: \theta = \theta_1 \quad (\theta_1 > \theta_0)$$

Solution:

$$L_0 = \theta_0^n e^{-\theta_0 \sum x_i}$$

$$L_1 = \theta_1^n e^{-\theta_1 \sum x_i}$$

According to NP lemma the MP test will have the form

Reject H_0 if $\lambda \leq K$

$$\frac{L_0}{L_1} \leq K$$

$$\frac{\theta_0^n e^{-\theta_0 \sum x_i}}{\theta_1^n e^{-\theta_1 \sum x_i}} \leq K$$

$$e^{-(\theta_0 - \theta_1) \sum x_i} \leq \left(\frac{\theta_1}{\theta_0}\right)^n K$$

Taking $\ln$

$$(\theta_1 - \theta_0) \sum x_i \leq \ln \left(\frac{\theta_1}{\theta_0}\right)^n K$$

$$\sum x_i \leq \frac{1}{(\theta_1 - \theta_0)} \ln \left(\frac{\theta_1}{\theta_0}\right)^n K = K' (\text{say})$$

$$\sum x_i \leq K'$$

where K' is just a constant

$$\alpha = P_{\theta=\theta_0} [\text{Reject } H_0]$$

$$\alpha = P_{\theta_0} \left[\sum_{i=1}^n x_i \leq K' \right]$$

We know $\sum x_i$ has a Gamma dist with parameter n and θ .

Therefore

$$\alpha = \int_0^{K'} \frac{1}{\Gamma(n)} \theta_0^n x^{n-1} e^{-\theta_0 x} dx$$

This is an equation in K' from which K' can be determined and NP test of size α of

$$H_0: \theta = \theta_0 \text{ vs } H_1: \theta = \theta_1$$

is this: Reject H_0 if

$$\sum x_i \leq K'$$

where K' is the α th quantile point of the Gamma dist with parameters n, θ_0 .

Example: let $x_1, \dots, x_n$ be a random sample from

$$f(x, \theta) = \theta^x (1-\theta)^{1-x} \quad x=0,1$$

Use NP lemma to test

$$H_0: \theta = \theta_0 \text{ vs } H_1: \theta = \theta_1 \quad (\theta_1 > \theta_0)$$

Sol:

$$L_0 = \theta_0^{\sum x_i} (1-\theta_0)^{n-\sum x_i}$$

$$L_1 = \theta_1^{\sum x_i} (1-\theta_1)^{n-\sum x_i}$$

According to NP lemma the MP test will have the form

Reject H_0 if $\sum x_i \leq K$

$$\frac{L_0}{L_1} \leq K$$

$$\frac{\theta_0^{\sum x_i} (1-\theta_0)^{n-\sum x_i}}{\theta_1^{\sum x_i} (1-\theta_1)^{n-\sum x_i}} \leq K$$

$$\left(\frac{\theta_0}{\theta_1}\right)^{\sum x_i} \left(\frac{1-\theta_0}{1-\theta_1}\right)^{n-\sum x_i} \leq K$$

$$\left(\frac{\theta_0}{\theta_1}\right)^{\sum x_i} \left(\frac{1-\theta_0}{1-\theta_1}\right)^n \left(\frac{1-\theta_0}{1-\theta_1}\right)^{-\sum x_i} \leq K$$

$$\left(\frac{\theta_0(1-\theta_1)}{\theta_1(1-\theta_0)}\right)^{\sum x_i} \leq \left(\frac{1-\theta_1}{1-\theta_0}\right)^n K$$

$$\left(\frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)}\right)^{\sum x_i} \geq \frac{1}{K} \left(\frac{1-\theta_0}{1-\theta_1}\right)^n$$

$$\sum x_i \ln \left(\frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} \right) \geq \ln \frac{(1-\theta_0)}{K(1-\theta_1)}$$

$$\sum x_i \geq K'$$

where K' is a constant

So a MP test would be of the form

Reject H_0 if $\sum x_i \geq K'$

Suppose $\theta_0 = 1/4$, $\theta_1 = 3/4$, $n = 10$

We must find K' so that

$$\alpha = P_{\theta_0}[\text{Reject } H_0]$$

$$= P_{\theta_0}[\sum x_i \geq K']$$

$$\alpha = \sum_{j=K'}^{10} \binom{10}{j} \left(\frac{1}{4}\right)^j \left(\frac{3}{4}\right)^{10-j}$$

$$\text{If } \alpha = 0.0197, K' = 6 \quad \left. \begin{array}{l} \text{From Cumulative Binomial} \\ \text{Probabilities (Table)} \end{array} \right\}$$

$$\text{If } \alpha = 0.0781, K' = 5$$

For $\alpha = 0.05$, there is no CR.

However, a MP test of size α does exist, the test would be a randomized test.

For $\alpha = 0.05$, the randomized test with critical function

$$\psi(x_1, \dots, x_n) = \begin{cases} 1 & \text{if } \sum x_i \geq 6 \\ \frac{0.05 - 0.0197}{0.0781 - 0.0197} & \text{if } \sum x_i = 5 \\ 0 & \text{if } \sum x_i \leq 4 \end{cases}$$

is the MP test of size $\alpha = 0.05$

$$\begin{array}{l} 0.0584 - 1 \\ 0.0584 - 1 \end{array} \times 0.0584$$

The Power Function of a Test

We have to consider the probability $\alpha(\theta)$ of committing type I error for all values of θ within the domain specified under H_0 and probabilities $\beta(\theta)$ of committing a type II error for all values of θ within the specified domain under H_1 .

The power function of a test of a statistical hypothesis H_0 against an AH H_1 is given by

$$\pi(\theta) = \begin{cases} \alpha(\theta) & \text{for values of } \theta \text{ assumed under } H_0 \\ 1 - \beta(\theta) & \text{for values of } \theta \text{ assumed under } H_1 \end{cases}$$

Where

$\alpha(\theta)$ = Probability of committing Type I error under H_0 .

$1 - \beta(\theta)$ = Probability of not committing type II error under H_1 .

Example : Suppose we want to test

$$H_0 : \theta \geq 0.90 \quad \text{against} \quad H_1 : \theta < 0.90$$

Estimate the power function for the test criterion according to which we reject H_0 when $x < 15$ and otherwise we accept it. where x is the required number of successes in $n = 20$ trials.

Sol:

Under H_0 : $\theta = 0.90, 0.95, 1.00$

$$\alpha(\theta) = P(\text{Reject } H_0 / H_0 \text{ is true}) = P_0(x < 15)$$

Under H_1 : $\theta = 0.85, 0.80, 0.75, 0.70, 0.65, 0.60, 0.55, 0.50$

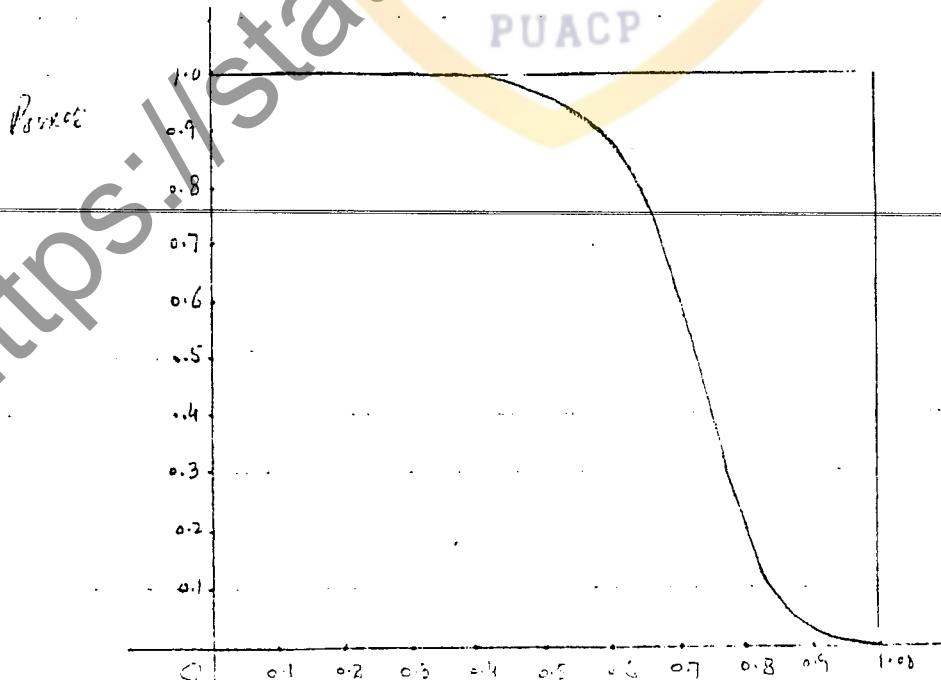
$$\beta(\theta) = P_1(\text{Accepting } H_0) = P(x \geq 15)$$

$$x \sim \text{bin}(n=20, \theta=0.90, 0.95, 1.00 \text{ for } \alpha(\theta))$$

$$x \sim \text{bin}(n=20, \theta=0.85, 0.80, 0.75, 0.70, 0.65, 0.60, 0.55, 0.50 \text{ for } \beta(\theta))$$

θ	$\alpha(\theta)$	$\beta(\theta)$	$1 - \beta(\theta)$
0.95	0.0003		
0.90	0.0113		
0.85		0.9327	0.0673
0.80		0.8042	0.1958
0.75		0.6172	0.3828
0.70		0.4164	0.5836
0.65		0.2454	0.7546
0.60		0.1256	0.8744
0.55		0.0553	0.9447
0.50		0.0207	0.9793

If we plot the probabilities of accepting H_0 (instead of probabilities of rejecting H_0), we will obtain the operating characteristic curve or simply OC curve of the given CR. In other words the values of the OC function are given by $(1 - \pi(\theta))$.



Most Powerful Test

A test Y^* of $H_0: \theta = \theta_0$ against $H_1: \theta \neq \theta_0$ is defined to be most powerful test of size α ($0 < \alpha < 1$) iff

(i) $\pi_{Y^*}(\theta_0) = \alpha$

(ii) $\pi_{Y^*}(\theta_1) \geq \pi_Y(\theta_1)$ for any other test Y for which $\pi_Y(\theta_0) \leq \alpha$

In other words a test Y^* is MP of size α if it has size α and if among all other tests of size α or less it has the greatest power.

Generalized Likelihood Ratio Tests

We shall present a general method for constructing CRs for tests of composite hypothesis. These tests are called generalized likelihood ratio tests and are based on a generalization of the simple H_0 vs simple H_1 method.

To illustrate the LR technique, let us suppose that $x_1, \dots, x_n$ constitute a random sample of size n from a population with density function $f(x; \theta)$, and that Ω is the set of values which can be taken on by the parameter θ , Ω is called the parameter space for θ .

We test

$$H_0: \theta \in \omega \text{ and } H_1: \theta \in \omega'$$

where ω is a subset of Ω and ω' is the complement of ω w.r.t Ω . Thus the parameter space Ω is partitioned into two disjoint sets ω and ω' . According to H_0 θ is an element of ω and according to H_1 θ is an element of ω' . In most problems Ω is the set of real numbers, the set of positive real numbers.

In the general case, where at least one of the hypotheses is composite, we compare the two quantities $\max L_0$ and $\max L$.

Where $\max L_0$ is the maximum value of the LF for all values of θ in ω , and $\max L$ is the maximum value of the LF for all values of θ in Ω . If we have a random sample of size n from a population with density function $f(x; \theta)$, $\hat{\theta}$ is the ML estimate of θ , subject to the restriction that θ must be an element of ω , and $\hat{\hat{\theta}}$ is the ML estimate of θ for all values of θ in Ω .

$$\max L_0 = \prod_{i=1}^n f(x_i; \hat{\theta}) \quad \text{and} \quad \max L = \prod_{i=1}^n f(x_i; \hat{\hat{\theta}})$$

The ratio $\lambda = \frac{\max L_0}{\max L}$ is referred to as the value of the LR statistic λ , where $0 \leq \lambda \leq 1$.

When H_0 is false, we would expect $\max L_0$ to be small compared to $\max L$, in which case λ would be close to zero. On the other hand when H_0 is true we would expect $\max L_0$ close to $\max L$, in which case λ would be close to 1.

A LR test states that the H_0 is rejected iff λ fall in the CR of the form $\lambda \leq K$ where $0 < K \leq 1$.

Definition:

If ω and ω' are complementary subsets of the parameter space Ω and if

$$\lambda = \frac{\max L_0}{\max L}$$

where $\max L_0$ and $\max L$ are the maximum values of LF for all values of θ in ω and Ω respectively, then the LR defined by $\lambda \leq K$, where $0 < K < 1$ defines a LR test of $H_0: \theta \in \omega$ against $H_1: \theta \in \omega'$.

If H_0 is a simple hypothesis K is chosen so that size of C.R equals α . If H_0 is composite K is chosen so that the probability of type I error is less than α or equal to α for all θ in ω and exactly equal to α if possible.

for at least one value of θ in ω . Thus if H_0 is a simple hypothesis and $g(x)$ is the density of Λ when H_0 is true, then K must be such that

$$P(\Lambda \leq K) = \int_0^K g(x) d\lambda = \alpha$$

In the discrete case integral is replaced by summation and K is taken to be the largest value for which sum is less than or equal to α .

Example: Find the CR of the LR test for testing

$$H_0: \mu = \mu_0 \text{ against } H_1: \mu \neq \mu_0$$

on the basis of random sample of size n from normal population with known variance σ^2 .

Solution:

Since ω contains only one value μ_0 , it follows that

$$\mu = \mu_0$$

Since $\mathcal{R}$ is the set of all real numbers, it follows that

$$\hat{\mu} = \bar{x}$$

$$\max L_0 = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right]$$

$$\max L = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2 \right]$$

$$\lambda = \frac{\max L_0}{\max L} = \frac{\exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \mu_0)^2 \right]}{\exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2 \right]}$$

$$= \exp \left[-\frac{1}{2\sigma^2} \left\{ \sum x_i^2 + n\mu_0^2 - 2\mu_0 \sum x_i - \sum x_i^2 + 2\bar{x} \sum x_i - n\bar{x}^2 \right\} \right]$$

$$\lambda = \exp \left[-\frac{1}{2\sigma^2} \left\{ n\mu_0^2 - 2n\mu_0\bar{x} + 2n\bar{x}^2 - n\bar{x}^2 \right\} \right]$$

$$\lambda = \exp \left[-\frac{n}{2\sigma^2} (\mu_0^2 - 2\mu_0\bar{x} + \bar{x}^2) \right]$$

$$\lambda = \frac{-\frac{n}{2\sigma^2} (\mu_0 - \bar{x})^2}{e^{\frac{-n}{2\sigma^2} (\bar{x} - \mu_0)^2}} = e^{\frac{-n}{2\sigma^2} (\bar{x} - \mu_0)^2}$$

The CR of the LR test is $\lambda \leq K$

$$\frac{-\frac{n}{2\sigma^2} (\bar{x} - \mu_0)^2}{e^{\frac{-n}{2\sigma^2} (\bar{x} - \mu_0)^2}} \leq K$$

Taking natural log

$$-\frac{n}{2\sigma^2} (\bar{x} - \mu_0)^2 \leq \ln K$$

Because $0 < K < 1$, therefore $\ln K$ would be negative

$$\therefore (\bar{x} - \mu_0)^2 \geq \frac{2\sigma^2}{n} \ln K$$

or

$$|(\bar{x} - \mu_0)| \geq K'$$

where K' will be determined so that size of the CR is α .

$$|\bar{x} - \mu_0| \geq Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

$$\text{or } |Z| \geq Z_{\frac{\alpha}{2}}$$

$$\text{where } Z_{\frac{\alpha}{2}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Thus H_0 is rejected when Z takes a value greater than or equal to $Z_{\frac{\alpha}{2}}$, or a value less than or equal to $-Z_{\frac{\alpha}{2}}$.

For large n , we compute $-2\ln \lambda$

$$\ln \lambda = -\frac{n}{2\sigma^2} (\bar{x} - \mu_0)^2$$

$$-2 \ln \lambda = \frac{n}{\sigma^2} (\bar{x} - \mu_0)^2$$

$$-2 \ln \lambda = \left(\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right)^2 \sim \chi^2_1$$

Theorem: For large n , the dist of $-2 \ln \lambda$ approaches under very general conditions, the χ^2 distribution with 1 d.f.

This theorem applies to the one parameter case. If the population involves more than one unknown parameter from which the H_0 imposes r restrictions, the number of d.f. in the χ^2 approximation of the distribution of $-2 \ln \lambda$ is equal to r . If we want to test H_0 that the unknown mean and variance of a normal distribution are respectively $\mu = \mu_0$ and $\sigma^2 = \sigma_0^2$ against the H_1 $\mu \neq \mu_0$ and $\sigma^2 \neq \sigma_0^2$, the number of d.f. in the χ^2 approximation of the dist of $-2 \ln \lambda$ will be 2.

Example: let $x_1, \dots, x_n$ be a random sample from

$$f(x; \theta) = \theta e^{-\theta x} \quad 0 \leq x < \infty$$

$$\text{where } \Omega = \{ \theta : \theta > 0 \}, \quad \omega = \{ \theta : \theta \leq \theta_0 \}$$

$$\text{Test } H_0: \theta \leq \theta_0 \quad \text{vs} \quad H_1: \theta > \theta_0$$

Solution:

$$\max_{\Omega} L = \max_{\theta > 0} \left[\theta^n \exp\{-\theta \sum x_i\} \right]$$

$$\text{MLE of } \theta \text{ is } 1/\bar{x} = n/\sum x_i$$

Thus

$$\begin{aligned} \max_{\Omega} L &= \left(\frac{n}{\sum x_i} \right)^n \exp \left[-\frac{n}{\sum x_i} (\sum x_i) \right] \\ &= \left(\frac{n}{\sum x_i} \right)^n e^{-n} \end{aligned}$$

$$\max_{0 \leq \theta \leq \theta_0} L_0 = \max_{0 \leq \theta \leq \theta_0} \left[\theta^n \exp\{-\theta \sum x_i\} \right]$$

$$\max L_0 = \begin{cases} \left(\frac{n}{\sum x_i} \right)^n e^{-n} & \text{if } \frac{n}{\sum x_i} \leq \theta_0 \\ \theta_0^n e^{-\theta_0 \sum x_i} & \text{if } \frac{n}{\sum x_i} > \theta_0 \end{cases}$$

Hence

$$\lambda = \begin{cases} 1 & \text{if } \frac{n}{\sum x_i} \leq \theta_0 \\ \frac{\theta_0^n e^{-\theta_0 \sum x_i}}{\left(\frac{n}{\sum x_i} \right)^n e^{-n}} & \text{if } \frac{n}{\sum x_i} > \theta_0 \end{cases}$$

If $0 < K < 1$, then LR test is given by the following

Reject H_0 if $\lambda \leq K$

or

Reject H_0 if $\frac{n}{\sum x_i} > \theta_0$ and $\left(\frac{\theta_0 \sum x_i}{n} \right)^n \exp[-\theta_0 \sum x_i + n] \leq K$

or if $\theta_0 \bar{x} < 1$ and $(\theta_0 \bar{x})^n \exp[-n\theta_0 \bar{x} + n] \leq K$

or

$\theta_0 \bar{x} < 1$ and $(\theta_0 \bar{x})^n \exp[-n(\theta_0 \bar{x} - 1)] \leq K$

putting $\theta_0 \bar{x} = y$

$$y < 1 \text{ and } y^n e^{-n(y-1)} \leq K$$

We note that $y^n e^{-n(y-1)}$ has a maximum value for $y=1$

Hence

$y < 1$ and $y^n e^{-n(y-1)} \leq K$ iff $y \leq K'$ where $0 < K' < 1$

Thus LR test reduces to following

Reject H_0 if $\theta_0 \bar{x} \leq K'$ or $\bar{x} \leq \frac{K'}{\theta_0}$

K' is obtained as a solution of

$$\alpha = P_{\theta_0}[\theta_0 \bar{x} \leq K'] \quad \text{for } 0 < \alpha < \alpha_0$$

$$\alpha = P_0 \left[\sum_{i=1}^n x_i \leq nk' \right]$$

$$\alpha = \int_0^{nk'} \frac{1}{\Gamma(n)} u^{n-1} e^{-u} du$$

Uniformly Most Powerful Tests

A test Y^* of $H_0: \theta \in \omega$ against $H_1: \theta \in \omega'$ is defined to be uniformly most powerful test of size α iff

(i) $\max_{\theta \in \omega} \pi_{Y^*}(\theta) = \alpha$

(ii) $\pi_{Y^*}(\theta) \geq \pi_Y(\theta)$ for all $\theta \in \omega'$ and for any test Y with size less than or equal to α .

A test Y^* is UMP of size α if it has size α and if among all tests of size less than or equal to α it has the largest power for all alternative values of θ . The adverb "uniformly" refers to "all" alternative θ values.

A UMP test does not exist for all testing problems. If it exists, it has the greatest chance of rejecting H_0 , whenever it should be rejected. UMP test can be obtained by NP lemma.

Example: Let $X_1, \dots, X_n$ be a random sample from

$$f(x; \theta) = \theta e^{-\theta x} \quad 0 \leq x < \infty$$

where $\Omega = \{\theta : \theta > 0\}$.

Find a UMP test of $H_0: \theta = \theta_0$ vs $H_1: \theta > \theta_0$.

Solution: For fixed $\theta_1 > \theta_0$, the MP test of

$$H_0: \theta = \theta_0 \quad \text{vs} \quad H_1: \theta = \theta_1$$

is given as

Reject H_0 if $\sum x_i \leq K'$, where K' was given as a solution of equation

$$\alpha = \int_0^{K'} \frac{1}{\Gamma(n)} \theta_0^n x^{n-1} e^{-\theta_0 x} dx$$

Such a test is given by Neyman Pearson Lemma. This test does not depend on θ_0 except that $\theta_0 > \infty$. Hence we would get the same MP test for any $\theta_0 > \infty$. Thus the test is actually UMP.

The following theorem gives us a technique which can be used to find UMP tests in most general situation.

Theorem:

Let $X_1, \dots, X_n$ be a random sample from the density $f(x; \theta)$, $\theta \in \Omega$, where Ω is some interval. Assume that

$$f(x; \theta) = a(\theta) b(x) \exp[c(\theta) d(x)]$$

and set $t(x_1, \dots, x_n) = \sum_{i=1}^n d(x_i)$

i) If $c(\theta)$ is a monotonic increasing function in θ and if there exists K^* such that

$$P_{\theta_0}[t(x_1, \dots, x_n) > K^*] = \alpha$$

then the test Y^* in a CR

$$C^* = \{(x_1, \dots, x_n) : t(x_1, \dots, x_n) > K^*\}$$

is a UMP test of size α of

$$H_0: \theta \leq \theta_0 \quad \text{vs} \quad H_1: \theta > \theta_0$$

or of

$$H_0: \theta = \theta_0 \quad \text{vs} \quad H_1: \theta > \theta_0$$

ii) If $c(\theta)$ is monotonic decreasing function in θ and if there exists K^* such that

$$P_{\theta_0}[t(x_1, \dots, x_n) < K^*] = \alpha$$

then the test Y^* with CR

$$C^* = \{(x_1, \dots, x_n) : t(x_1, \dots, x_n) < K^*\}$$

is a UMP test of size α of

$$H_0: \theta \leq \theta_0 \quad \text{vs} \quad H_1: \theta > \theta_0$$

or of

$$H_0: \theta = \theta_0 \quad \text{vs} \quad H_1: \theta > \theta_0$$

Example: Let $X_1, \dots, X_n$ be independent random variable each normally distributed with mean μ and variance σ^2 . Where μ and σ^2 both unknown.

Test $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$

$$\Omega = \{ \theta = (\mu, \sigma^2) : -\infty < \mu < \infty, \sigma^2 > 0 \}$$

and

$$\omega = \{ \theta = (\mu, \sigma^2) : \mu = \mu_0, \sigma^2 > 0 \}$$

Solution:

Under H_0 the maximum of the function $f(x; \theta)$ over ω occurs at $\hat{\mu} = \mu_0, \hat{\sigma}^2 = \frac{\sum_{i=1}^n (x_i - \mu_0)^2}{n}$

Maximum of $f(x; \theta)$ over Ω occurs at $\hat{\mu} = \bar{x}, \hat{\sigma}^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$

$$\lambda = \frac{\max_{\omega} f(x_1, \dots, x_n; \theta)}{\max_{\Omega} f(x_1, \dots, x_n; \theta)} = \frac{\max_{\omega} L_0}{\max_{\Omega} L}$$

$$\max_{\Omega} L = \max_{-\infty < \mu < \infty, \sigma^2 > 0} \left[\left(\frac{1}{2\pi\sigma^2} \right)^{n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\} \right]$$

$$= \left(\frac{1}{2\pi \frac{\sum (x_i - \bar{x})^2}{n}} \right)^{n/2} \exp \left\{ -\frac{\sum (x_i - \bar{x})^2}{2 \frac{\sum (x_i - \bar{x})^2}{n}} \right\}$$

$$\max_{\Omega} L = \frac{(ne)^{n/2}}{(2\pi)^{n/2} \left(\sum (x_i - \bar{x})^2 \right)^{n/2}}$$

$$\max_{\omega} L_0 = \max_{\mu = \mu_0, \sigma^2 > 0} \left[\left(\frac{1}{2\pi\sigma^2} \right)^{n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\} \right]$$

$$= \left(\frac{1}{2\pi \frac{\sum (x_i - \mu_0)^2}{n}} \right)^{n/2} \exp \left\{ -\frac{\sum (x_i - \mu_0)^2}{\sum (x_i - \mu_0)^2} \right\}$$

$$\max_{\omega} L_0 = \frac{(ne^{-1})^{n/2}}{(2\pi)^{n/2} (\sum (x_i - \mu_0)^2)^{n/2}}$$

$$\lambda = \left[\frac{\sum (x_i - \bar{x})^2}{\sum (x_i - \mu_0)^2} \right]^{n/2} = \left[\frac{\sum (x_i - \bar{x})^2}{\sum \{(x_i - \bar{x}) + (\bar{x} - \mu_0)\}^2} \right]^{n/2}$$

$$= \left[\frac{\sum (x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2} \right]^{n/2}$$

$$\lambda = \left[\frac{1}{1 + \frac{n(\bar{x} - \mu_0)^2}{\sum (x_i - \bar{x})^2}} \right]^{n/2}$$

The CR of the LR test is

Reject H_0 if $\lambda \leq K$ where $0 \leq K \leq 1$

$$\frac{1}{\left[1 + \frac{n(\bar{x} - \mu_0)^2}{\sum (x_i - \bar{x})^2} \right]^{n/2}} \leq K \quad \text{or} \quad \frac{1}{1 + \frac{n(\bar{x} - \mu_0)^2}{\sum (x_i - \bar{x})^2}} \leq K^{2/n}$$

$$1 + \frac{n(\bar{x} - \mu_0)^2}{\sum (x_i - \bar{x})^2} \geq K^{-\frac{2}{n}}$$

$$\frac{n(\bar{x} - \mu_0)^2}{\sum (x_i - \bar{x})^2} \geq (K^{-\frac{2}{n}} - 1) \quad \text{--- ①}$$

$$\frac{n(\bar{x} - \mu_0)^2}{\sum (x_i - \bar{x})^2} \geq \frac{(K^{-\frac{2}{n}} - 1)}{(n-1)}$$

$$\frac{\sqrt{n} |\bar{x} - \mu_0|}{\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}} \geq \sqrt{(n-1)(K^{-\frac{2}{n}} - 1)}$$

$$\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} \geq c$$

$$|t| \geq c$$

If $H_0: \mu = \mu_0$ is true then

$$\frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{(n-1)}$$

Value of c will be determined so that size of CR is α .

$$\alpha = P\{t \geq c\}$$

$$|t| \geq t_{\frac{\alpha}{2}}$$

Thus H_0 is rejected if t takes a value greater than or equal to $t_{\frac{\alpha}{2}}$, or a value less than or equal to $-t_{\frac{\alpha}{2}}$.

OR

After Eq ① we can proceed as

$$\begin{aligned} \frac{n(\bar{x} - \mu_0)^2}{\frac{\sum (x_i - \bar{x})^2}{(n-1)}} &= \frac{n(\bar{x} - \mu_0)^2 / \sigma^2}{\left[\frac{(n-1)s^2 / \sigma^2}{1} \right] \frac{1}{(n-1)}} \\ &= \frac{\chi^2_{(1)}}{\chi^2_{(n-1)}} \end{aligned}$$

$$\therefore \frac{n(\bar{x} - \mu_0)^2}{s^2} \sim F(1, n-1)$$

UNBIASED TESTS

There are many hypotheses-testing problems for which a uniformly most powerful test does not exist. In these cases it may be possible to restrict the class of tests and find a UMP test in the restricted class. One such class that has some merit is the class of Unbiased tests.

Definition:

A test γ of the $H_0: \theta \in \omega$ against $H_1: \theta \in \omega'$ is an unbiased test iff

$$\alpha \quad \max_{\theta \in \omega} \pi_{\gamma}(\theta) \leq \min_{\theta \in \omega'} \pi_{\gamma}(\theta) \quad (1-\beta)$$

In an unbiased test the probability of rejecting H_0 when it is false is at least as large as the probability of rejecting H_0 when it is true. If within this restricted class, a test exists i.e. UMP, then we have a UMPU test.

Confidence Intervals

Let $x_1, \dots, x_n$ be a random sample from the density $f(x; \theta)$. Let $T_1 = t_1(x_1, \dots, x_n)$ and $T_2 = t_2(x_1, \dots, x_n)$ be two statistics satisfying $T_1 \leq T_2$ for which

$$P_\theta[T_1 < \tau(\theta) < T_2] = \gamma$$

where γ does not depend on θ ; then the random interval (T_1, T_2) is called a 100γ percent confidence interval for $\tau(\theta)$; γ is called the confidence coefficient; and T_1 and T_2 are called lower and upper confidence limits respectively, for $\tau(\theta)$.

A value (t_1, t_2) of the random interval is also called a 100γ percent confidence interval for $\tau(\theta)$.

One Sided Confidence Interval

Let $x_1, x_2, \dots, x_n$ be a random sample from the density $f(u; \theta)$. Let $T_1 = t_1(x_1, \dots, x_n)$ be a statistic for which

$P_\theta[T_1 < \tau(\theta)] \equiv \gamma$; then T_1 is called a one-sided lower confidence limit for $\tau(\theta)$.

Similarly, let $T_2 = t_2(x_1, \dots, x_n)$ be a statistic for which

$$P_\theta[\tau(\theta) < T_2] \equiv \gamma$$

Then T_2 is called a one-sided upper confidence limit for $\tau(\theta)$. (γ does not depend on θ).

Example: Let $x_1, \dots, x_n$ be a random sample from $f(x; \theta) = N(\theta, 1)$

Set $T_1 = \bar{x} - \frac{6}{\sqrt{n}}$ and $T_2 = \bar{x} + \frac{6}{\sqrt{n}}$; then (T_1, T_2) constitute a random interval and is a confidence interval for $\tau(\theta) = \theta$ with confidence coefficient

$$\gamma = P_\theta\left[\bar{x} - \frac{6}{\sqrt{n}} < \theta < \bar{x} + \frac{6}{\sqrt{n}}\right]$$

Subtracting $\bar{x}$ from each term

$$\gamma = P_0 \left[-\frac{6}{\sqrt{n}} < \theta - \bar{x} < \frac{6}{\sqrt{n}} \right]$$

Multiplying by (-1)

$$\gamma = P_0 \left[-\frac{6}{\sqrt{n}} < \bar{x} - \theta < \frac{6}{\sqrt{n}} \right]$$

$$\gamma = P_0 \left[-2 < \frac{\bar{x} - \theta}{3/\sqrt{n}} < 2 \right]$$

$$= P_0 \left[-2 < Z < 2 \right]$$

$$= \phi(2) - \phi(-2)$$

$$\gamma = 0.9772 - 0.0228$$

$$\gamma = 0.9544$$

For $n=25$, $\bar{x}=17.5$, the interval

$\left(17.5 - \frac{6}{\sqrt{25}}, 17.5 + \frac{6}{\sqrt{25}} \right)$ is called a 95.44% CI for θ

Remarks: If a Confidence interval for θ has been determined, then a whole family of Confidence intervals has been determined.

For a given 100 γ percent confidence interval estimator of θ ; a 100 γ percent confidence interval estimator of $\tau(\theta)$ can be obtained. Where $\tau(\cdot)$ is any strictly monotone function. For example if $\tau(\cdot)$ is a monotone, increasing function and (T_1, T_2) is a 100 γ percent confidence interval for θ , then

$(\tau(T_1), \tau(T_2))$ is a 100 γ percent confidence interval for $\tau(\theta)$, since

$$P_0[\tau(T_1) < \tau(\theta) < \tau(T_2)] = P_0[T_1 < \theta < T_2] = \gamma$$

Pivotal Quantity

Let $x_1, \dots, x_n$ be a random sample from the density $f(x; \theta)$. Let Q be a function of $x_1, x_2, \dots, x_n$ and θ . If Q has a distribution that does not depend on θ , then Q is defined to be a pivotal quantity. Pivotal quantity method is a method of finding confidence intervals.

Example: Let $x_1, \dots, x_n$ be a random sample from $f(x; \theta) = N(\theta, \sigma^2)$. $\bar{x} - \theta$ is a pivotal quantity since $\bar{x} - \theta$ is normally distributed with mean 0 and variance σ^2/n . Also $\frac{\bar{x} - \theta}{\sigma/\sqrt{n}}$ has SND and, hence, is a pivotal quantity. On the other hand, $\frac{\bar{x}}{\theta}$ is not a pivotal quantity since $\frac{\bar{x}}{\theta}$ is $N(1, \frac{\sigma^2}{n\theta^2})$, which depends on θ .

Large Sample Confidence Intervals

It is sometimes possible to find a sequence of estimators, say $T_n = t_n(x_1, \dots, x_n)$ of θ in a density $f(x; \theta)$ that are asymptotically normally distributed about θ ; that is, T_n is approximately normally distributed with mean θ and variance, say, $\sigma_n^2(\theta)$, where $\sigma_n^2(\theta)$ indicates that the variance is a function of θ and the sample size n .

In particular for large samples the M.L estimator, say $\hat{\theta} = \hat{\theta}(x_1, \dots, x_n)$ for a parameter θ in a density $f(x; \theta)$ is approximately normally distributed about θ under rather general conditions. The large sample variance of the maximum likelihood estimator is

$$\sigma_n^2(\theta) = \frac{1}{n E_{\theta} \left[\left\{ \left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right)^2 \right\} \right]} = \frac{-1}{n E_{\theta} \left[\frac{\partial^2}{\partial \theta^2} \log f(x; \theta) \right]}$$

$\frac{T_n - \theta}{\sigma_n(\theta)}$ can be treated as an approximate pivotal quantity and therefore for large sample size, a confidence interval with an approximate confidence coefficient γ may be determined by converting the inequalities

$$-Z < \frac{T_n - \theta}{\sigma_n(\theta)} < Z$$

where $Z = Z_{\frac{1+\gamma}{2}}$ is defined by

$$\phi\left(Z_{\frac{1+\gamma}{2}}\right) = \frac{1+\gamma}{2} \text{ or } \phi(Z) - \phi(-Z) = \gamma$$

Example: Let $x_1, \dots, x_n$ be a random sample from the density

$$f(x; \theta) = \theta e^{-\theta x} \quad 0 < x < \infty$$

Find a large sample confidence interval for θ .

Sol: Likelihood Function of the sample is

$$L(\theta) = \theta^n e^{-\theta \sum x_i}$$

$$\ln L = n \ln \theta - \theta \sum x_i$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta} - \sum x_i \quad \frac{n}{\theta} = \sum x_i$$

M.L.E of θ is $\hat{\theta} = \frac{1}{\bar{x}}$, it has asymptotic normal dist with mean θ and variance is

$$\sigma_n^2(\theta) = \frac{-1}{n E_{\theta} \left[\frac{\partial^2}{\partial \theta^2} \ln f(x; \theta) \right]} = \frac{\theta^2}{n}$$

Therefore

$$P \left[-Z < \frac{\ln - \theta}{\sigma_n(\theta)} < Z \right] \approx \gamma$$

$$P \left[-Z < \frac{\frac{1}{\bar{x}} - \theta}{\theta/\sqrt{n}} < Z \right] \approx \gamma$$

$$P \left[\frac{\frac{1}{\bar{x}}}{1 + Z/\sqrt{n}} < \theta < \frac{\frac{1}{\bar{x}}}{1 - Z/\sqrt{n}} \right] \approx \gamma$$

Hence $\left(\frac{1/\bar{x}}{1 + Z/\sqrt{n}}, \frac{1/\bar{x}}{1 - Z/\sqrt{n}} \right)$ is a large sample confidence interval

for θ with an approximate confidence coefficient γ , where Z is given by $\phi(Z) - \phi(-Z) = \gamma$

Example: Consider sampling from the Bernoulli distribution with parameter

$$\theta = P[X=1] = 1 - P[X=0]$$

The MLE of θ is

$$\hat{\theta} = \bar{X}_n \text{ and it has variance}$$

$$\sigma_n^2(\theta) = \frac{\theta(1-\theta)}{n}$$

An approximate 100γ percent confidence interval for θ is obtained by converting the inequalities in

$$P\left[-Z < \frac{\hat{\theta} - \theta}{\sqrt{\frac{\theta(1-\theta)}{n}}} < Z\right] \cong \gamma$$

$$\begin{aligned} (\hat{\theta} - \theta)^2 &= Z^2 \frac{\theta(1-\theta)}{n} \\ \hat{\theta}^2 + \theta^2 - 2\theta\hat{\theta} &= \frac{\theta Z^2 - \theta^2 Z^2}{n} \end{aligned}$$

$$\begin{aligned} n\hat{\theta}^2 + n\theta^2 - 2n\theta\hat{\theta} &= \theta Z^2 - \theta^2 Z^2 \\ (n + Z^2)\theta^2 - (2n\hat{\theta} + Z^2)\theta + n\hat{\theta}^2 &= 0 \\ \theta &= \frac{(2n\hat{\theta} + Z^2) \pm \sqrt{(2n\hat{\theta} + Z^2)^2 - 4(n + Z^2)n\hat{\theta}^2}}{2(n + Z^2)} \end{aligned}$$

$$\begin{aligned} \theta &= \frac{(2n\hat{\theta} + Z^2) \pm \sqrt{4n^2\hat{\theta}^2 + Z^4 + 4n\hat{\theta}Z^2 - 4n\hat{\theta}^2 - 4n\hat{\theta}^2 Z^2}}{2(n + Z^2)} \\ &= \frac{(2n\hat{\theta} + Z^2) \pm \sqrt{Z^4 + 4n\hat{\theta}Z^2 - 4n\hat{\theta}^2 Z^2}}{2(n + Z^2)} \\ &= \frac{(2n\hat{\theta} + Z^2) \pm Z\sqrt{Z^2 + 4n\hat{\theta} - 4n\hat{\theta}^2}}{2(n + Z^2)} \end{aligned}$$

So we get

$$P\left[\frac{(2n\hat{\theta} + Z^2) - Z\sqrt{Z^2 + 4n\hat{\theta} - 4n\hat{\theta}^2}}{2(n + Z^2)} < \theta < \frac{(2n\hat{\theta} + Z^2) + Z\sqrt{Z^2 + 4n\hat{\theta} - 4n\hat{\theta}^2}}{2(n + Z^2)}\right] \cong \gamma \quad (1)$$

These expressions for the limits may be simplified since in deriving the large-sample distribution certain terms containing the factor $1/\sqrt{n}$ are neglected. We therefore neglect terms of this order in the limits in (1) without affecting the accuracy of the approximation.

So we may omit out the Z^2 terms in Eq (1)

Now Eq (1) may be written as

$$P\left[\frac{2\hat{\theta} - Z\sqrt{\frac{4\hat{\theta} - 4\hat{\theta}^2}{n}}}{2} < \theta < \frac{2\hat{\theta} + Z\sqrt{\frac{4\hat{\theta} - 4\hat{\theta}^2}{n}}}{2}\right] \cong \gamma$$

$$P\left[\hat{\theta} - z\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} < \theta < \hat{\theta} + z\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}\right] \approx \gamma \quad (2)$$

In particular

$$P\left[\hat{\theta} - 1.96\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} < \theta < \hat{\theta} + 1.96\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}\right] \approx 0.95 \quad (2A)$$

gives an approximate 95 percent Confidence interval for θ for large samples.

We may observe that $E_{\theta}(z)$ is just the expression that would have been obtained had $\hat{\theta}$ been substituted for θ in $\sigma_n^2(\theta)$. The substitution would imply that

$$\frac{\hat{\theta} - \theta}{\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}} \text{ is approximately } N(0, 1). \text{ This is true in general.}$$

Optimum Property:

An optimum property of large Sample Confidence intervals based on M.L.E is the following

"Large Sample Confidence intervals based on the M.L.E will be shorter, on the average, than intervals determined by any other estimator."

Tests of Hypotheses and Confidence Intervals

- Likelihood ratio principle provides us a method of obtaining tests of hypotheses. Another method, called the C.I method, is to use a confidence interval to obtain a test of hypothesis. For instance, if it is desired to test $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$, then we might compute a confidence interval estimate of θ from the data and if the interval contains θ_0 , accept H_0 , and otherwise reject H_0 . If the confidence interval had a confidence coefficient γ , then the resulting test would have size $1-\gamma$. We can reverse the operation; that is, one can use a family of tests of

$H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$ (the family is generated by varying θ_0) to obtain a confidence interval for θ .

Example:

Let $x_1, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$. We know that

$$\bar{X} - Z_{\frac{\alpha}{2}} \frac{\sigma_0}{\sqrt{n}}, \bar{X} + Z_{\frac{\alpha}{2}} \frac{\sigma_0}{\sqrt{n}} \text{ is a } \gamma = 1-\alpha \text{ level}$$

Confidence interval for μ .

If we define a test γ that reject H_0 : a value $\mu = \mu_0$, iff μ_0 lies outside this interval, that is

$$\text{iff } \frac{|\bar{X} - \mu_0|}{\sigma_0/\sqrt{n}} \geq Z_{\frac{\alpha}{2}}, \text{ then}$$

$$\left(P_{\mu_0} \left\{ \frac{\sqrt{n} |\bar{X} - \mu_0|}{\sigma_0} \geq Z_{\frac{\alpha}{2}} \right\} = \alpha \right)$$

and the test γ is a size α test of $\mu = \mu_0$ vs $\mu \neq \mu_0$

Conversely, the family of α level tests for the hypothesis $\mu = \mu_0$ generates a family of CI for μ by simply taking the region of acceptance of the test.

Similarly we can generate a family of α level tests from $(1-\alpha)$ lower (or upper) confidence bound.

Suppose we start with the $(1-\alpha)$ level lower bound $\bar{X} - Z_{\frac{\alpha}{2}} \frac{\sigma_0}{\sqrt{n}}$

then by defining a test γ that reject $\mu \leq \mu_0$ if

$$\mu_0 \leq \bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

We get an α level test for a hypothesis of the form $\mu \leq \mu_0$.

Confidence Sets

Let $\mathcal{X}$ denote the sample space, Ω the parameter space and $x_1, \dots, x_n$ observed sample.

A family of subsets of the parameter space Ω indexed by $(x_1, \dots, x_n) \in \mathcal{X}$, denoted by

$$\Phi = \{ \Omega(x_1, \dots, x_n) : \Omega(x_1, \dots, x_n) \in \Omega : (x_1, \dots, x_n) \in \mathcal{X} \}$$

is defined to be a family of Confidence Sets with Confidence Coefficient γ iff

$$P_{\theta}[\Omega(x_1, \dots, x_n) \text{ contains } \theta] = \gamma \quad \text{for all } \theta \in \Omega$$

$\Omega(x_1, \dots, x_n)$ is a random subset of the family of Confidence sets is a subset of Ω , the parameter space.

$\Omega(x_1, \dots, x_n)$ is a random subset; for any possible value, say $(x_1, \dots, x_n)$, of $(X_1, \dots, X_n)$, $\Omega(x_1, \dots, x_n)$ takes on the value $\Omega(x_1, \dots, x_n)$.

Example: Suppose $(x_1, \dots, x_n)$ is a random sample from $N(\theta, 1)$

$$\Omega = \{ \theta : -\infty < \theta < \infty \}$$

Let the subset $\Omega(x_1, \dots, x_n)$ be the interval $(\bar{x} - \frac{z}{\sqrt{n}}, \bar{x} + \frac{z}{\sqrt{n}})$

where z is given by

$$\Phi(z) - \Phi(-z) = \gamma$$

Then the family of subsets

$$\Phi = \{ \Omega(x_1, \dots, x_n) ; \Omega(x_1, \dots, x_n) = (\bar{x} - \frac{z}{\sqrt{n}}, \bar{x} + \frac{z}{\sqrt{n}}) \}$$

is a family of Confidence Sets with a Confidence Coefficient γ .

Since

$$P_{\theta}[\Omega(x_1, \dots, x_n) \text{ contains } \theta] = P\left[\bar{x} - \frac{z}{\sqrt{n}} < \theta < \bar{x} + \frac{z}{\sqrt{n}}\right]$$

$$= P\left[-z < \frac{\bar{x} - \theta}{1/\sqrt{n}} < z\right] = \gamma$$

for all $\theta \in \Omega$

In general, a Confidence interval is an example of a Confidence Set.

Construction of Confidence Sets

Confidence Sets can be constructed from the tests of hypothesis. Let γ_{θ_0} be a test of $H_0: \theta = \theta_0$, and let $\mathcal{K}(\theta_0)$ be the acceptance region of the test γ_{θ_0} . [The acceptance region is the set of complement of the critical region; that is if CR is given by $C(\theta_0)$, then $\mathcal{K}(\theta_0) = \mathcal{K} - C(\theta_0)$] $\mathcal{K}(\theta_0)$ is a subset of the $\mathcal{K}$ denoted by θ_0 . Since the test γ_{θ_0} has the size α

$$P_{\theta_0}[(X_1, \dots, X_n) \in \mathcal{K}(\theta_0)] = 1 - \alpha$$

$$P_{\theta_0}[\mathcal{K}(X_1, \dots, X_n) \text{ contains } \theta_0] = P_{\theta_0}[(X_1, \dots, X_n) \in \mathcal{K}(\theta_0)] = 1 - \alpha$$

Example: let $X_1, \dots, X_n$ be a random sample from $N(\theta, 1)$ and consider testing $H_0: \theta = \theta_0$. A test of size α is given by the following, Reject H_0 iff

$$\frac{|\bar{x} - \theta_0|}{1/\sqrt{n}} \geq Z \quad \text{where } Z \text{ is defined by}$$

$$\Phi(Z) - \Phi(-Z) = 1 - \alpha$$

The acceptance region of this test is given by

$$\mathcal{K}(\theta_0) = \left\{ (x_1, \dots, x_n) : \theta_0 - \frac{Z}{\sqrt{n}} < \bar{x} < \theta_0 + \frac{Z}{\sqrt{n}} \right\}$$

$$\mathcal{K}(x_1, \dots, x_n) = \left\{ \theta_0 : (x_1, \dots, x_n) \in \mathcal{K}(\theta_0) \right\}$$

$$= \left\{ \theta_0 : \theta_0 - \frac{Z}{\sqrt{n}} < \bar{x} < \theta_0 + \frac{Z}{\sqrt{n}} \right\}$$

$$= \left\{ \theta_0 : \bar{x} - \frac{Z}{\sqrt{n}} < \theta_0 < \bar{x} + \frac{Z}{\sqrt{n}} \right\}$$

$\mathcal{K}(x_1, \dots, x_n)$ is a Confidence Set with a Confidence Coeff $\gamma = 1 - \alpha$.

Sequential Analysis

Sequential analysis refers to techniques for testing hypotheses or estimating parameters when the sample size is not fixed in advance but is determined during the course of experiment by criteria which depend on the observations as they occur.

In testing the simple null hypothesis $H_0: \theta = \theta_0$ vs the simple alternative hypothesis $H_1: \theta = \theta_1$, we know that for sample of fixed size n , the test which minimizes the size, say β , of the type II error for fixed size, say α , of type I error was a simple likelihood-ratio test. That is for fixed α and n , β was minimized.

Suppose it is desired to fix both α and β in advance and then find that simple likelihood-ratio test having minimum sample size n and having size of type I error equal to α and size of type II error β .

We study one form of sequential analysis, namely sequential probability ratio test SPRT.

Definition of (SPRT):

Consider testing a simple H_0 against simple H_1 , suppose a sample of size n is drawn from one of the two distributions (it is not known from which one) and it is desired to test that the sample came from population one against the possibility that it came from the other. If $X_1, X_2, \dots$ denote r.v's, we want to test.

$$H_0: X_i \sim f_0(\cdot) \quad \text{vs} \quad H_1: X_i \sim f_1(\cdot)$$

The SLR test was of the form

$$\text{Reject } H_0 \text{ if } \lambda = \frac{L_0}{L_1} \leq K \text{ for some constant } K > 0$$

The Sequential test employs the likelihood-ratios sequentially.

Define

$$\lambda_m = \frac{L_0(X_1, \dots, X_m)}{L_1(X_1, \dots, X_m)} = \frac{L_0(m)}{L_1(m)} = \frac{\prod_{i=1}^m f_0(X_i)}{\prod_{i=1}^m f_1(X_i)}$$

for $m=1, 2, \dots$, and compute sequentially $\lambda_1, \lambda_2, \dots$.

For fixed K_0 and K_1 . We adopt the following procedure

Take observation x_1 and compute λ_1 ,

If $\lambda_1 \leq K_0$ reject H_0

If $\lambda_1 \geq K_1$ accept H_0

If $K_0 < \lambda_1 < K_1$, take second observation x_2 and compute

λ_2

If $\lambda_2 \leq K_0$ reject H_0

If $\lambda_2 \geq K_1$ accept H_0

If $K_0 < \lambda_2 < K_1$, take observation x_3 and so on.

We continue sampling as long as

$K_0 < \lambda_j < K_1$, rejecting H_0 if

$\lambda_m \leq K_0$

accepting H_0 if

$\lambda_m \geq K_1$

Critical Region:

The critical region of the sequential test is defined as

$$C = \bigcup_{n=1}^{\infty} C_n, \text{ where}$$

$$C_n = \{(x_1, \dots, x_n) : K_0 < \lambda_j(x_1, \dots, x_j) < K_1, j=1, \dots, n-1, \lambda_n(x_1, \dots, x_n) \leq K_0\} \quad (1)$$

A point in C_n indicates that H_0 is to be rejected for a sample of size n .

Acceptance Region:

The acceptance region can be defined as

$$A = \bigcup_{n=1}^{\infty} A_n, \text{ where}$$

$$A_n = \{(x_1, \dots, x_n) : K_0 < \lambda_j(x_1, \dots, x_j) < K_1, j=1, \dots, n-1, \lambda_n(x_1, \dots, x_n) \geq K_1\} \quad (2)$$

For fixed $0 < K_0 < K_1$, a test described above is defined to be a SPRT.

In this test we want to determine K_0 and K_1 so that the sequential probability ratio test will have preassigned α and β for its respective sizes α of type I and type II errors.

$$\alpha = [P(\text{reject } H_0 / H_0 \text{ is true})] = \sum_{n=1}^{\infty} \left(\frac{L_0(n)}{C_n} \right) \quad (3)$$

$$\beta = P(\text{accept } H_0 / H_0 \text{ is false}) = \sum_{n=1}^{\infty} \left(\frac{L_1(n)}{A_n} \right) \quad (4)$$

for fixed α and β .

Eq (3) and Eq (4) are two eq in two unknowns K_0 and K_1 . Both A_n and C_n are defined in terms of K_0 and K_1 . A solution of these two equations would give the sequential probability ratio test having the desired preassigned α and β .

The sample size of a SPRT is a random variable. The actual sample size depends upon which x_i are observed. It is a function of x vis $x_1, \dots, x_n$ and itself is a random variable. Denote it by N . One way of assessing the performance of the SPRT would be to evaluate the expected sample size that is required under each hypothesis.

The following theorem states that the SPRT is an optimal test if the performance is measured using expected sample size.

Theorem: The SPRT with error size α and β minimizes both $E[N / H_0 \text{ is true}]$ and $E[N / H_1 \text{ is true}]$ among all tests (sequential or not) which satisfy the following

$$P[H_0 \text{ is rejected} / H_0 \text{ is true}] \leq \alpha$$

$$P[H_0 \text{ is accepted} / H_0 \text{ is false}] \leq \beta$$

and the expected sample size is finite.

The SPRT requires fewer observations on the average than does the fixed sample size test that has the same error size.

Approximate SPRT,

The determination of K_0 and K_1 that defines that particular SPRT which has error sizes α and β and in general computationally quite difficult.

The following result gives approximate values for K_0 and K_1 .

Result: let K_0 and K_1 be defined so that the SPRT corresponding to K_0 and K_1 has error sizes α and β . then K_0 and K_1 can be approximated by, say, K'_0 and K'_1 , where

$$K'_0 = \frac{\alpha}{1-\beta}, \quad K'_1 = \frac{1-\alpha}{\beta}$$

Proof: Assume that

$$\sum_{n=1}^{\infty} P[N=n/H_i] = 1 \quad \text{for } i=0,1$$

$$\alpha = P[\text{reject } H_0 / H_0 \text{ is true}]$$

$$= \sum_{n=1}^{\infty} \left(\sum_{C_n} L_0(n) \right) \leq \sum_{n=1}^{\infty} \left(\sum_{C_n} K_0 L_1(n) \right) \quad \because \frac{L_0}{L_1} \leq K_0$$

$$= K_0 \sum_{n=1}^{\infty} \left(\sum_{C_n} L_1(n) \right)$$

$$\text{hence } \alpha \leq K_0 P[\text{reject } H_0 / H_1 \text{ is true}] = K_0(1-\beta)$$

$$K_0 \geq \frac{\alpha}{1-\beta}$$

Similarly

$$1-\alpha = P[\text{Accept } H_0 / H_0 \text{ is true}]$$

$$\frac{L_0}{L_1} \geq K_1$$

$$= \sum_{n=1}^{\infty} \left(\sum_{A_n} L_0(n) \right) \geq \sum_{n=1}^{\infty} \left(\sum_{A_n} K_1 L_1(n) \right)$$

$$= K_1 \sum_{n=1}^{\infty} \left(\sum_{A_n} L_1(n) \right) = K_1 P[\text{accept } H_0 / H_1 \text{ is true}]$$

$$= K_1 \beta$$

Hence $K_0 \beta \leq (1-\alpha) \Rightarrow R_1 \leq \frac{(1-\alpha)}{\beta}$.

The approximation $K'_0 = \frac{\alpha}{1-\beta}$, $K'_1 = \frac{1-\alpha}{\beta}$ satisfy

$$K'_0 = \frac{\alpha}{1-\beta} \leq K_0 \leq K_1 \leq \frac{1-\alpha}{\beta} = K'_1$$

Result: let α' and β' be the error sizes of the SPRT defined by K'_0 and K'_1 , where $K'_0 = \frac{\alpha}{1-\beta}$, $K'_1 = \frac{1-\alpha}{\beta}$.

Then $\alpha' + \beta' \leq \alpha + \beta$

Proof: let A' and C' (with corresponding A'_n and C'_n) denote the acceptance and critical regions of SPRT defined by K'_0 and K'_1 . Then

$$\begin{aligned} \alpha' &= \sum_{n=1}^{\infty} \int_{C'_n} L_0(n) \leq \sum_{n=1}^{\infty} \int_{C'_n} K'_0 L_1(n) = \frac{\alpha}{1-\beta} \sum_{n=1}^{\infty} \int_{C'_n} L_1(n) \\ &= \frac{\alpha}{(1-\beta)} P[\text{reject } H_0 / H_1 \text{ is true}] \\ &= \frac{\alpha}{(1-\beta)} (1-\beta') \\ \alpha' &\leq \frac{\alpha}{(1-\beta)} (1-\beta') \quad \text{--- ①} \end{aligned}$$

and

$$\begin{aligned} 1-\alpha' &= P[\text{accept } H_0 / H_0 \text{ is true}] = \sum_{n=1}^{\infty} \int_{A'_n} L_0(n) \geq \sum_{n=1}^{\infty} \int_{A'_n} K'_1 L_1(n) \\ &= \frac{(1-\alpha) \beta'}{\beta} \\ \frac{\beta'}{(1-\alpha')} &\geq \frac{(1-\alpha) \beta'}{\beta} \quad \text{--- ②} \end{aligned}$$

From ① and ②

$$\alpha'(1-\beta) \leq \alpha(1-\beta')$$

and $\beta(1-\alpha') \geq \beta'(1-\alpha)$

which together implies that

$$\alpha(1-\beta') \geq \alpha'(1-\beta) \quad \text{--- (3)}$$

$$\beta(1-\alpha') \geq \beta'(1-\alpha) \quad \text{--- (4)}$$

Adding (3) and (4)

$$\alpha(1-\beta') + (1-\alpha)\beta \geq \alpha'(1-\beta) + (1-\alpha')\beta'$$

$$\alpha - \alpha\beta' + \beta - \alpha'\beta \geq \alpha' - \alpha'\beta + \beta' - \alpha\beta'$$

$$(\alpha + \beta) \geq (\alpha' + \beta') \quad \text{Hence the result.}$$

Approximate Expected Sample Size of SPRT

The procedure used in performing a SPRT is to Continue Sampling as long as $K_0 < \lambda_n < K_1$ and stop Sampling as soon as $\lambda_n \leq K_0$ or $\lambda_n \geq K_1$.

If $Z_i = \ln \left[\frac{f_0(x_i)}{f_1(x_i)} \right]$ an equivalent test is given by as follows.

Continue Sampling as long as

$$\ln K_0 < \sum_{i=1}^n Z_i < \ln K_1$$

and stop Sampling as soon as

$$\sum_{i=1}^n Z_i \leq \ln K_0 \text{ or } \sum_{i=1}^n Z_i \geq \ln K_1.$$

Let N be t.v. denoting the sample size of the SPRT, and let

$$Z_i = \ln \left[\frac{f_0(x_i)}{f_1(x_i)} \right]$$

The following theorem is used for finding the approximate expected sample size of the SPRT

Theorem: (Wald's Equation): Let $Z_1, \dots, Z_n, \dots$ be independent identically distributed random variables satisfying

$$E[|Z_i|] < \infty$$

Let N be an integer-valued random variable whose value n depends only on the values of the first n, Z_i 's.

Suppose

$$E[N] < \infty, \text{ Then}$$

$$E[Z_1 + \dots + Z_N] = E[N] \cdot E[Z_i]$$

Proof: $E[Z_1 + \dots + Z_N] = E\left[E\left\{Z_1 + \dots + Z_N \middle| N\right\}\right]$

$$= \sum_{n=1}^{\infty} E[Z_1 + \dots + Z_n / N=n] P[N=n]$$

$$= \sum_{n=1}^{\infty} \sum_{i=1}^n E[Z_i / N=n] P[N=n]$$

$$= \sum_{i=1}^{\infty} \sum_{n=i}^{\infty} E[Z_i / N=n] P[N=n]$$

$$= \sum_{i=1}^{\infty} E[Z_i / N \geq i] P[N \geq i]$$

$E(Z_i) = E[Z_i / N \geq i]$ since $(N \geq i)$ depends only on $Z_1, \dots, Z_{i-1}$, hence independent of Z_i

$$= \sum_{i=1}^{\infty} E[Z_i] P[N \geq i]$$

$$= E(Z_i) \sum_{i=1}^{\infty} P[N \geq i]$$

$$E(Z_1 + \dots + Z_N) = E(Z_i) E(N)$$

If the SPRT leads to rejection of H_0 , then the E.V

$$Z_1 + Z_2 + \dots + Z_N \leq \ln K_0$$

$$E[Z_1 + \dots + Z_N] \leq \ln K_0 \quad \text{--- (1)}$$

Similarly if the test leads to the acceptance of H_0 , then the E.V

$$Z_1 + \dots + Z_N \geq \ln K_1$$

$$E[Z_1 + \dots + Z_N] \geq \ln K_1 \quad \text{--- (2)}$$

Hence

$$E[Z_1 + \dots + Z_N] \leq P \ln K_0 + (1-P) \ln K_1 \quad \text{--- (3)}$$

where $P = P[H_0 \text{ is rejected}]$

now by using the result

$$E[Z_1 + \dots + Z_N] = E[Z_i] E[N]$$

$$E[N] = \frac{E[Z_1 + \dots + Z_N]}{E[Z_i]} = \frac{P \ln K_0 + (1-P) \ln K_1}{E[Z_i]}$$

we get

$$E[N/H_0 \text{ is true}] \cong \frac{\alpha \ln K_0 + (1-\alpha) \ln K_1}{E[Z_i/H_0 \text{ is true}]} \quad \text{--- (4)}$$

$$\cong \frac{\alpha \ln \left(\frac{\alpha}{1-\beta} \right) + (1-\alpha) \ln \left(\frac{1-\alpha}{\beta} \right)}{E[Z_i/H_0 \text{ is true}]}$$

$$E[N/H_0 \text{ is false}] \cong \frac{(1-\beta) \ln K_0 + \beta \ln K_1}{E[Z_i/H_0 \text{ is false}]}$$

$$\cong \frac{(1-\beta) \ln \left(\frac{\alpha}{1-\beta} \right) + \beta \ln \left(\frac{1-\alpha}{\beta} \right)}{E[Z_i/H_0 \text{ is false}]} \quad \text{--- (5)}$$

Example: Consider Sampling from $N(\theta, \sigma^2)$, where σ^2 is assumed known. Test $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$

Now

$$Z_i = \ln \left(\frac{f_0(x_i)}{f_1(x_i)} \right) = \ln \left(\frac{\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta_0)^2}{2\sigma^2}}}{\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta_1)^2}{2\sigma^2}}} \right)$$

$$= \ln \left(\frac{e^{-\frac{(x-\theta_0)^2}{2\sigma^2}}}{e^{-\frac{(x-\theta_1)^2}{2\sigma^2}}} \right)$$

$$= -\frac{1}{2\sigma^2} \left[(x_i - \theta_0)^2 - (x_i - \theta_1)^2 \right]$$

$$= -\frac{1}{2\sigma^2} \left[x_i^2 + \theta_0^2 - 2x_i\theta_0 - x_i^2 - \theta_1^2 + 2x_i\theta_1 \right]$$

$$= -\frac{1}{2\sigma^2} \left[(\theta_0^2 - \theta_1^2) - 2x_i(\theta_0 - \theta_1) \right]$$

Hence

$$E[Z_i/H_0 \text{ is true}] = -\frac{1}{2\sigma^2} \left[(\theta_0^2 - \theta_1^2) - 2\theta_0(\theta_0 - \theta_1) \right]$$

$$= -\frac{1}{2\sigma^2} \left[\theta_0^2 - \theta_1^2 - 2\theta_0^2 + 2\theta_0\theta_1 \right]$$

$$= \frac{1}{2\sigma^2} \left[\theta_0^2 + \theta_1^2 - 2\theta_0\theta_1 \right]$$

$E(x_i) = \theta_0$
if H_0 is true

$$E[Z_i/H_0 \text{ is true}] = \frac{1}{2\sigma^2} (\theta_1 - \theta_0)^2$$

$$E[Z_i / H_0 \text{ is false}] = \frac{-1}{2\sigma^2} \left[(\theta_0 - \theta_1)^2 - 2\theta_1(\theta_0 - \theta_1) \right]$$

$$\bar{X}(x_i) = \theta_1 \\ H_1 \text{ is true}$$

$$= -\frac{1}{2\sigma^2} (\theta_1 - \theta_0)^2$$

Now for numerical values

$$\alpha = 0.01, \beta = 0.05, \sigma^2 = 100, \theta_0 = 100, \theta_1 = 105$$

$$E[Z_i / H_0 \text{ is false}] = -\frac{1}{2(100)} [5]^2 = -0.125$$

$$E[Z_i / H_0 \text{ is true}] = 0.125$$

and

$$E[N / H_0 \text{ is true}] \approx \frac{\alpha \ln\left(\frac{\alpha}{1-\beta}\right) + (1-\alpha) \ln\left(\frac{1-\alpha}{\beta}\right)}{0.125}$$

$$\approx 24$$

$$E[N / H_0 \text{ is false}] \approx \frac{(1-\beta) \ln\left(\frac{\alpha}{1-\beta}\right) + \beta \ln\left(\frac{1-\alpha}{\beta}\right)}{E[Z_i / H_0 \text{ is false}]}$$

$$\approx \frac{-4.1769}{-0.125} \approx 34$$

Now the sample size for fixed sample size test is obtained as

$$\alpha = P[\text{Reject } H_0 / H_0 \text{ is true}]$$

$$0.01 = P[\bar{X} > K]$$

$$= 1 - P[\bar{X} \leq K]$$

$$\text{--- } \bar{X} \text{ in SNU is}$$

$$\text{for } H_0: \theta \leq 100 \text{ vs } H_1: \theta > 100$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$0.01 = 1 - \Phi\left[\frac{K - 100}{10/\sqrt{n}}\right] \quad \text{--- } \textcircled{1}$$

$$\beta = P[\text{accept } H_0 / H_1 \text{ is true}]$$

$$0.05 = P[\bar{X} \leq K : H_1 \text{ is true}]$$

we get

$$E[N/H_0 \text{ is true}] \cong \frac{\alpha \ln K_0 + (1-\alpha) \ln K_1}{E[Z_i/H_0 \text{ is true}]} \quad \text{--- (4)}$$

$$\cong \frac{\alpha \ln \left(\frac{\alpha}{1-\beta} \right) + (1-\alpha) \ln \left(\frac{1-\alpha}{\beta} \right)}{E[Z_i/H_0 \text{ is true}]}$$

$$E[N/H_0 \text{ is false}] \cong \frac{(1-\beta) \ln K_0 + \beta \ln K_1}{E[Z_i/H_0 \text{ is false}]}$$

$$\cong \frac{(1-\beta) \ln \left(\frac{\alpha}{1-\beta} \right) + \beta \ln \left(\frac{1-\alpha}{\beta} \right)}{E[Z_i/H_0 \text{ is false}]} \quad \text{--- (5)}$$

Example: Consider sampling from $N(\theta, \sigma^2)$, where σ^2 is assumed known. Test $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$

Now

$$Z_i = \ln \left(\frac{f_0(x_i)}{f_1(x_i)} \right) = \ln \left(\frac{\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta_0)^2}{2\sigma^2}}}{\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta_1)^2}{2\sigma^2}}} \right)$$

$$= \ln \left[\frac{e^{-\frac{(x-\theta_0)^2}{2\sigma^2}}}{e^{-\frac{(x-\theta_1)^2}{2\sigma^2}}} \right]$$

$$= -\frac{1}{2\sigma^2} \left[(x_i - \theta_0)^2 - (x_i - \theta_1)^2 \right]$$

$$= -\frac{1}{2\sigma^2} \left[x_i^2 + \theta_0^2 - 2x_i\theta_0 - x_i^2 - \theta_1^2 + 2x_i\theta_1 \right]$$

$$= -\frac{1}{2\sigma^2} \left[(\theta_0^2 - \theta_1^2) - 2x_i(\theta_0 - \theta_1) \right]$$

Hence

$$E[Z_i/H_0 \text{ is true}] = -\frac{1}{2\sigma^2} \left[(\theta_0^2 - \theta_1^2) - 2\theta_0(\theta_0 - \theta_1) \right]$$

$E(x_i) = \theta_0$
if H_0 is true

$$= -\frac{1}{2\sigma^2} \left[\theta_0^2 - \theta_1^2 - 2\theta_0^2 + 2\theta_0\theta_1 \right]$$

$$= \frac{1}{2\sigma^2} \left[\theta_0^2 + \theta_1^2 - 2\theta_0\theta_1 \right]$$

$$E[Z_i/H_0 \text{ is true}] = \frac{1}{2\sigma^2} \left[\theta_1 - \theta_0 \right]^2$$

$$E[Z_i / H_0 \text{ is false}] = \frac{-1}{2\sigma^2} \left[(\theta_0 - \theta_1)^2 - 2\theta_1(\theta_0 - \theta_1) \right]$$

$$\bar{X}(H_1) = \theta_1$$

$H_1 \text{ is true}$

$$= -\frac{1}{2\sigma^2} (\theta_1 - \theta_0)^2$$

Now for numerical values

$$\alpha = 0.01, \beta = 0.05, \sigma^2 = 100, \theta_0 = 100, \theta_1 = 105$$

$$E[Z_i / H_0 \text{ is false}] = -\frac{1}{2(100)} [5]^2 = -0.125$$

$$E[Z_i / H_0 \text{ is true}] = 0.125$$

and

$$E[N / H_0 \text{ is true}] \approx \frac{\alpha \ln\left(\frac{\alpha}{1-\beta}\right) + (1-\alpha) \ln\left(\frac{1-\alpha}{\beta}\right)}{0.125}$$

$$\approx 24$$

$$E[N / H_0 \text{ is false}] \approx \frac{(1-\beta) \ln\left(\frac{\alpha}{1-\beta}\right) + \beta \ln\left(\frac{1-\alpha}{\beta}\right)}{E[Z_i / H_0 \text{ is false}]}$$

$$\approx \frac{-4.1769}{-0.125} \approx 34$$

Now the sample size for fixed sample size test is obtained as

$$\alpha = P[\text{Reject } H_0 / H_0 \text{ is true}]$$

$$0.01 = P[\bar{X} > K]$$

$$= 1 - P[\bar{X} \leq K]$$

$$\Rightarrow \bar{X} \text{ in SNU is}$$

$$\text{for } H_0: \theta \leq 100 \text{ vs } H_1: \theta > 100$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$0.01 = 1 - \Phi\left[\frac{K - 100}{10/\sqrt{n}}\right] \quad \text{--- (1)}$$

$$\beta = P[\text{accept } H_0 / H_1 \text{ is true}]$$

$$0.05 = P[\bar{X} \leq K : H_1 \text{ is true}]$$

$$\begin{aligned}
 E[N/H_0 \text{ is false}] &\cong \frac{0.99 \ln\left(\frac{0.05}{0.99}\right) + (0.01) \ln\left(\frac{0.95}{0.01}\right)}{\left(-\frac{1}{2\sigma^2}\right)} \\
 &\cong \frac{0.99(-2.9857) + (0.01)(4.5539)}{-\frac{1}{2\sigma^2}} \\
 &\cong (-2\sigma^2)(-2.9103)
 \end{aligned}$$

$$E[N/H_0 \text{ is false}] = 5.8206 \sigma^2$$

Q: Obtain average sample size for normal population with variance σ^2 for testing

$H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$
and in particular if $\theta_0 = 0, \theta_1 = 1, \sigma = 1, \alpha = 0.01, \beta = 0.05$
Sol;

$$E[N/H_0 \text{ is true}] = \frac{\alpha \ln\left(\frac{\alpha}{1-\beta}\right) + \beta(1-\alpha) \ln\left(\frac{1-\alpha}{\beta}\right)}{E[Z_i/H_0 \text{ is true}]}$$

$$E[N/H_0 \text{ is false}] = \frac{(1-\beta) \ln\left(\frac{\alpha}{1-\beta}\right) + \beta \ln\left(\frac{1-\alpha}{\beta}\right)}{E[Z_i/H_0 \text{ is false}]}$$

$$Z_i = \ln\left[\frac{f_0(x_i)}{f_1(x_i)}\right]$$

$$= \ln\left[\frac{\frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \theta_0)^2\right\}}{\frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \theta_1)^2\right\}}\right]$$

$$= \ln\left[\exp\left\{-\frac{1}{2\sigma^2}\left((x_i - \theta_0)^2 - (x_i - \theta_1)^2\right)\right\}\right]$$

$$= \ln\left[\exp\left\{-\frac{1}{2\sigma^2}\left(x_i^2 + \theta_0^2 - 2\theta_0 x_i - x_i^2 - \theta_1^2 + 2\theta_1 x_i\right)\right\}\right]$$

$$Z_i = \ln \left\{ \exp \left\{ -\frac{1}{2\sigma^2} \left(\theta_0^2 - \theta_1^2 - 2\theta_0 x_i + 2\theta_1 x_i \right) \right\} \right\}$$

$$Z_i = -\frac{1}{2\sigma^2} \left((\theta_0^2 - \theta_1^2) - 2x_i(\theta_0 - \theta_1) \right)$$

under H_0 $E(x_i) = \theta_0$; under H_1 : $E(x_i) = \theta_1$

$$\begin{aligned} E[Z_i / H_0 \text{ is true}] &= -\frac{1}{2\sigma^2} \left[(\theta_0^2 - \theta_1^2) - 2\theta_0(\theta_0 - \theta_1) \right] \\ &= -\frac{1}{2\sigma^2} \left[\theta_0^2 - \theta_1^2 - 2\theta_0^2 + 2\theta_0\theta_1 \right] \end{aligned}$$

$$E[Z_i / H_0 \text{ is true}] = -\frac{1}{2\sigma^2} \left[\theta_0 - \theta_1 \right]^2$$

$$E[Z_i / H_0 \text{ is false}] = -\frac{1}{2\sigma^2} \left[(\theta_0^2 - \theta_1^2) - 2\theta_1(\theta_0 - \theta_1) \right]$$

$$E[Z_i / H_0 \text{ is false}] = -\frac{1}{2\sigma^2} (\theta_0 - \theta_1)^2$$

In particular for $\theta_0 = 0$, $\theta_1 = 1$, $\sigma = 1$, $\alpha = 0.01$, $\beta = 0.05$

$$E[Z_i / H_0 \text{ is true}] = -\frac{1}{2}$$

$$E[Z_i / H_0 \text{ is false}] = -\frac{1}{2}$$

$$E[N / H_0 \text{ is true}] \approx \frac{(0.01) \ln\left(\frac{0.01}{0.95}\right) + (0.99) \ln\left(\frac{0.99}{0.05}\right)}{0.5}$$

$$\approx 39.11$$

$$E[N / H_0 \text{ is true}] \approx 40$$

$$E[N / H_0 \text{ is false}] \approx \frac{0.95(-4.5539) + (0.05)(19.8)}{-0.5}$$

$$\approx 6.67$$

$$\approx 7$$

Truncated Distributions

Suppose a random variable X defined over a sample space Ω then for any subspace $\Omega_0 \subset \Omega$. then the probability dist of X defined over Ω_0 is

$$P(X \in \Omega_0 / X \in \Omega) = \frac{P(X \in \Omega_0)}{P(X \in \Omega)}$$

If Ω_0 is the interval $a < X \leq b$, then the distribution having distribution function

$$F(x/a < X \leq b) = \begin{cases} 0 & \text{for } x \leq a \\ \frac{F(x) - F(a)}{F(b) - F(a)} & \text{for } a < x \leq b \\ 1 & \text{for } x > b \end{cases}$$

is called truncated distribution.

If $F(x)$ is differentiable, then the probability density function of the conditional distribution of X is

$$f(x/a < X \leq b) = \frac{f(x)}{\int_a^b f(t) dt} \quad \text{for all } a < x \leq b$$

$$= 0 \quad \text{otherwise.}$$

Truncation is a property of the distribution. A distribution is said to be truncated when some of its portion is missing. A distribution is truncated at points 'a' and 'b' if nothing is known about the values below 'a' and 'b'. This type of truncation is called two sided or doubly truncation. Portion of the area below 'a' and above 'b' is called degree of truncation.

The probability density of a singly truncated distribution at point 'b' ie nothing is known about 'b' is

$$f(x/x < b) = \frac{f(x)}{\int_{-\infty}^b f(u) du} \quad -\infty < x < b$$

This is right truncated distribution at point 'b'. Similarly the left truncated dist at point 'a' is

$$f(x/x > a) = \frac{f(x)}{\int_a^{\infty} f(x) dx} \quad a \leq x \leq \infty$$

Example 1; Find the probability distribution of a binomially dist random variable (based on n independent trials), truncated at the right of $X=n$, that is $X=n$ cannot be observed. Obtain the expected value and variance of this distribution.

Sol; The prob dist of truncated variable at $x=n$ is given as

x	$p(x)$
0	q^n/k
1	$\binom{n}{1} q^{n-1} p/k$
2	$\binom{n}{2} q^{n-2} p^2/k$
$\vdots$	$\vdots$
$n-1$	$n q p^{n-1}/k$

$$\frac{1}{k} \left(q^n + \binom{n}{1} q^{n-1} p + \dots + n q p^{n-1} \right) = 1$$

$$\frac{1}{k} \left(q^n + \binom{n}{1} q^{n-1} p + \dots + n q p^{n-1} + p^n - p^n \right) = 1$$

$$\frac{1}{k} (1 - p^n) = 1 \Rightarrow 1 - p^n = k$$

$$E(X) = \sum_{x=0}^{n-1} x \frac{\binom{n}{x} p^x q^{n-x}}{1 - p^n}$$

$$= \frac{1}{(1 - p^n)} \sum_{x=1}^{n-1} n \binom{n-1}{x-1} q^{n-x} p^x$$

$$E(X) = \frac{np}{1-p^n} \sum_{x=1}^{n-1} \binom{n-1}{x-1} q^{n-x} p^{x-1}$$

$$= \frac{np}{1-p^n} \left[\sum_{x=1}^{n-1} \binom{n-1}{x-1} q^{n-x} p^{x-1} - p^{n-1} \right]$$

$$E(X) = \frac{np}{1-p^n} (1 - p^{n-1})$$

Similarly

$$\text{Var}(X) = \frac{np}{1-p^n} \left[(n-1)p(1-p^{n-2}) + (1-p)^{n-1} - \frac{np(1-p^{n-1})^2}{1-p^n} \right]$$

Example:2, A random variable X has Poisson distribution

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Find the Poisson dist. truncated at the left of $x=1$. Find the mean and variance of this dist.

Sol; The truncated ^{prob} dist is given as

$$P(X/x \geq 1) = \frac{\lambda^x e^{-\lambda}}{x!} \bigg/ \sum_{x=1}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \frac{\lambda^x e^{-\lambda}}{x! \left[\sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} + e^{-\lambda} - e^{-\lambda} \right]}$$

$$= \frac{\lambda^x e^{-\lambda}}{x! \left[\sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} - e^{-\lambda} \right]}$$

$$P(X/x \geq 1) = \frac{\lambda^x e^{-\lambda}}{x! (1 - e^{-\lambda})}$$

$$x = 1, 2, \dots, \infty$$

$$E(X) = \sum_{x=1}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x! (1 - e^{-\lambda})}$$

$$= \frac{e^{-\lambda}}{1 - e^{-\lambda}} \sum_{x=1}^{\infty} x \frac{\lambda^x}{x!}$$

$$= \frac{e^{-\lambda}}{(1 - e^{-\lambda})} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$$

$$E(X) = \frac{e^{-\lambda} \lambda}{1 - e^{-\lambda}} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$\sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} = e^{\lambda}$$

$$E(X) = \frac{\lambda}{1 - e^{-\lambda}}$$

$$\begin{aligned} E(X^2) &= \sum_{x=1}^{\infty} x^2 \frac{\lambda^x e^{-\lambda}}{x! (1 - e^{-\lambda})} \\ &= \frac{\lambda^2}{1 - e^{-\lambda}} + \frac{\lambda}{1 - e^{-\lambda}} \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \frac{\lambda^2}{1 - e^{-\lambda}} + \frac{\lambda}{1 - e^{-\lambda}} - \frac{\lambda^2}{(1 - e^{-\lambda})^2} \end{aligned}$$

$$= \frac{\lambda}{1 - e^{-\lambda}} \left[\lambda + 1 - \frac{\lambda}{(1 - e^{-\lambda})} \right]$$

$$= \frac{\lambda}{1 - e^{-\lambda}} \left[\frac{\lambda - \lambda e^{-\lambda} + 1 - e^{-\lambda} - \lambda}{(1 - e^{-\lambda})} \right]$$

$$= \frac{\lambda}{1 - e^{-\lambda}} \left[\frac{(1 - e^{-\lambda}) - \lambda e^{-\lambda}}{1 - e^{-\lambda}} \right]$$

$$\text{Var}(X) = \frac{\lambda}{1 - e^{-\lambda}} - \frac{\lambda^2 e^{-\lambda}}{(1 - e^{-\lambda})^2}$$

Continued to Previous Example:

Example 3: If a random sample of n values contains y successes, then show that if

$\lambda^* = \sum_{x=2}^{\infty} \frac{x y}{n}$ then λ^* is an unbiased estimate of λ i.e.

$$E(\lambda^*) = \sum_{x=2}^{\infty} \frac{x}{n} E(y) = \lambda$$

Sol;
$$E(\lambda^*) = \sum_{x=2}^{\infty} \frac{x E(y)}{n}$$

$$E(\lambda^*) = \sum_{x=2}^{\infty} \frac{x}{n} \sum_{y=0}^n y \binom{n}{y} p^y q^{n-y}$$

where p is the prob of success. As $y \sim \text{bin dist}$

$$E(\lambda^*) = \sum_{x=2}^{\infty} \frac{x}{n} np$$

$$E(\lambda^*) = \sum_{x=2}^{\infty} xp$$

Since sample is taken from truncated poisson dist

$$P(x/x \geq 1) = \frac{\lambda^x e^{-\lambda}}{x! (1 - e^{-\lambda})}$$

Thus probability of success is

$$p = \frac{\lambda^x e^{-\lambda}}{x! (1 - e^{-\lambda})}$$

Therefore

$$\begin{aligned} E(\lambda^*) &= \sum_{x=2}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x! (1 - e^{-\lambda})} \\ &= \frac{e^{-\lambda}}{1 - e^{-\lambda}} \sum_{x=2}^{\infty} \frac{\lambda^x}{x!} \lambda \\ &= \frac{e^{-\lambda}}{1 - e^{-\lambda}} \left[\sum_{x=2}^{\infty} \frac{\lambda^x}{(x-1)!} + \lambda \right] - \lambda \\ &= \frac{e^{-\lambda}}{1 - e^{-\lambda}} \left[\sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} - \lambda \right] \\ &= \frac{e^{-\lambda}}{1 - e^{-\lambda}} \left[\lambda e^{-\lambda} - \lambda \right] \\ &= \frac{\lambda e^{-\lambda}}{1 - e^{-\lambda}} \left[e^{-\lambda} - 1 \right] \\ &= \frac{(1 - e^{-\lambda}) \lambda}{(1 - e^{-\lambda})} \end{aligned}$$

$$E(\lambda^*) = \lambda$$

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Exmpl; If the poisson dist is truncated at the left of $x=1$, then show that the unbiased estimator of $1-\bar{e}^\theta$ take values 0 when x is odd and 2 when x is even. Where θ is the parameter of Poisson dist. Sol;

The truncated Poisson dist is

$$P(x/x \geq 1) = \frac{\bar{e}^\theta \theta^x}{x!(1-\bar{e}^\theta)} \quad x=1, 2, \dots$$

let θ^* be the estimator of $(1-\bar{e}^\theta)$ and it take values zero when x is odd and 2 when x is even.

Now

$$\begin{aligned} E(\theta^*) &= \sum_{x=1}^{\infty} \theta^* \frac{\bar{e}^\theta \theta^x}{x!(1-\bar{e}^\theta)} \\ &= \sum_{y=0}^{\infty} 0 \frac{\bar{e}^\theta \theta^{2y+1}}{(2y+1)!(1-\bar{e}^\theta)} + \sum_{y=1}^{\infty} 2 \frac{\bar{e}^\theta \theta^{2y}}{(2y)!(1-\bar{e}^\theta)} \\ &= 0 + \frac{2\bar{e}^\theta}{1-\bar{e}^\theta} \sum_{y=1}^{\infty} \frac{\theta^{2y}}{(2y)!} \\ &= \frac{2\bar{e}^\theta}{1-\bar{e}^\theta} \left(\frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \frac{\theta^6}{6!} + \dots \right) \end{aligned}$$

$$e^\theta = 1 + \theta + \frac{\theta^2}{2!} + \frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \dots$$

$$\bar{e}^\theta = 1 - \theta + \frac{\theta^2}{2!} - \frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \dots$$

$$e^\theta + \bar{e}^\theta = 2 \left(1 + \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots \right)$$

Thus $\frac{e^\theta + \bar{e}^\theta}{2} = \left(1 + \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots \right)$

$$\frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots = \frac{e^\theta + \bar{e}^\theta}{2} - 1$$

$$\therefore E(\theta^*) = \frac{2\bar{e}^\theta}{1-\bar{e}^\theta} \left(\frac{e^\theta + \bar{e}^\theta}{2} - 1 \right)$$

$$E(\theta^*) = \frac{2\bar{e}^\theta}{1-\bar{e}^\theta} \left[\frac{e^\theta + \bar{e}^\theta - 2}{2} \right]$$

$$E(\theta^*) = \frac{1 + \bar{e}^{2\theta} - 2\bar{e}^{\theta}}{1 - \bar{e}^{\theta}}$$

$$= \frac{(1 - \bar{e}^{\theta})^2}{1 - \bar{e}^{\theta}}$$

$$E(\theta^*) = 1 - \bar{e}^{\theta}$$

Censored Data

Censoring is a property of the sample. By censoring we mean that the count of the sample is known but not their individual values in the truncated position. e.g: we know that 10 students have height more than six feet we don't know the actual height. Similarly 15 students have height less than 5 feet but we don't know the actual height. This is called double censoring. The samples can be singly or doubly censored. Singly censoring is a special case of double censoring.

Since in case of censored dist though we don't know the individual values of the count but we know the no of observations, hence we have some extra information in case of censored dist, then true called dist. This extra information adds efficiency in our estimation process.

Types of Censoring

Censoring is of two types. Gupta discriminated between two types of censored samples.

Type I Censoring, Type II Censoring

TYPE I Censoring:

In type I censoring we fix the time (suppose we say that we will stop the experiment after a fixed time) but the observations are r v's.

TYPE II Censoring:

In type II censoring we fix the ~~time~~ no of obs (e.g: we will stop the experiment after burning 20 bulbs) before the experiment starts.

The theory of type II censoring is simpler than type I censoring but as the sample size increases the two types of censoring are equivalent.

Maximum Likelihood Estimation

(Truncated and Censored data).

The p.d.f of a truncated distribution truncated at points 'a' and 'b' ($a < b$) is

$$p(x) = \frac{f(x)}{\int_a^b f(x) dx}$$

If n independent observations are taken from the doubly truncated dist, then the likelihood function is

$$L = \frac{\prod_{i=1}^n f(x_i)}{\int_a^b f(x) dx}$$

If the variable is doubly censored at the fixed point 'a' and 'b' with t_1 observations missing in the left tail and t_2 observations missing in the right tail, then for type I censoring likelihood function is

$$L \propto \left\{ \int_{-\infty}^b f(x) dx \right\}^{t_1} \frac{\prod_{i=1}^{n-t_1-t_2} f(x_i)}{\int_a^b f(x) dx} \left\{ \int_b^{\infty} f(x) dx \right\}^{t_2}$$

For type II censoring it is

$$L \propto \left\{ \int_{-\infty}^{x(t_1+1)} f(x) dx \right\}^{t_1} \frac{\prod_{i=t_1+1}^{n-t_2} f(x_i)}{\int_a^b f(x) dx} \left\{ \int_b^{\infty} f(x) dx \right\}^{t_2}$$

The MLE can be obtained by minimizing the LF or log LF.

Non Central Chi-Square Distribution

Let $x_1, x_2, \dots, x_n$ be independent normal variates such that

$$E(x_i) = \mu_i \quad \text{and} \quad \text{Var}(x_i) = 1$$

We consider the dist of the statistic

$$Z = \sum_{i=1}^n x_i^2$$

and we write $\lambda = \sum \mu_i^2 = \mu' \mu$

where λ is defined as the noncentrality parameter (if $\lambda = 0$, then it is central χ^2).

We write the joint dist of the x_i as

$$f(x_1) \dots f(x_n) = K e^{-\frac{1}{2}[(x-\mu)'(x-\mu)]}$$

Consider an orthogonal transformation on X as

$$Y = BX$$

where B is orthogonal matrix such that $B'B = I$

$$E(Y) = BE(X)$$

$$= B\mu$$

$$= 0 \quad (\text{say})$$

$$Q'Q = (B\mu)'(B\mu)$$

$$= \mu' B' B \mu$$

$$= \mu' \mu$$

$$= \lambda = \sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2$$

We now make first $(n-1)$ components of $Q = 0$, then

$$Q'Q = \lambda = \sigma_n^2$$

$$\text{Thus } Z = \sum x_i^2$$

$$= X'X$$

$$= (B'Y)'(B'Y)$$

$$= Y'(B')'B'Y = Y'(BB')'Y$$

$$= Y'Y$$

$$Z = \sum y_i^2$$

is the sum of square of n independent normal variates. The first $(n-1)$ of which are standardized and

the last of which has got mean

$$\sigma^2 = \sigma_n^2 = \lambda$$

$$\sigma_n = \lambda^{1/2} \text{ and Var } 1.$$

Therefore

$$U = \sum_{i=1}^{n-1} y_i^2 \sim \chi_{(n-1)}^2$$

and $V = y_n^2$ is normal with mean $\lambda^{1/2}$

So

$$dF(u) = K_2 e^{-\frac{1}{2}u} u^{\frac{n-3}{2}} du$$

where K_2 is a constant.

The dist of y_n is

$$dF(y_n) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(y_n - \lambda^{1/2})^2\right] dy_n$$

$$-\infty < y_n < \infty$$

$$\text{Since } V = y_n^2 \Rightarrow dV = 2y_n dy_n \quad 0 < V < \infty$$

$$dy_n = \frac{dV}{2\sqrt{V}}$$

The dist of V is

$$\begin{aligned} dF(V) &= \frac{1}{\sqrt{2\pi}} \left[\exp\left\{-\frac{1}{2}(V^{1/2} - \lambda^{1/2})^2\right\} + \exp\left\{-\frac{1}{2}(-V^{1/2} - \lambda^{1/2})^2\right\} \right] \frac{dV}{2\sqrt{V}} \\ &= K_3 V^{-1/2} e^{-\frac{1}{2}(V+\lambda)} \sum_{\xi=0}^{\infty} \frac{(V\lambda)^{\xi}}{(2\xi)!} dV \end{aligned}$$

The joint dist of U and V is then

$$\begin{aligned} dF(U, V) &= f_1(u) f_2(v) du dv \\ &= K_2 K_3 e^{-\frac{1}{2}u} u^{\frac{n-3}{2}} V^{-1/2} e^{-\frac{1}{2}(V+\lambda)} \sum_{\xi=0}^{\infty} \frac{(V\lambda)^{\xi}}{(2\xi)!} du dv \\ &= K_2 K_3 e^{-\frac{1}{2}(U+V+\lambda)} u^{\frac{n-3}{2}} V^{-1/2} \sum_{\xi=0}^{\infty} \frac{(V\lambda)^{\xi}}{(2\xi)!} du dv \end{aligned}$$

Make the following transformation

$$Z = U + V \Rightarrow U = Z - V \rightarrow V = Z - WZ$$

$$\text{and } W = \frac{U}{U+V} \Rightarrow U = WZ$$

Jacobian of transformation is

$$|J| = \begin{vmatrix} \frac{\partial u}{\partial z} & \frac{\partial u}{\partial w} \\ \frac{\partial v}{\partial z} & \frac{\partial v}{\partial w} \end{vmatrix} = \begin{vmatrix} w & z \\ 1-w & -z \end{vmatrix} = |-wz - z + zw|$$

$$J = |-z| = -z$$

Thus the joint dist of z and w is

$$\begin{aligned} dF(z, w) &= K_4 e^{-\frac{1}{z}(z+\lambda)} (zw)^{\frac{\eta-3}{2}} (z(1-w))^{-\frac{1}{2}} \\ &\quad \times \sum_{\epsilon=0}^{\infty} \frac{\lambda^{\epsilon}}{\epsilon!} \frac{(z(1-w))^{\epsilon}}{(2\epsilon)!} z dz dw \\ &= K_4 e^{-\frac{1}{z}(z+\lambda)} z^{\frac{\eta-2}{2}} w^{\frac{\eta-3}{2}} (1-w)^{-1/2} \sum_{\epsilon=0}^{\infty} \frac{\lambda^{\epsilon} z^{\epsilon} (1-w)^{\epsilon}}{(2\epsilon)!} dz dw \end{aligned}$$

The marginal dist of z is thus

$$\begin{aligned} h(z) &= K \int_0^1 e^{-\frac{1}{z}(z+\lambda)} z^{\frac{\eta-2}{2}} w^{\frac{\eta-3}{2}} (1-w)^{-1/2} \sum_{\epsilon=0}^{\infty} \frac{\lambda^{\epsilon} z^{\epsilon} (1-w)^{\epsilon}}{(2\epsilon)!} dw \\ &= K e^{-\frac{1}{z}(z+\lambda)} z^{\frac{\eta-2}{2}} \sum_{\epsilon=0}^{\infty} \frac{\lambda^{\epsilon} z^{\epsilon}}{(2\epsilon)!} \int_0^1 w^{\frac{\eta-3}{2}} (1-w)^{\epsilon-1/2} dw \end{aligned}$$

$$h(z) = K e^{-\frac{1}{z}(z+\lambda)} z^{\frac{\eta-2}{2}} \sum_{\epsilon=0}^{\infty} \frac{\lambda^{\epsilon} z^{\epsilon}}{(2\epsilon)!} \beta\left(\frac{\eta-1}{2}, \epsilon + \frac{1}{2}\right)$$

To find the value of K solve $\int_0^{\infty} h(z) dz = 1$

$$\frac{1}{K} = \int_0^{\infty} e^{-\frac{1}{z}z} e^{-\lambda/z} z^{\frac{\eta-2}{2}} \sum_{\epsilon=0}^{\infty} \frac{\lambda^{\epsilon} z^{\epsilon}}{(2\epsilon)!} \beta\left(\frac{\eta-1}{2}, \epsilon + \frac{1}{2}\right) dz$$

$$\begin{aligned}
 \frac{1}{K} &= e^{-\lambda/2} \int_0^{\infty} e^{-\frac{1}{2}z} z^{\frac{n-2}{2}} \left[\beta\left(\frac{n-1}{2}, \frac{1}{2}\right) + \frac{\lambda z}{2} \beta\left(\frac{n-1}{2}, \frac{3}{2}\right) \right. \\
 &\quad \left. + \frac{(\lambda z)^2}{4!} \beta\left(\frac{n-1}{2}, \frac{5}{2}\right) + \dots \right] dz \\
 &= e^{-\lambda/2} \left[\beta\left(\frac{n-1}{2}, \frac{1}{2}\right) \int_0^{\infty} e^{-\frac{1}{2}z} z^{\frac{n-2}{2}} dz + \frac{\lambda}{2} \beta\left(\frac{n-1}{2}, \frac{3}{2}\right) \int_0^{\infty} e^{-\frac{1}{2}z} z^{\frac{n-2}{2}} dz \right. \\
 &\quad \left. + \dots \right] \\
 &= e^{-\lambda/2} 2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right) \Gamma\left(\frac{1}{2}\right) \left[1 + \frac{\lambda}{2} + \frac{(\lambda^2)}{2!} \frac{1}{2!} + \frac{(\lambda^3)}{3!} \frac{1}{3!} + \dots \right] \\
 \frac{1}{K} &= e^{-\lambda/2} 2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right) \Gamma\left(\frac{1}{2}\right) e^{\lambda/2} \\
 \Rightarrow K &= \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right) \Gamma\left(\frac{1}{2}\right)}
 \end{aligned}$$

$$\text{Thus } dF(z) = \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right) \Gamma\left(\frac{1}{2}\right)} e^{-\frac{1}{2}(z+\lambda)} z^{\frac{n-2}{2}} \times \sum_{t=0}^{\infty} \frac{(\lambda z)^t}{(2t)!} \beta\left(\frac{n-1}{2}, t + \frac{1}{2}\right)$$

This distribution is called non-central chi-square with n.d.f and non centrality parameter λ .

The m.g.f of non-central chi-square is

$$M_0(t) = (1-2t)^{-\frac{n}{2}} \exp\left(\frac{\lambda t}{1-2t}\right)$$

$$\text{Mean} = n + \lambda$$

$$\text{Variance} = 2(n + 2\lambda)$$

$$\begin{aligned}
 M_{\chi_1^2}(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left[t x_1^2 - \frac{1}{2}(x_1 - \mu)^2\right] dx_1 \\
 &= (1-2t)^{-1/2} \exp\left(\frac{\mu^2 t}{1-2t}\right)
 \end{aligned}$$

$$\text{Since } \chi^2 = \sum x_i^2$$

$$\begin{aligned}
 M_{\chi^2}(t) &= \prod_{i=1}^n (1-2t)^{-1/2} \exp\left\{\frac{\mu_i^2 t}{1-2t}\right\} \\
 &= (1-2t)^{-n/2} \exp\left(\sum_{i=1}^n \frac{\mu_i^2 t}{1-2t}\right) \\
 &= (1-2t)^{-n/2} \exp\left(\frac{\lambda t}{1-2t}\right)
 \end{aligned}$$

Non Central F-Distribution

The ratio of two non-central χ^2 variates with n_1 and n_2 degrees of freedom is non-central F, with non-centrality parameters λ_1 and λ_2 .

If Z_1 is non-central χ^2 with n_1 df and λ_1 is non-centrality parameter and

Z_2 is another non-central χ^2 with n_2 df and λ_2 is non-centrality parameter, then make transformation

$$U = \frac{Z_1}{Z_2} \quad \text{and} \quad V = Z_2 \quad (\text{just for convenience})$$

$$\text{then } U = \frac{Z_1}{V}, \quad Z_1 = UV, \quad Z_2 = V$$

$$J = \begin{vmatrix} \frac{\partial Z_1}{\partial V} & \frac{\partial Z_1}{\partial U} \\ \frac{\partial Z_2}{\partial V} & \frac{\partial Z_2}{\partial U} \end{vmatrix} = \begin{vmatrix} U & V \\ 1 & 0 \end{vmatrix} = -V$$

$$dF(u, v) = f_1(uv) f_2(v) v du dv$$

$$dH(u) = \left[\int_{-\infty}^{\infty} f_1(uv) f_2(v) v dv \right] du$$

$$= \left[\int_0^{\infty} \frac{1}{2^{\frac{n_1}{2}} \sqrt{\frac{n_1-1}{2}} \sqrt{\frac{1}{2}}} e^{-\frac{1}{2}(uv+\lambda_1)} (uv)^{\frac{n_1-2}{2}} \sum_{s=0}^{\infty} \frac{(\lambda_1 uv)^s}{(2s)!} \right. \\ \left. \beta\left(\frac{n_1-1}{2}, s+\frac{1}{2}\right) \times \right. \\ \left. \frac{1}{2^{\frac{n_2}{2}} \sqrt{\frac{n_2-1}{2}} \sqrt{\frac{1}{2}}} e^{-\frac{1}{2}(v+\lambda_2)} v^{\frac{n_2-2}{2}} \sum_{s=0}^{\infty} \frac{(\lambda_2 v)^s}{(2s)!} \beta\left(\frac{n_2-1}{2}, s+\frac{1}{2}\right) \right. \\ \left. v dv \right] du$$

$$\text{let } n_1 + n_2 = n, \quad \lambda_1 + \lambda_2 = \lambda$$

So that

$$dH(u)du = \frac{e^{-1/2}}{2^{1/2}} \sum_{\epsilon=0}^{\infty} \sum_{\delta=0}^{\infty} \frac{\lambda_1^{\epsilon}}{(2\epsilon)!} \frac{\lambda_2^{\delta}}{(2\delta)!} \frac{|\epsilon+1/2| |\delta+1/2|}{\left(\frac{1}{2}\right)^{\frac{\eta_1}{2}+\epsilon} \left(\frac{1}{2}\right)^{\frac{\eta_2}{2}+\delta}} \\ \times \left[\int_0^{\infty} e^{-\frac{1}{2}v(1+u)} v^{\frac{\eta_1}{2}+\epsilon+\delta-1} dv \right] u^{\frac{\eta_1}{2}+\epsilon-1} du$$

let

$$I = \int_0^{\infty} e^{-\frac{1}{2}v(1+u)} v^{\frac{\eta_1}{2}+\epsilon+\delta-1} dv$$

putting $x = \frac{1}{2}v(1+u)$

$$dx = \frac{1}{2}(1+u) dv$$

$$dv = \frac{2dx}{1+u}$$

$$v = \frac{2x}{1+u}$$

$$\Rightarrow I = \int_0^{\infty} e^{-x} \left(\frac{2x}{1+u} \right)^{\frac{\eta_1}{2}+\epsilon+\delta-1} \frac{2dx}{(1+u)}$$

$$= \frac{2^{\frac{\eta_1}{2}+\epsilon+\delta-1+1}}{(1+u)^{\frac{\eta_1}{2}+\epsilon+\delta}} \int_0^{\infty} e^{-x} x^{\frac{\eta_1}{2}+\epsilon+\delta-1} dx$$

$$I = \frac{2^{\frac{\eta_1}{2}+\epsilon+\delta}}{(1+u)^{\frac{\eta_1}{2}+\epsilon+\delta}} \left| \frac{\eta_1}{2} + \delta + \epsilon \right|$$

Thus

$$dH(u)du = \frac{e^{-1/2}}{2^{1/2}} \sum_{\epsilon=0}^{\infty} \sum_{\delta=0}^{\infty} \frac{\lambda_1^{\epsilon}}{(2\epsilon)!} \frac{\lambda_2^{\delta}}{(2\delta)!} \frac{|\epsilon+1/2| |\delta+1/2|}{\left(\frac{1}{2}\right)^{\frac{\eta_1}{2}+\epsilon} \left(\frac{1}{2}\right)^{\frac{\eta_2}{2}+\delta}} \frac{\left|\frac{\eta_1}{2} + \delta + \epsilon\right|}{\left(\frac{1+u}{2}\right)^{\frac{\eta_1}{2}+\delta+\epsilon}} u^{\frac{\eta_1}{2}+\epsilon-1} du$$

Now using the formula

$$\sqrt{z}^p = \frac{z^{p-1}}{\Gamma(1/2)} \sqrt{p} \sqrt{p+1/2}$$

$$\sqrt{p+1/2} = \frac{\sqrt{z}^p \Gamma(1/2)}{z^{p-1} \sqrt{p}}$$

$$\begin{aligned}
 \text{Now } \frac{\sqrt{t+\frac{1}{2}} \sqrt{b+\frac{1}{2}} 2^{t+b}}{(2t)!(2b)!(\sqrt{1/2})^2} &= \left(\frac{2t}{2} \sqrt{1/2} \right) \frac{2b}{2} \sqrt{1/2} \left(\frac{2^{t+b}}{(2t)!(2b)!(\sqrt{1/2})^2} \right) \\
 &= \frac{(2t-1)(2b-1) 2^{t+b}}{2^{2(t+b)-2} (t-1)(b-1)(2t)(2b)(2b-1)} \\
 &= \frac{1}{2^{t+b} t b}
 \end{aligned}$$

So

$$dH(u) du = e^{-\frac{1}{2}\lambda} \sum_{t=0}^{\infty} \sum_{b=0}^{\infty} \frac{(\frac{1}{2}\lambda)^t}{t!} \frac{(\frac{1}{2}\lambda)^b}{b!} u^{\frac{n_1}{2}+t-1} \frac{1}{(1+u)^{\frac{n_2}{2}+t+b}}$$

$$\times \frac{du}{\beta\left(\frac{n_1}{2}+t, \frac{n_2}{2}+b\right)}$$

$$\text{Since } F = \frac{Z_1/n_1}{Z_2/n_2} = \frac{n_2}{n_1} u$$

if $\lambda_1 = \lambda$ and $\lambda_2 = 0$, we have

$$dH(F) dF = e^{-\frac{\lambda}{2}} \sum_{t=0}^{\infty} \sum_{b=0}^{\infty} \frac{(\frac{1}{2}\lambda)^t}{t!} \left(\frac{n_1}{n_2}\right)^{\frac{n_2}{2}+t-1} \times \frac{1}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_2}{2}+t+b}}$$

$$\times \frac{dF}{\beta\left(\frac{n_1}{2}+t, \frac{n_2}{2}+b\right)} \quad \circ \text{CF of } F$$

This is non central F-dist

with λ as non-centrality parameter.

When $n_1=1$, this dist changes to t^2 -dist (non-central) and hence non central t-dist can be derived from it.

Mean of non central F is

$$= \frac{(n_1 + \lambda) n_2}{(n_2 - 2) n_1} \quad n_2 > 2$$

$$\begin{aligned}
 \text{Variance} &= \frac{(n_1 + \lambda)^2 + 2(n_1 + \lambda) n_2}{(n_2 - 2)(n_2 - 4) n_1^2} - \frac{(n_1 + \lambda)^2 n_2}{(n_2 - 2)^2 n_1^2} \\
 &\quad n_2 > 4
 \end{aligned}$$

Ex: let n_1 and n_2 be the sizes of the samples from two normal populations with means μ_1 and μ_2 respectively and variances σ^2 . Test the equality of two normal populations with the same variance.

Sol: We wish to test the hypothesis

$$H_0: \mu_1 = \mu_2, \sigma^2 > 0 \quad H_1: \mu_1 \neq \mu_2, \sigma^2 > 0$$

$$\Omega = \{(\mu_1, \mu_2, \sigma^2); -\infty < \mu_1 < \infty, -\infty < \mu_2 < \infty, 0 < \sigma^2 < \infty\}$$

$$\text{and } \omega = \{(\mu_1, \mu_2, \sigma^2); -\infty < \mu_1 = \mu_2 < \infty, 0 < \sigma^2 < \infty\}$$

$$\lambda = \frac{\max_{\omega} L(\omega)}{\max_{\Omega} L(\omega)}$$

$$L(\omega) = \left(\frac{1}{(2\pi)\sigma^2} \right)^{\frac{n_1+n_2}{2}} \exp \left[-\frac{\sum_{i=1}^{n_1} (x_i - \mu_1)^2 + \sum_{j=1}^{n_2} (y_j - \mu_1)^2}{2\sigma^2} \right]$$

MLE of μ_1, σ^2 are solution of $\frac{\partial \ln L(\omega)}{\partial \mu_1} = 0$ and $\frac{\partial \ln L(\omega)}{\partial \sigma^2} = 0$, (under H_0)

$$\ln L(\omega) = \left(\frac{n_1+n_2}{2} \right) \ln \left[\frac{1}{(2\pi)\sigma^2} \right] - \frac{1}{2\sigma^2} \left[\sum_{i=1}^{n_1} (x_i - \mu_1)^2 + \sum_{j=1}^{n_2} (y_j - \mu_1)^2 \right]$$

$$\frac{\partial \ln L(\omega)}{\partial \mu_1} = -\frac{1}{2\sigma^2} \left[2 \sum_{i=1}^{n_1} (x_i - \mu_1)(-1) + 2 \sum_{j=1}^{n_2} (y_j - \mu_1)(-1) \right] = 0$$

$$\sum_{i=1}^{n_1} (x_i - \hat{\mu}_1) + \sum_{j=1}^{n_2} (y_j - \hat{\mu}_1) = 0$$

$$\Rightarrow \hat{\mu}_1 = \frac{\sum x + \sum y}{n_1 + n_2} = \hat{\mu}(\text{say})$$

$$\frac{\partial \ln L(\omega)}{\partial \sigma^2} = -\frac{(n_1+n_2)}{2} \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} \left[\sum_{i=1}^{n_1} (x_i - \hat{\mu}_1)^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu}_1)^2 \right]$$

$$\Rightarrow -(n_1+n_2) + \frac{1}{\sigma^4} \left[\sum_{i=1}^{n_1} (x_i - \hat{\mu})^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu})^2 \right] = 0$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n_1} (x_i - \hat{\mu})^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu})^2}{n_1 + n_2} = W$$

Now

$$\max_{\omega} L(\omega) = \left(\frac{1}{2\pi W} \right)^{\frac{n_1+n_2}{2}} \exp \left[- \frac{\sum_{i=1}^{n_1} (x_i - \hat{\mu})^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu})^2}{2 \frac{\sum_{i=1}^{n_1} (x_i - \hat{\mu})^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu})^2}{n_1 + n_2}} \right]$$

$$\boxed{\max_{\omega} L(\omega) = \left(\frac{e^{-1}}{2\pi W} \right)^{\frac{n_1+n_2}{2}}} \quad \text{--- (1)}$$

Now

$$L(\omega) = \left(\frac{1}{2\pi \sigma^2} \right)^{\frac{n_1+n_2}{2}} \exp \left[- \left(\frac{\sum_{i=1}^{n_1} (x_i - \mu_1)^2 + \sum_{j=1}^{n_2} (y_j - \mu_2)^2}{2\sigma^2} \right) \right]$$

MLE of μ_1, μ_2, σ^2 over ω are

$$\hat{\mu}_1 = \frac{\sum_{i=1}^{n_1} x_i}{n_1}, \quad \hat{\mu}_2 = \frac{\sum_{j=1}^{n_2} y_j}{n_2}$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n_1} (x_i - \hat{\mu}_1)^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu}_2)^2}{n_1 + n_2} = W'$$

And

$$\max_{\omega} L(\omega) = \left(\frac{e^{-1}}{2\pi W'} \right)^{\frac{n_1+n_2}{2}} \quad \text{--- (2)}$$

Now

$$\lambda = \left(\frac{W'}{W} \right)^{\frac{n_1+n_2}{2}}$$

$$\frac{2}{\lambda^{n_1+n_2}}$$

$$= \frac{W'}{W}$$

$$= \frac{\sum_{i=1}^{n_1} (x_i - \hat{\mu}_1)^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu}_2)^2}{\sum_{i=1}^{n_1} (x_i - \hat{\mu})^2 + \sum_{j=1}^{n_2} (y_j - \hat{\mu})^2}$$

Putting the values of $\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}$

$$\frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{\sum_{i=1}^{n_1} \left[x_i - \left\{ \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right\} \right]^2 + \sum_{j=1}^{n_2} \left[y_j - \left\{ \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right\} \right]^2}$$

Now

$$\sum_{i=1}^{n_1} \left[x_i - \left\{ \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right\} \right]^2 = \sum_{i=1}^{n_1} \left[(x_i - \bar{x}) + \left(\bar{x} - \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right) \right]^2$$

$$= \sum_{i=1}^{n_1} (x_i - \bar{x})^2 + n_1 \left(\bar{x} - \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right)^2$$

Similarly

$$\sum_{j=1}^{n_2} \left[y_j - \left\{ \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right\} \right]^2 = \sum_{j=1}^{n_2} (y_j - \bar{y})^2 + n_2 \left(\bar{y} - \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right)^2$$

But

$$n_1 \left(\bar{x} - \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right)^2 = \frac{n_1 n_2}{(n_1 + n_2)^2} (\bar{x} - \bar{y})^2$$

$$n_2 \left(\bar{y} - \frac{n_1 \bar{x} + n_2 \bar{y}}{n_1 + n_2} \right)^2 = \frac{n_1 n_2}{(n_1 + n_2)^2} (\bar{x} - \bar{y})^2$$

Hence

$$\frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2 + \frac{(\bar{x} - \bar{y})^2}{\left(\frac{n_1 + n_2}{n_1 n_2} \right)}} = \frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2 + \frac{n_1 n_2}{(n_1 + n_2)} (\bar{x} - \bar{y})^2}$$

$$\frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}{1 + \frac{\frac{n_1 n_2}{n_1 + n_2} (\bar{x} - \bar{y})^2}{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{j=1}^{n_2} (y_j - \bar{y})^2}}$$

The CR of the LR test is

Reject H_0 if $\lambda \leq K$

where $0 < K < 1$

$$R = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2}$$

$$ic \frac{1}{1 + \frac{(n_1 n_2 / (n_1 + n_2)) (\bar{x} - \bar{y})^2}{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}} \leq K^{\frac{2}{n_1 + n_2}}$$

$$oc \frac{n_1 n_2 (\bar{x} - \bar{y})^2}{(n_1 + n_2) [\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2]} \geq K^{\frac{-2}{n_1 + n_2} - 1}$$

$$\sqrt{\frac{n_1 n_2}{n_1 + n_2}} (\bar{x} - \bar{y}) \geq \sqrt{\frac{2}{n_1 + n_2} (K^{\frac{-2}{n_1 + n_2} - 1}) (n_1 + n_2 - 2)} = C$$

Under $H_0: \mu_1 = \mu_2$

$$T = \frac{\sqrt{\frac{n_1 n_2}{n_1 + n_2}} (\bar{x} - \bar{y})}{\sqrt{\frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2}}} \sim T_{(n_1 + n_2 - 2)}$$

Rejection region is

$$|T| \geq C$$

$$\alpha = P(|T| \geq C; H_0)$$

BEHRENS - FISHER TEST

Students t-test cannot be used to compare the means of two normal populations with unequal standard deviations, as in this case $n_1 S_1^2 + n_2 S_2^2$ is not considered to be distributed as $\sigma^2 \chi^2$ variate.

In such cases, Behrens and Fisher suggested an alternative method. Let $\bar{x}_1, \bar{x}_2$ be the sample means and S_1^2, S_2^2 be the sample variances based on n_1 and n_2 observations from the two normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively with $\sigma_1^2 \neq \sigma_2^2$ (unknown).

Consider the inequality

$$K_1 < C_1 \frac{(\bar{x}_1 - \mu_1)}{S_1} + C_2 \frac{(\bar{x}_2 - \mu_2)}{S_2} < K_2 \quad \text{--- (1)}$$

where C_1, C_2, K_1 and K_2 are constants

$$\text{let } t_1 = \frac{(\bar{x}_1 - \mu_1) \sqrt{n_1 - 1}}{S_1} \quad \text{and} \quad t_2 = \frac{(\bar{x}_2 - \mu_2) \sqrt{n_2 - 1}}{S_2}$$

are independently dist^d as Student's t-dist with $n_1 - 1, n_2 - 1$ d.f respectively.

$$K_1 < C_1 \frac{t_1}{\sqrt{n_1 - 1}} + C_2 \frac{t_2}{\sqrt{n_2 - 1}} < K_2 \quad \text{--- (2)}$$

$$\text{Now } P \left[K_1 < \frac{C_1 t_1}{\sqrt{n_1 - 1}} + \frac{C_2 t_2}{\sqrt{n_2 - 1}} < K_2 \right] \text{ can be obtained through}$$

$$\int \int f_{n_1 - 1}(t_1) f_{n_2 - 1}(t_2) dt_1 dt_2 \quad \text{--- (3)}$$

let $C_1 = S_1, C_2 = S_2$ So that equation (2) can be written as

$$K_1 < \frac{S_1 t_1}{\sqrt{n_1 - 1}} + \frac{S_2 t_2}{\sqrt{n_2 - 1}} < K_2$$

$$\text{put } \frac{S_1}{\sqrt{n_1 - 1}} = r \sin \theta \quad \text{and} \quad \frac{S_2}{\sqrt{n_2 - 1}} = r \cos \theta$$

So that

$$K_1 < t_1 \epsilon \sin \theta + t_2 \epsilon \cos \theta < K_2$$

Since

$$\begin{aligned} \epsilon^2 \sin^2 \theta + \epsilon^2 \cos^2 \theta &= \epsilon^2 \\ &= \frac{S_1^2}{n_1-1} + \frac{S_2^2}{n_2-1} \end{aligned}$$

$$\Rightarrow \epsilon = \sqrt{\frac{S_1^2}{n_1-1} + \frac{S_2^2}{n_2-1}}$$

$$\text{Also } \frac{\epsilon \sin \theta}{\epsilon \cos \theta} = \tan \theta = \frac{S_1}{S_2} \sqrt{\frac{n_2-1}{n_1-1}}$$

$$\text{Let } P[(t_1 \sin \theta - t_2 \cos \theta) > d] = \alpha$$

$$P\left[\left(t_1 \frac{S_1}{\epsilon \sqrt{n_1-1}} - t_2 \frac{S_2}{\epsilon \sqrt{n_2-1}}\right) > d\right] = \alpha$$

$$P\left[\left(\frac{\bar{x}_1 - \mu_1}{\epsilon} - \frac{\bar{x}_2 - \mu_2}{\epsilon}\right) > d\right] = \alpha$$

$$P[(\bar{x}_1 - \mu_1) - (\bar{x}_2 - \mu_2) > d\epsilon] = \alpha$$

$$P[(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2) > d\epsilon] = \alpha$$

$$P[(\mu_1 - \mu_2) < (\bar{x}_1 - \bar{x}_2) - d\epsilon] = \alpha$$

So that if $(\bar{x}_1 - \bar{x}_2) > d\epsilon$, the region $\mu_1 \leq \mu_2$ will have a confidence coefficient less than α , and if $(\bar{x}_1 - \bar{x}_2) < d\epsilon$ the region $\mu_1 > \mu_2$ will have a confidence coefficient less than α .

Thus if a α is for sufficient means μ_1 and μ_2 are sufficiently different

$$|\bar{x}_1 - \bar{x}_2| > d\epsilon$$

Reject $H_0: \mu_1 = \mu_2$ if $|\bar{x}_1 - \bar{x}_2| >$

The values of ϵ and d are given in the tables.

The chief objection raised to Fisher-Behrens test is that we assume $\frac{S_1^2}{S_2^2}$ as a constant. Therefore various other approximate tests are proposed by different statisticians, one of them is due to Cochran and Cox.

Cochran Cox Test

The Cochran-Cox test can be described as follows,

The unbiased estimate of the variance of $\bar{X}_1 - \bar{X}_2$ is

$$\frac{S_1^2}{n_1-1} + \frac{S_2^2}{n_2-1}$$

Compute

$$t_c = \frac{\bar{X}_1 - \bar{X}_2}{\left(\frac{S_1^2}{n_1-1} + \frac{S_2^2}{n_2-1} \right)^{1/2}}$$

define

$$t' = \frac{t_{1\alpha} w_1 + t_{2\alpha} w_2}{w_1 + w_2}$$

$$\text{where } w_1 = \frac{S_1^2}{n_1-1}, w_2 = \frac{S_2^2}{n_2-1}$$

$$t_{1\alpha} = t_{\frac{\alpha}{2}, n_1-1}, t_{2\alpha} = t_{\frac{\alpha}{2}, n_2-1}$$

The decision rule is

accept $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 \neq \mu_2$

if $t_c < t'$

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